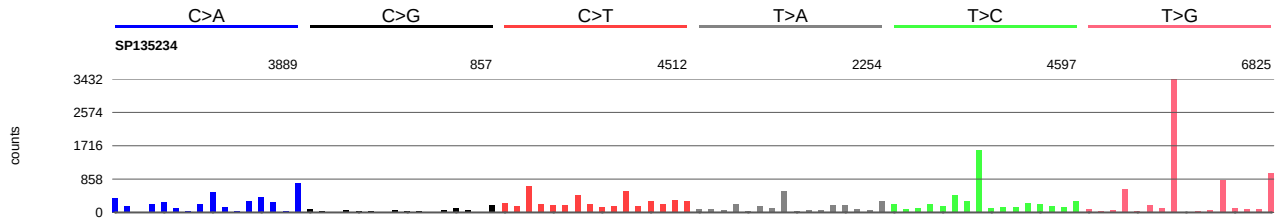
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SP135234
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857
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6825
counts
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858
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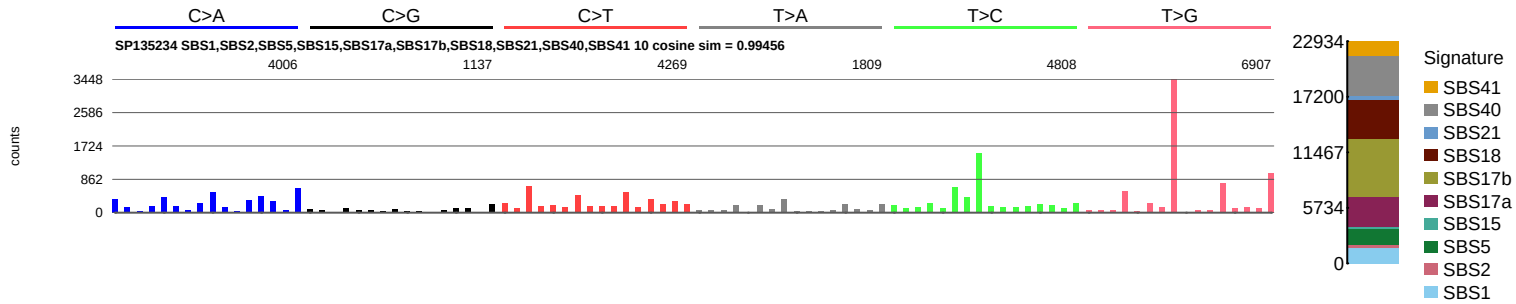
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SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99456
4006
1137
4269
1809
4808
6907
counts
3448
2586
1724
862
0
22934
17200
11467
5734
0
Signature
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SBS18
SBS17b
SBS17a
SBS15
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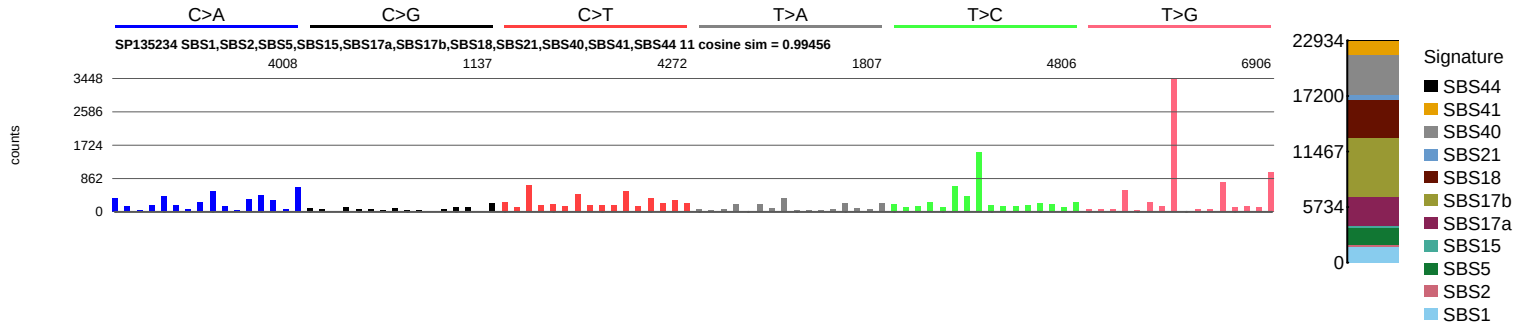
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SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99456
4008
1137
4272
1807
4806
6906
counts
3448
2586
1724
862
0
22934
17200
11467
5734
0
Signature
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SBS5
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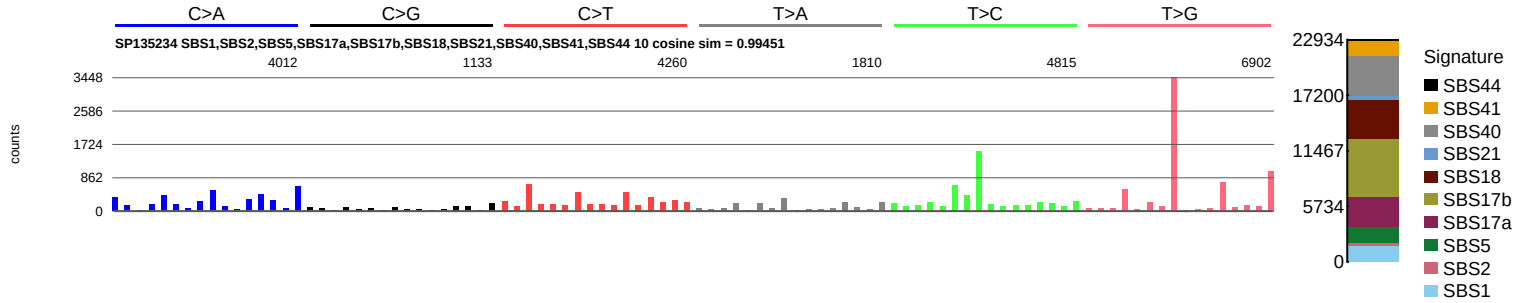
SBS1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4012
1133
4260
1810
4815
6902
counts
3448
2586
1724
862
0
22934
17200
11467
5734
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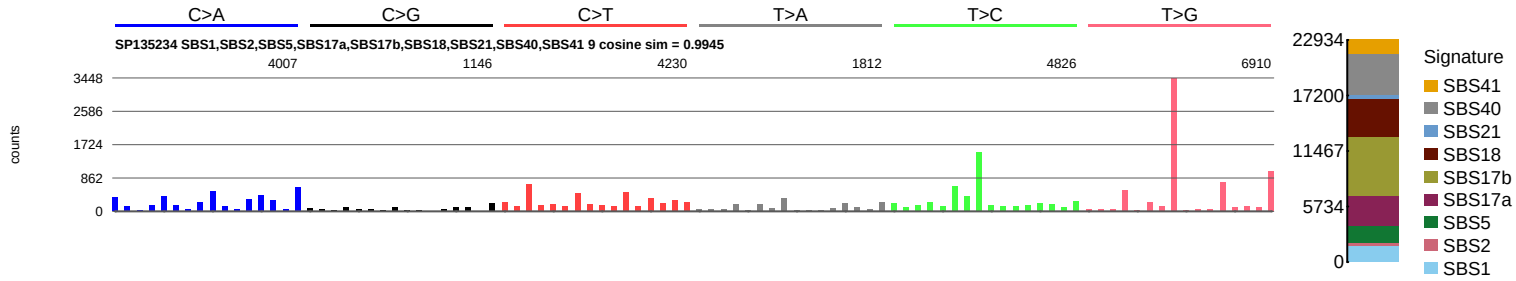
SBS2
SBS1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SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.9945
4007
1146
4230
1812
4826
6910
counts
3448
2586
1724
862
0
22934
17200
11467
5734
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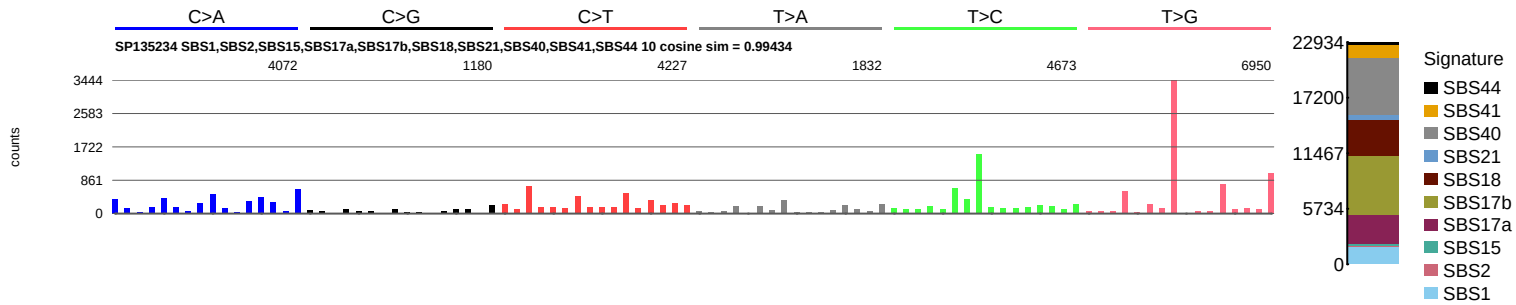
SBS2
SBS1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SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99434
4072
1180
4227
1832
4673
6950
counts
3444
2583
1722
861
0
22934
17200
11467
5734
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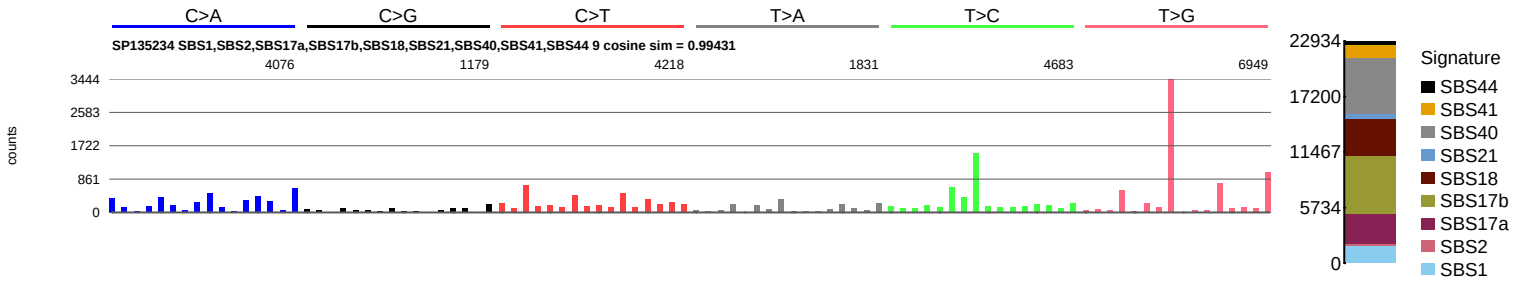
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
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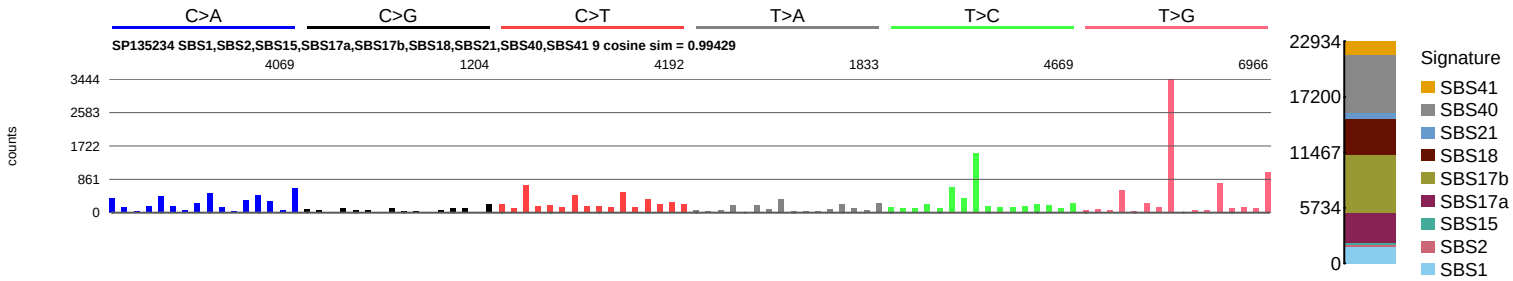
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99431
4076
1179
4218
1831
4683
6949
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99429
4069
1204
4192
1833
4669
6966
counts
3444
2583
1722
861
0
22934
17200
11467
5734
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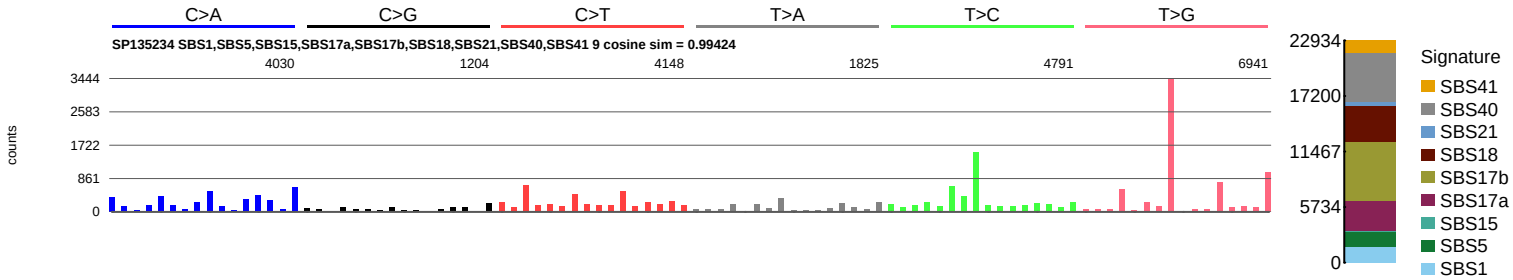
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
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T>C
T>G
SP135234 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99424
4030
1204
4148
1825
4791
6941
counts
3444
2583
1722
861
0
22934
17200
11467
5734
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Signature
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SBS18
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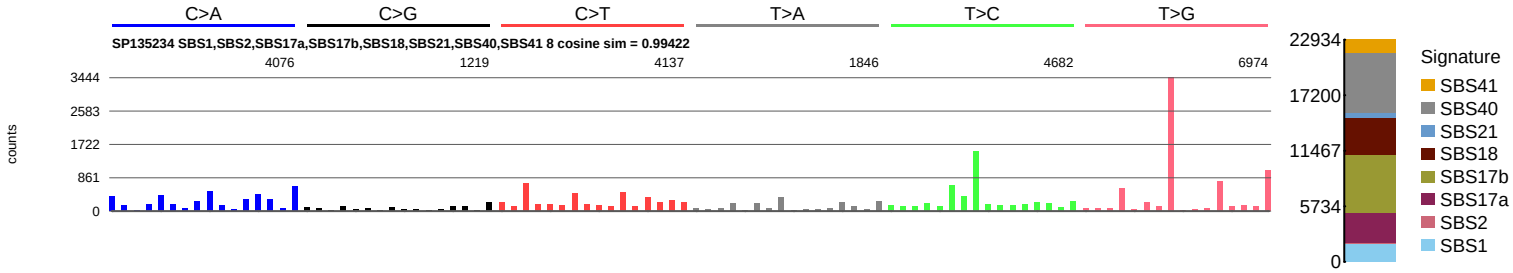
SBS15
SBS5
SBS1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C>G
C>T
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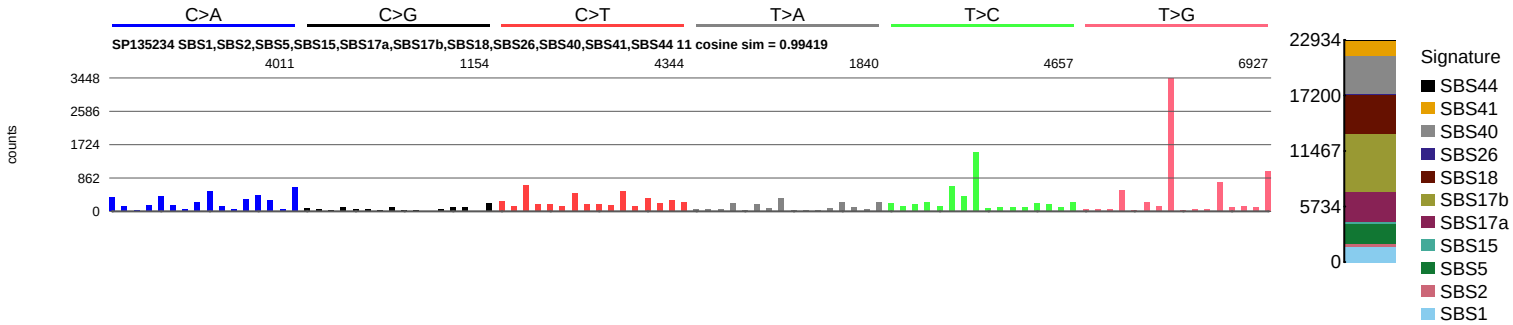
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99422
4076
1219
4137
1846
4682
6974
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99419
4011
1154
4344
1840
4657
6927
counts
3448
2586
1724
862
0
22934
17200
11467
5734
0
Signature
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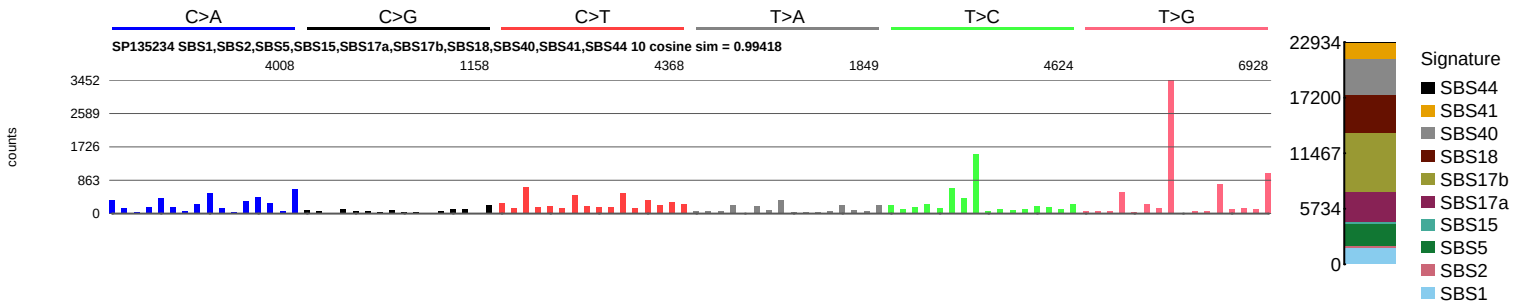
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99418
4008
1158
4368
1849
4624
6928
counts
3452
2589
1726
863
0
22934
17200
11467
5734
0
Signature
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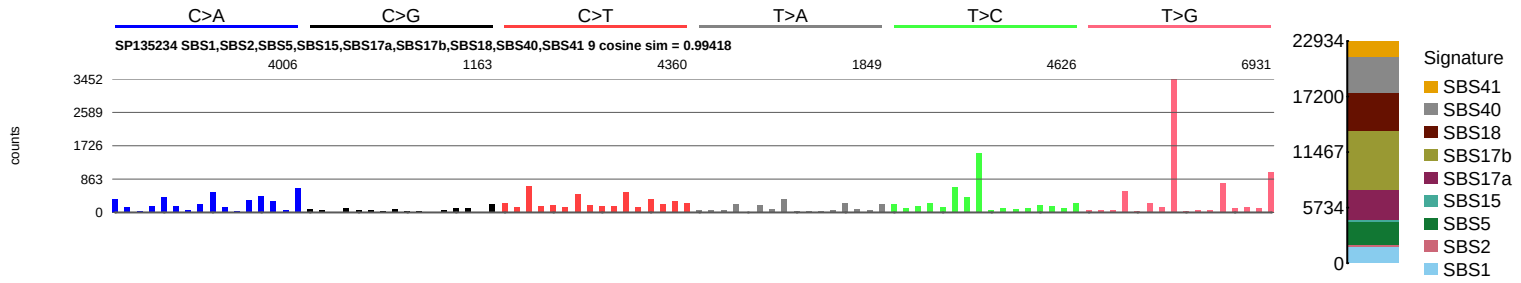
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99418
4006
1163
4360
1849
4626
6931
counts
3452
2589
1726
863
0
22934
17200
11467
5734
0
Signature
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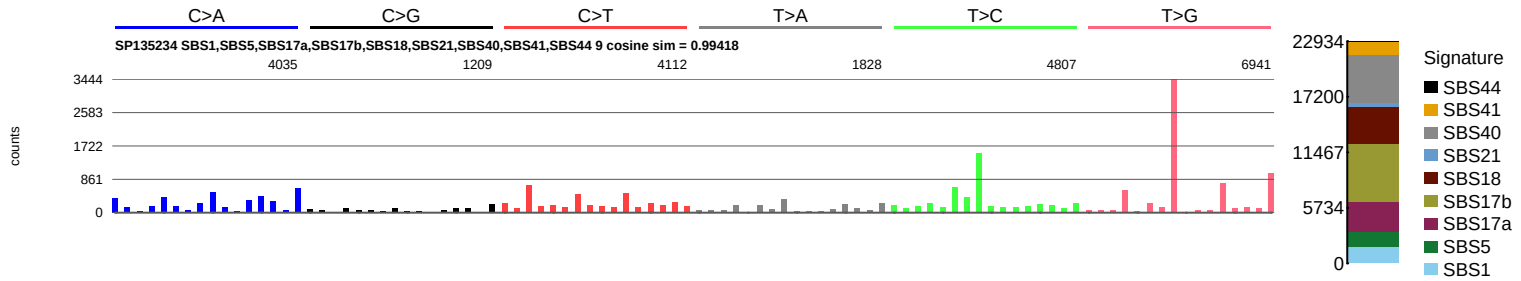
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
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T>C
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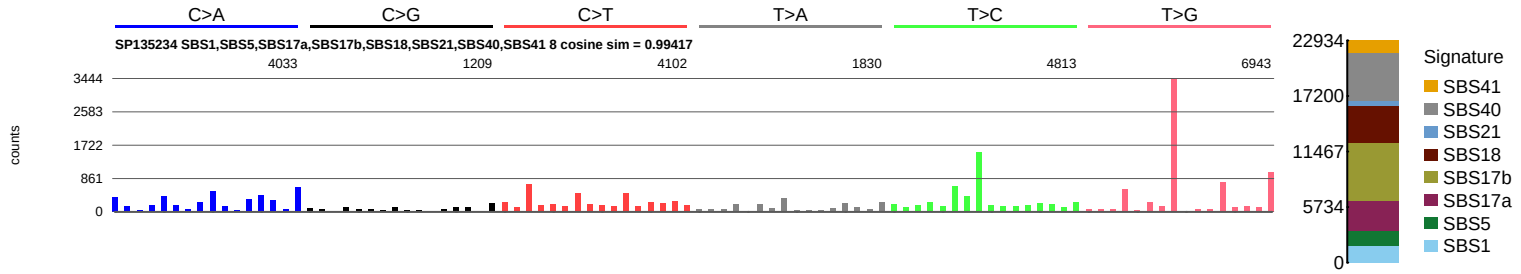
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99418
4035
1209
4112
1828
4807
6941
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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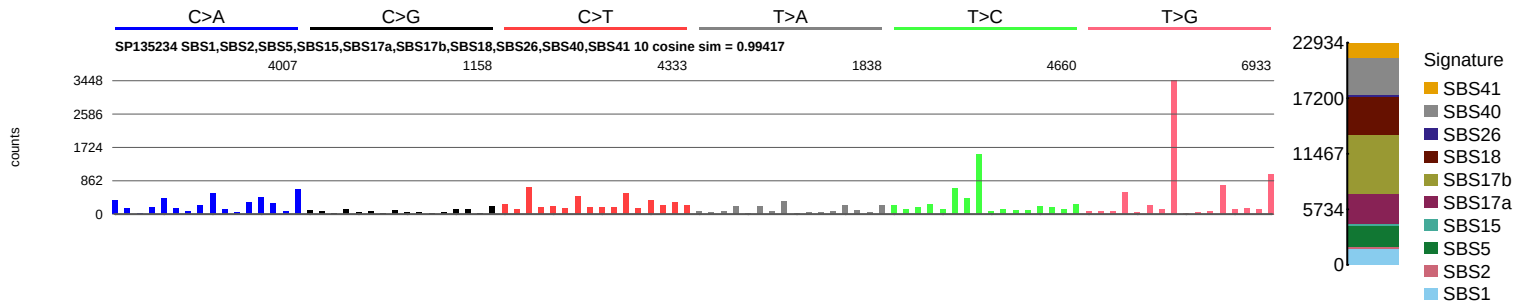
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99417
4033
1209
4102
1830
4813
6943
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99417
4007
1158
4333
1838
4660
6933
counts
3448
2586
1724
862
0
22934
17200
11467
5734
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Signature
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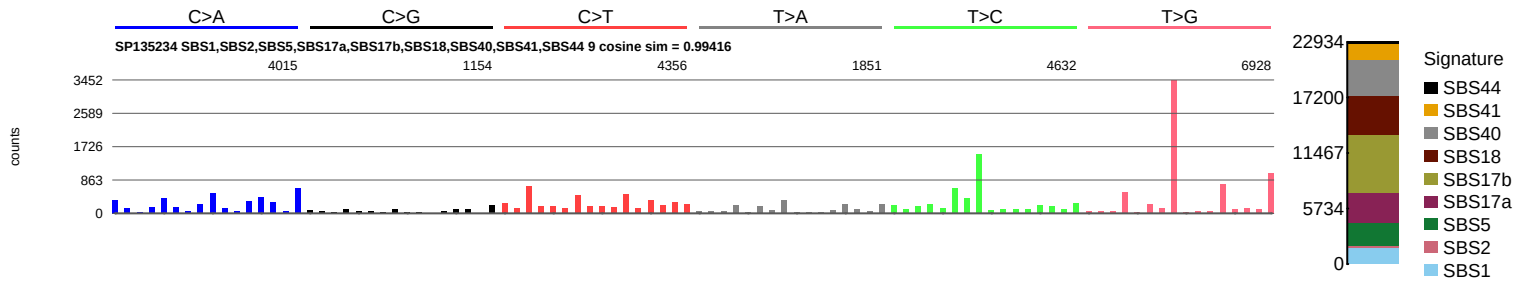
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
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T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99416
4015
1154
4356
1851
4632
6928
counts
3452
2589
1726
863
0
22934
17200
11467
5734
0
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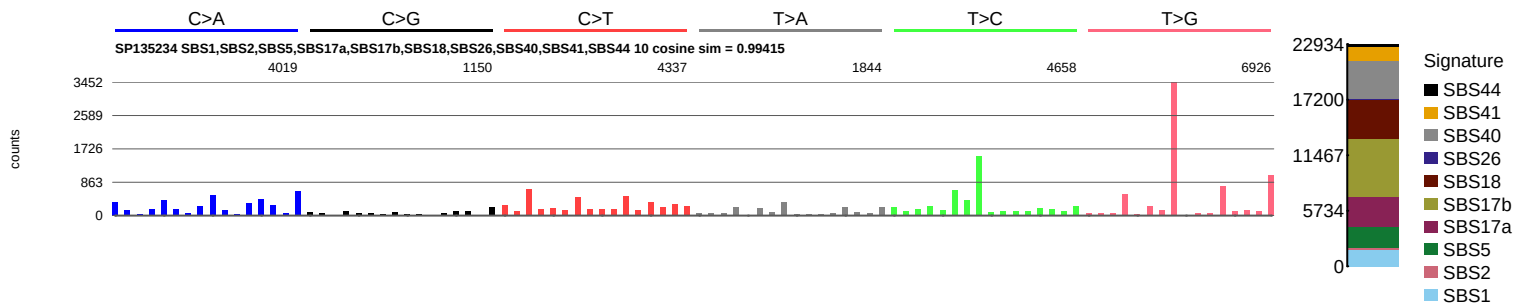
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99415
4019
1150
4337
1844
4658
6926
counts
3452
2589
1726
863
0
22934
17200
11467
5734
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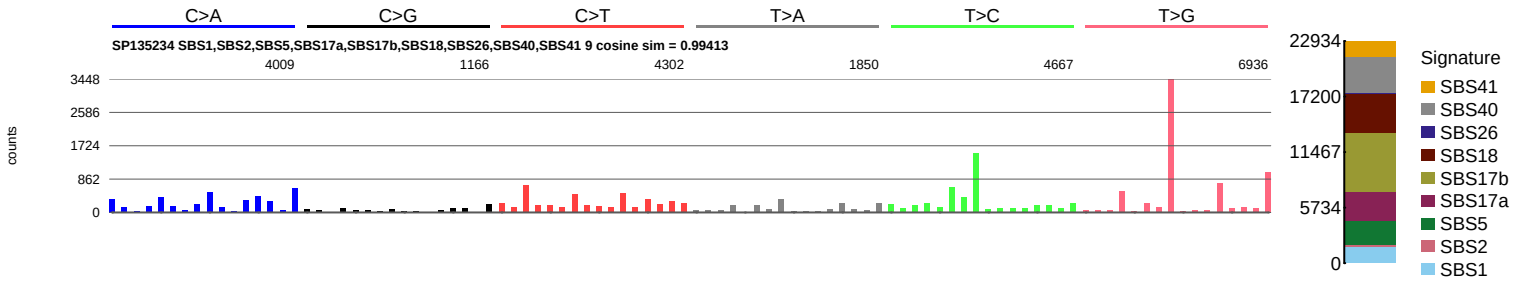
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99413
4009
1166
4302
1850
4667
6936
counts
3448
2586
1724
862
0
22934
17200
11467
5734
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SBS40
SBS26
SBS18
SBS17b
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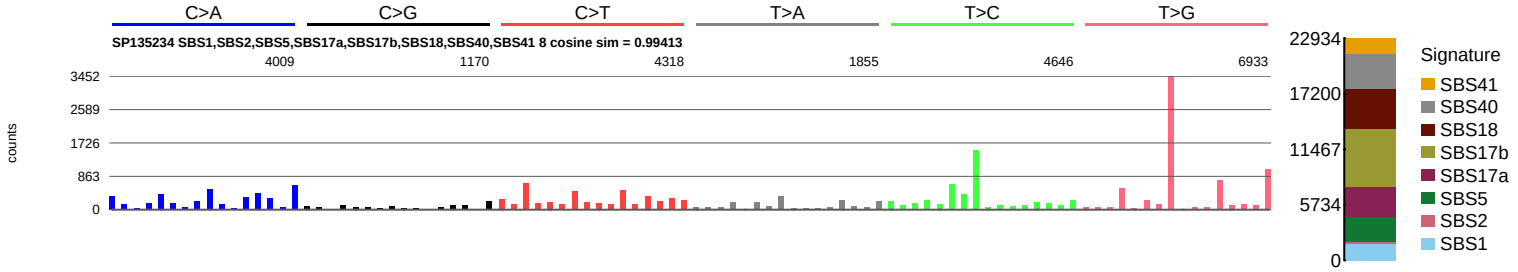
SBS5
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99413
4009
1170
4318
1855
4646
6933
counts
3452
2589
1726
863
0
22934
17200
11467
5734
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SBS40
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SBS17b
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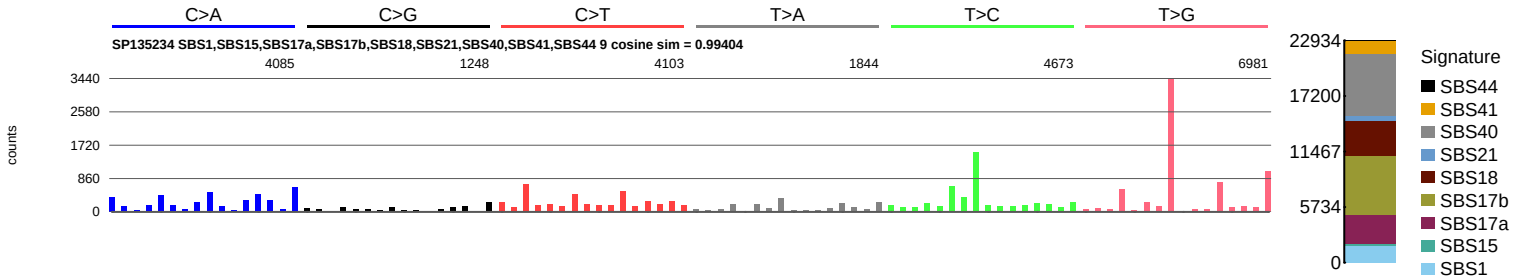
SBS5
SBS2
SBS1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C>G
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T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99404
4085
1248
4103
1844
4673
6981
counts
3440
2580
1720
860
0
22934
17200
11467
5734
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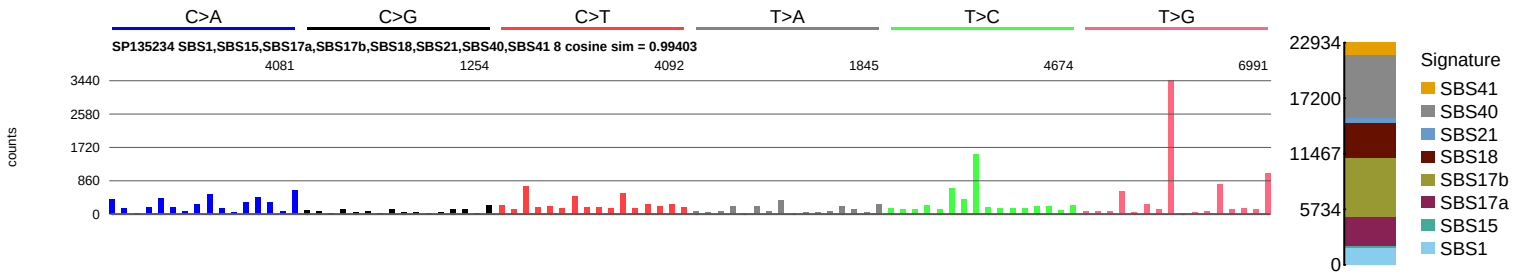
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99403
4081
1254
4092
1845
4674
6991
counts
3440
2580
1720
860
0
22934
17200
11467
5734
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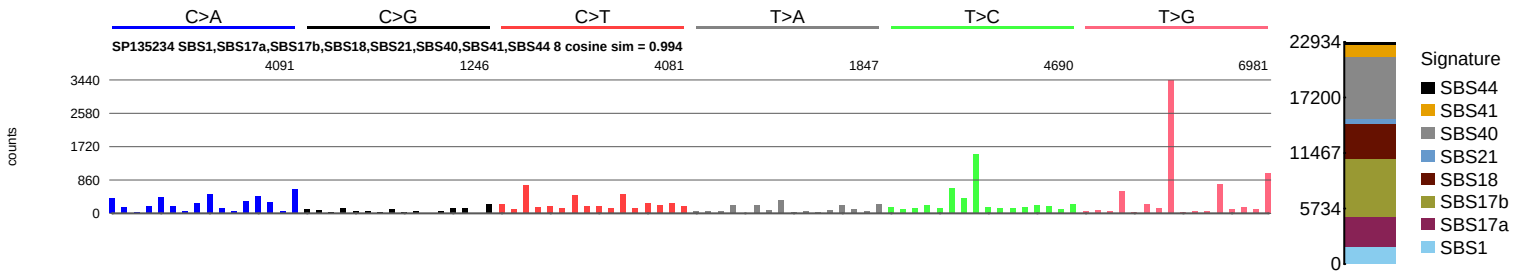
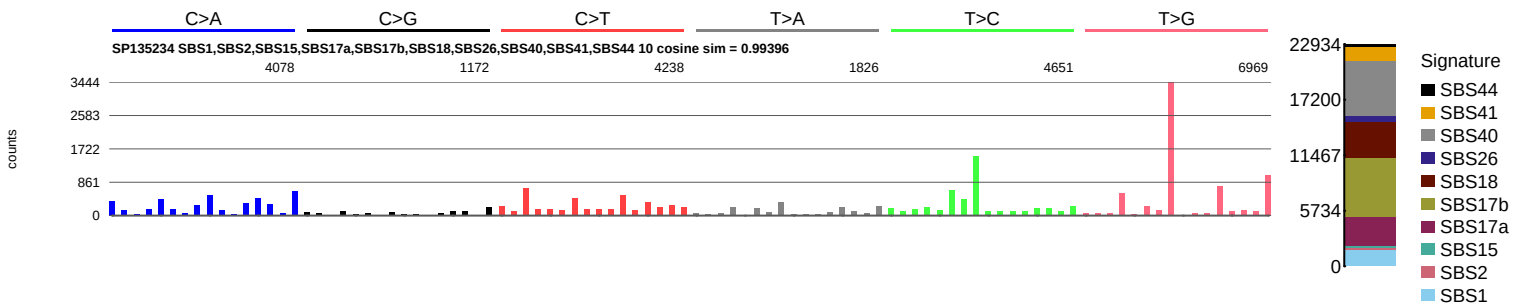
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.994
4091
1246
4081
1847
4690
6981
counts
3440
2580
1720
860
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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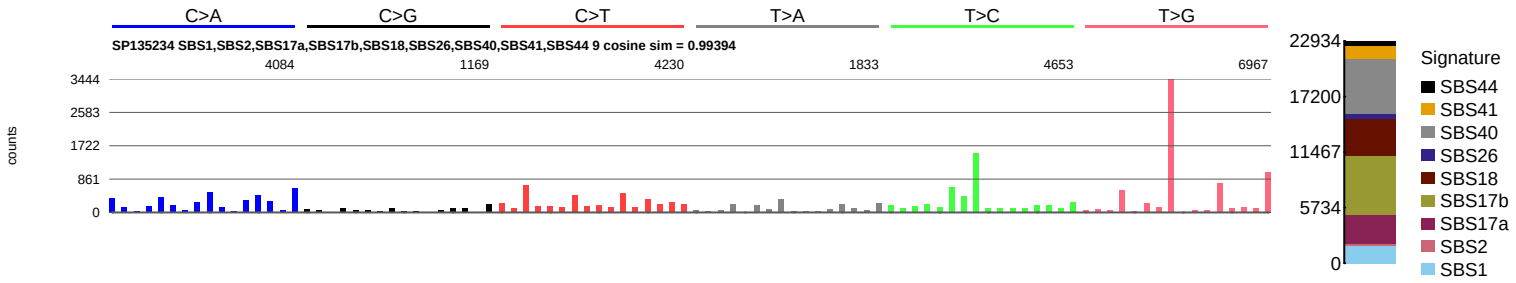
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99396
4078
1172
4238
1826
4651
6969
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>T
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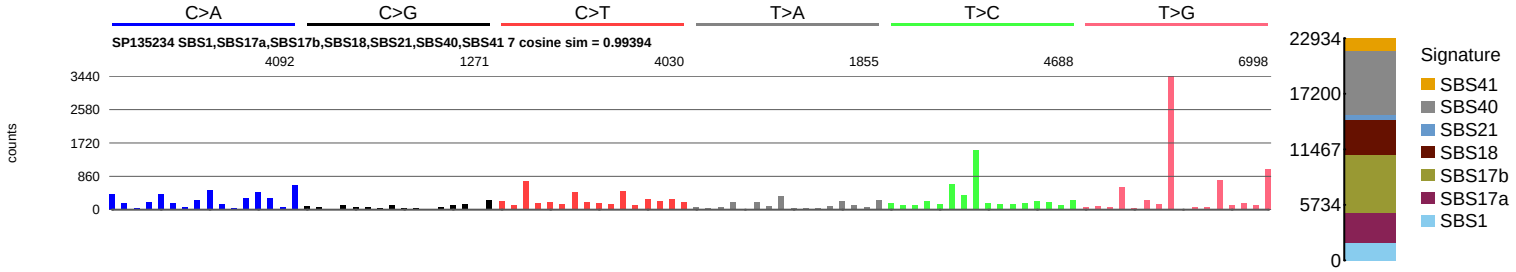
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99394
4084
1169
4230
1833
4653
6967
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
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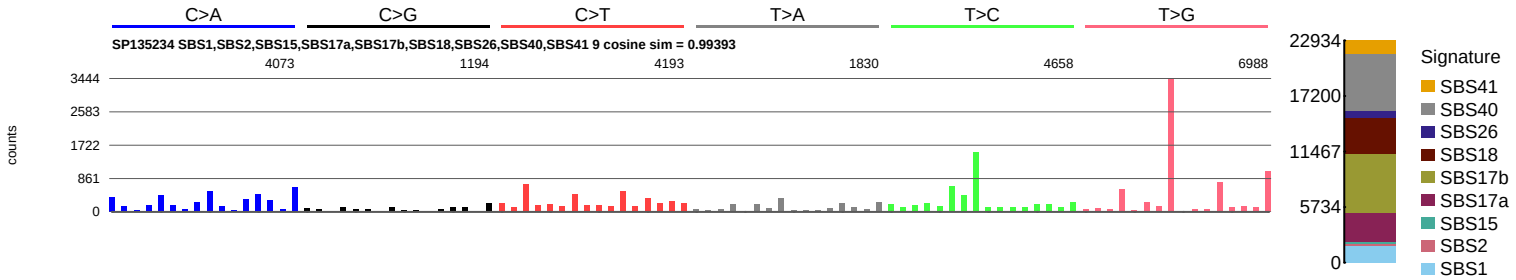
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.99394
4092
1271
4030
1855
4688
6998
counts
3440
2580
1720
860
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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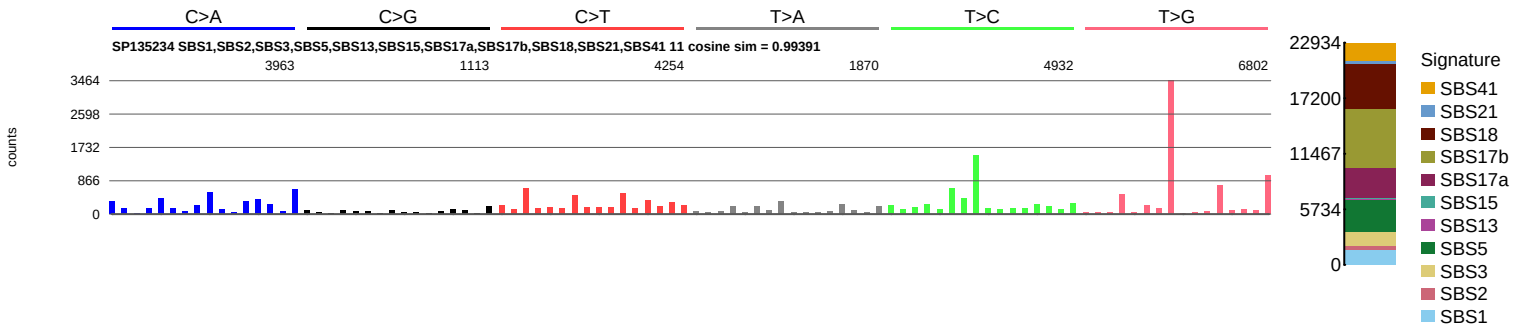
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99393
4073
1194
4193
1830
4658
6988
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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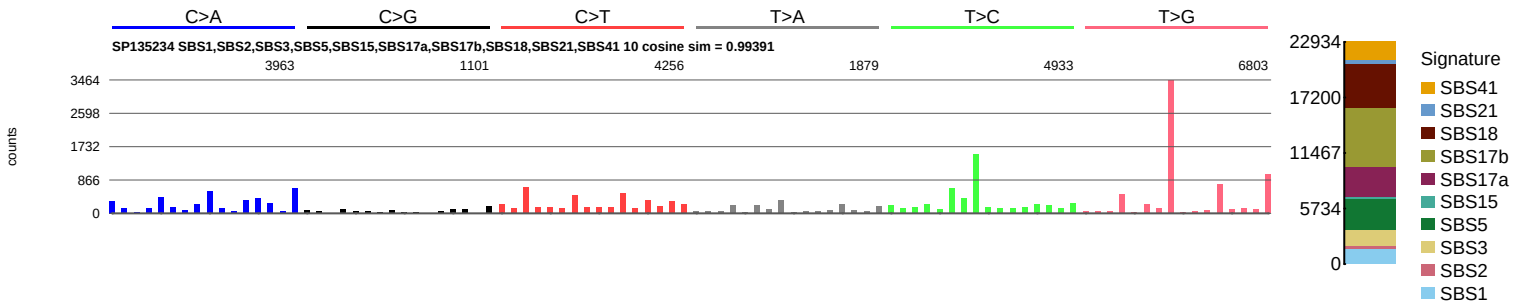
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.99391
3963
1113
4254
1870
4932
6802
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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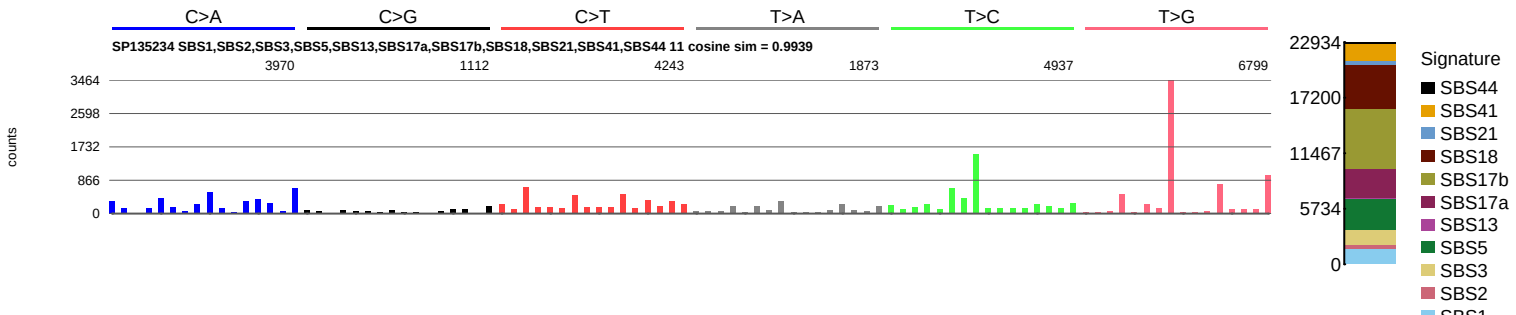
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99391
3963
1101
4256
1879
4933
6803
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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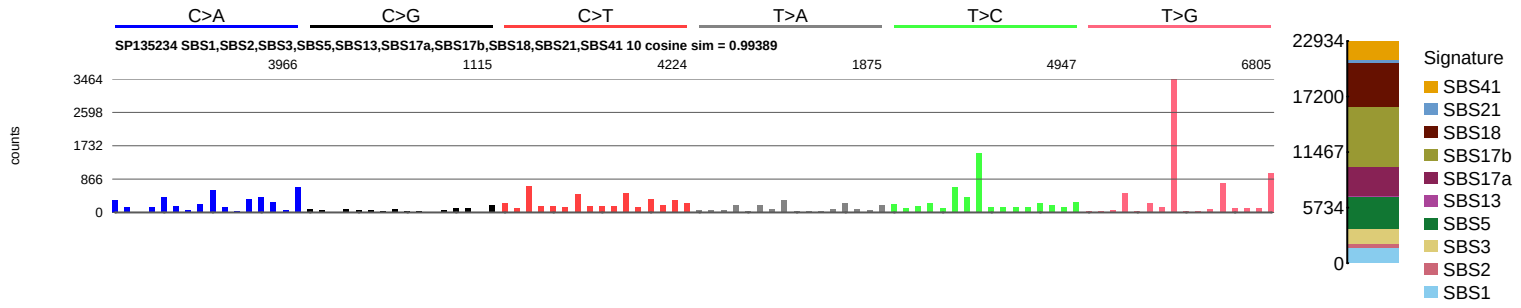
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.9939
3970
1112
4243
1873
4937
6799
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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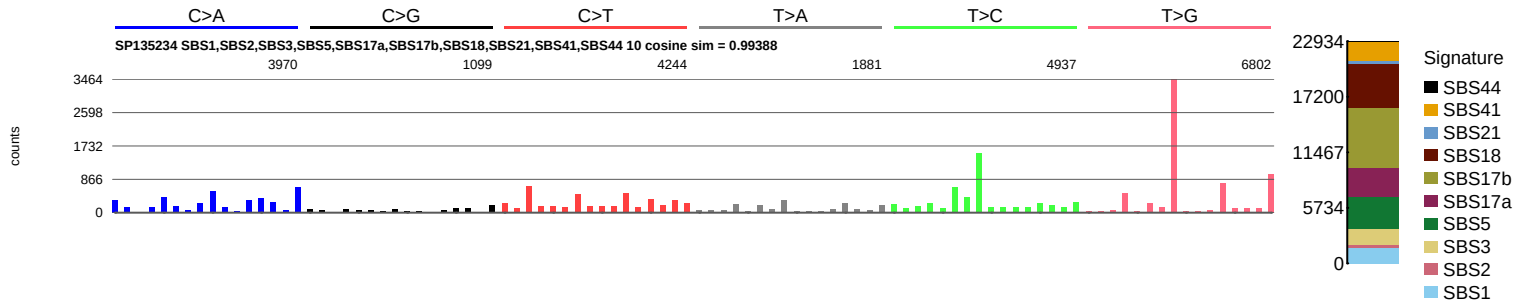
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99389
3966
1115
4224
1875
4947
6805
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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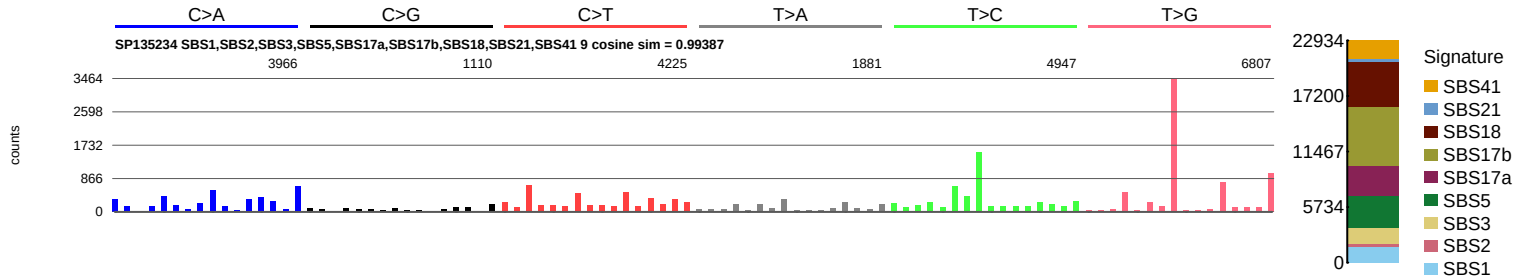
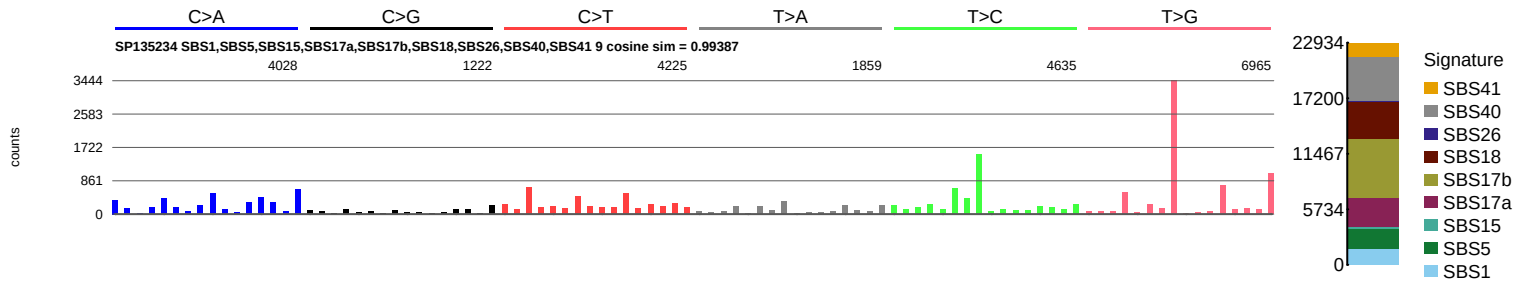
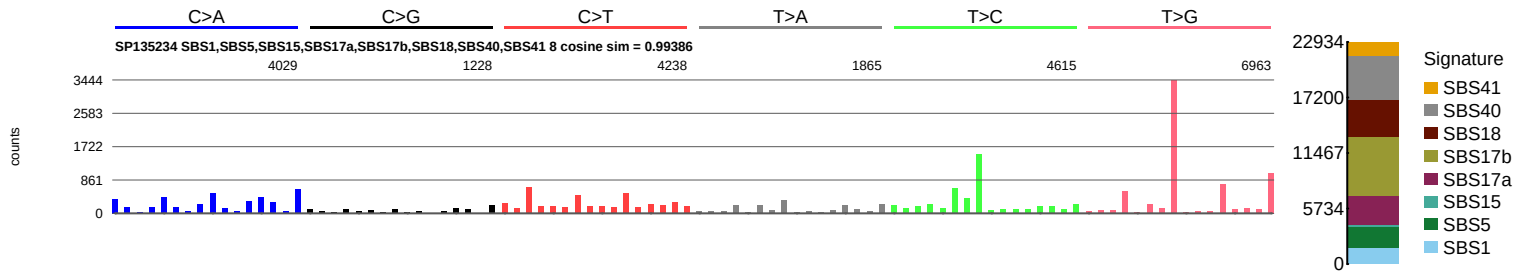
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.99388
3970
1099
4244
1881
4937
6802
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99387
3966
1110
4225
1881
4947
6807
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99387
4028
1222
4225
1859
4635
6965
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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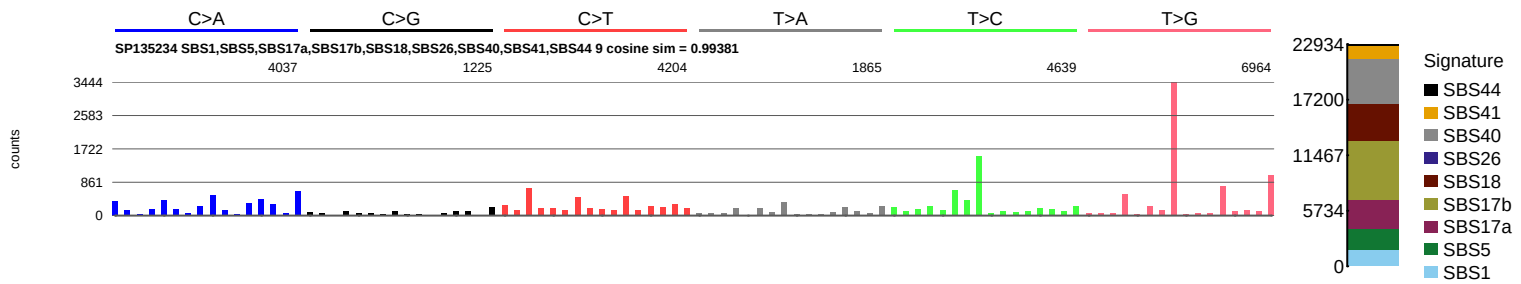
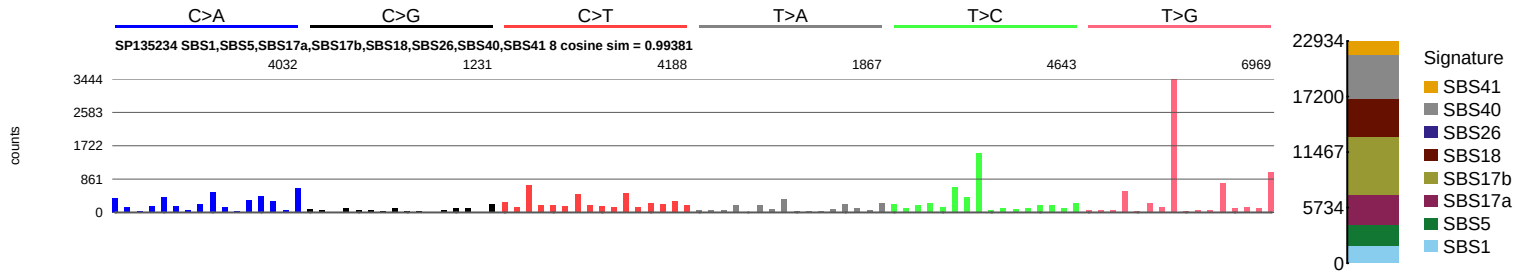
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99386
4029
1228
4238
1865
4615
6963
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99381
4037
1225
4204
1865
4639
6964
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1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C>G
C>T
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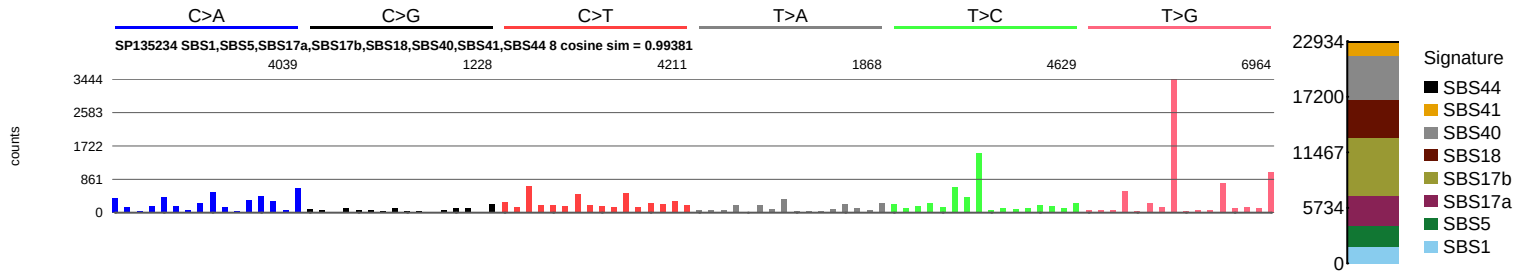
T>C
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.99381
4032
1231
4188
1867
4643
6969
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1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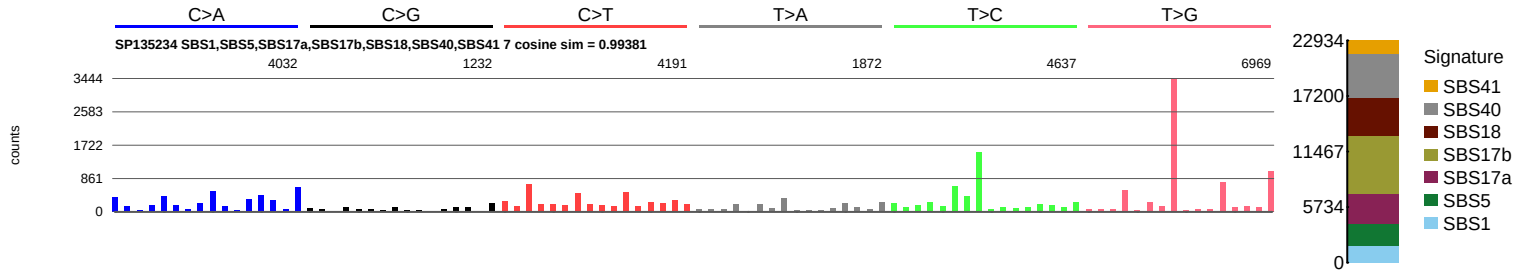
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.99381
4039
1228
4211
1868
4629
6964
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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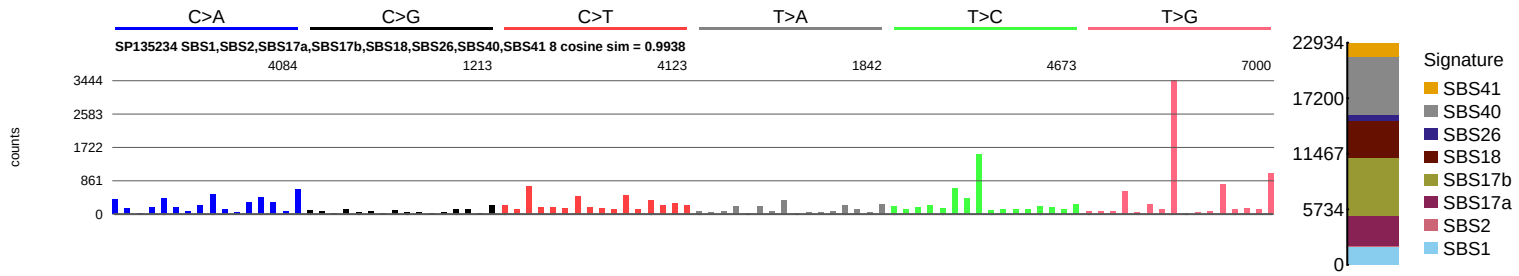
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99381
4032
1232
4191
1872
4637
6969
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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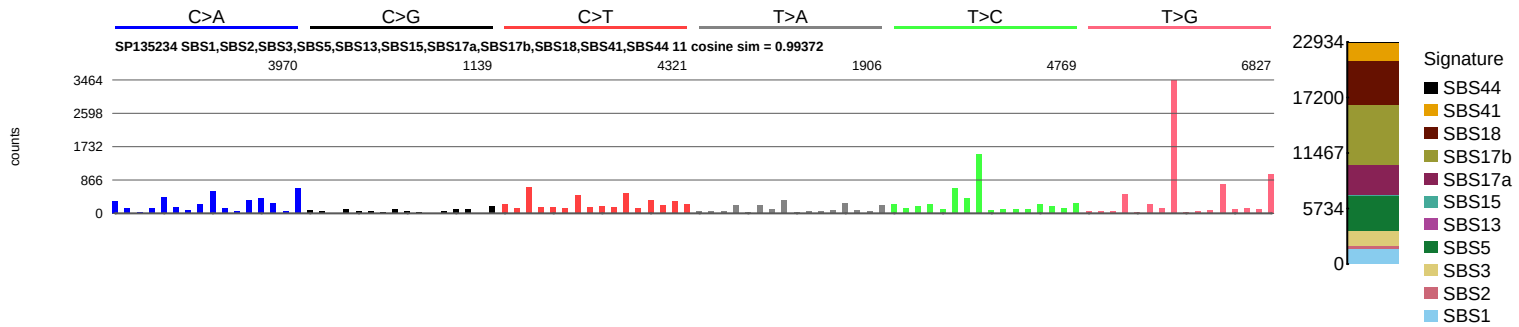
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.9938
4084
1213
4123
1842
4673
7000
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
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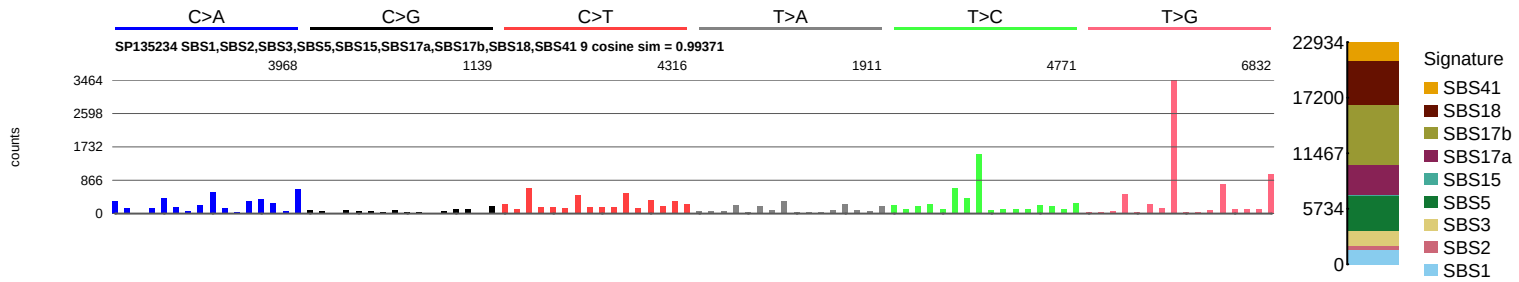
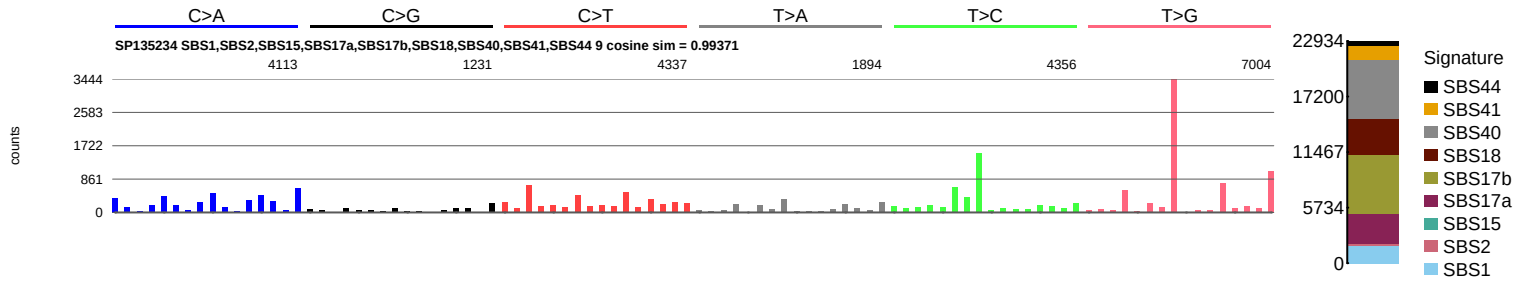
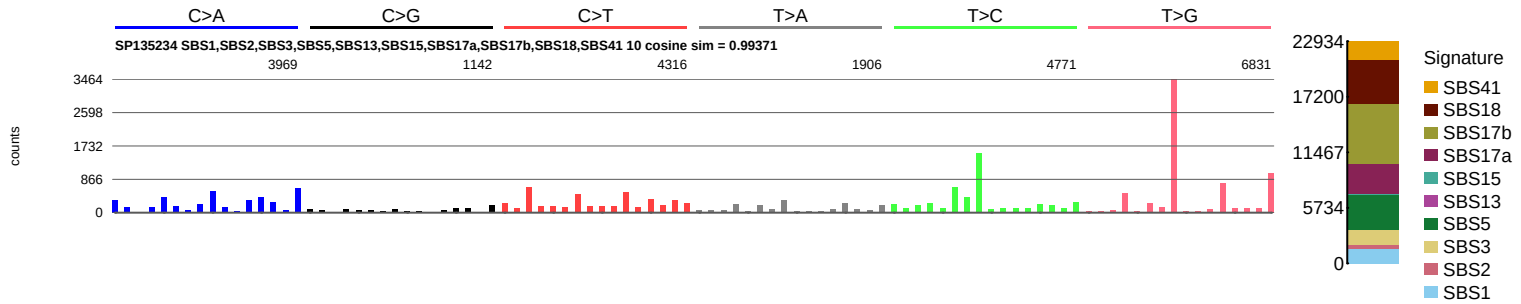
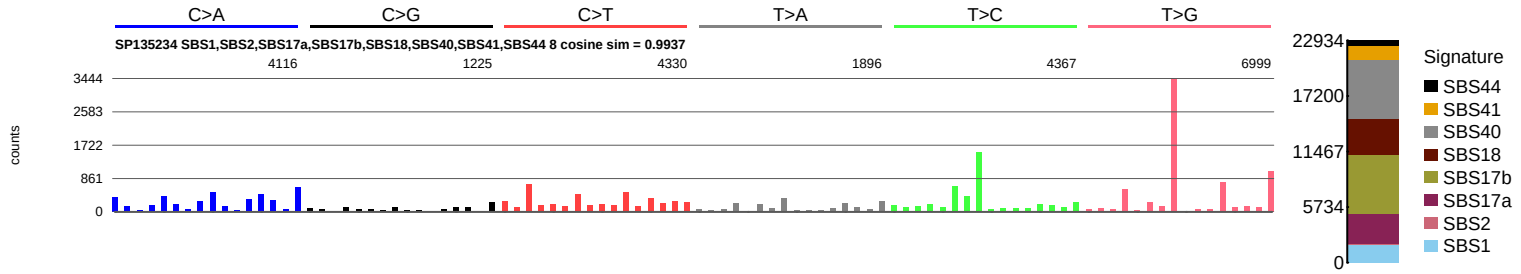
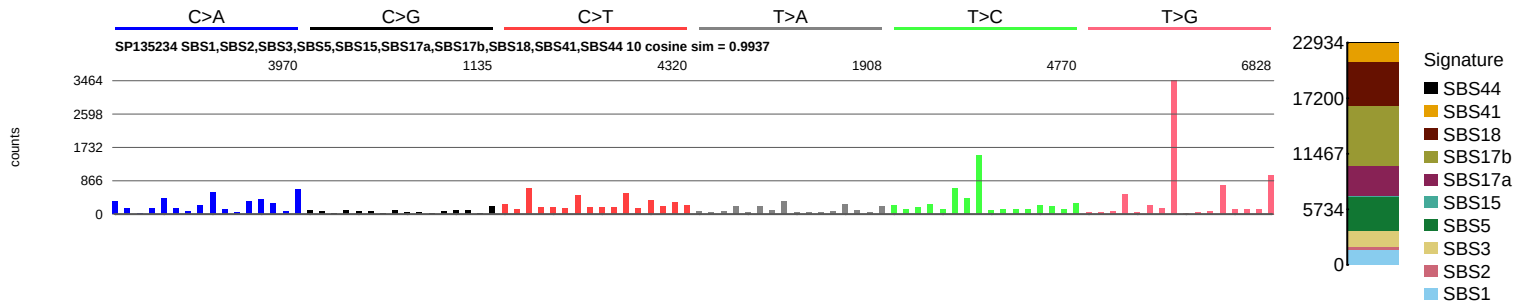
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 11 cosine sim = 0.99372
3970
1139
4321
1906
4769
6827
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

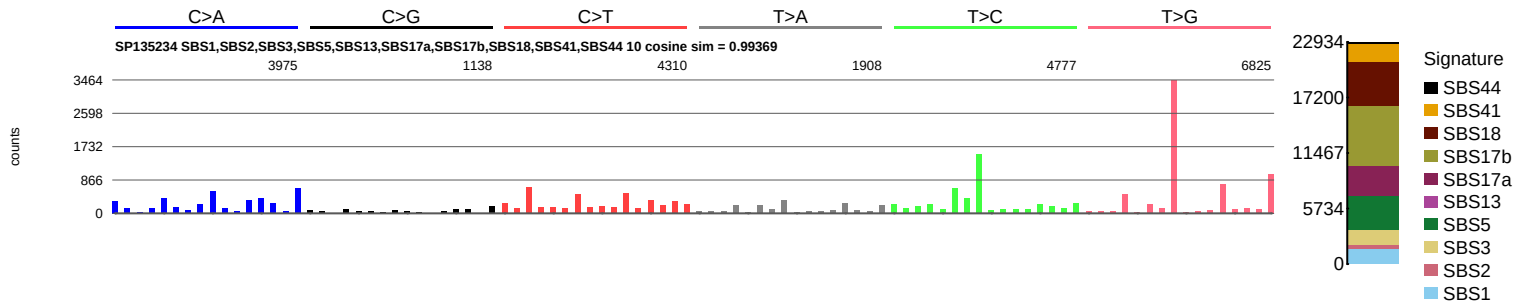

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99371
3968
1139
4316
1911
4771
6832
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

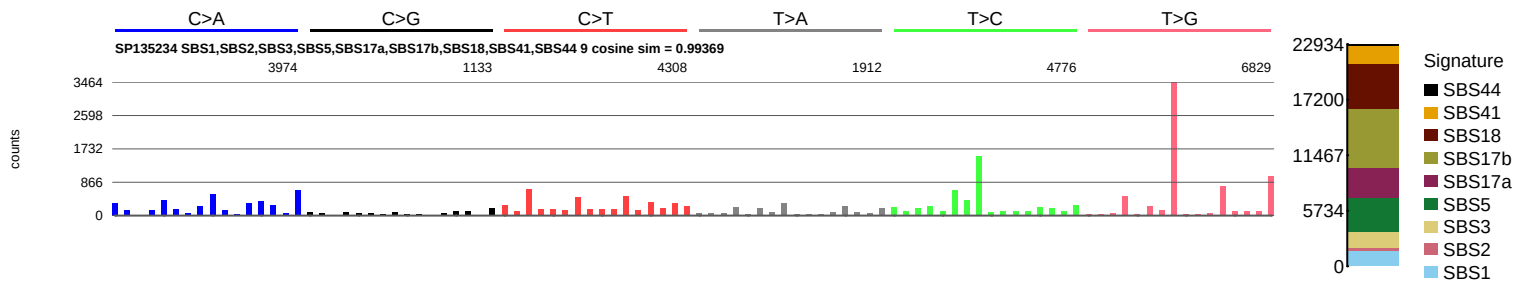

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99371
4113
1231
4337
1894
4356
7004
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

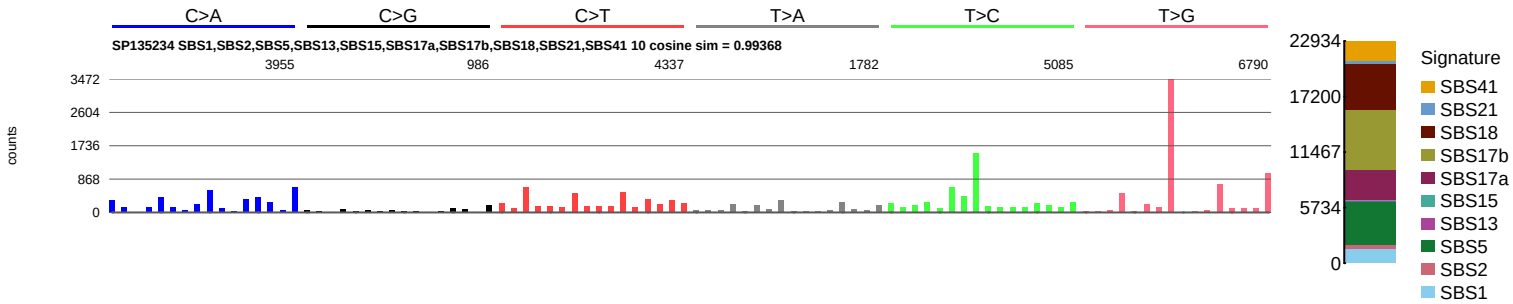

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.99371
3969
1142
4316
1906
4771
6831
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9937
4116
1225
4330
1896
4367
6999
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.9937
3970
1135
4320
1908
4770
6828
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.99369
3975
1138
4310
1908
4777
6825
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99369
3974
1133
4308
1912
4776
6829
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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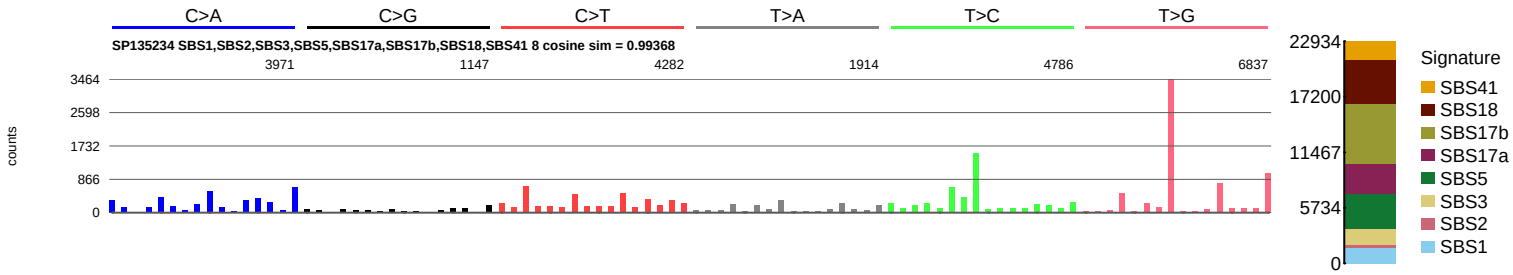
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99368
3955
986
4337
1782
5085
6790
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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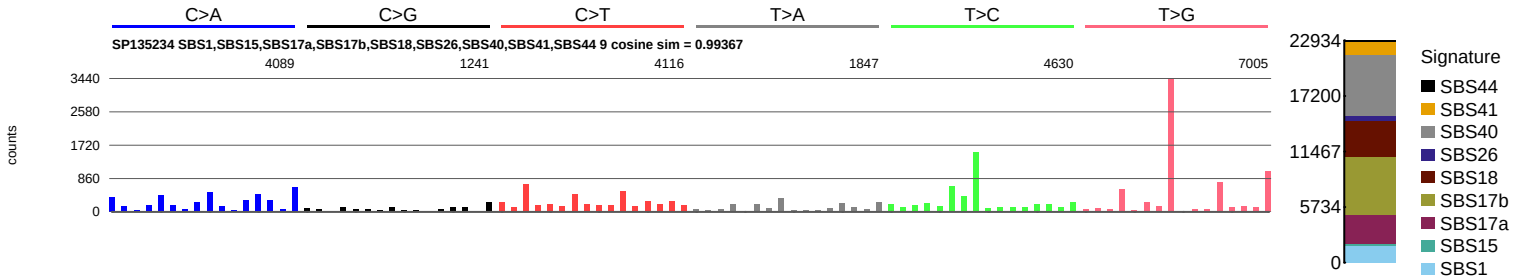
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99368
3971
1147
4282
1914
4786
6837
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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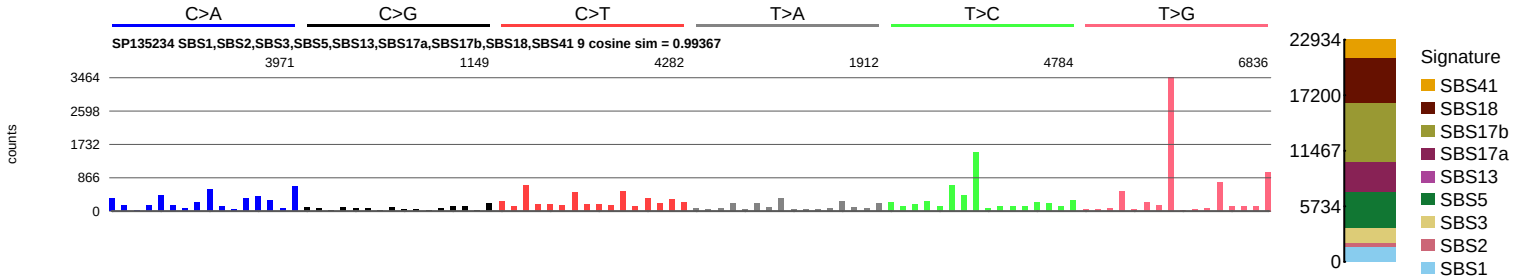
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99367
4089
1241
4116
1847
4630
7005
counts
3440
2580
1720
860
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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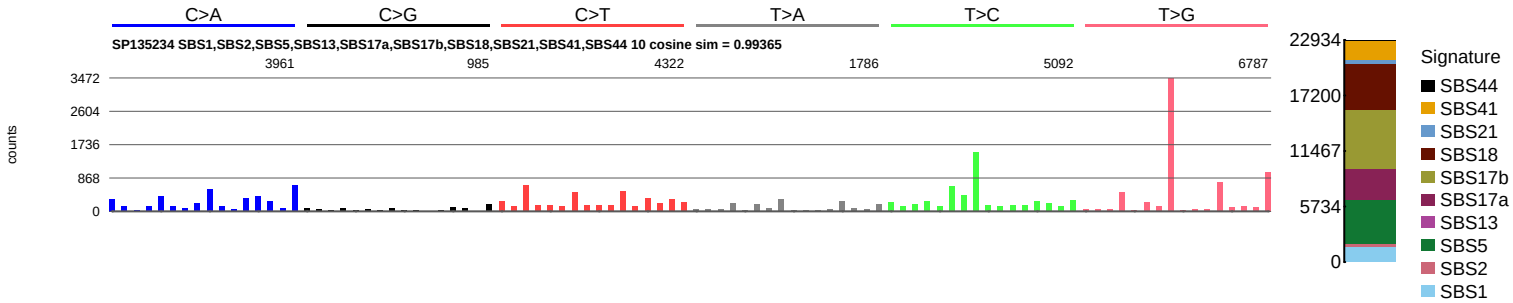
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99367
3971
1149
4282
1912
4784
6836
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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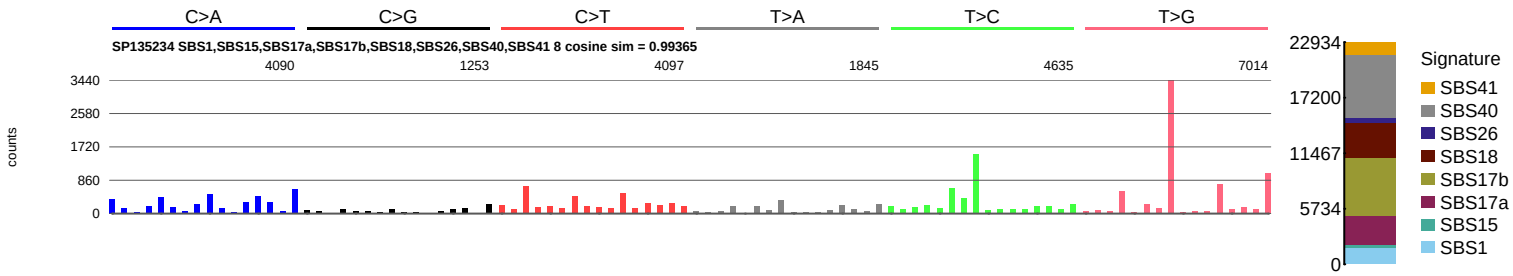
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.99365
3961
985
4322
1786
5092
6787
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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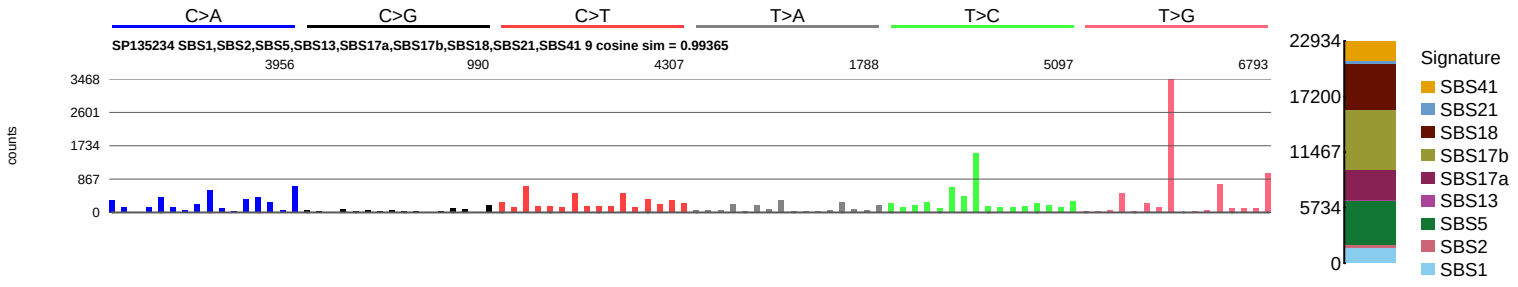
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.99365
4090
1253
4097
1845
4635
7014
counts
3440
2580
1720
860
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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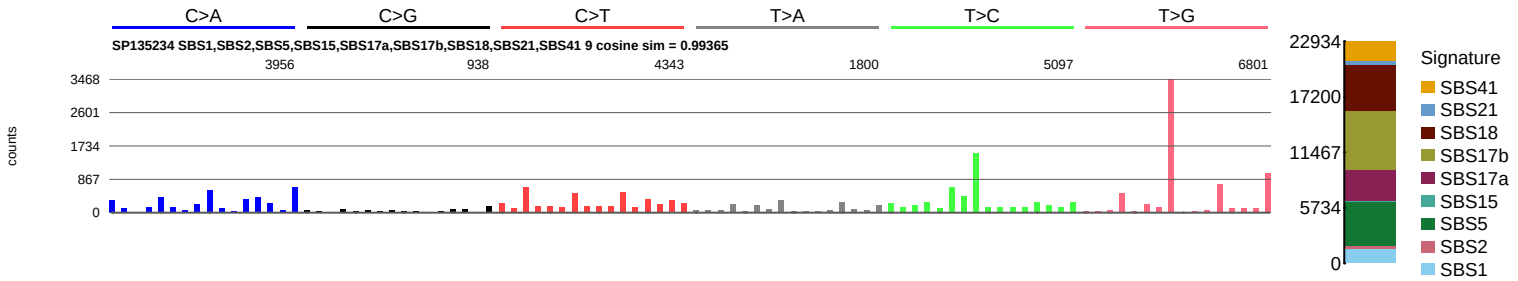
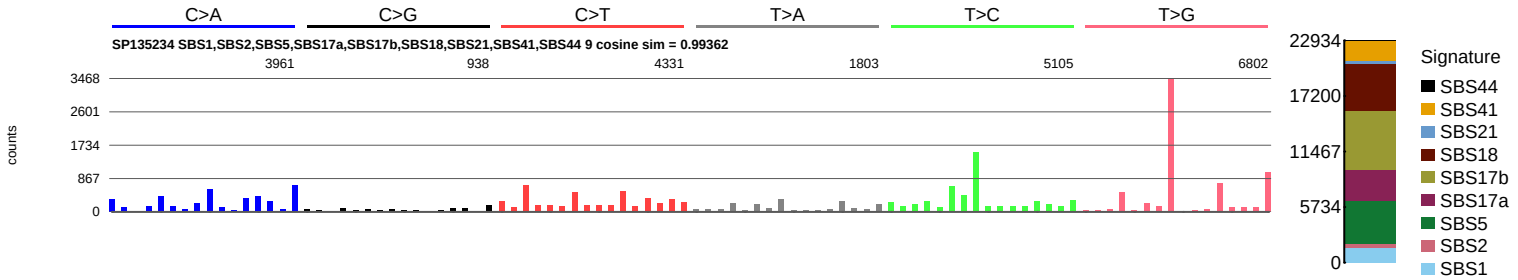
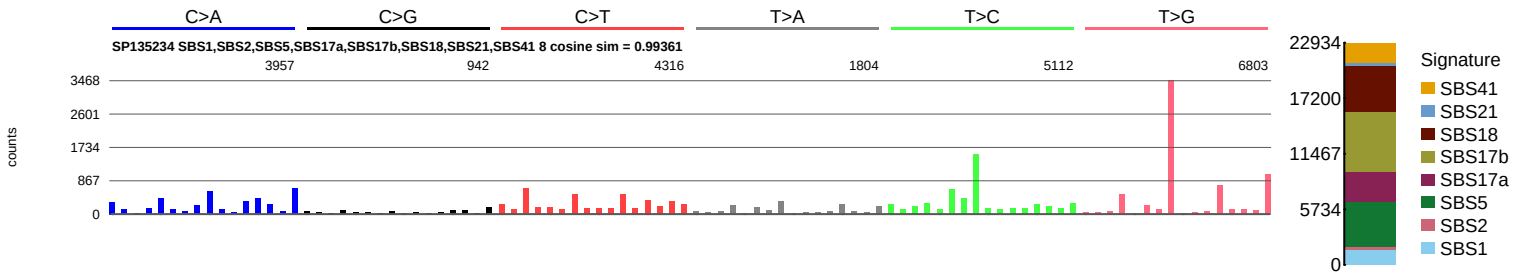
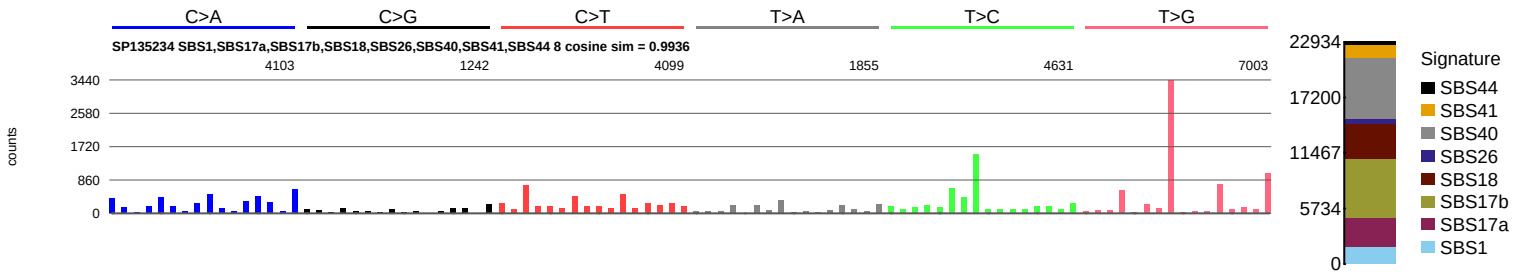
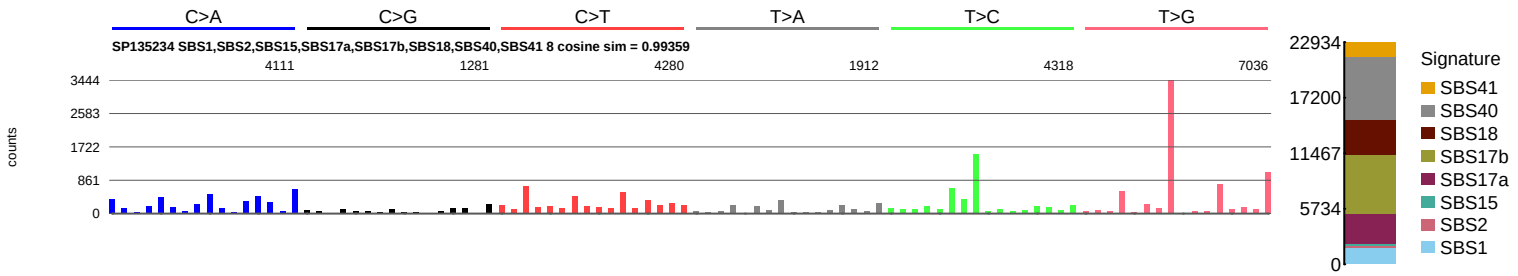
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99365
3956
990
4307
1788
5097
6793
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

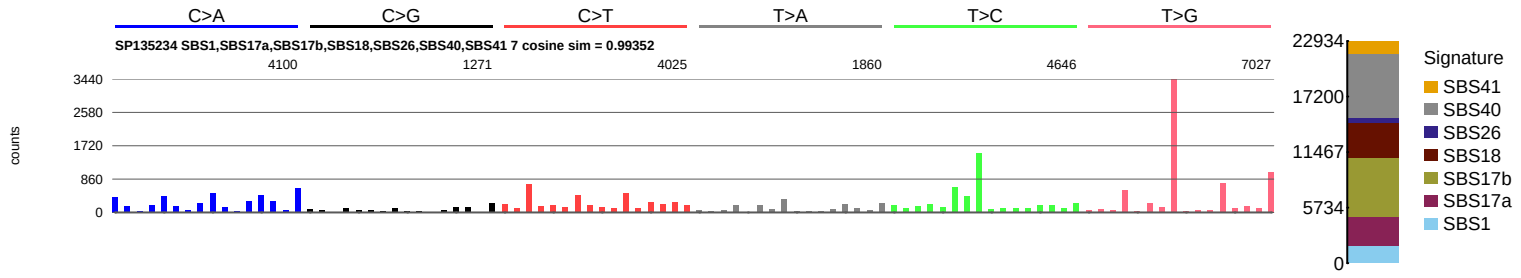

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99365
3956
938
4343
1800
5097
6801
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

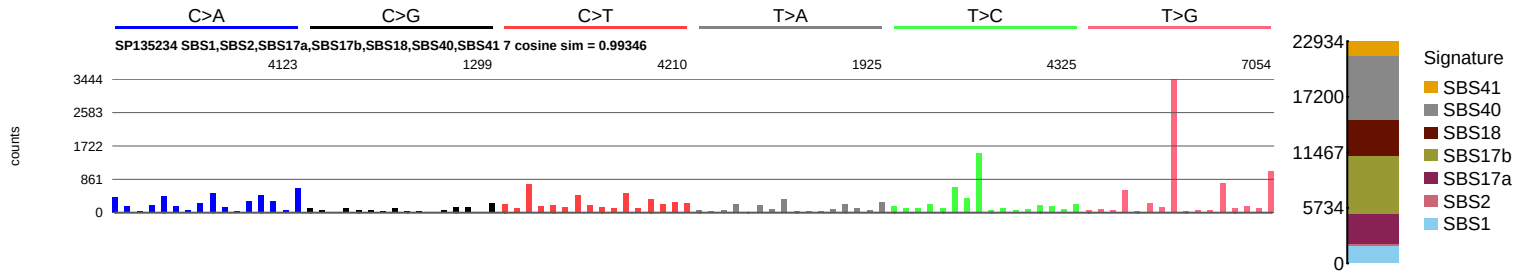

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.99362
3961
938
4331
1803
5105
6802
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

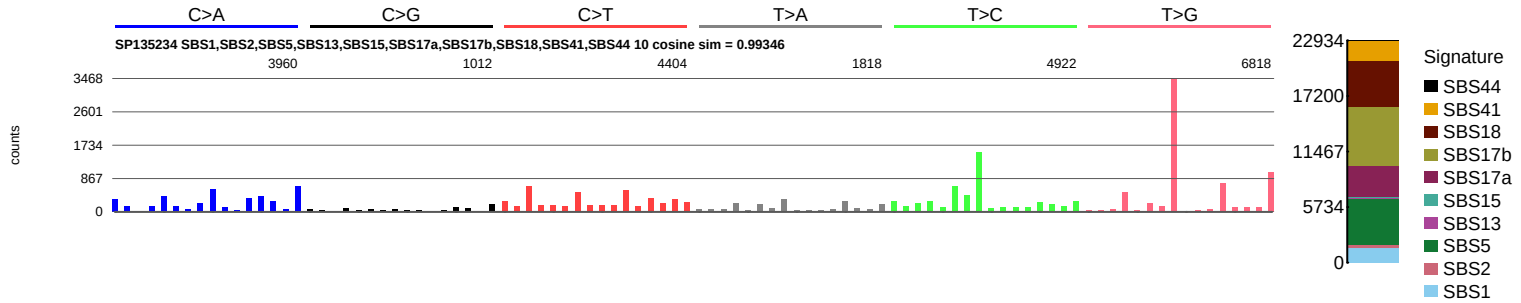

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99361
3957
942
4316
1804
5112
6803
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

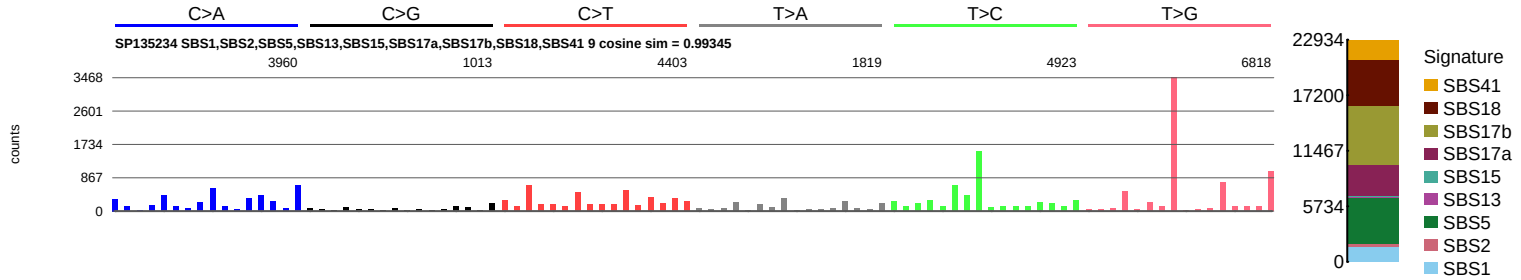

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.9936
4103
1242
4099
1855
4631
7003
counts
3440
2580
1720
860
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99359
4111
1281
4280
1912
4318
7036
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.99352
4100
1271
4025
1860
4646
7027
counts
3440
2580
1720
860
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99346
4123
1299
4210
1925
4325
7054
counts
3444
2583
1722
861
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.99346
3960
1012
4404
1818
4922
6818
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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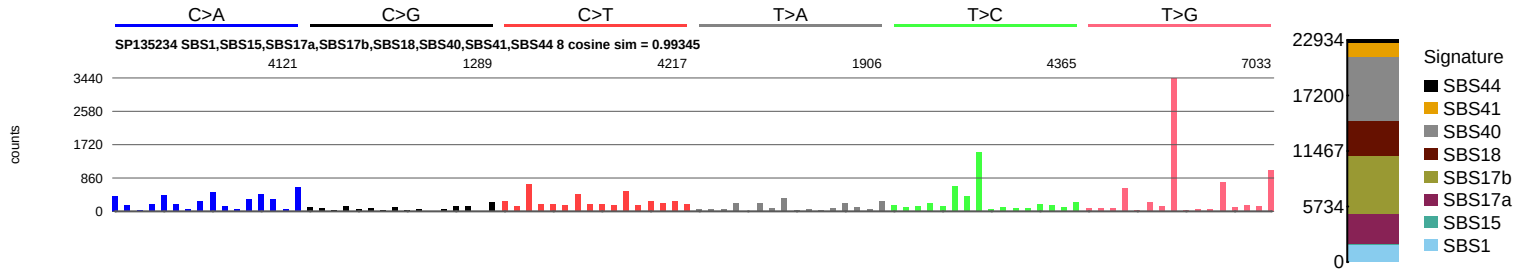
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99345
3960
1013
4403
1819
4923
6818
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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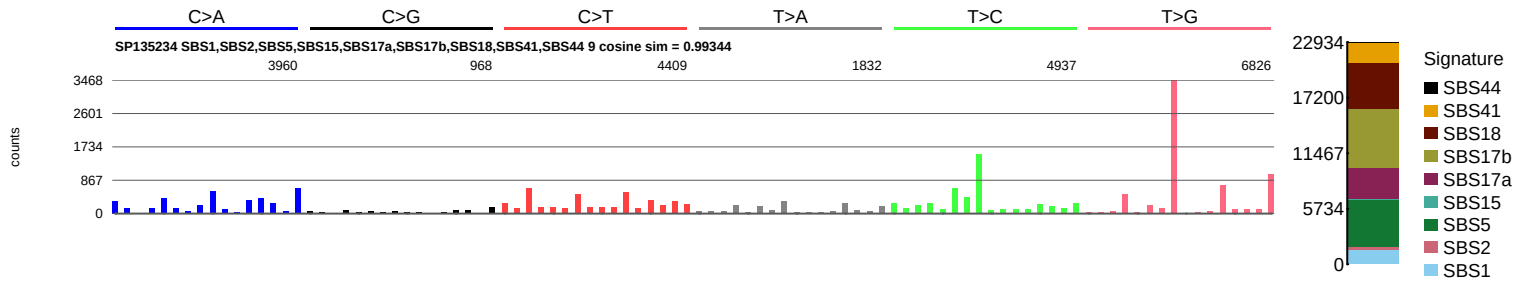
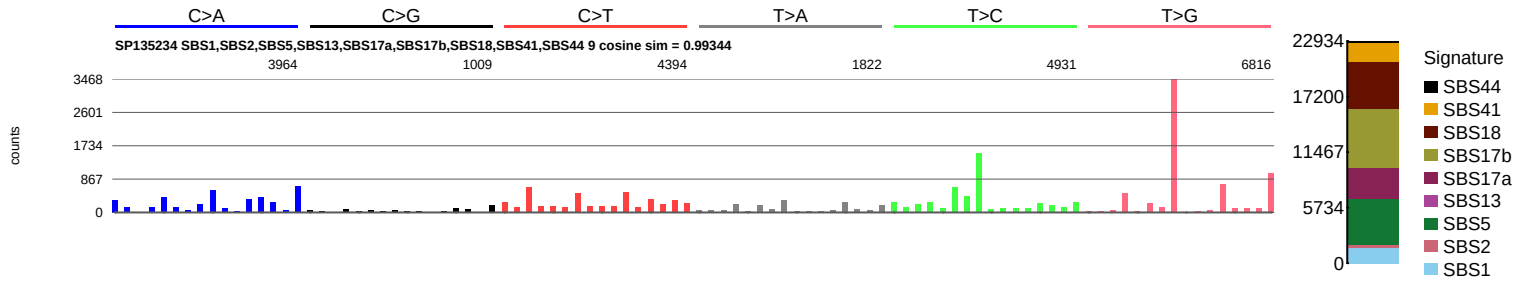
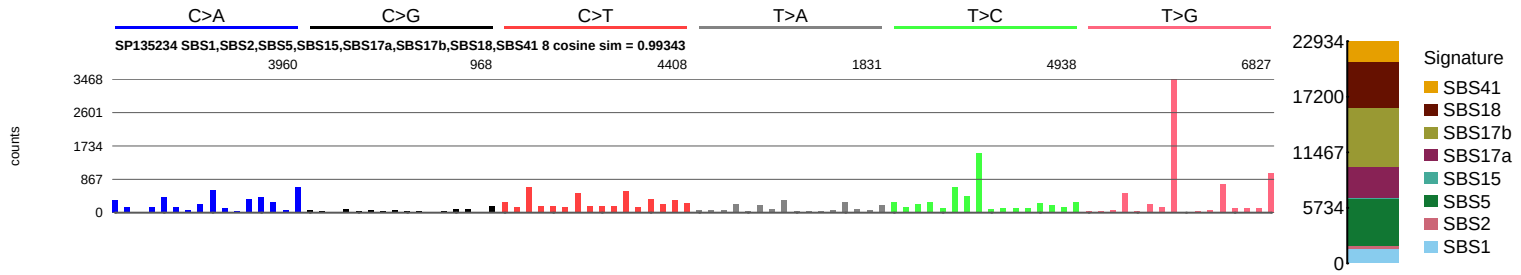
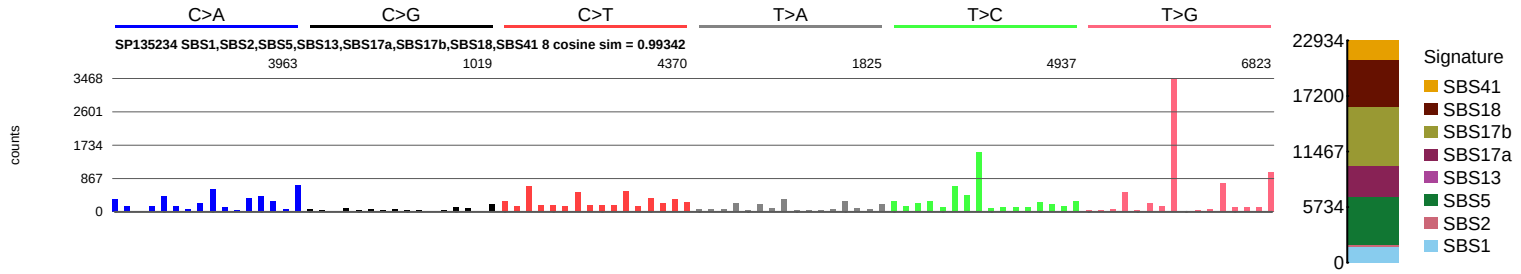
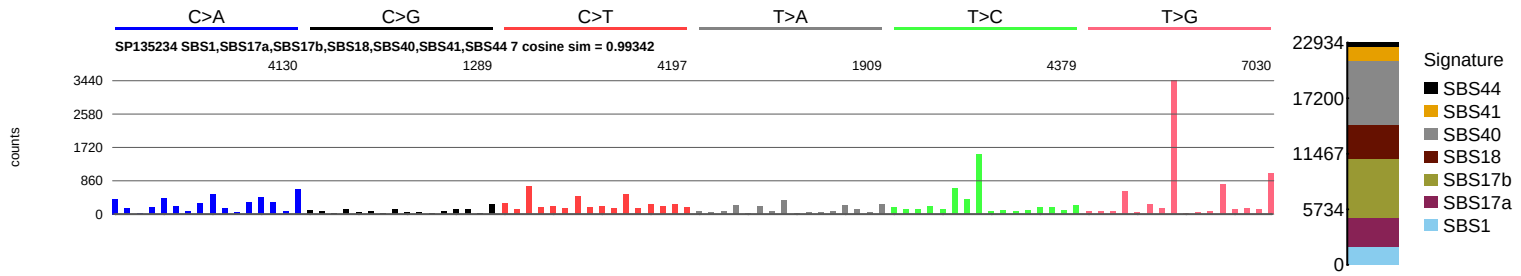
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.99345
4121
1289
4217
1906
4365
7033
counts
3440
2580
1720
860
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

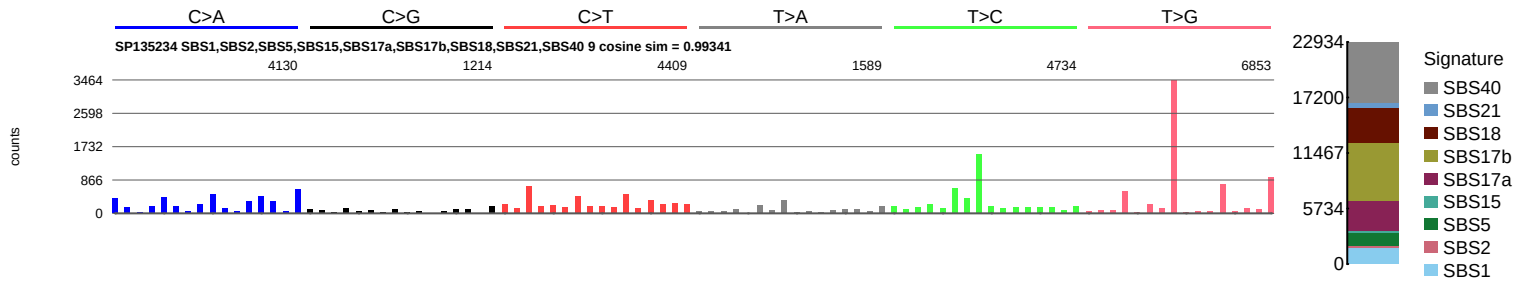

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99344
3960
968
4409
1832
4937
6826
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

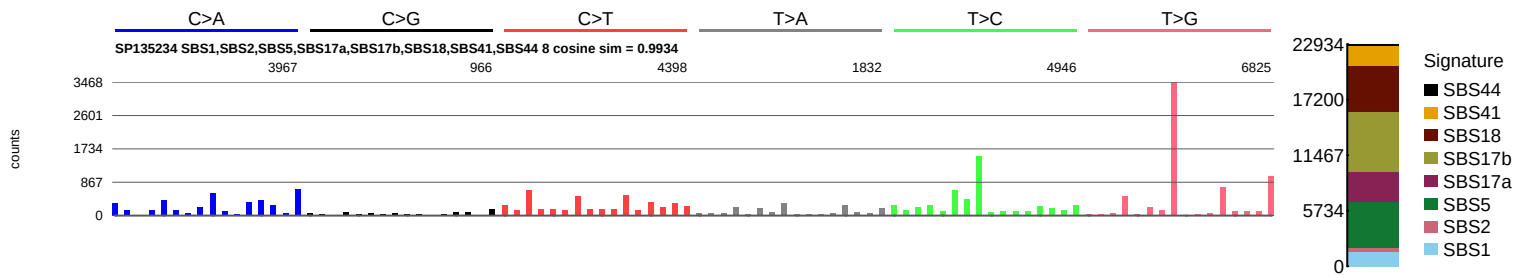

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99344
3964
1009
4394
1822
4931
6816
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

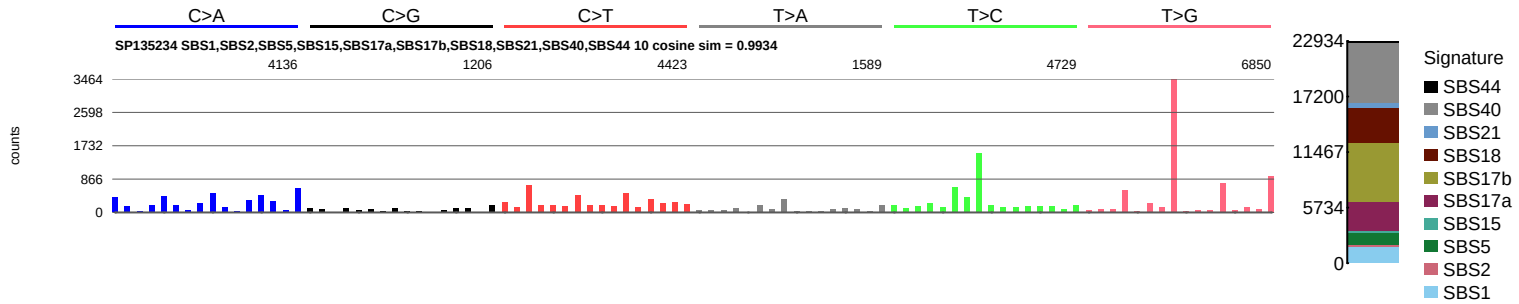

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99343
3960
968
4408
1831
4938
6827
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99342
3963
1019
4370
1825
4937
6823
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.99342
4130
1289
4197
1909
4379
7030
counts
3440
2580
1720
860
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.99341
4130
1214
4409
1589
4734
6853
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.9934
3967
966
4398
1832
4946
6825
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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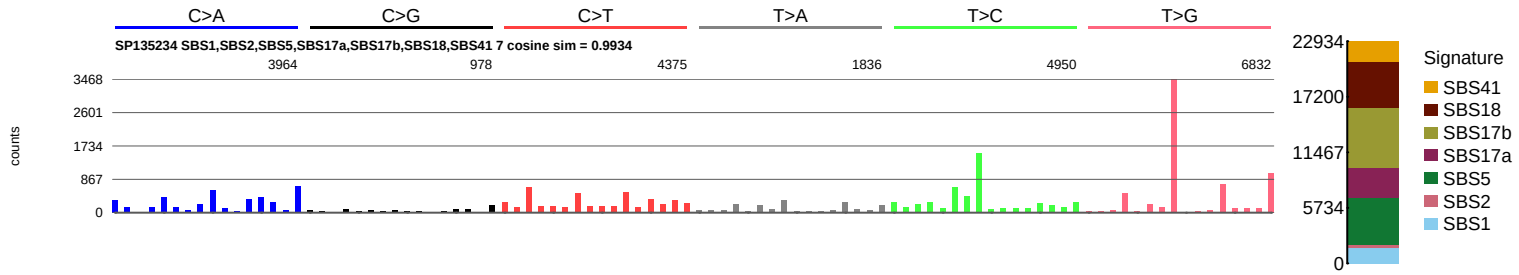
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.9934
4136
1206
4423
1589
4729
6850
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
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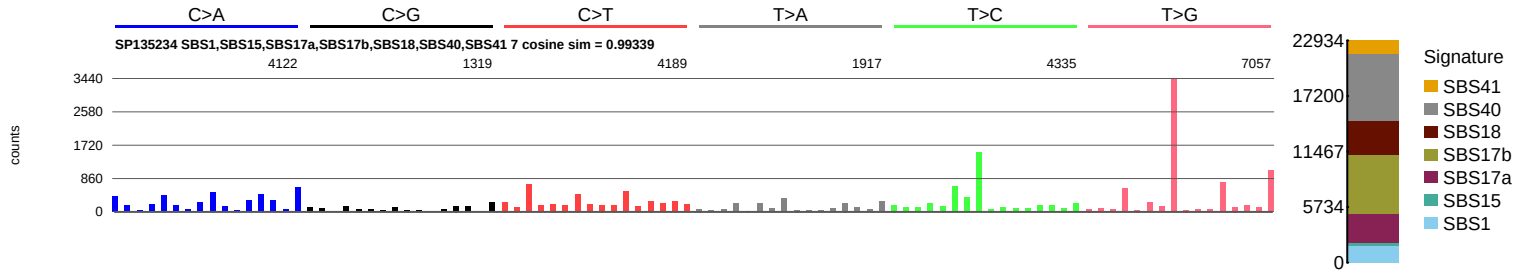
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9934
3964
978
4375
1836
4950
6832
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
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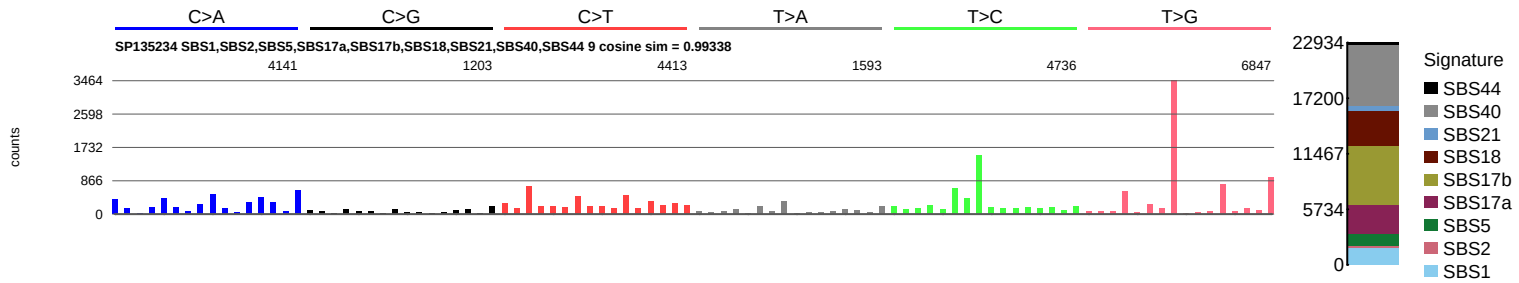
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99339
4122
1319
4189
1917
4335
7057
counts
3440
2580
1720
860
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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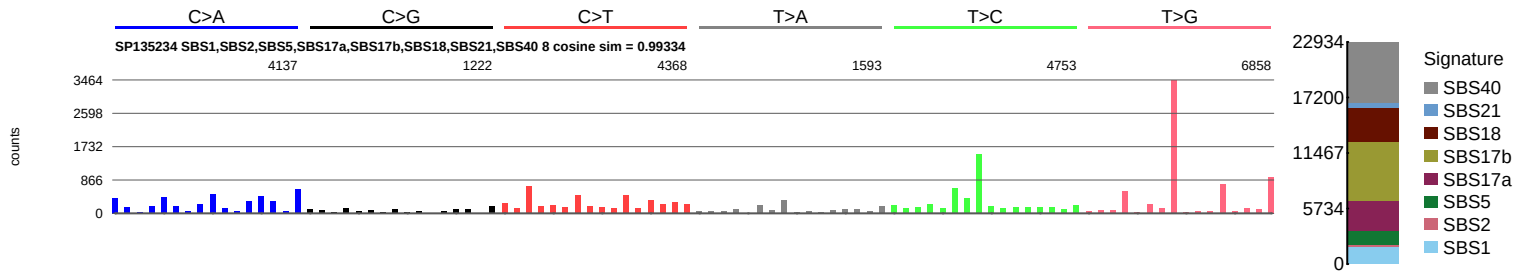
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.99338
4141
1203
4413
1593
4736
6847
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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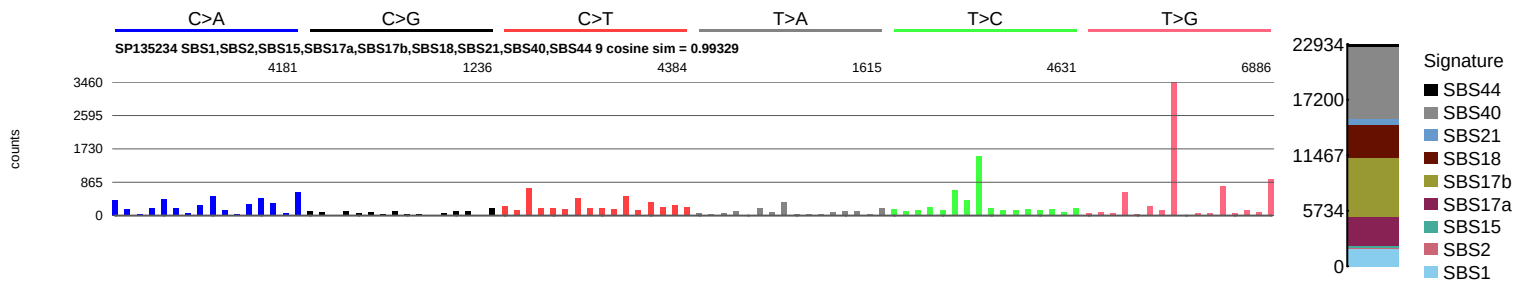
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.99334
4137
1222
4368
1593
4753
6858
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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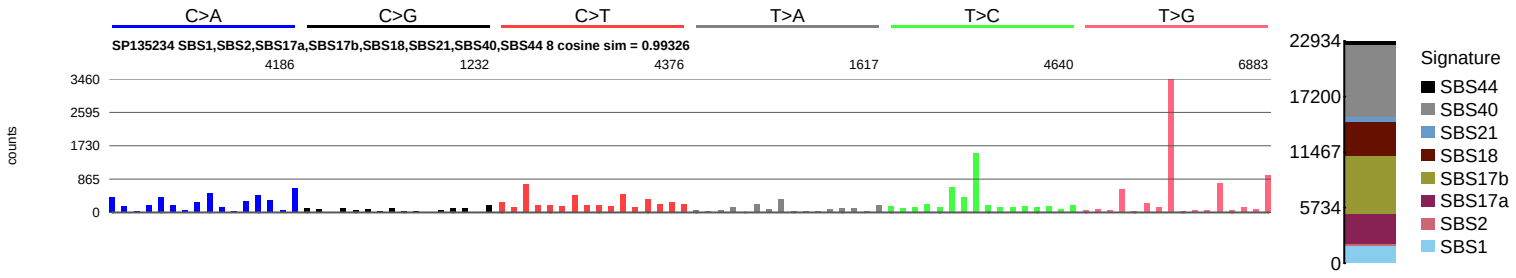
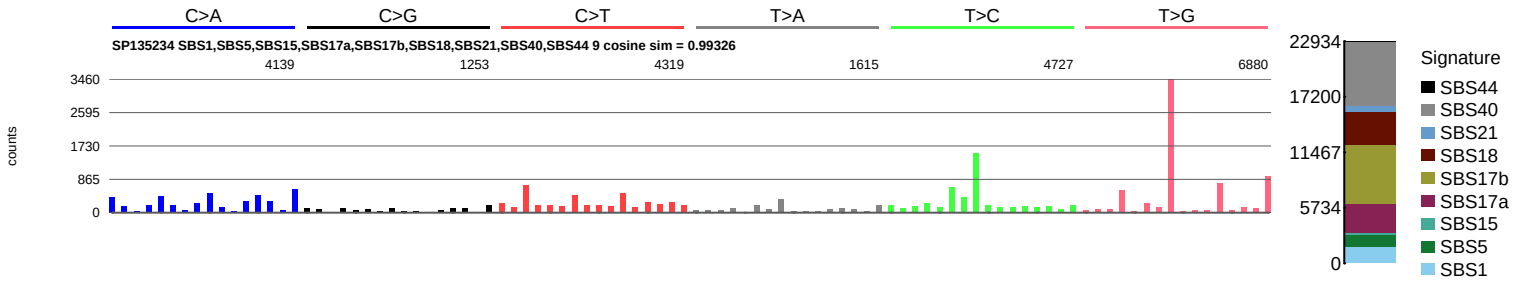
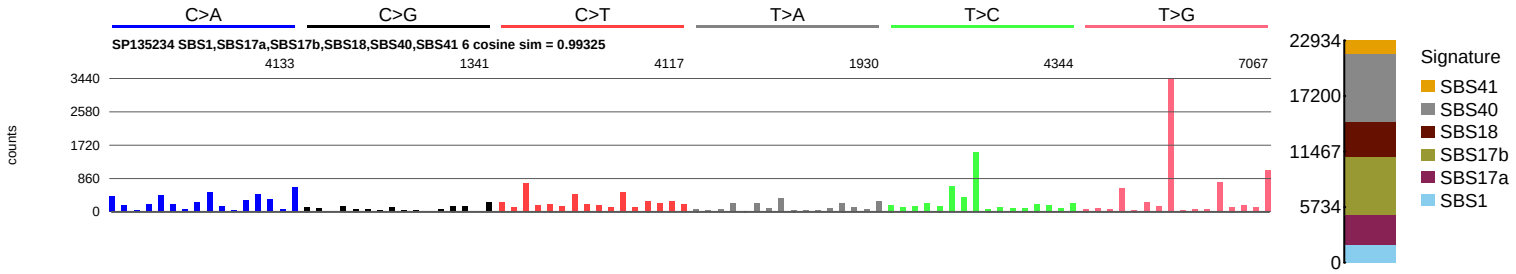
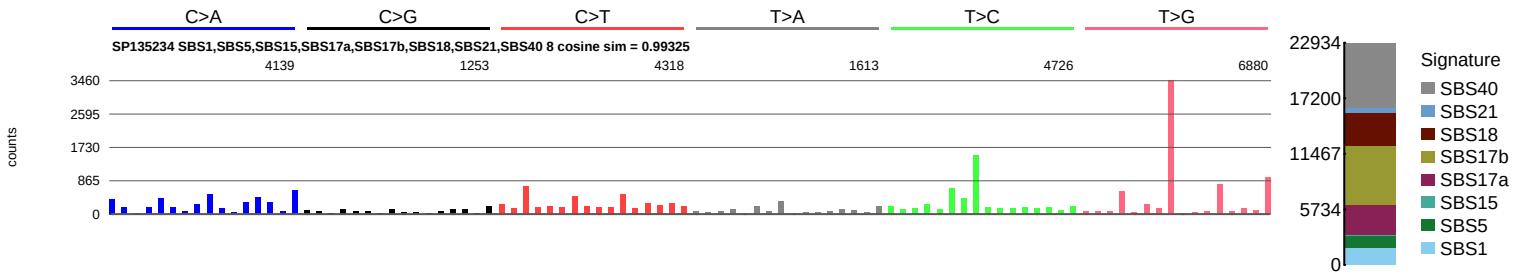
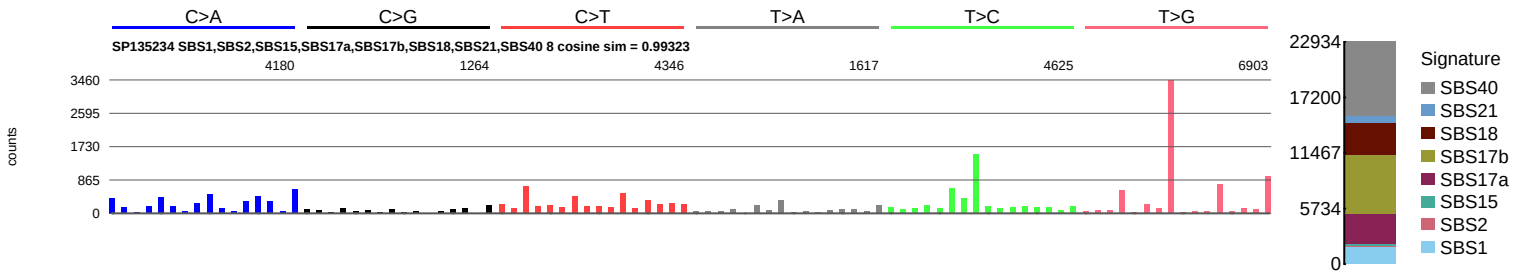
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.99329
4181
1236
4384
1615
4631
6886
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

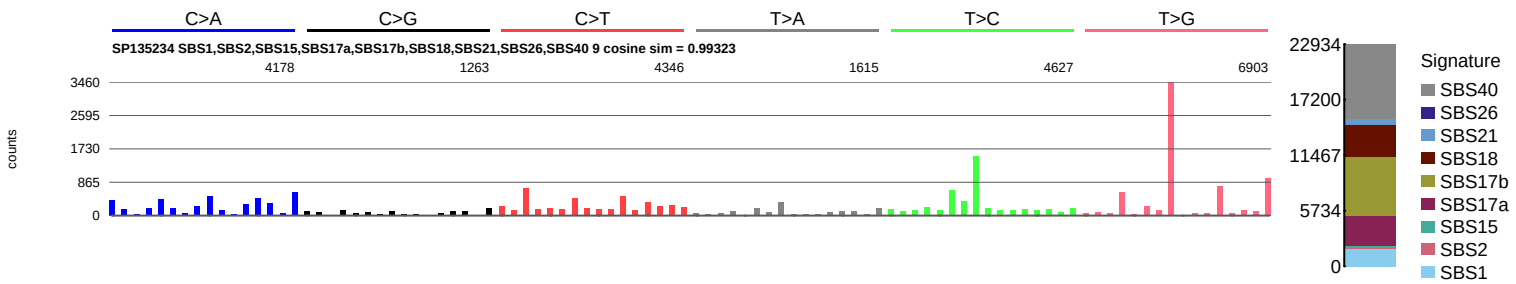

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.99326
4186
1232
4376
1617
4640
6883
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

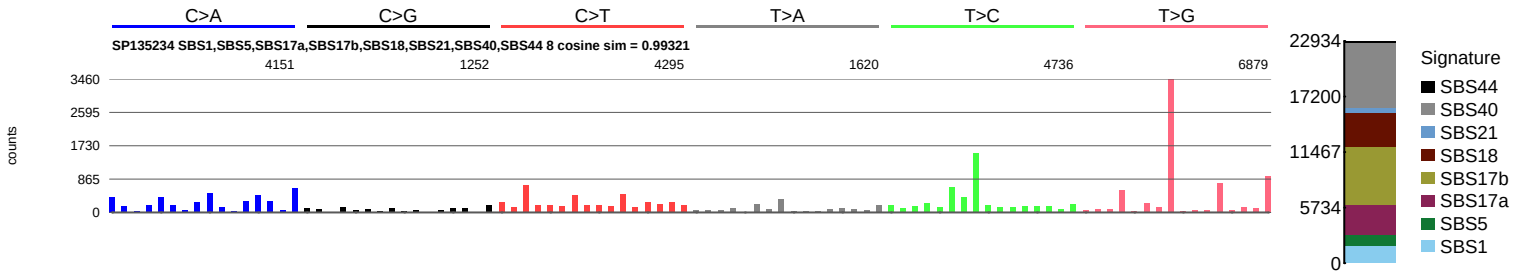

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.99326
4139
1253
4319
1615
4727
6880
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

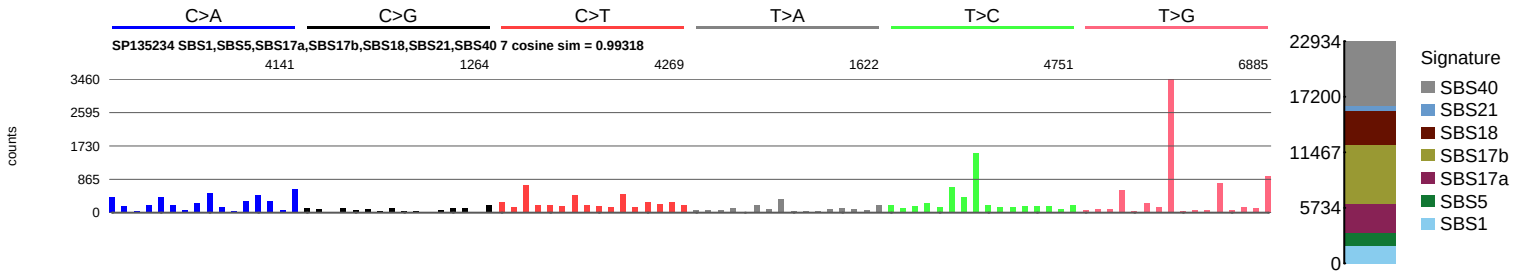

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.99325
4133
1341
4117
1930
4344
7067
counts
3440
2580
1720
860
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1

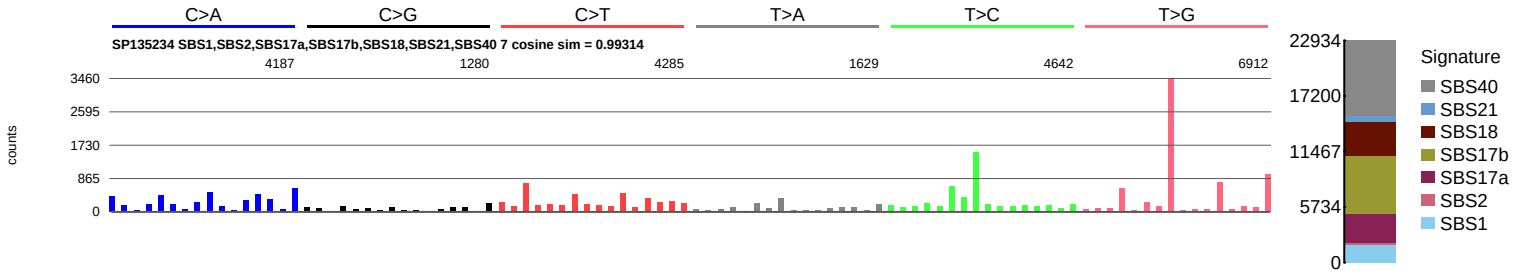

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.99325
4139
1253
4318
1613
4726
6880
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

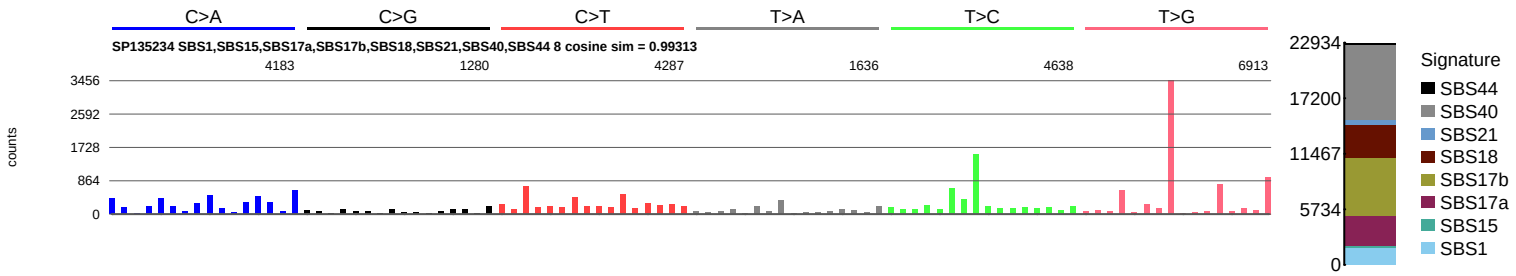

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.99323
4180
1264
4346
1617
4625
6903
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

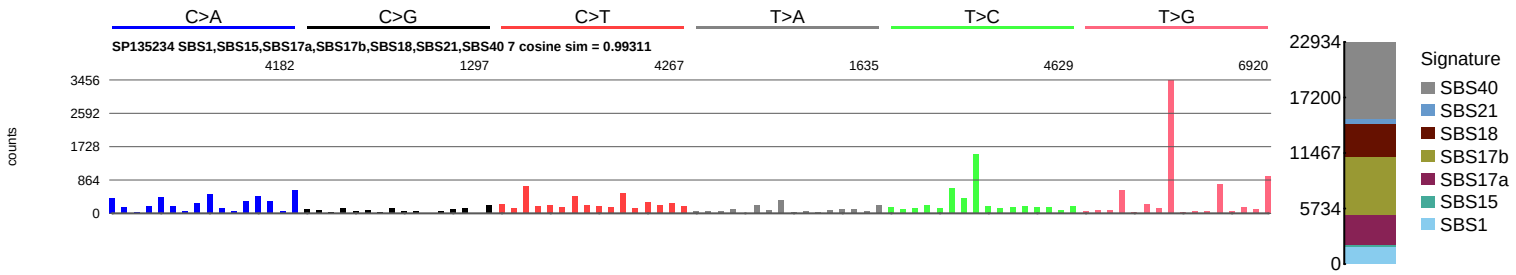

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.99323
4178
1263
4346
1615
4627
6903
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.99321
4151
1252
4295
1620
4736
6879
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.99318
4141
1264
4269
1622
4751
6885
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.99314
4187
1280
4285
1629
4642
6912
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.99313
4183
1280
4287
1636
4638
6913
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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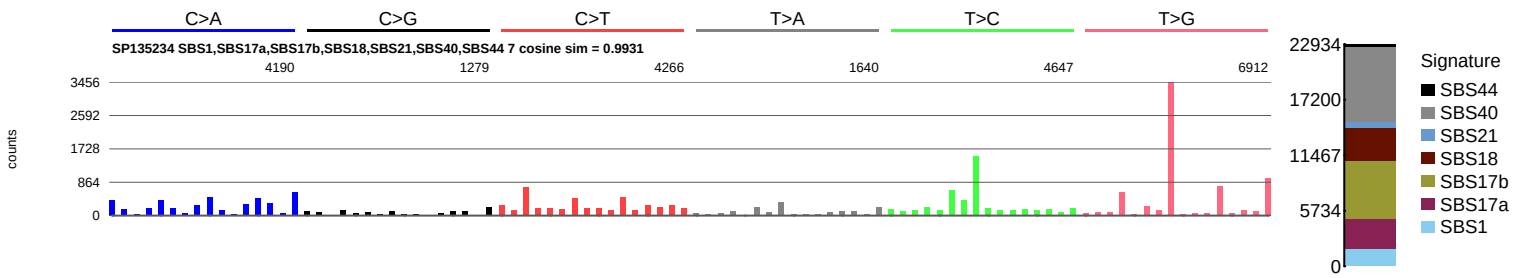
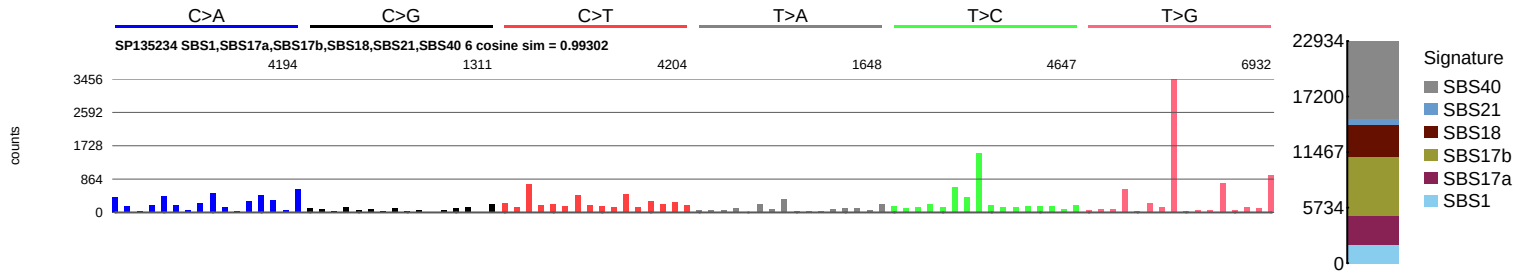
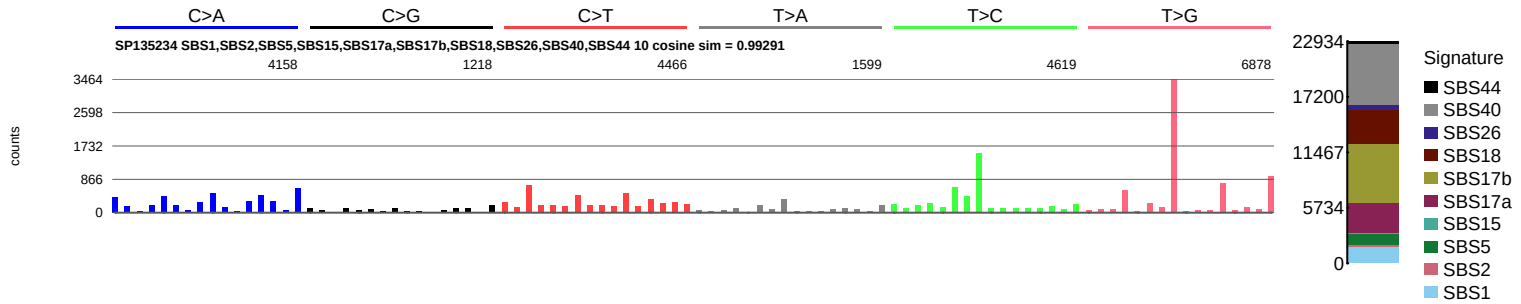
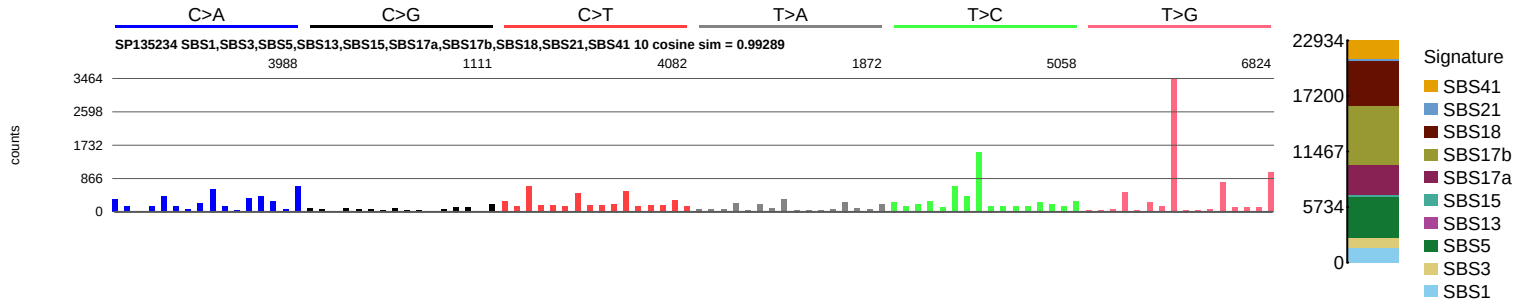
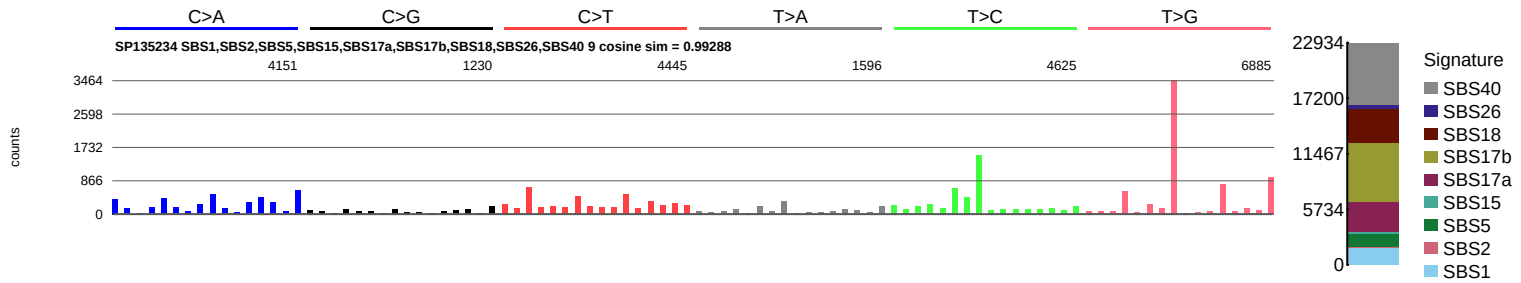
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.99311
4182
1297
4267
1635
4629
6920
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

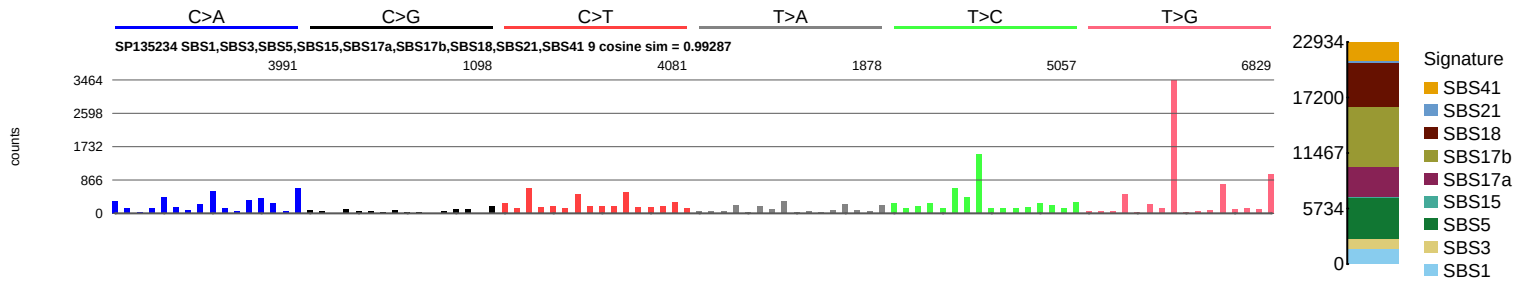

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.9931
4190
1279
4266
1640
4647
6912
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

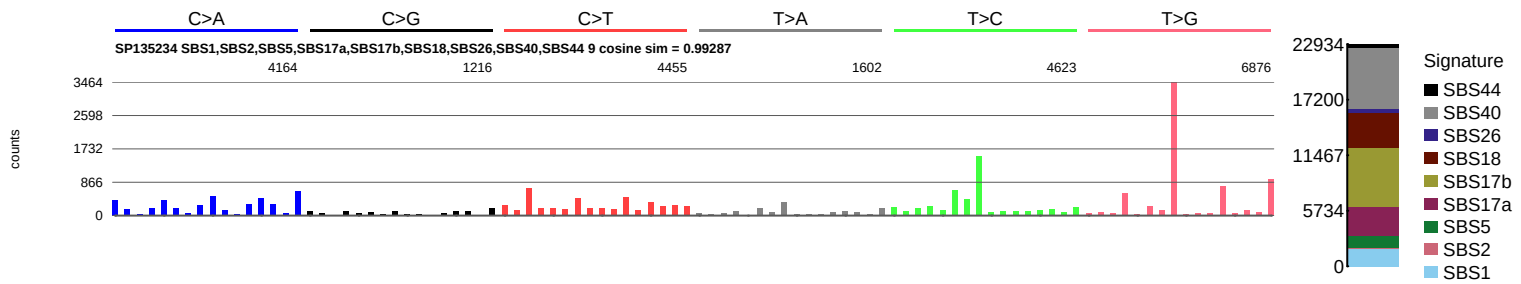

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.99302
4194
1311
4204
1648
4647
6932
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

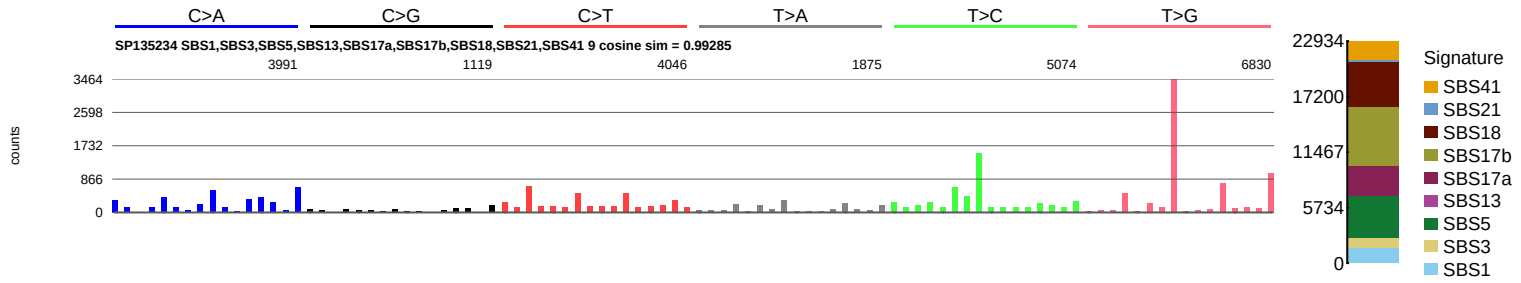

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.99291
4158
1218
4466
1599
4619
6878
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

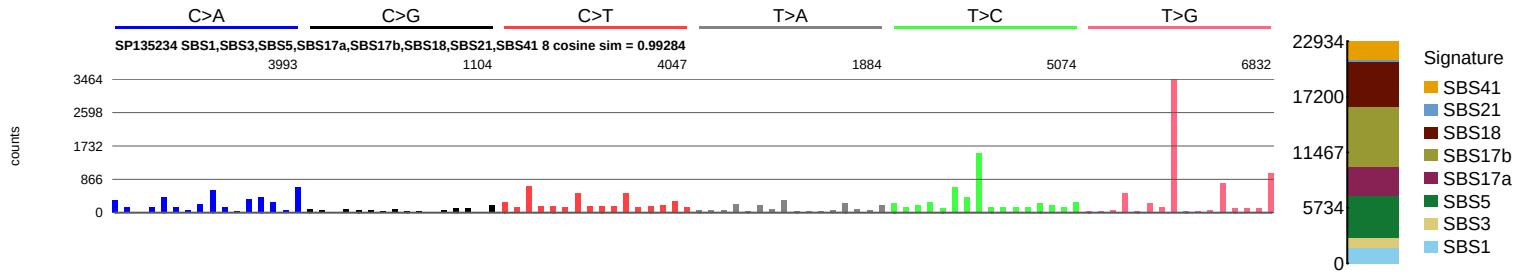

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99289
3988
1111
4082
1872
5058
6824
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.99288
4151
1230
4445
1596
4625
6885
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99287
3991
1098
4081
1878
5057
6829
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.99287
4164
1216
4455
1602
4623
6876
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99285
3991
1119
4046
1875
5074
6830
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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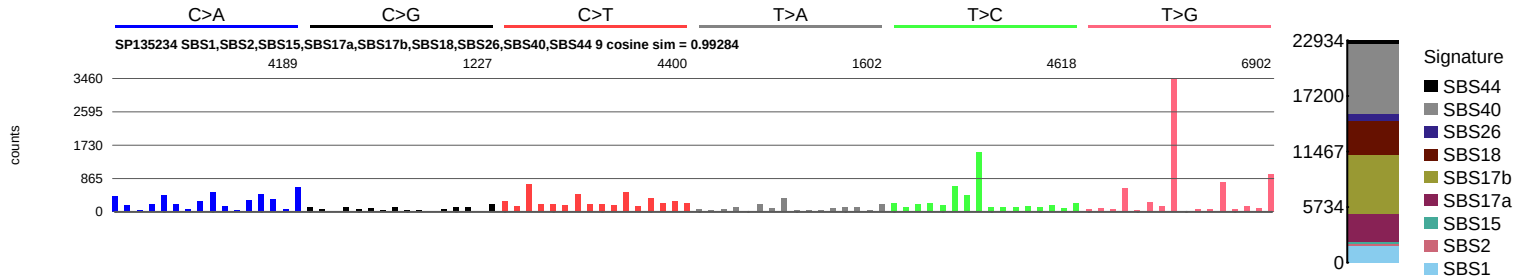
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99284
3993
1104
4047
1884
5074
6832
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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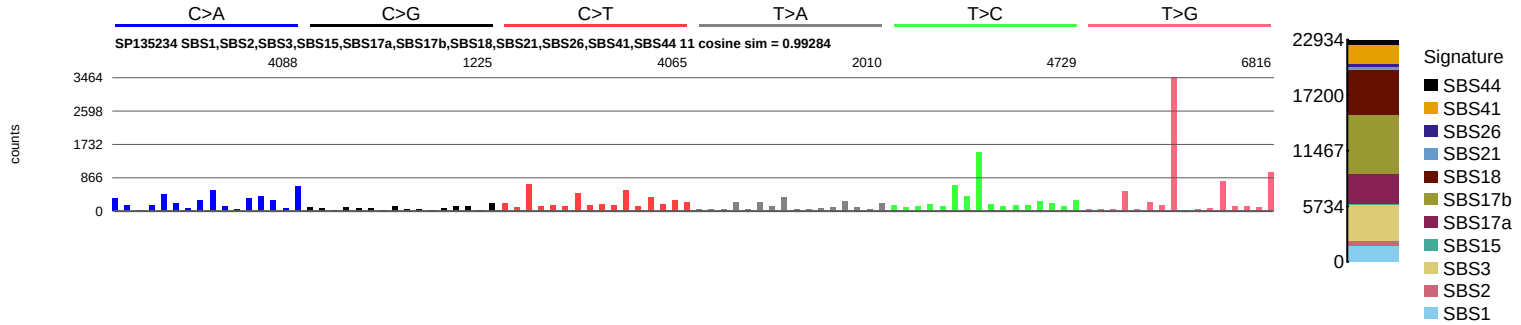
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.99284
4189
1227
4400
1602
4618
6902
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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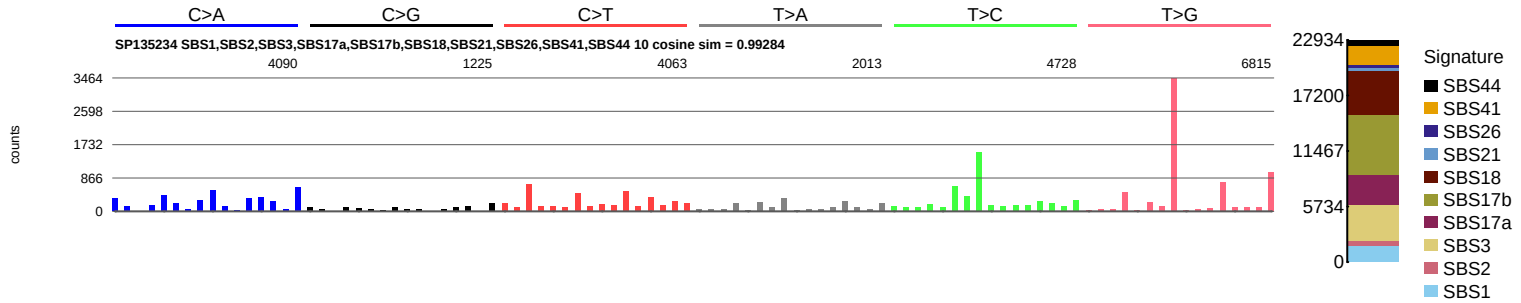
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.99284
4088
1225
4065
2010
4729
6816
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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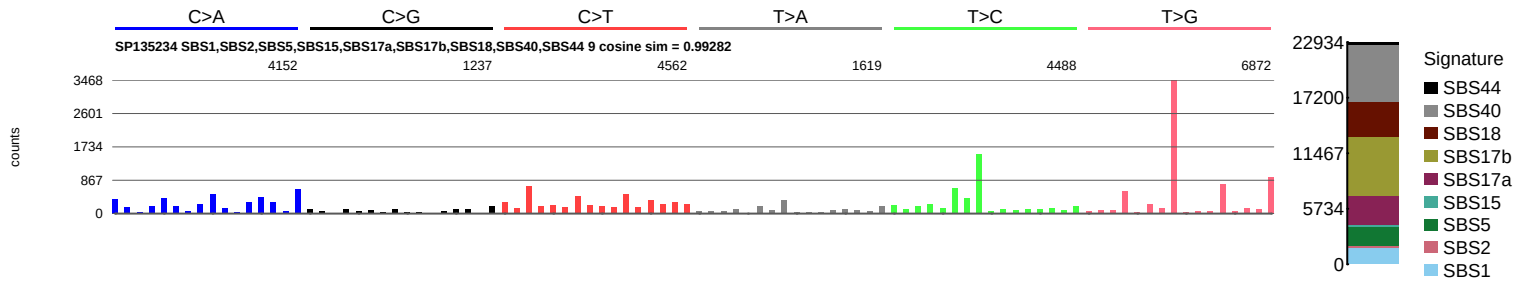
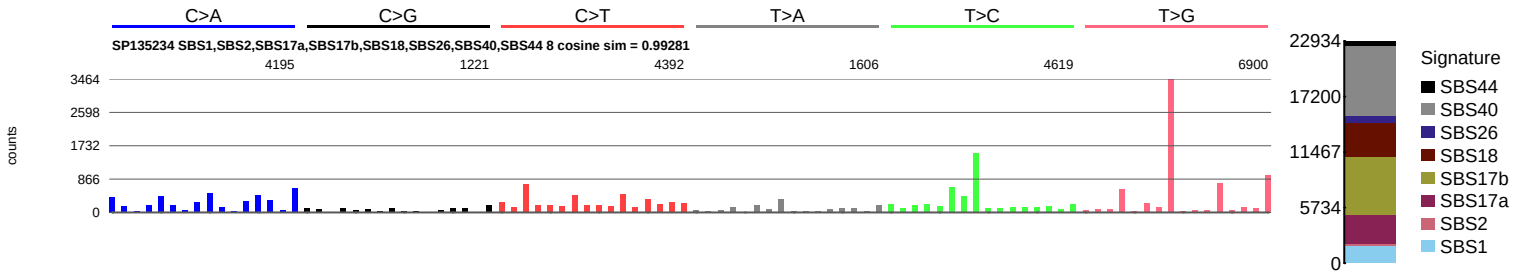
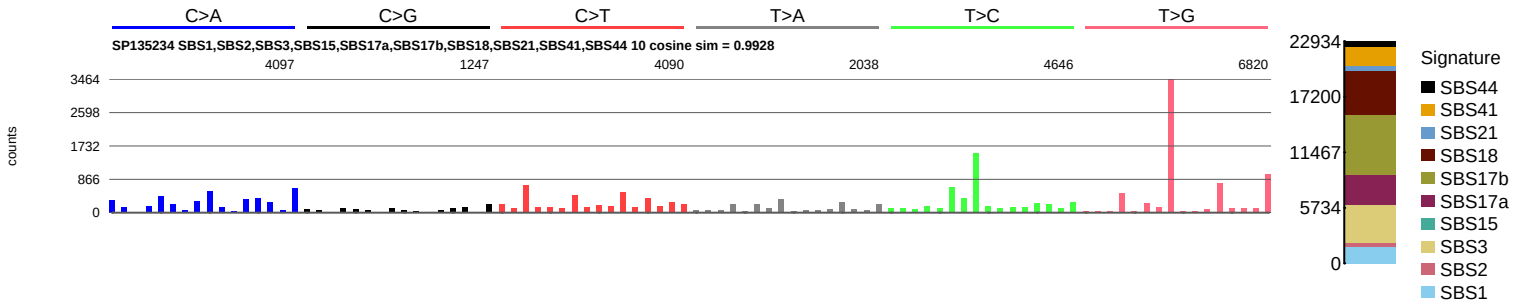
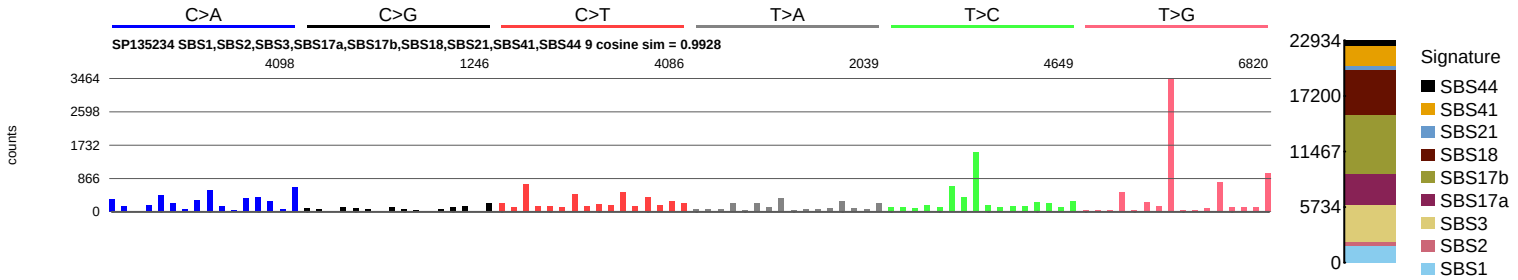
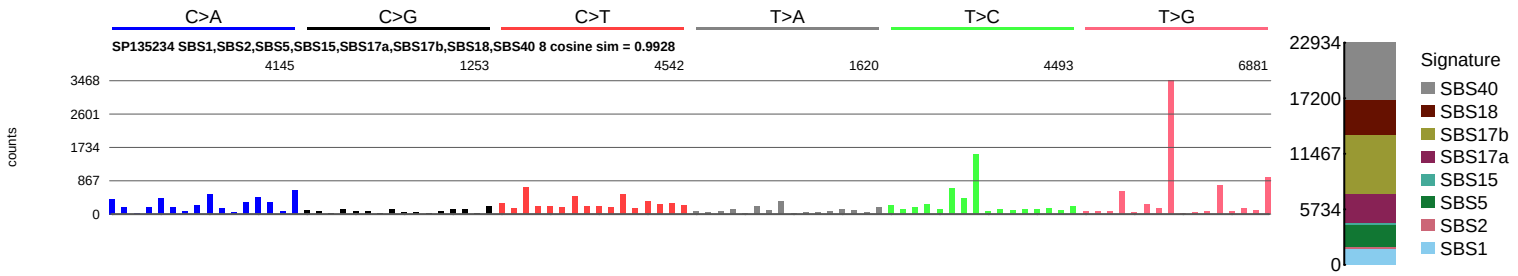
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99284
4090
1225
4063
2013
4728
6815
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

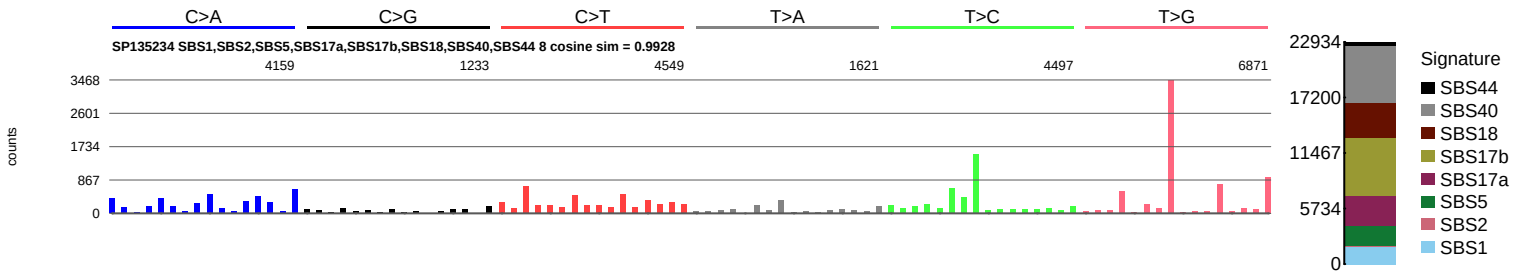

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.99282
4152
1237
4562
1619
4488
6872
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

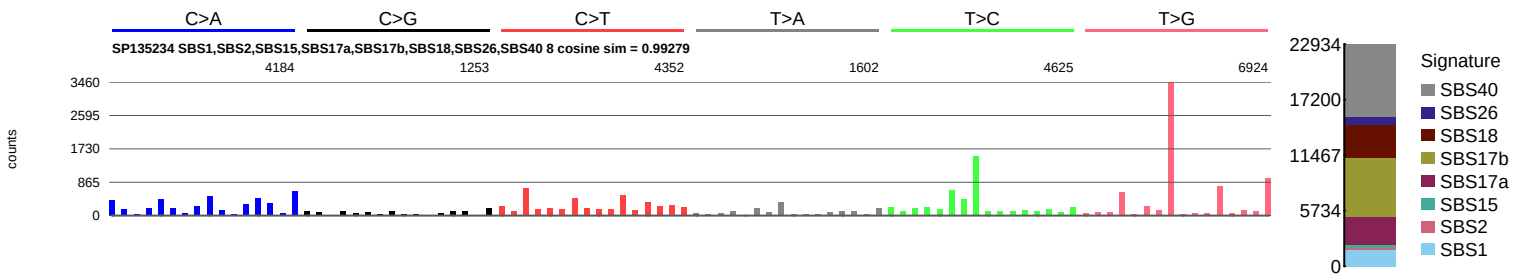

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.99281
4195
1221
4392
1606
4619
6900
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

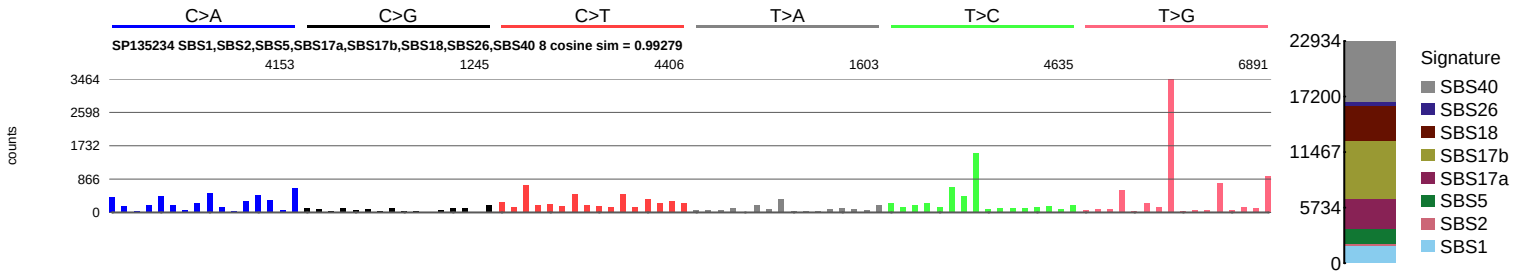

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.9928
4097
1247
4090
2038
4646
6820
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

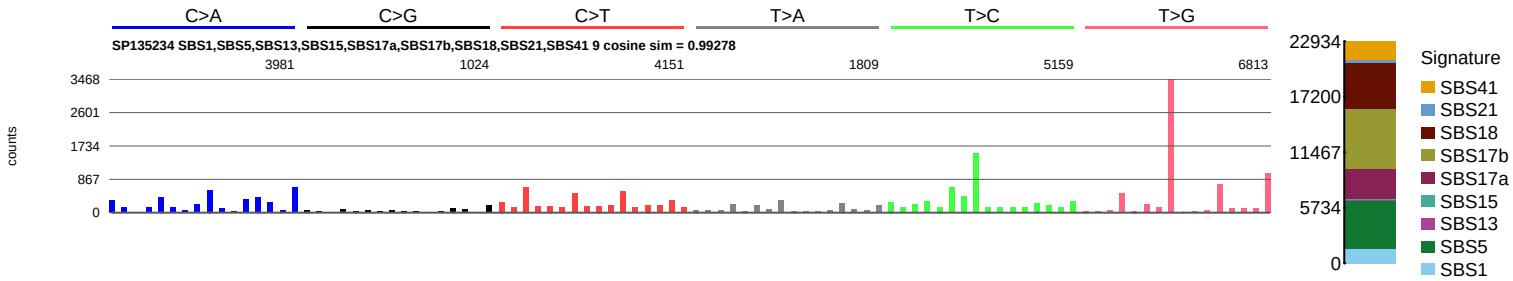

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.9928
4098
1246
4086
2039
4649
6820
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

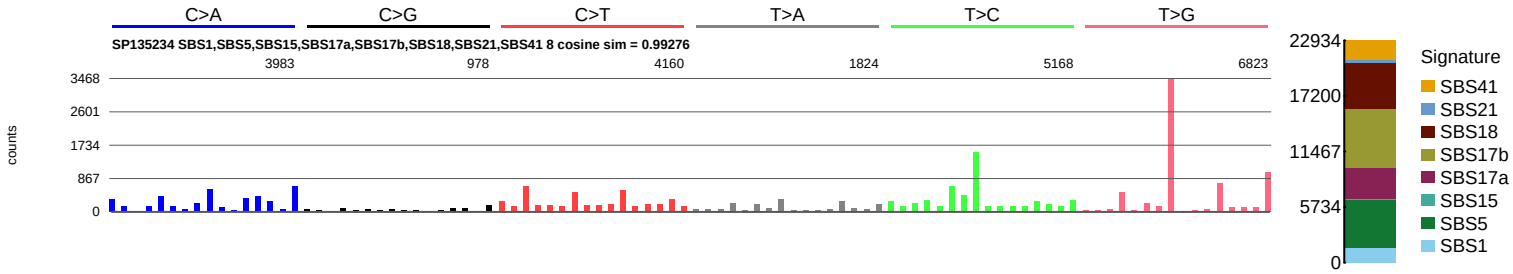

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.9928
4145
1253
4542
1620
4493
6881
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

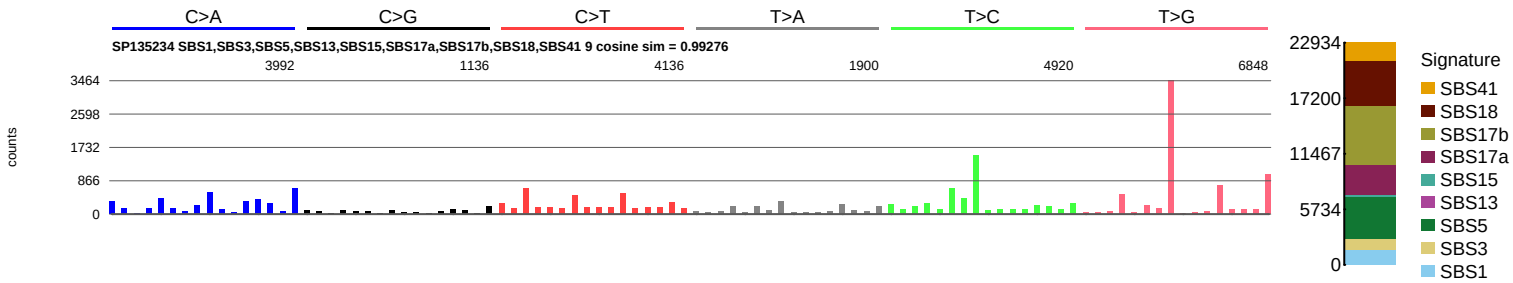

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.9928
4159
1233
4549
1621
4497
6871
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

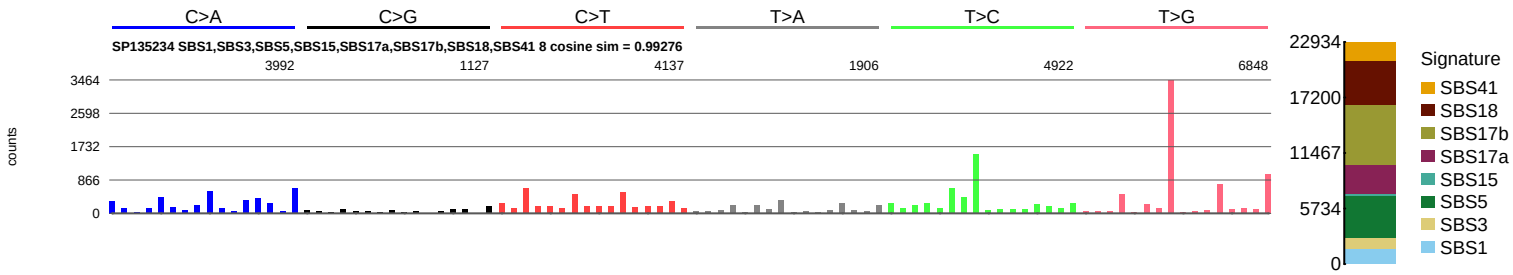

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99279
4184
1253
4352
1602
4625
6924
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

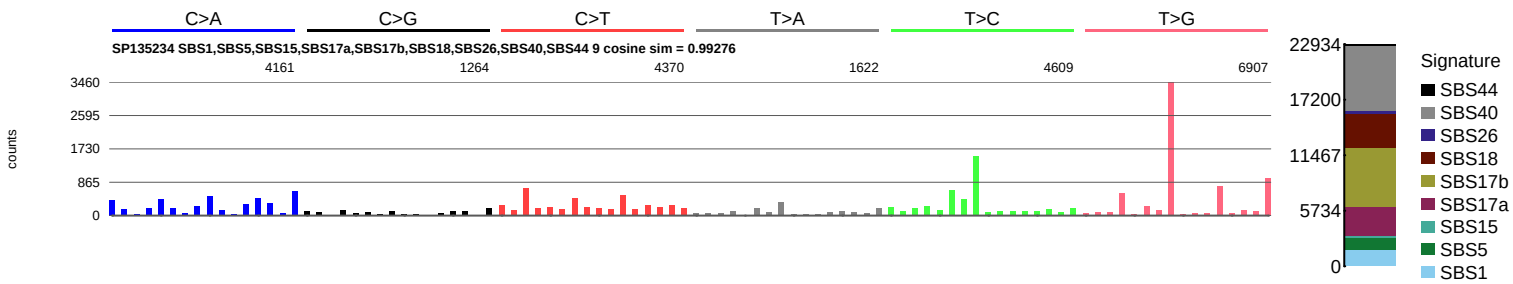

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99279
4153
1245
4406
1603
4635
6891
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

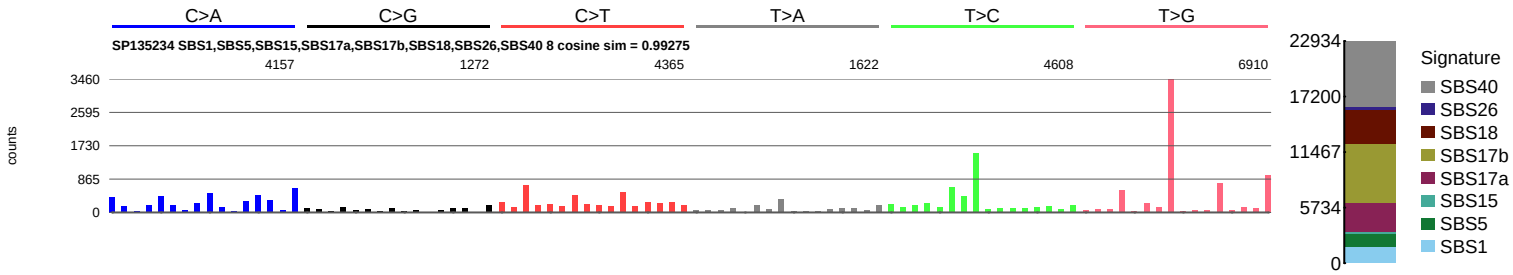

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99278
3981
1024
4151
1809
5159
6813
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

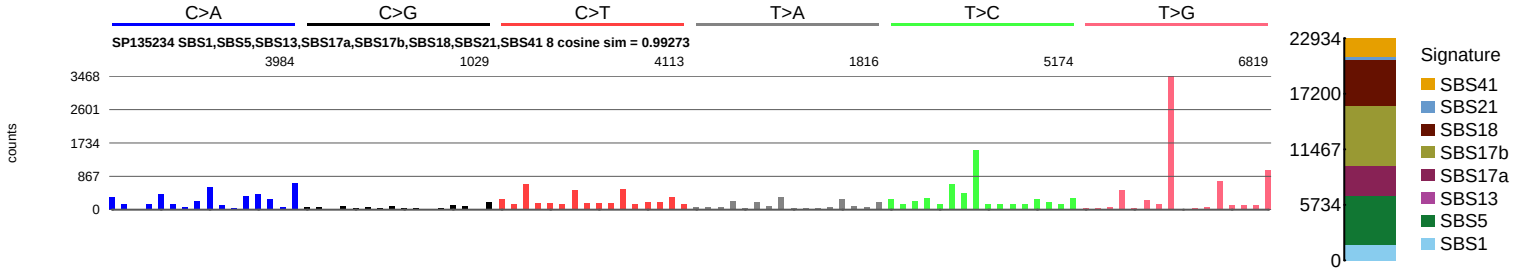

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99276
3983
978
4160
1824
5168
6823
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

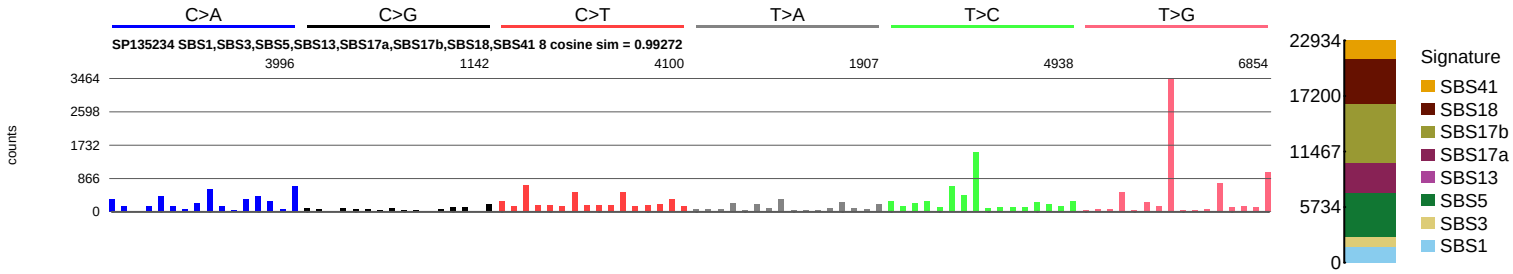

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99276
3992
1136
4136
1900
4920
6848
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

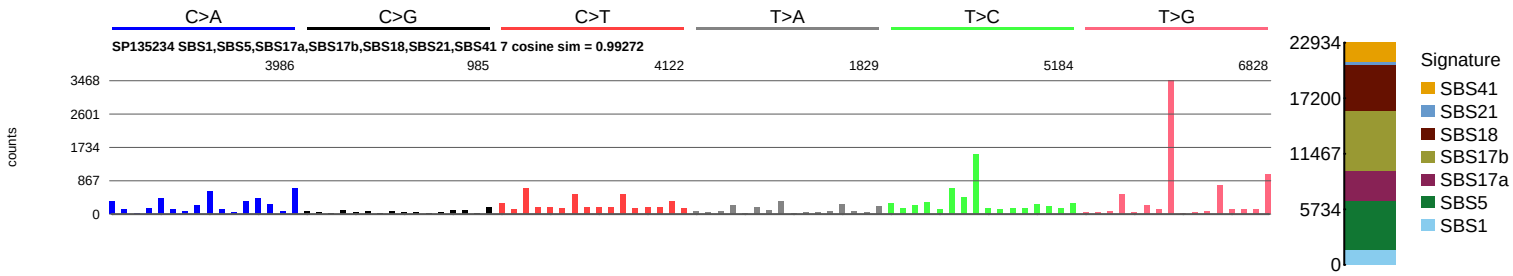

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99276
3992
1127
4137
1906
4922
6848
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

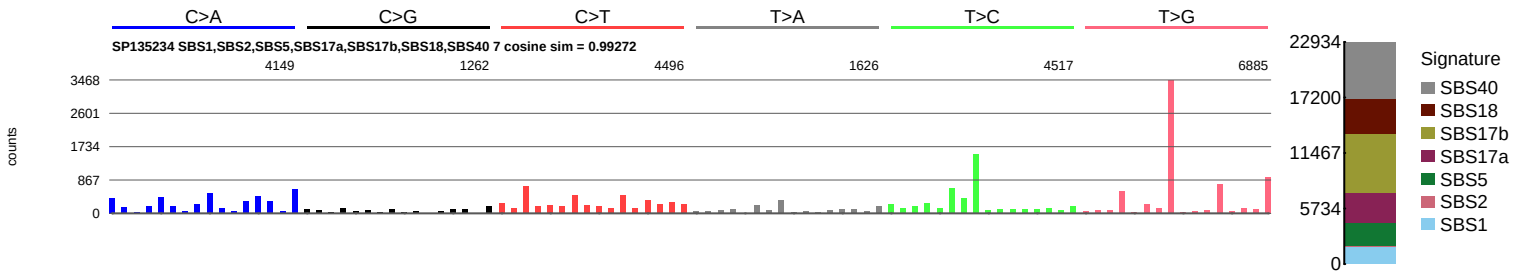

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.99276
4161
1264
4370
1622
4609
6907
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

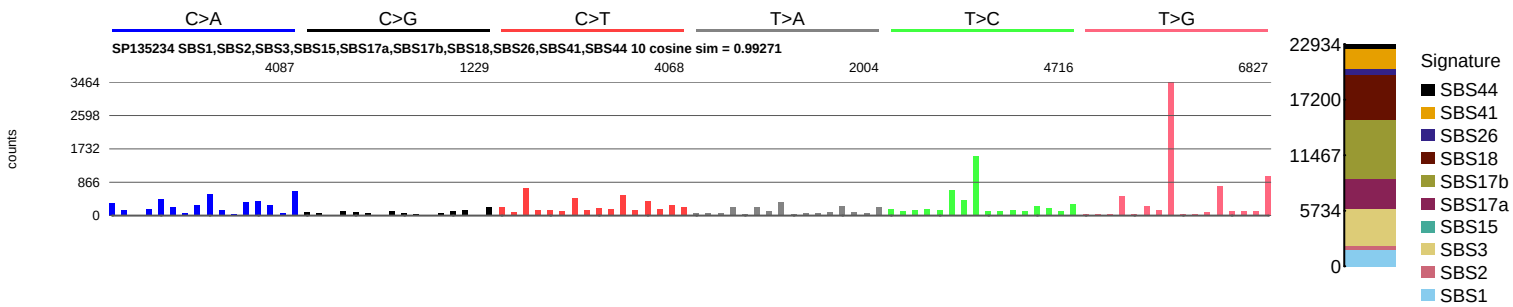

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99275
4157
1272
4365
1622
4608
6910
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

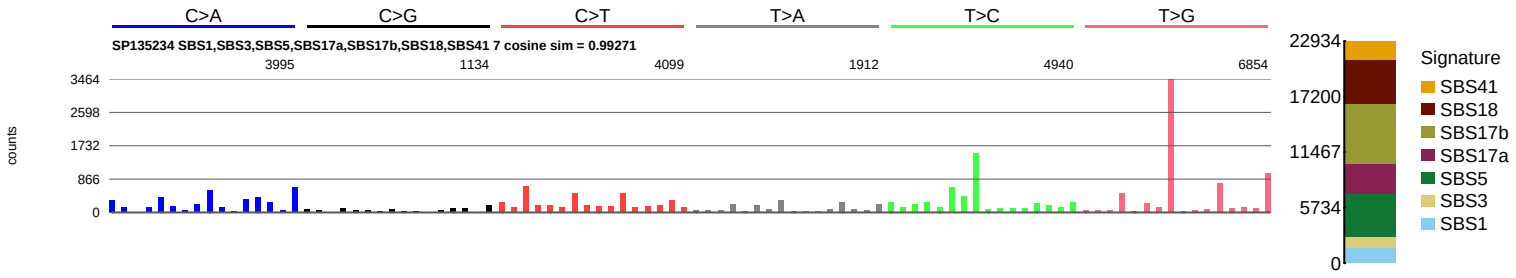

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99273
3984
1029
4113
1816
5174
6819
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99272
3996
1142
4100
1907
4938
6854
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99272
3986
985
4122
1829
5184
6828
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.99272
4149
1262
4496
1626
4517
6885
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.99271
4087
1229
4068
2004
4716
6827
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99271
3995
1134
4099
1912
4940
6854
counts
3464
2598
1732
866
0
22934
17200
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SBS41
SBS18
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SBS17a
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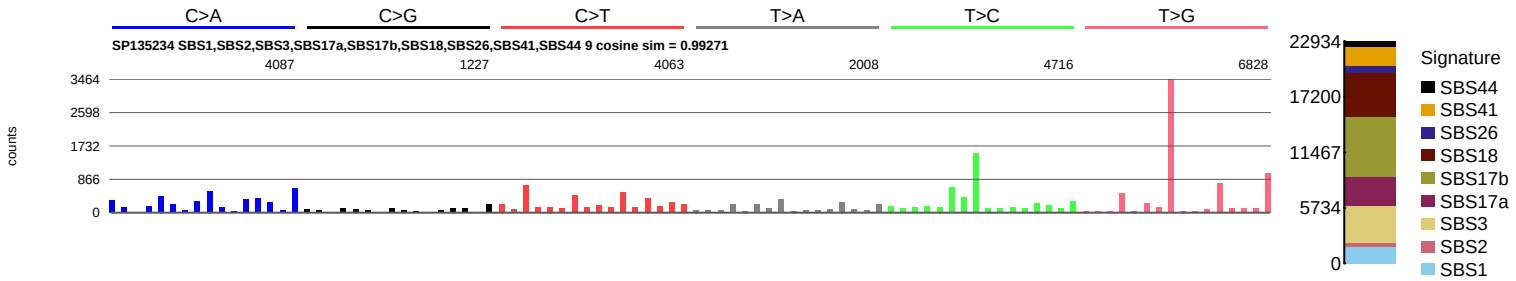
SBS3
SBS1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SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99271
4087
1227
4063
2008
4716
6828
counts
3464
2598
1732
866
0
22934
17200
11467
5734
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SBS41
SBS26
SBS18
SBS17b
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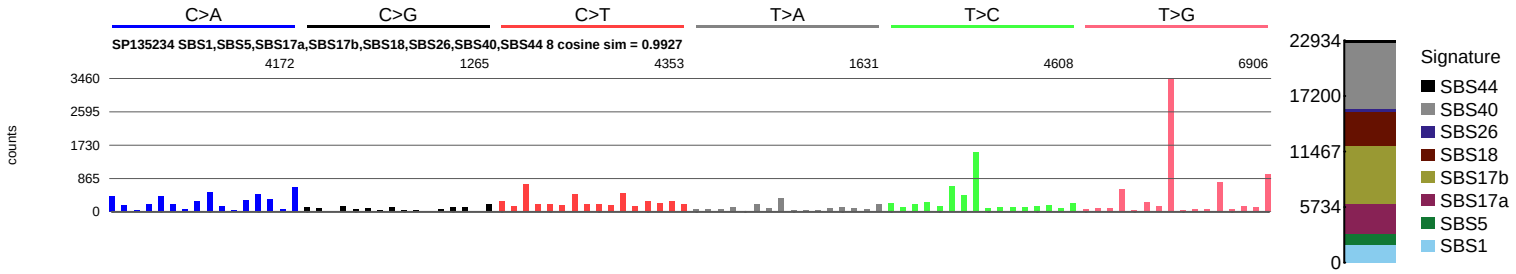
SBS3
SBS2
SBS1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C>G
C>T
T>A
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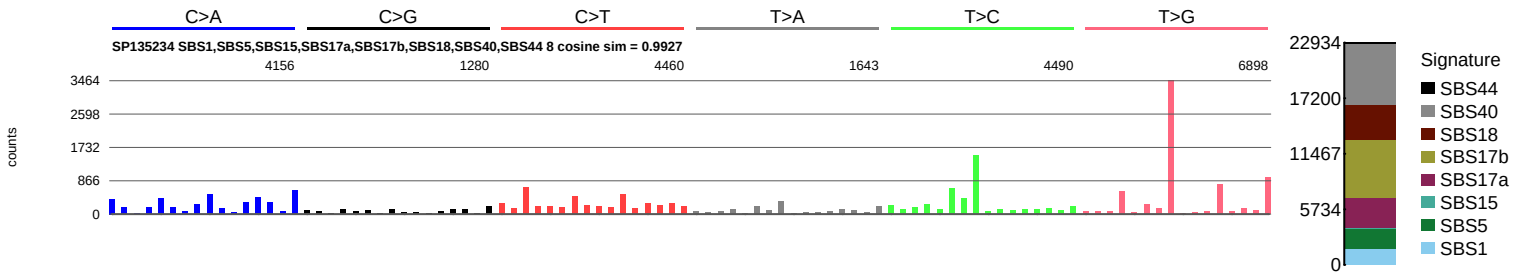
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.9927
4172
1265
4353
1631
4608
6906
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1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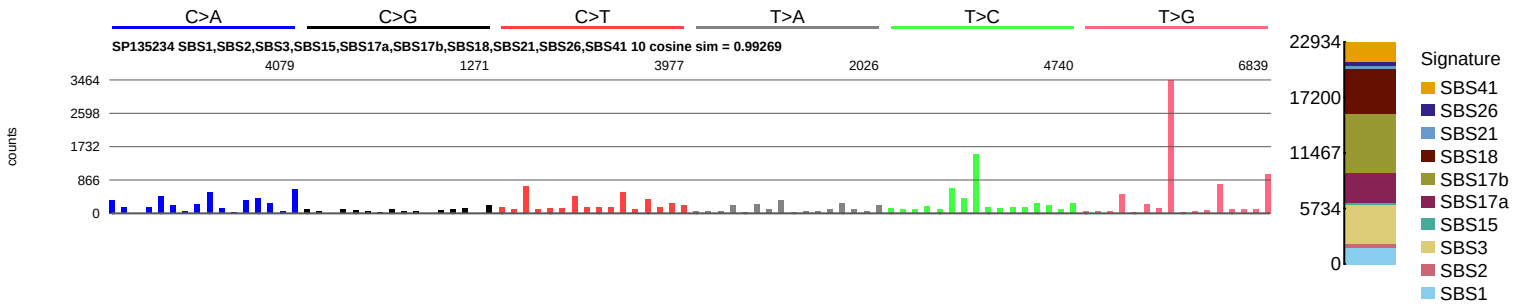
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.9927
4156
1280
4460
1643
4490
6898
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
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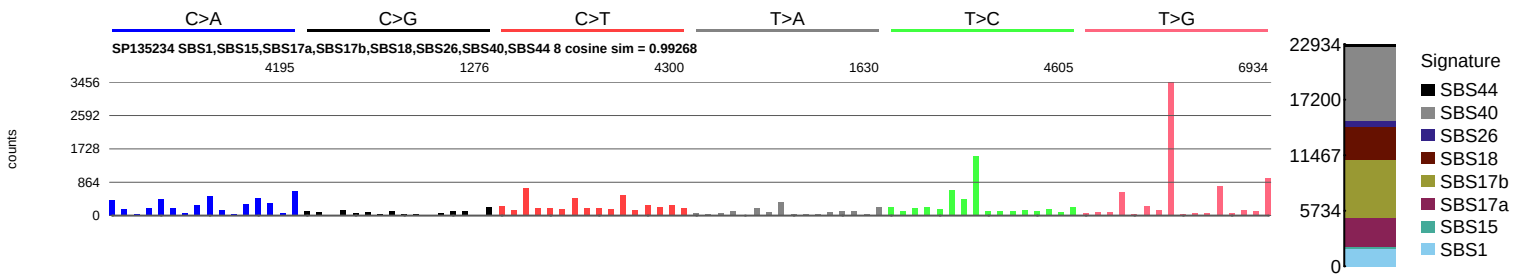
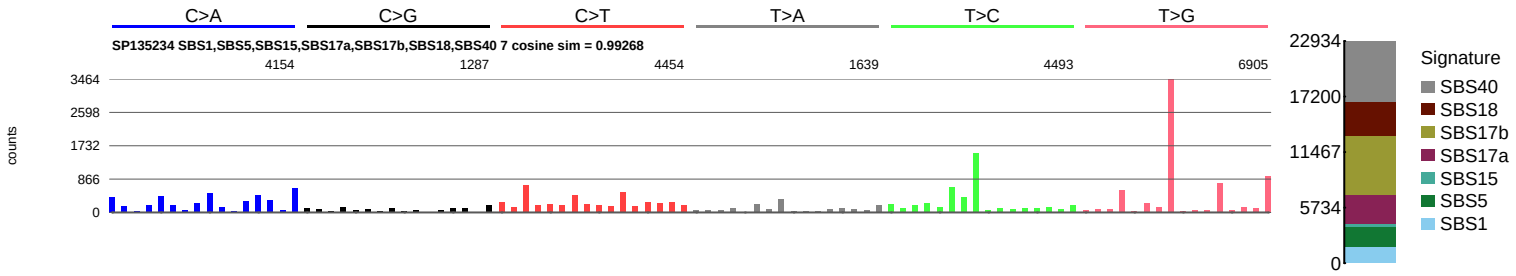
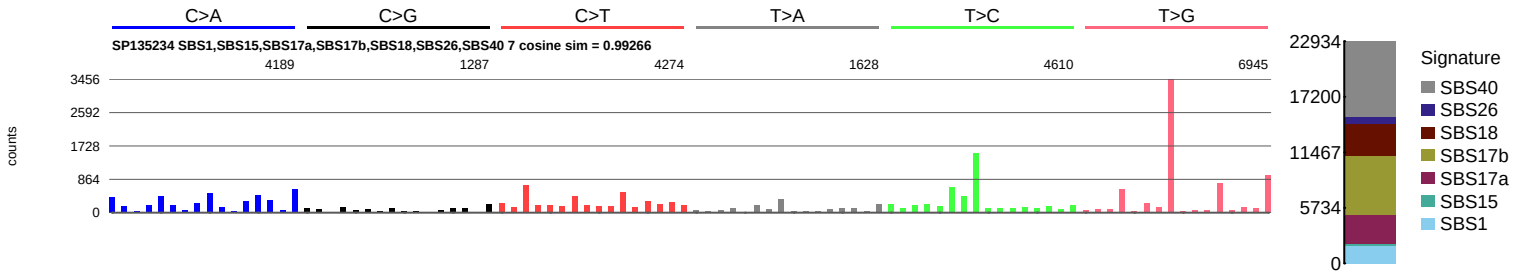
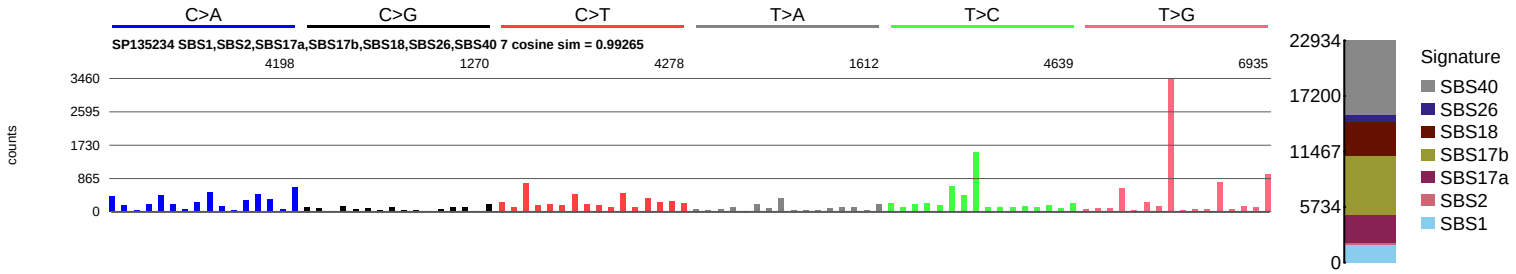
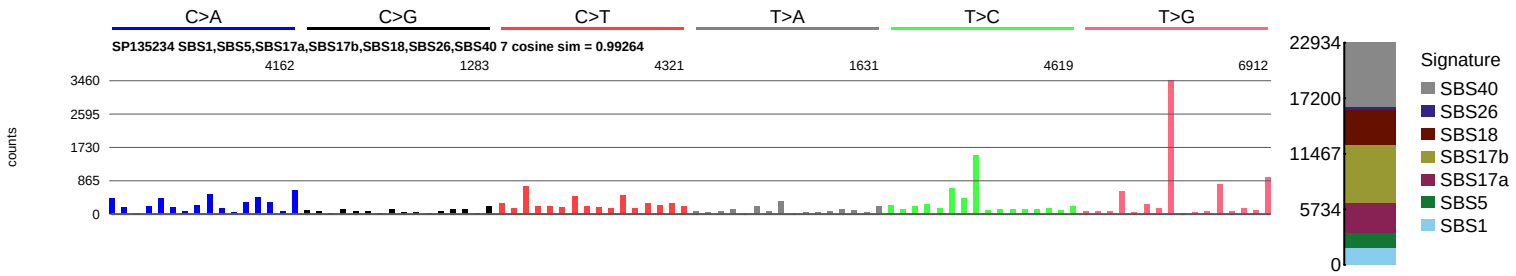
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99269
4079
1271
3977
2026
4740
6839
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

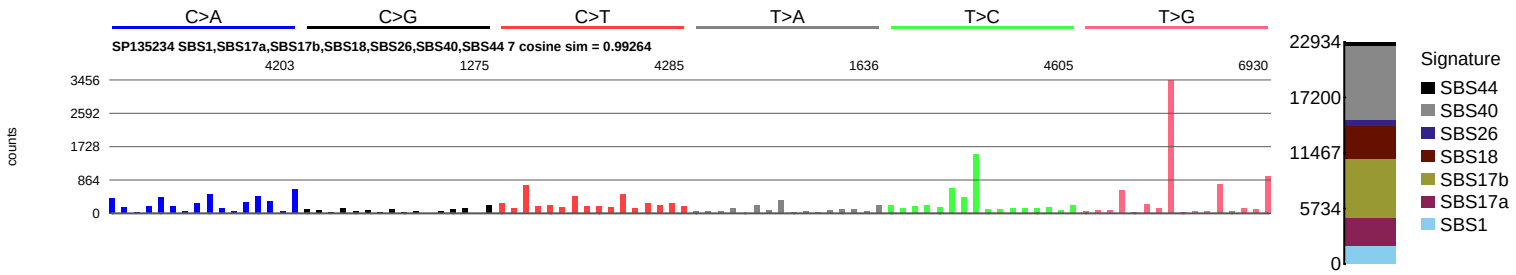

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.99268
4195
1276
4300
1630
4605
6934
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

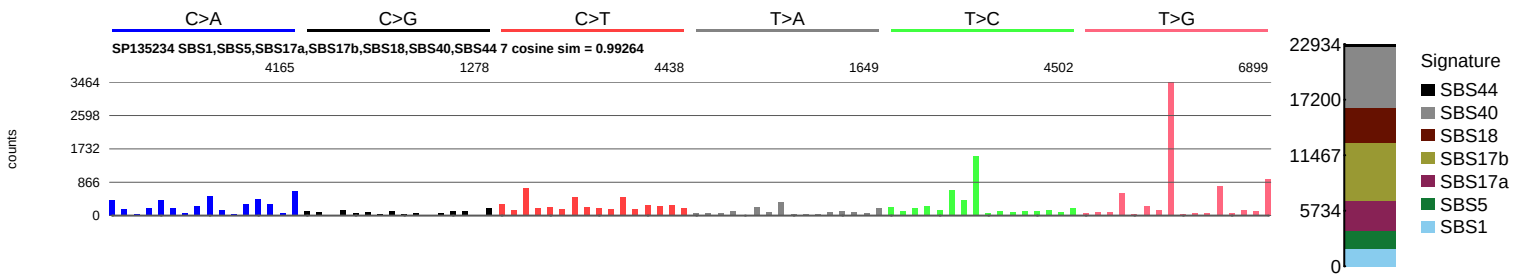

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.99268
4154
1287
4454
1639
4493
6905
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

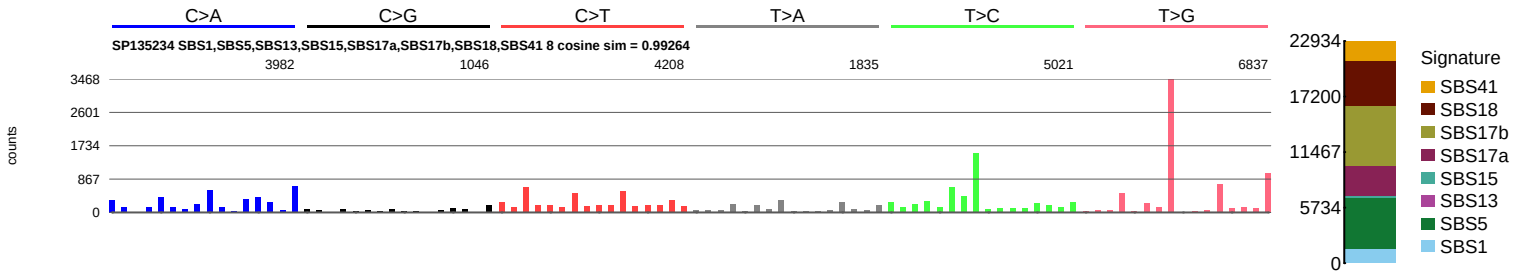

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.99266
4189
1287
4274
1628
4610
6945
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

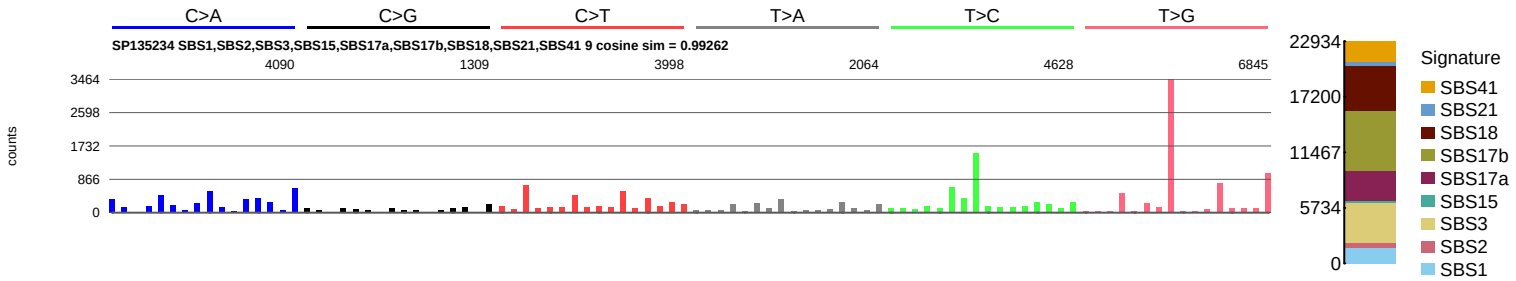

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.99265
4198
1270
4278
1612
4639
6935
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

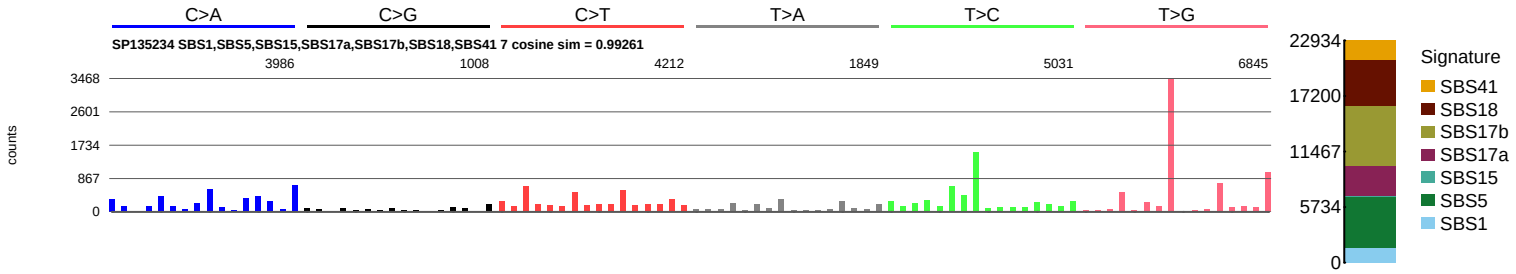

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.99264
4162
1283
4321
1631
4619
6912
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1

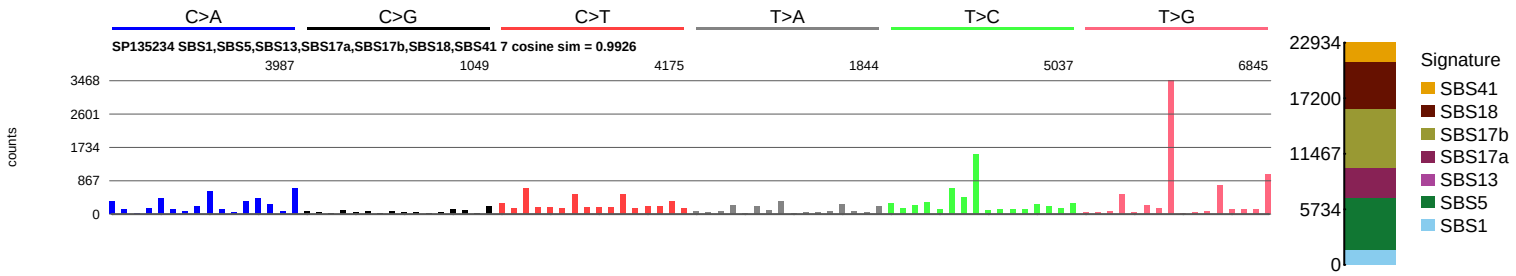

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.99264
4203
1275
4285
1636
4605
6930
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

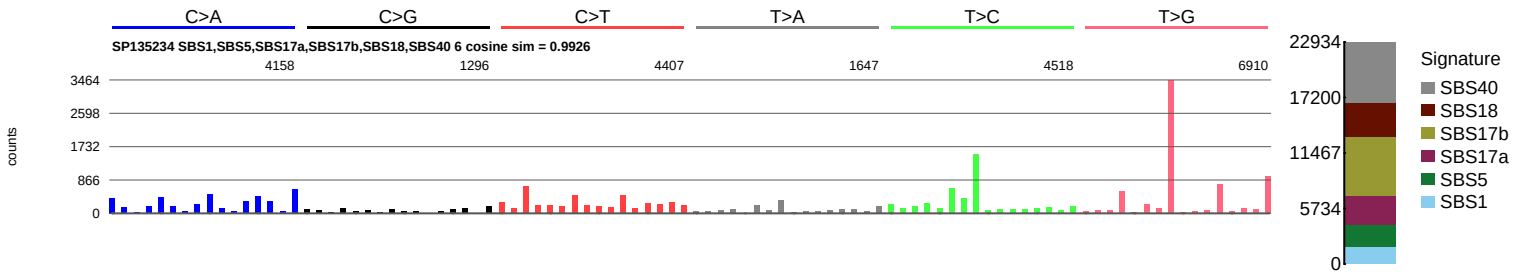

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.99264
4165
1278
4438
1649
4502
6899
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

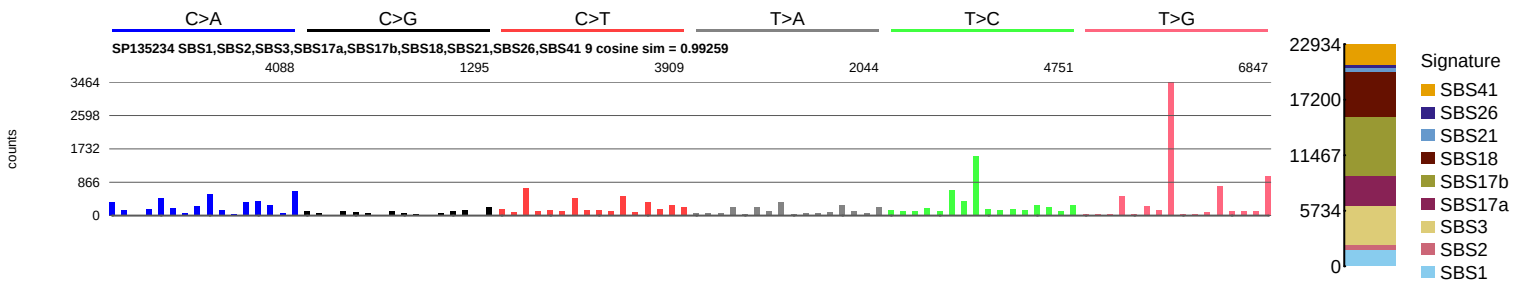

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99264
3982
1046
4208
1835
5021
6837
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

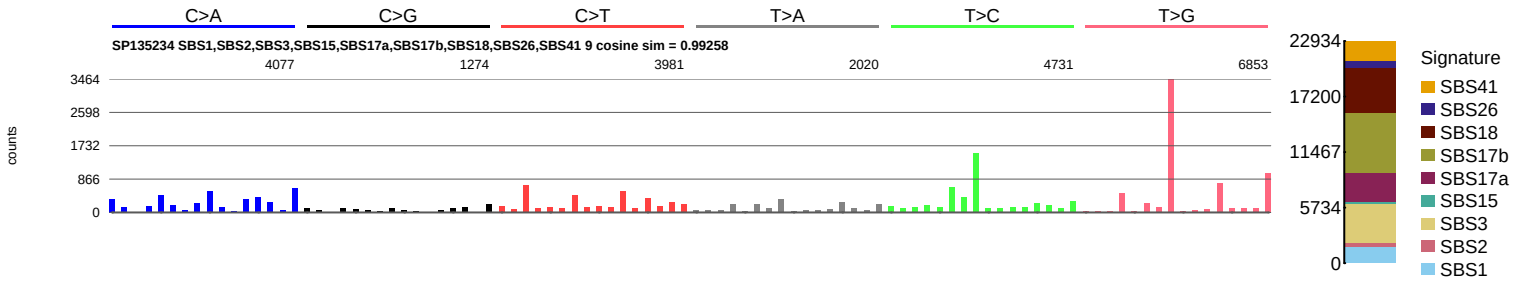

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99262
4090
1309
3998
2064
4628
6845
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

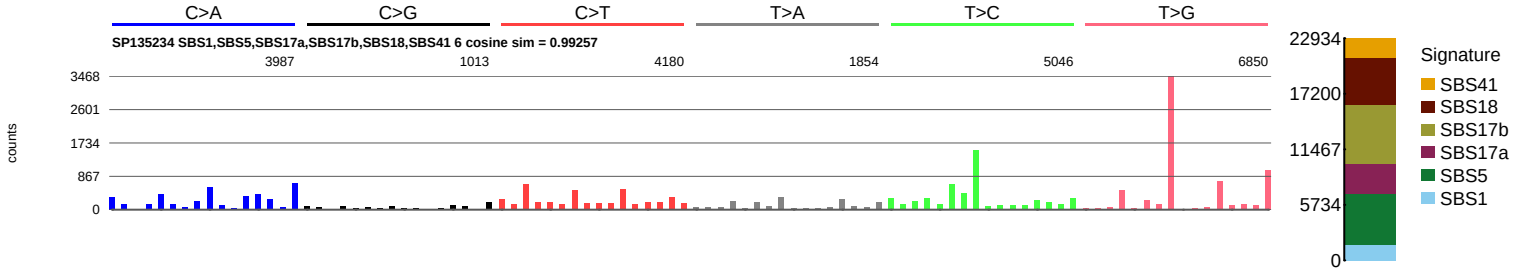

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99261
3986
1008
4212
1849
5031
6845
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

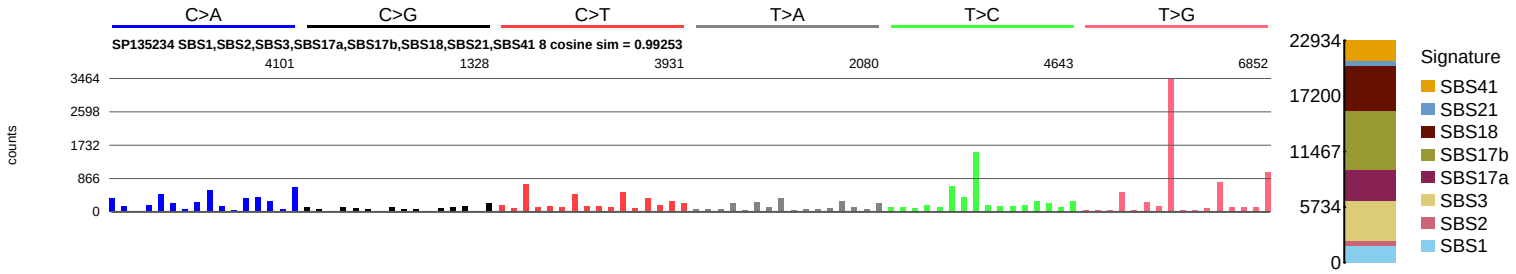

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9926
3987
1049
4175
1844
5037
6845
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.9926
4158
1296
4407
1647
4518
6910
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99259
4088
1295
3909
2044
4751
6847
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99258
4077
1274
3981
2020
4731
6853
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.99257
3987
1013
4180
1854
5046
6850
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99253
4101
1328
3931
2080
4643
6852
counts
3464
2598
1732
866
0
22934
17200
11467
5734
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Signature
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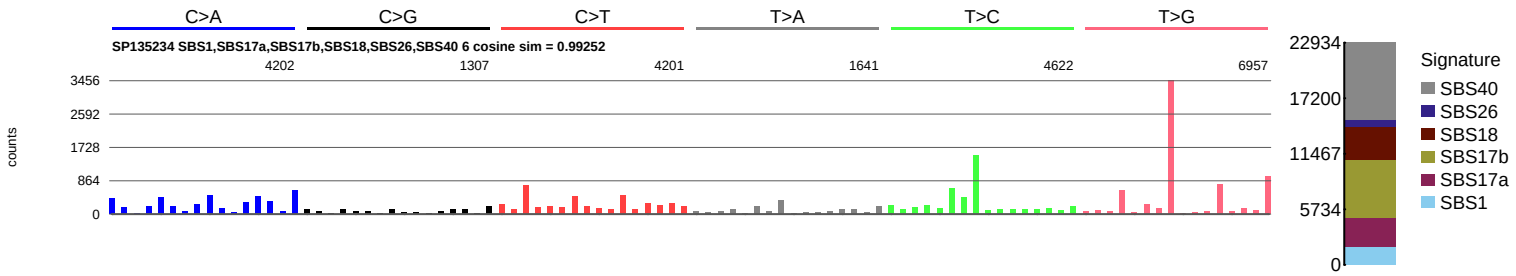
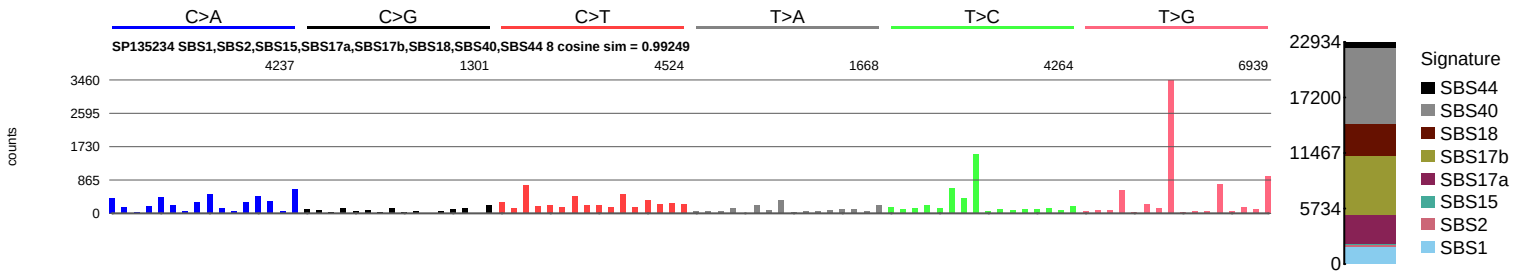
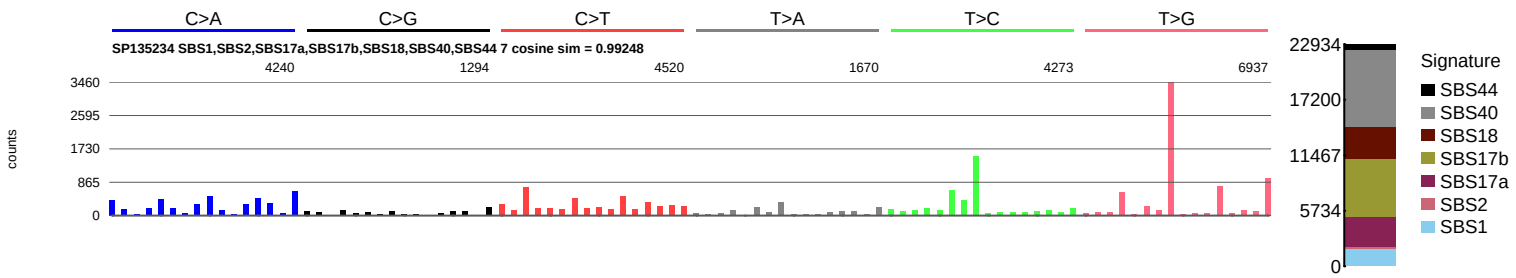
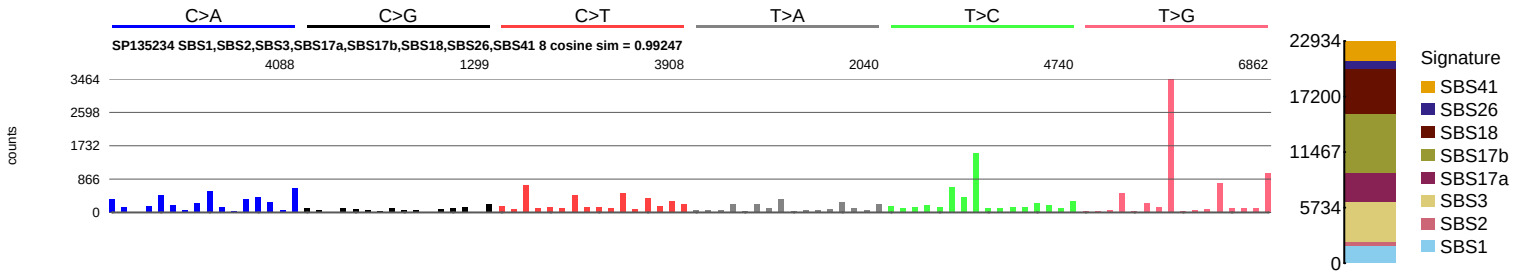
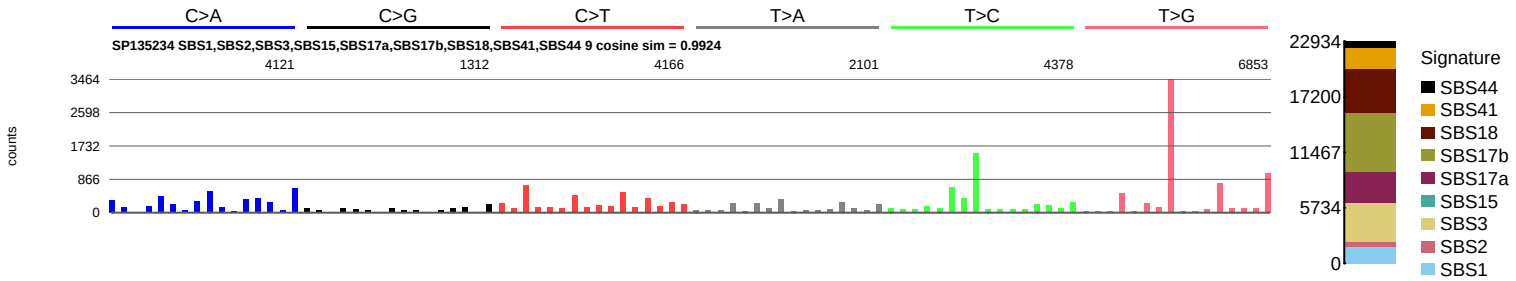
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.99252
4202
1307
4201
1641
4622
6957
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.99249
4237
1301
4524
1668
4264
6939
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.99248
4240
1294
4520
1670
4273
6937
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99247
4088
1299
3908
2040
4740
6862
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.9924
4121
1312
4166
2101
4378
6853
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
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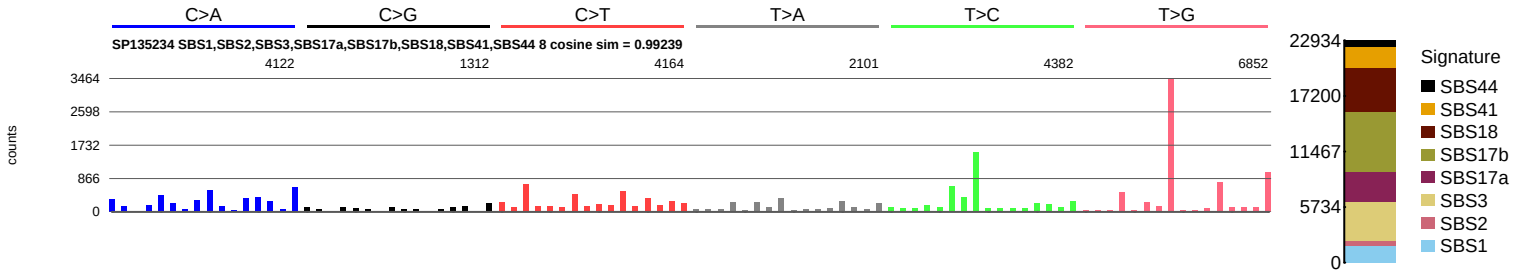
SBS1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SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99239
4122
1312
4164
2101
4382
6852
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
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SBS2
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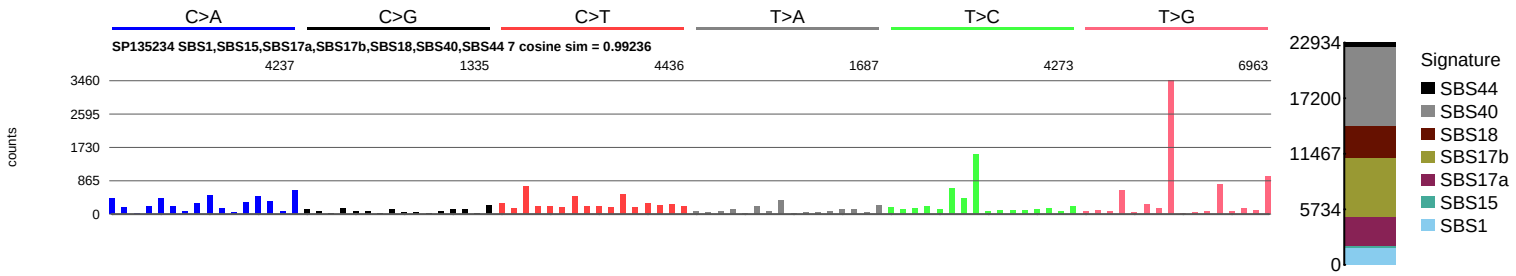
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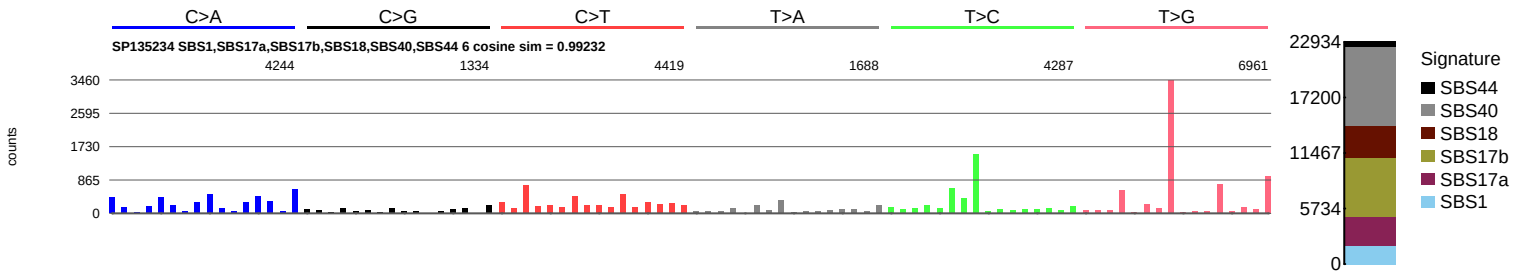
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.99236
4237
1335
4436
1687
4273
6963
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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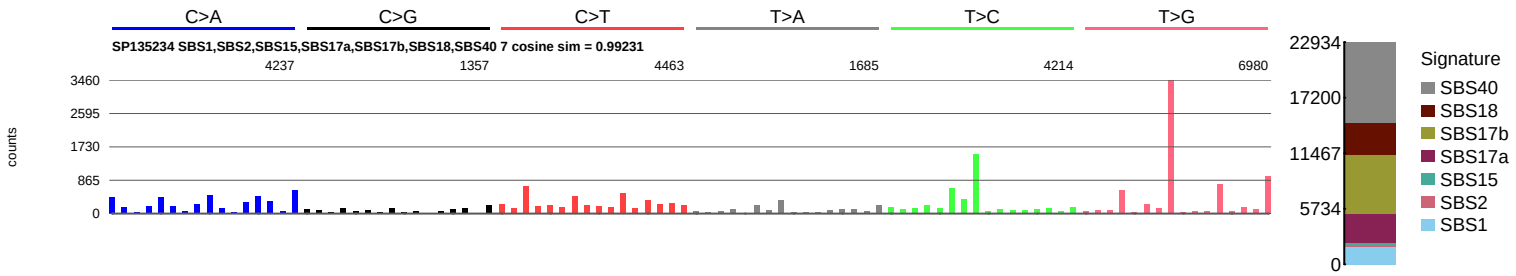
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.99232
4244
1334
4419
1688
4287
6961
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS1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C>G
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SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.99231
4237
1357
4463
1685
4214
6980
counts
3460
2595
1730
865
0
22934
17200
11467
5734
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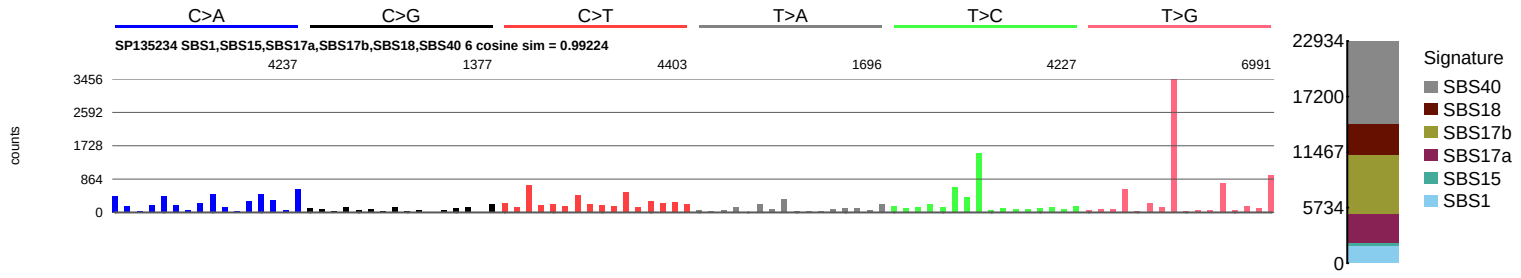
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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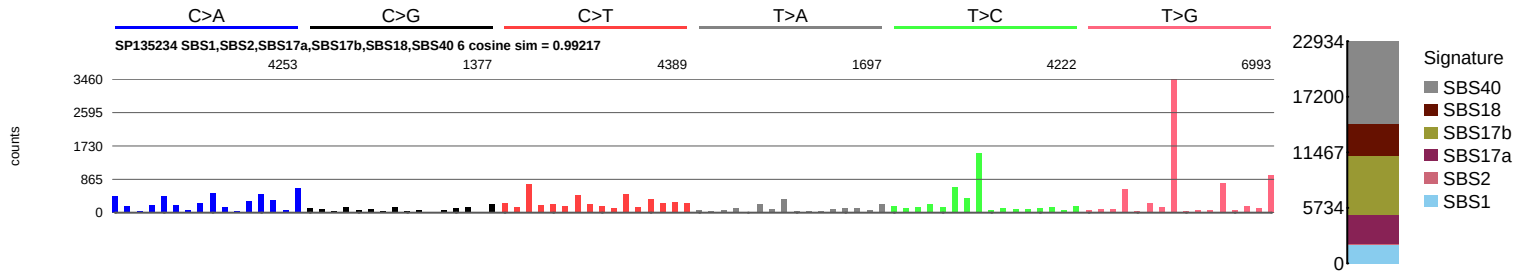
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.99224
4237
1377
4403
1696
4227
6991
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
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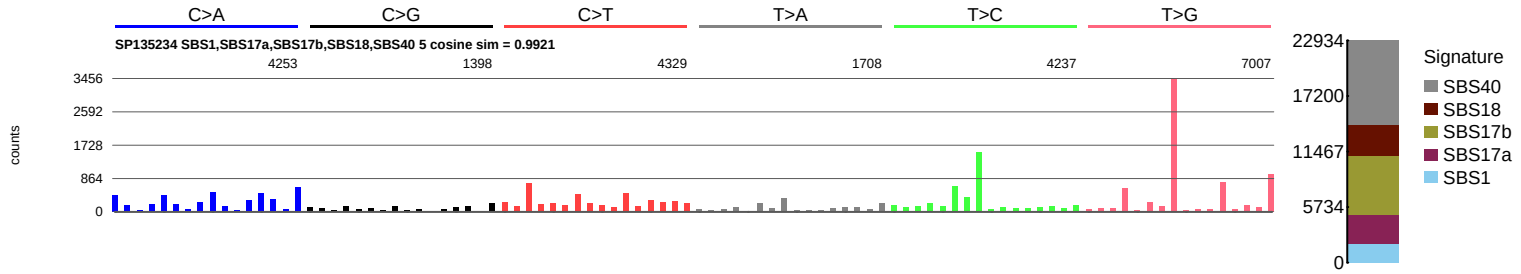
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.99217
4253
1377
4389
1697
4222
6993
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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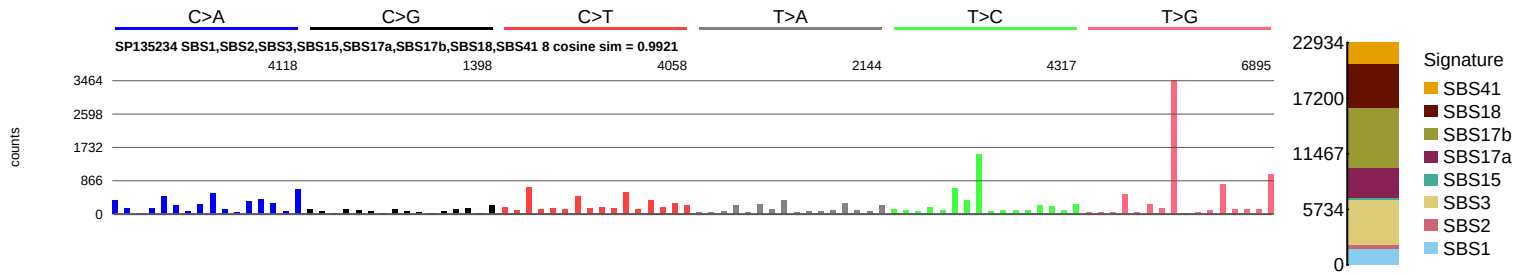
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.9921
4253
1398
4329
1708
4237
7007
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS1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C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9921
4118
1398
4058
2144
4317
6895
counts
3464
2598
1732
866
0
22934
17200
11467
5734
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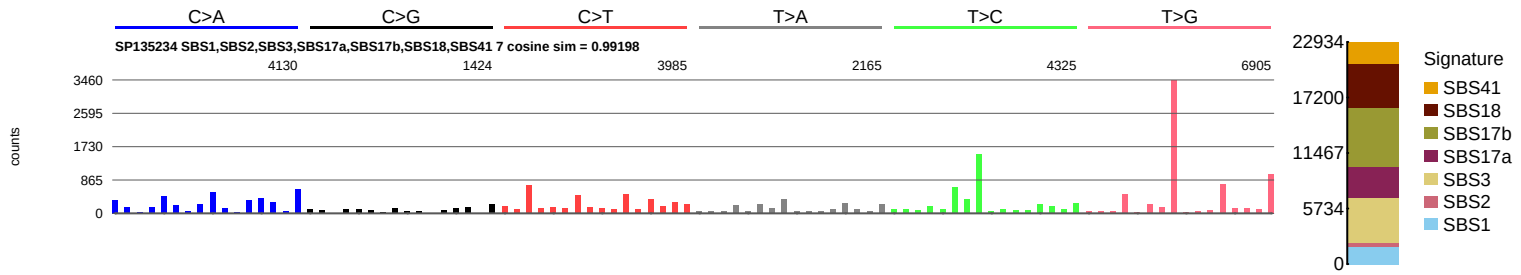
SBS2
SBS1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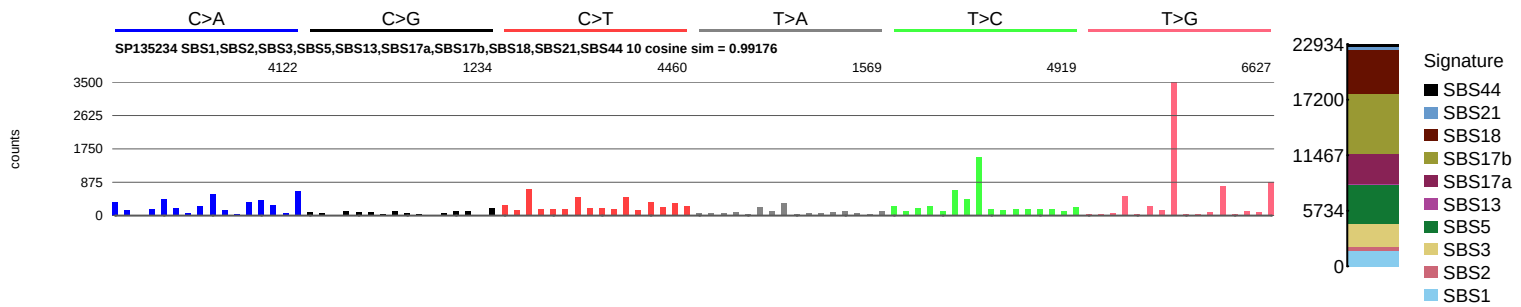
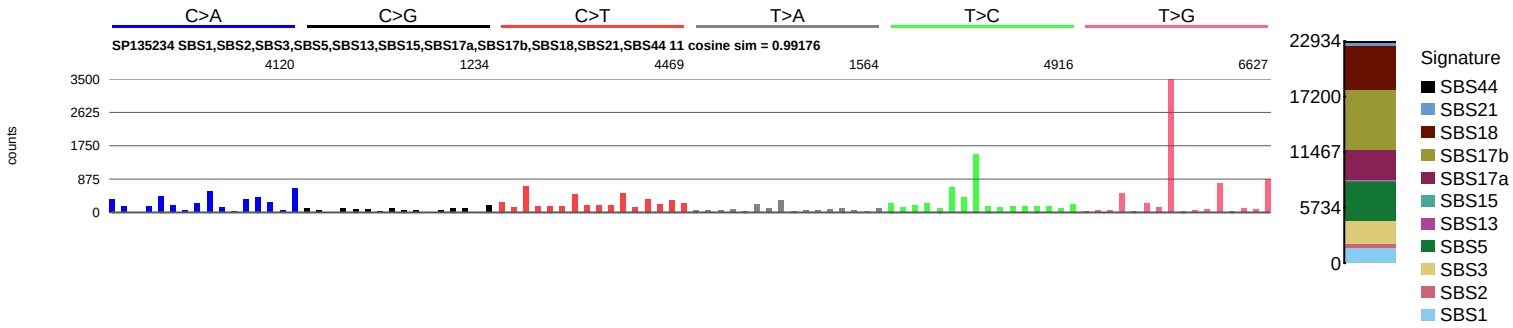
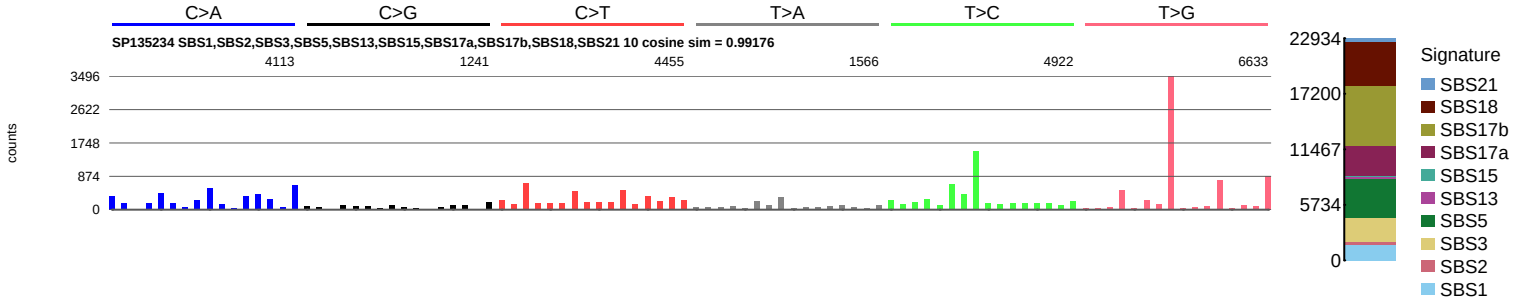
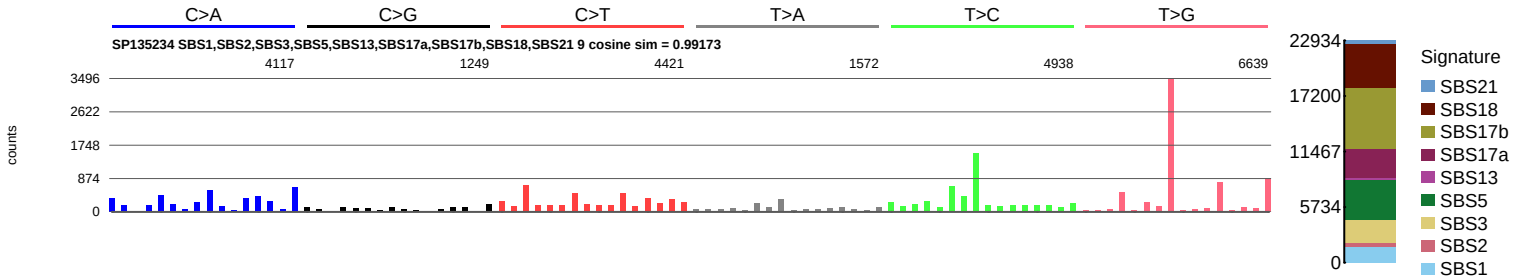
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99198
4130
1424
3985
2165
4325
6905
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.99176
4122
1234
4460
1569
4919
6627
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.99176
4120
1234
4469
1564
4916
6627
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.99176
4113
1241
4455
1566
4922
6633
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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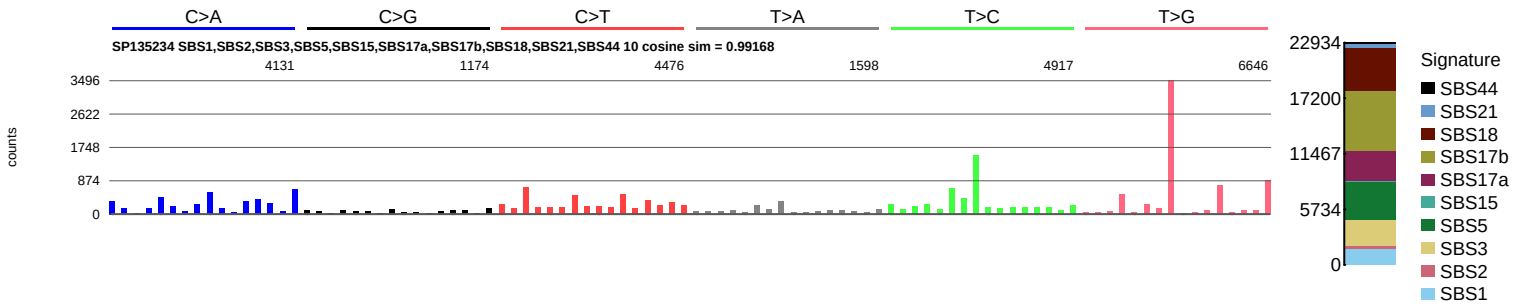
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.99173
4117
1249
4421
1572
4938
6639
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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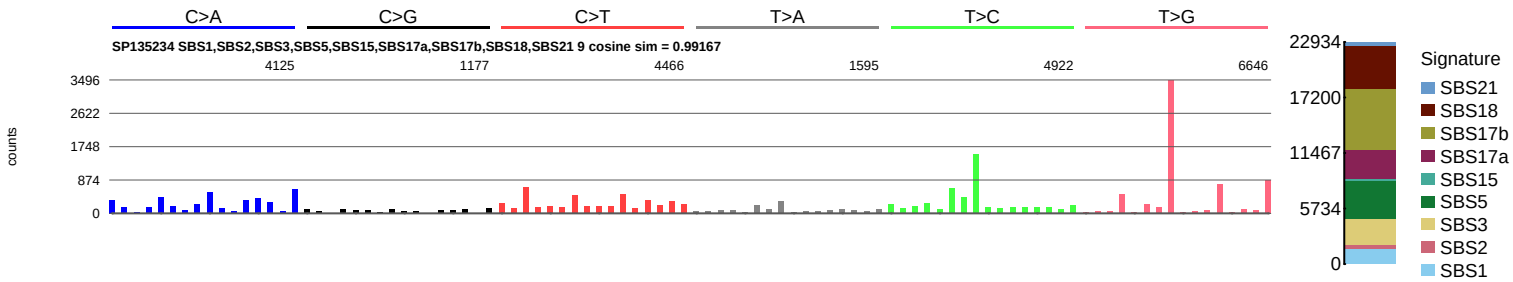
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.99168
4131
1174
4476
1598
4917
6646
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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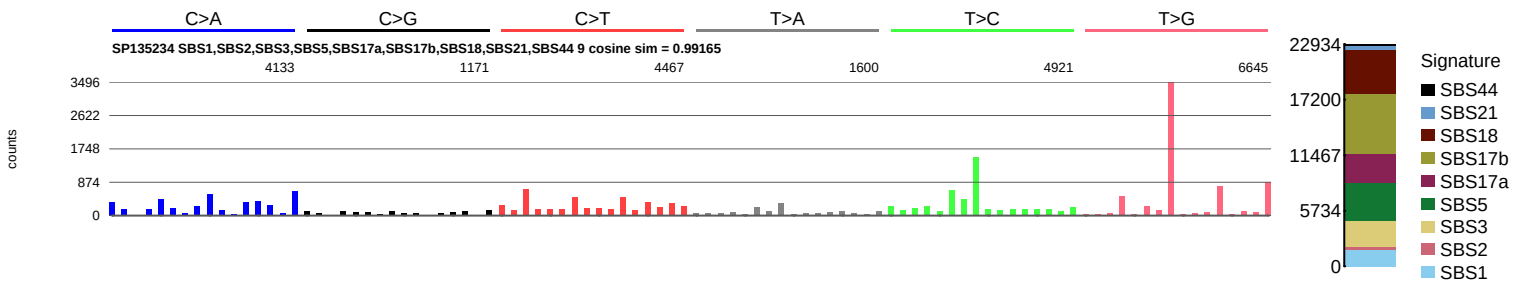
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.99167
4125
1177
4466
1595
4922
6646
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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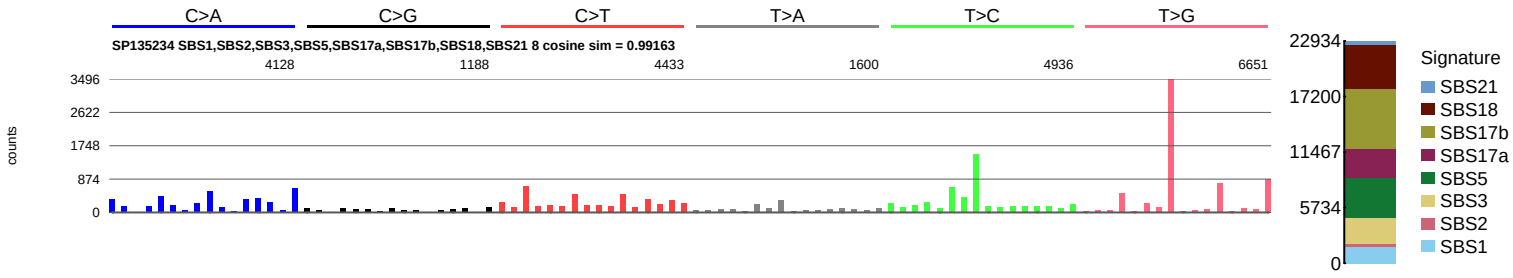
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.99165
4133
1171
4467
1600
4921
6645
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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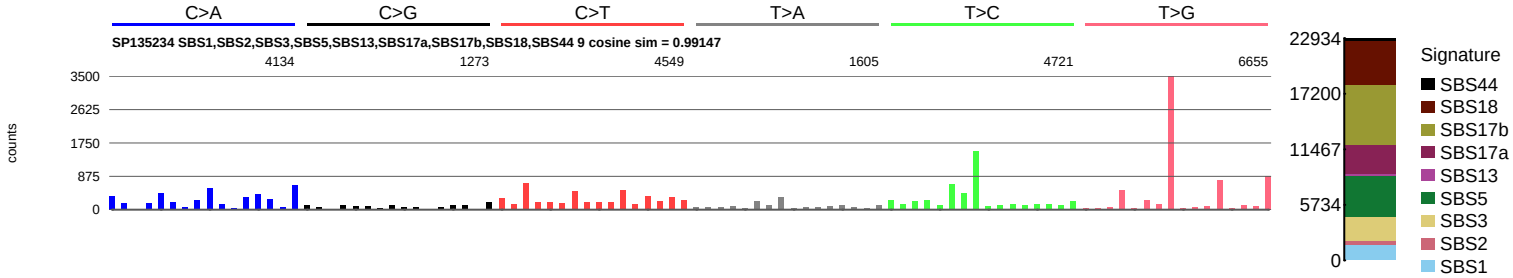
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.99163
4128
1188
4433
1600
4936
6651
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
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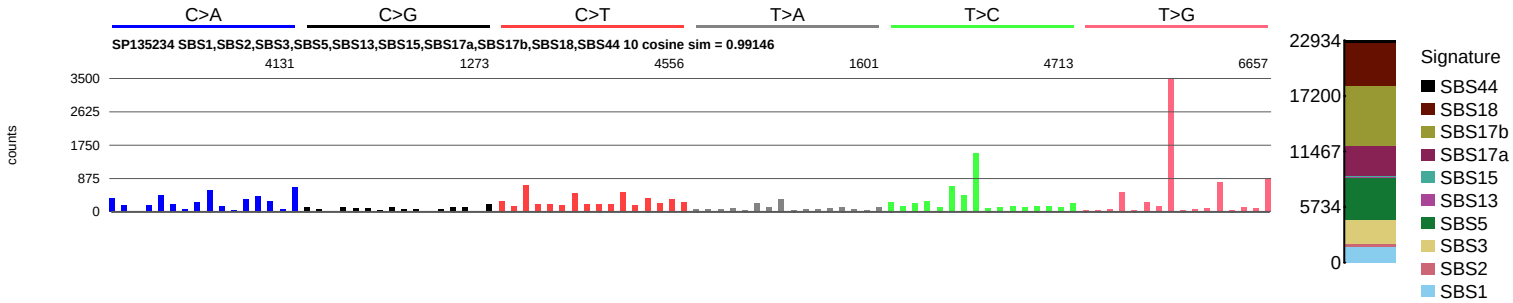
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.99147
4134
1273
4549
1605
4721
6655
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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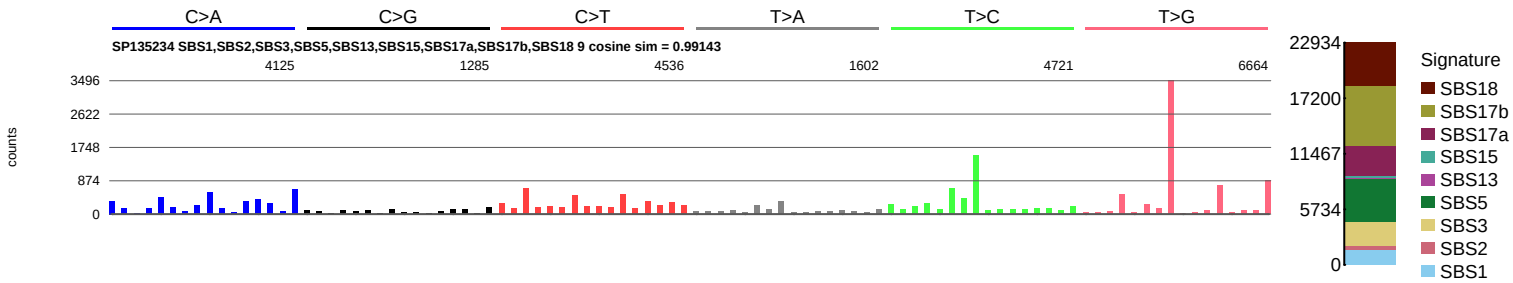
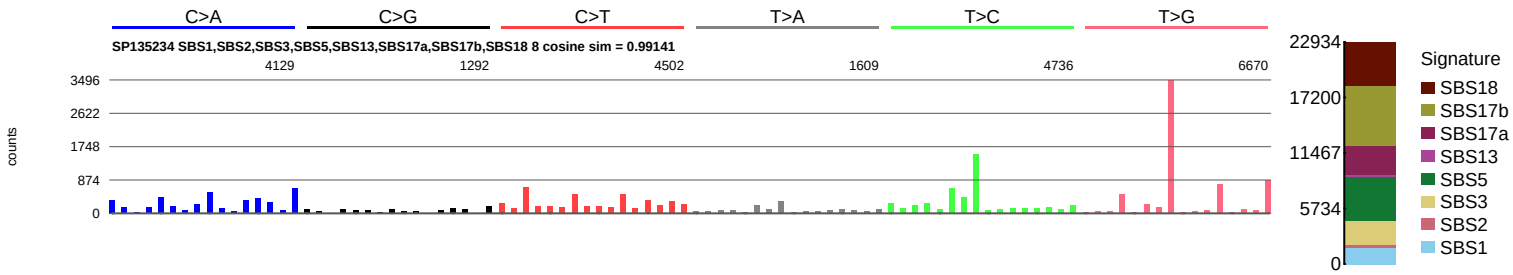
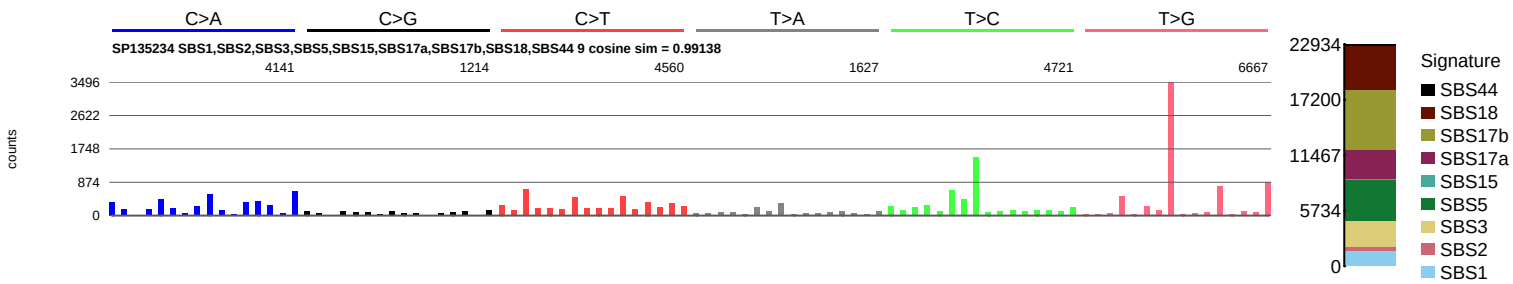
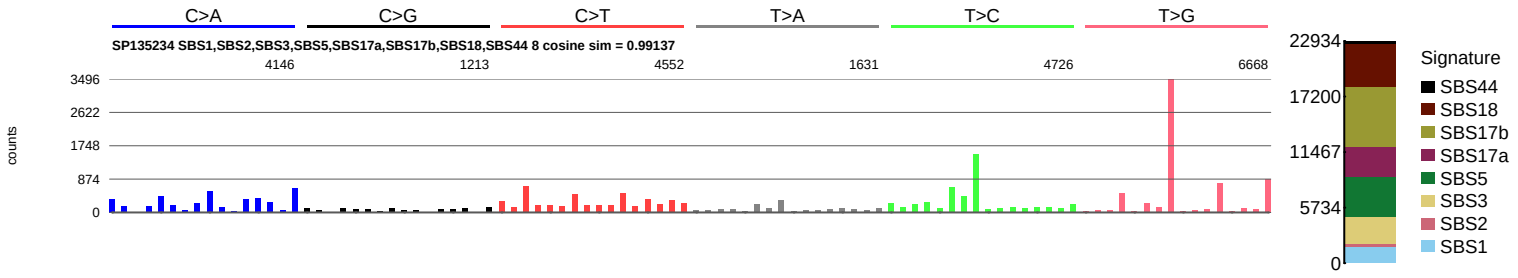
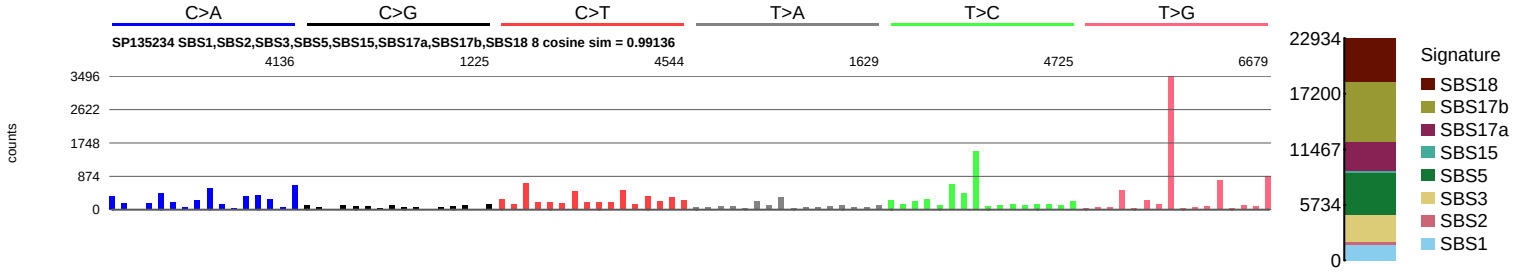
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.99146
4131
1273
4556
1601
4713
6657
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

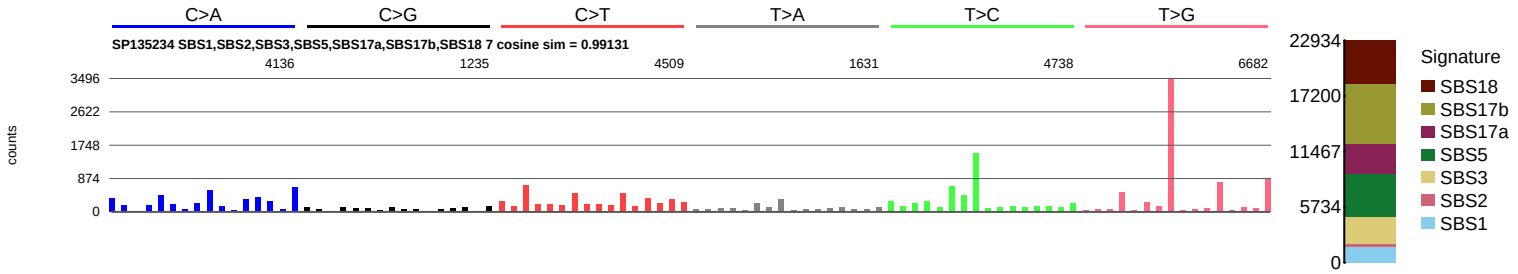

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.99143
4125
1285
4536
1602
4721
6664
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

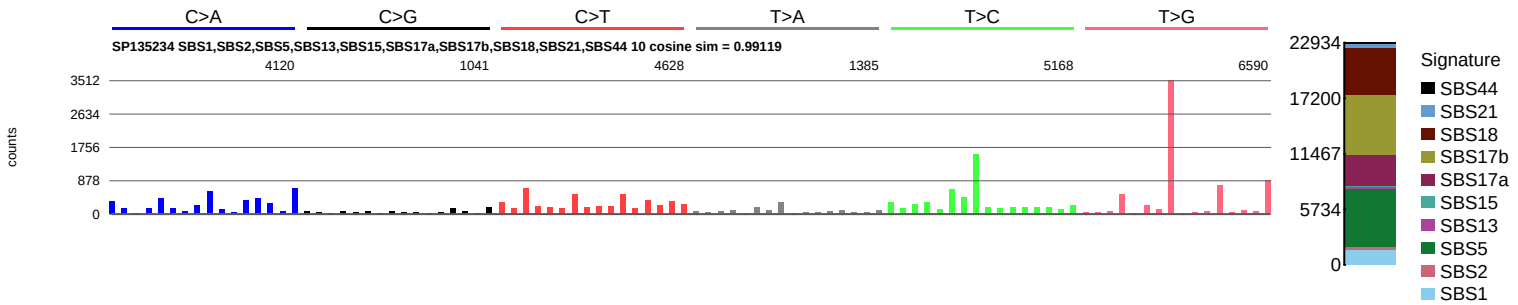

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.99141
4129
1292
4502
1609
4736
6670
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.99138
4141
1214
4560
1627
4721
6667
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.99137
4146
1213
4552
1631
4726
6668
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.99136
4136
1225
4544
1629
4725
6679
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.99131
4136
1235
4509
1631
4738
6682
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.99119
4120
1041
4628
1385
5168
6590
counts
3512
2634
1756
878
0
22934
17200
11467
5734
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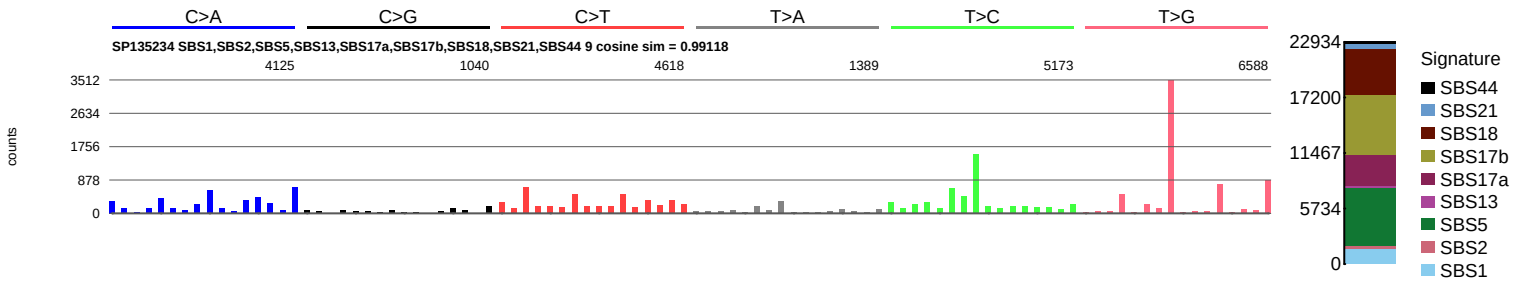
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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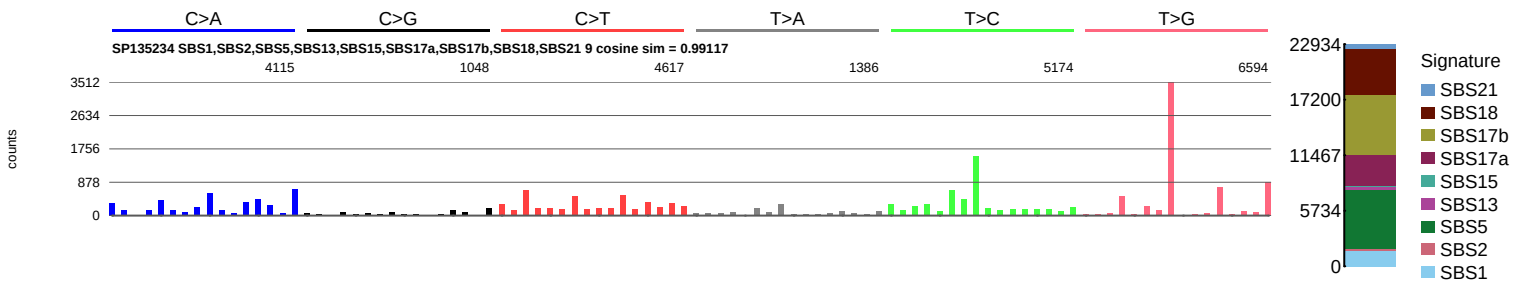
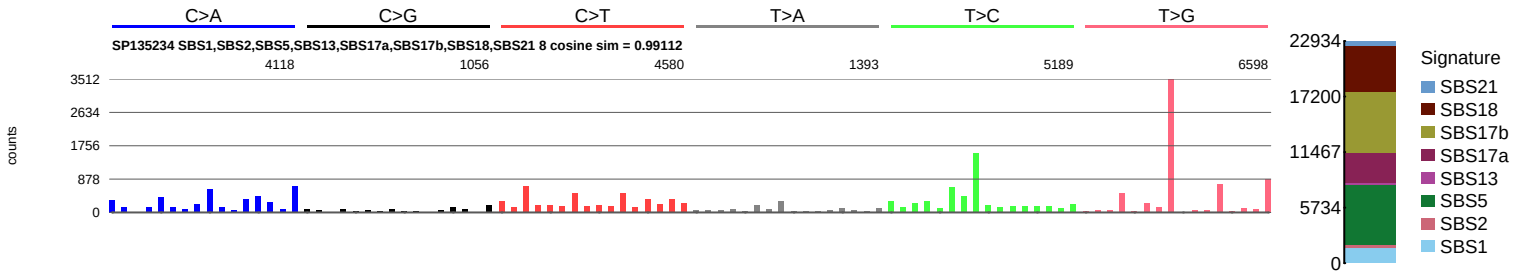
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.99118
4125
1040
4618
1389
5173
6588
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.99117
4115
1048
4617
1386
5174
6594
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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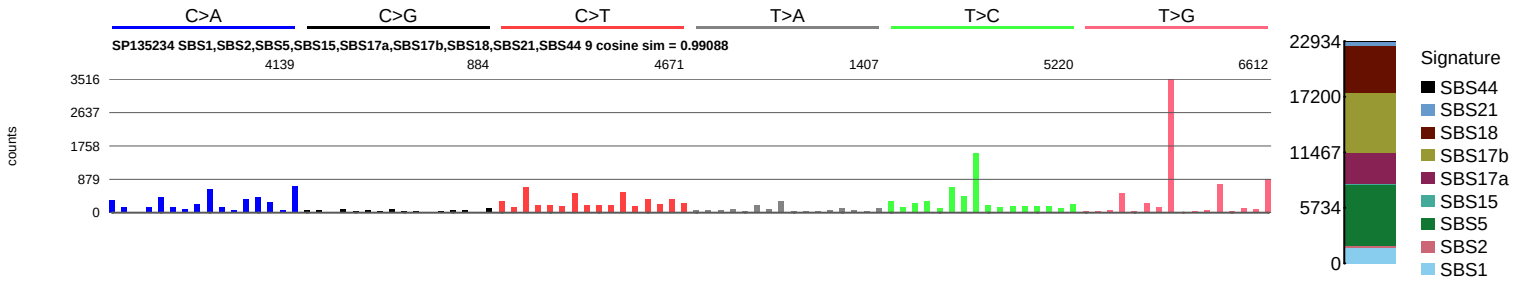
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.99112
4118
1056
4580
1393
5189
6598
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.99088
4139
884
4671
1407
5220
6612
counts
3516
2637
1758
879
0
22934
17200
11467
5734
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Signature
SBS44
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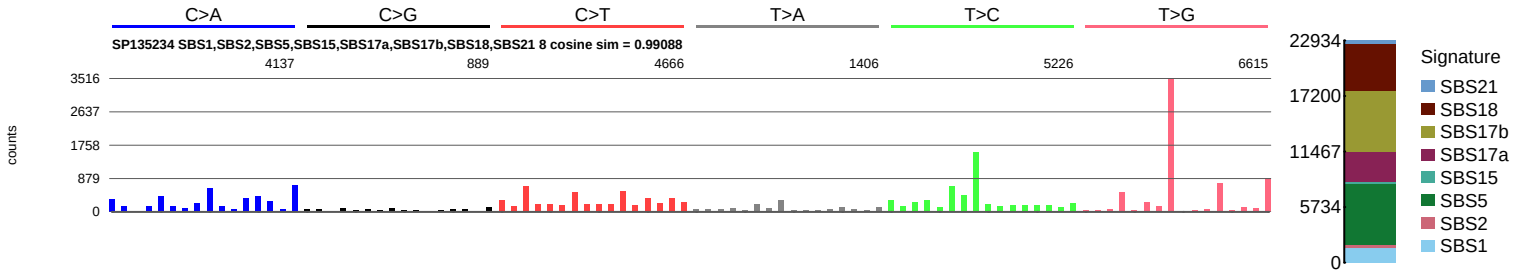
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.99088
4137
889
4666
1406
5226
6615
counts
3516
2637
1758
879
0
22934
17200
11467
5734
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Signature
SBS21
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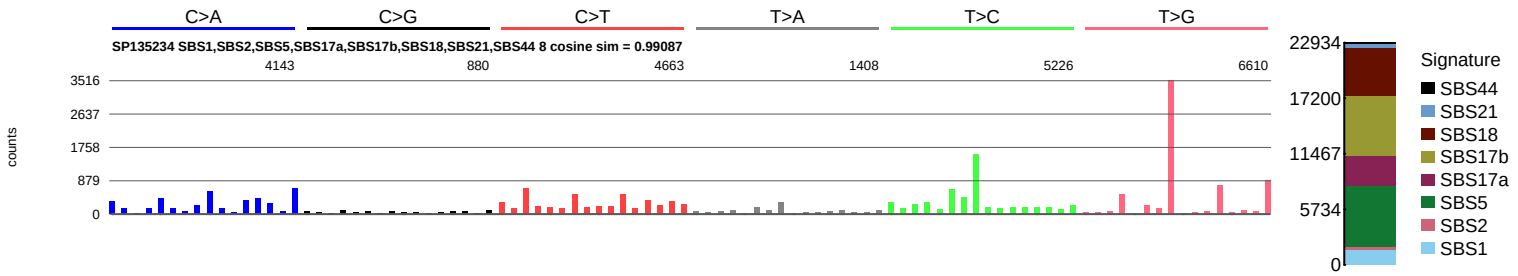
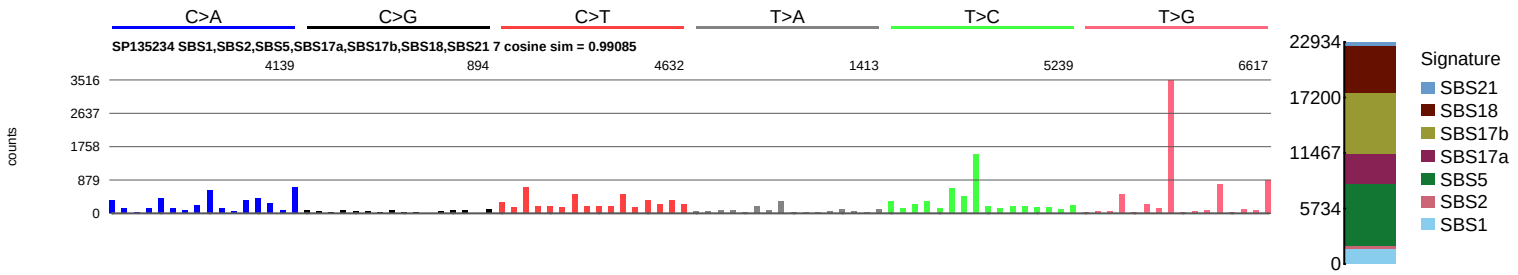
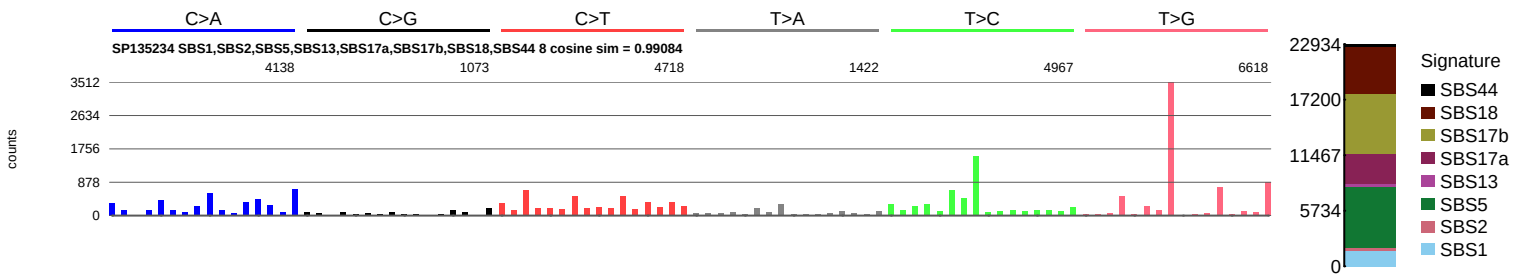
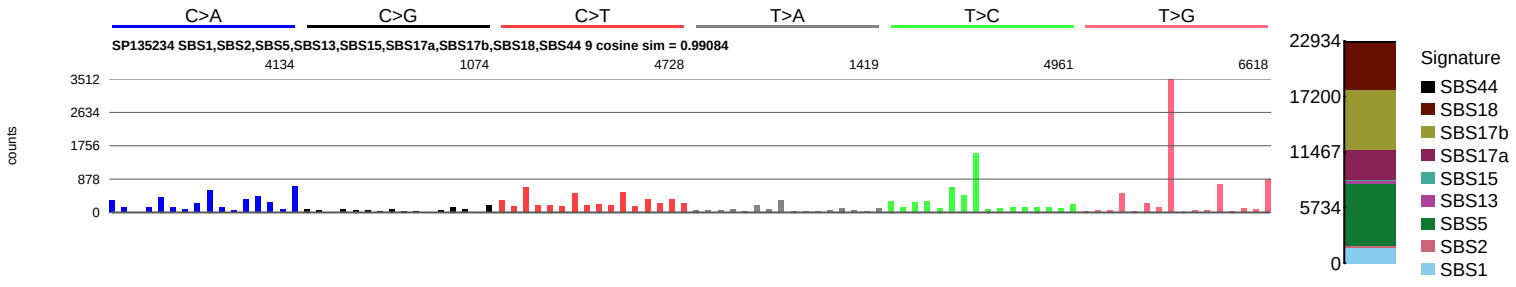
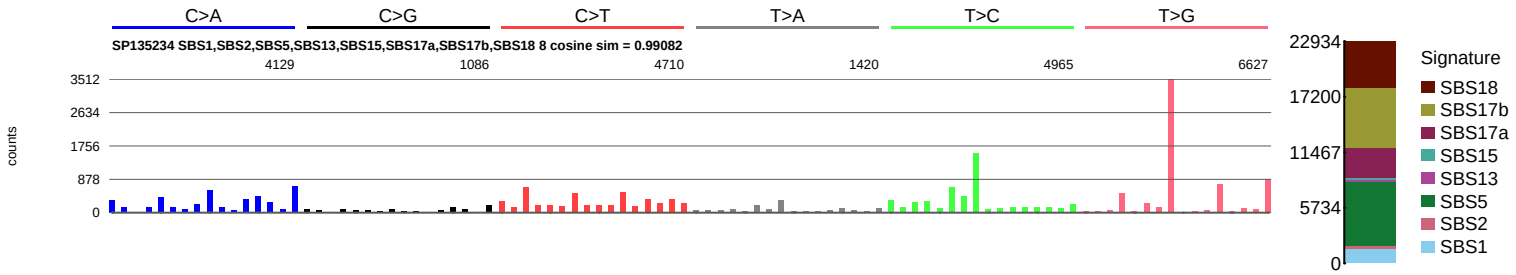
SBS5
SBS2
SBS1

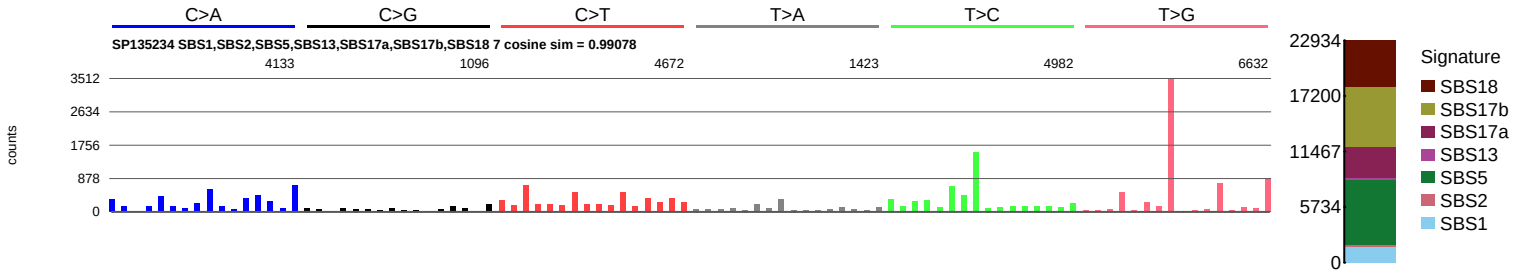

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.99087
4143
880
4663
1408
5226
6610
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

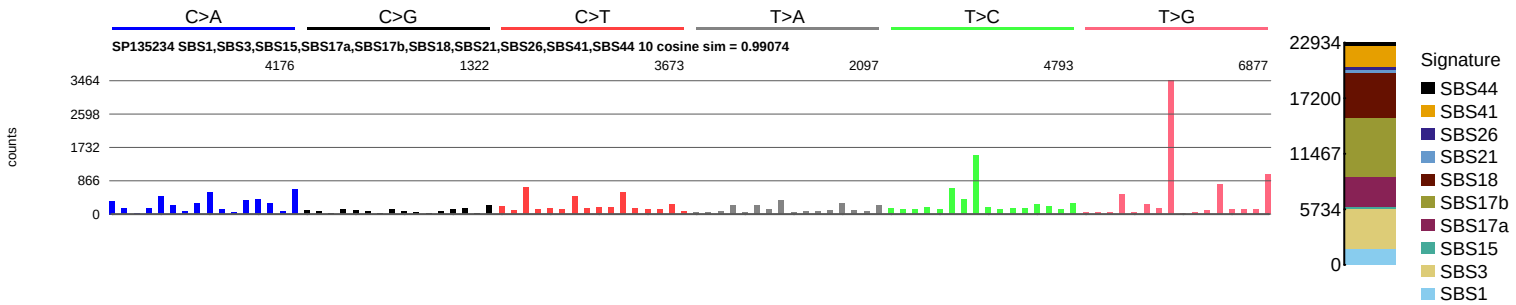

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.99085
4139
894
4632
1413
5239
6617
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

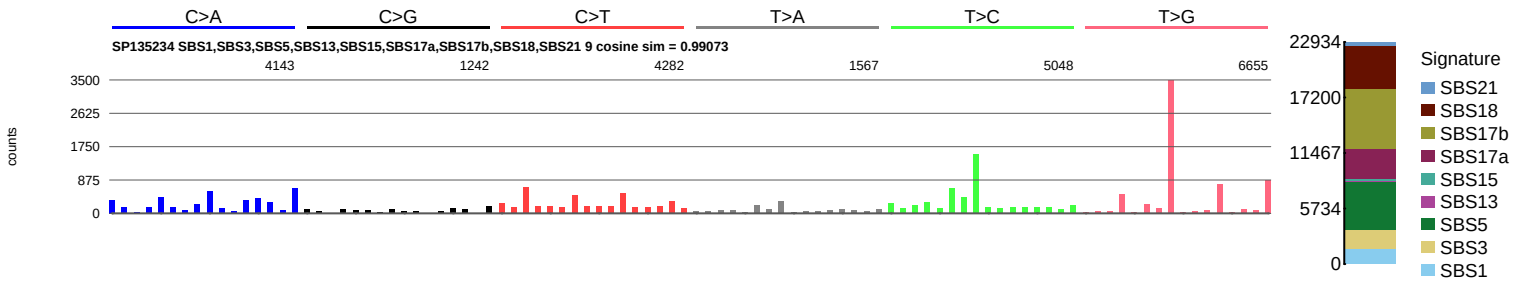

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.99084
4138
1073
4718
1422
4967
6618
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

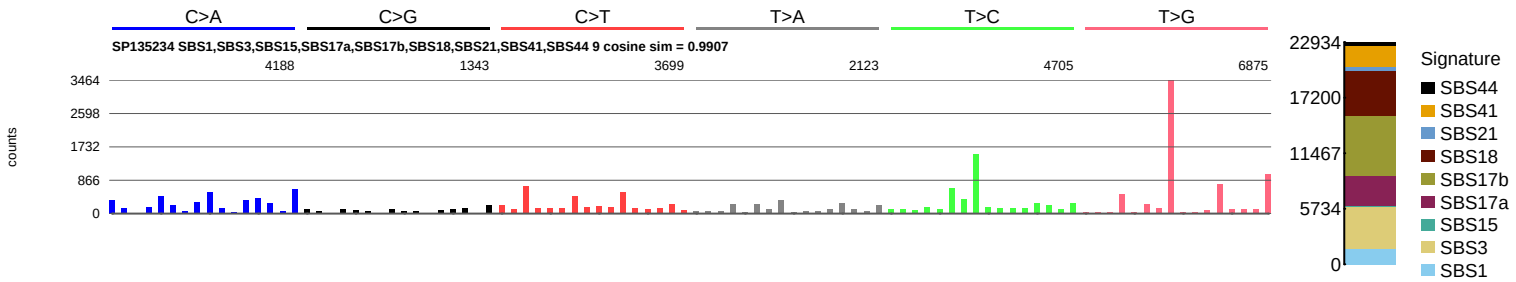

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.99084
4134
1074
4728
1419
4961
6618
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

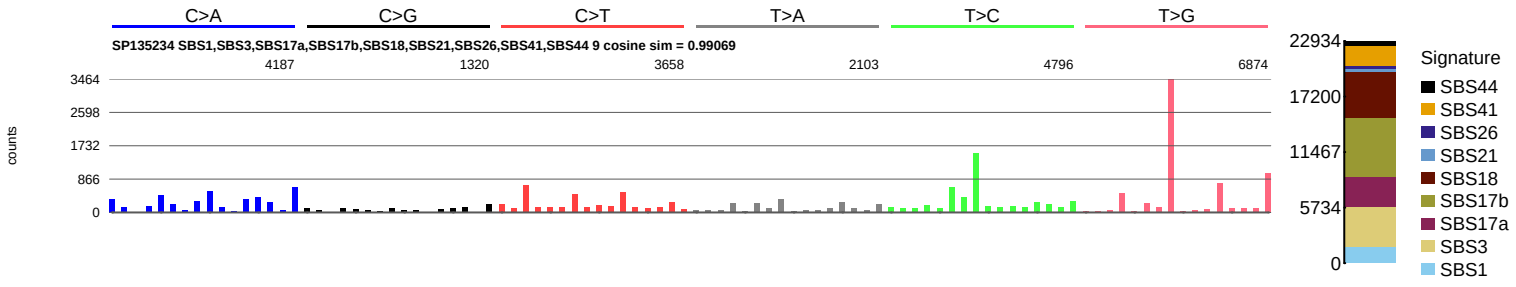

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.99082
4129
1086
4710
1420
4965
6627
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

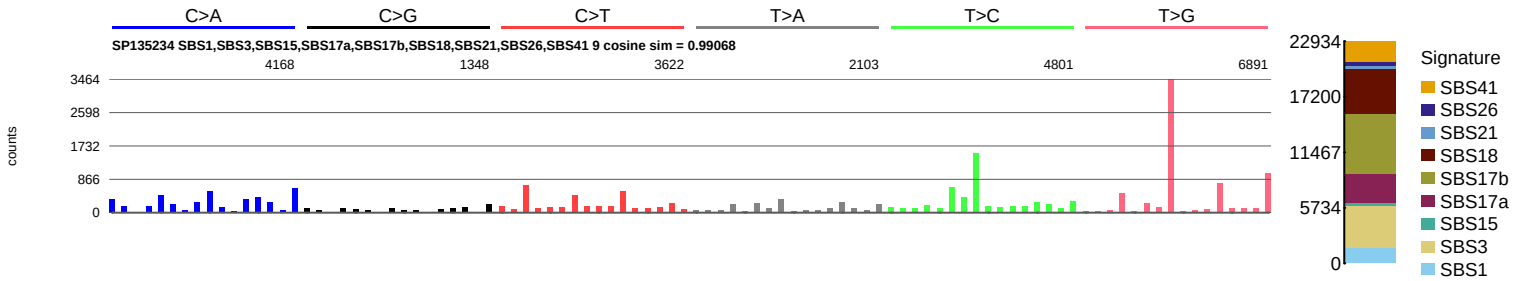

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.99078
4133
1096
4672
1423
4982
6632
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

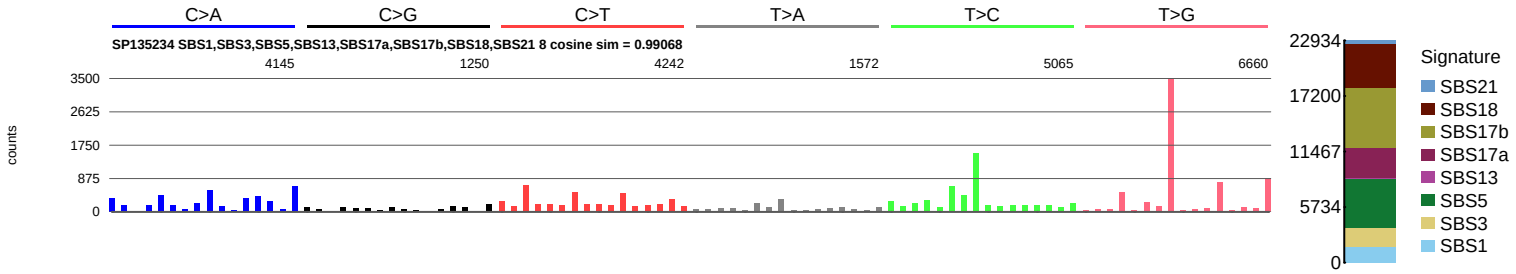

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99074
4176
1322
3673
2097
4793
6877
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

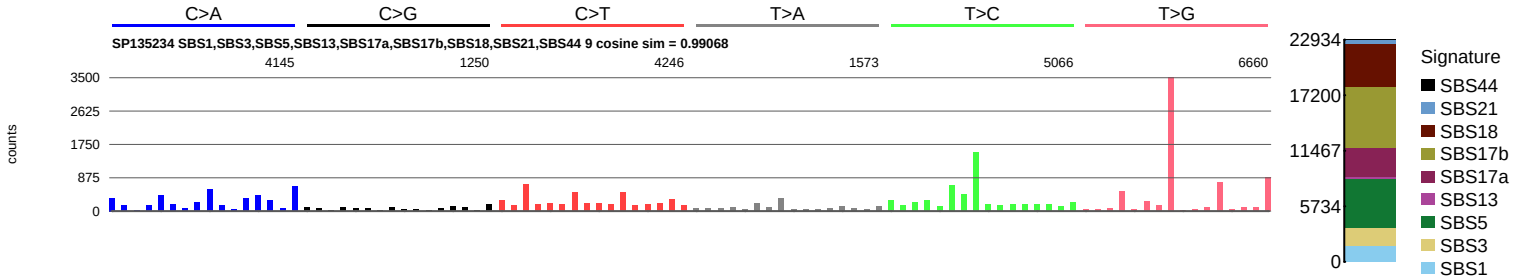

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.99073
4143
1242
4282
1567
5048
6655
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

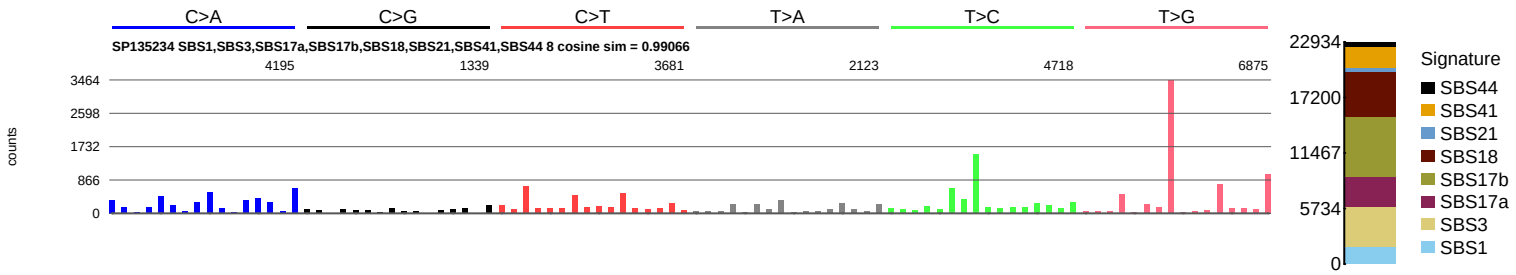

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.9907
4188
1343
3699
2123
4705
6875
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

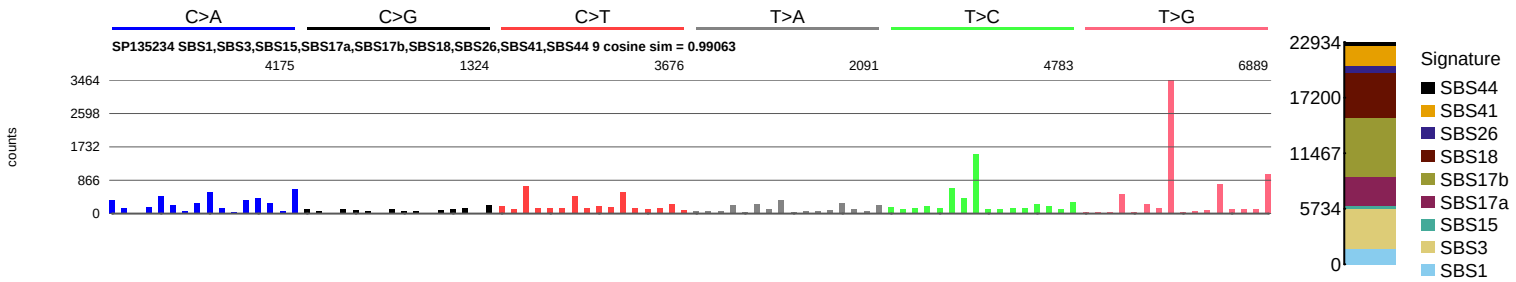

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.99069
4187
1320
3658
2103
4796
6874
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

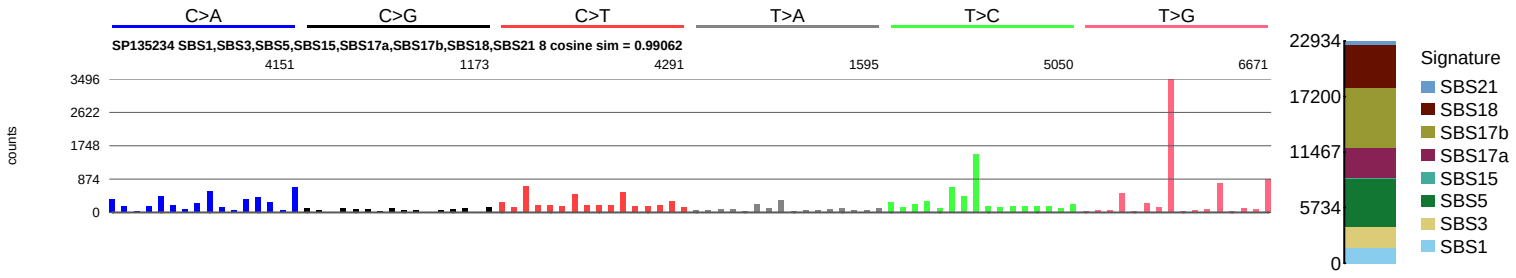

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99068
4168
1348
3622
2103
4801
6891
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

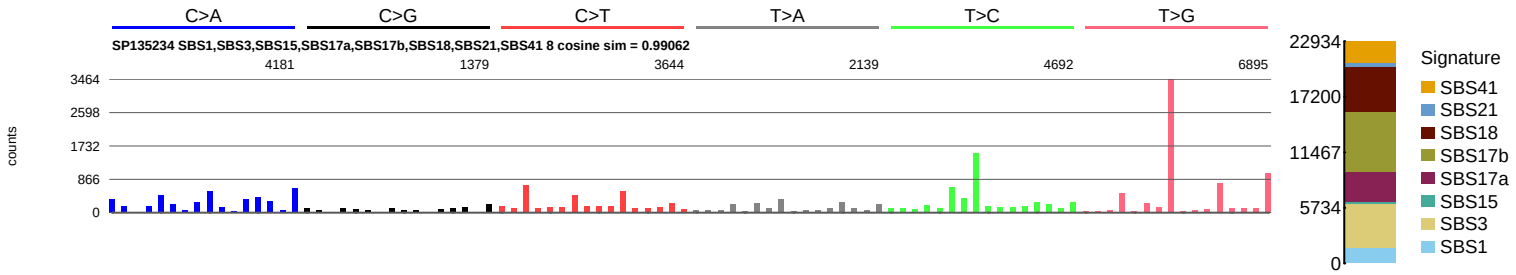

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.99068
4145
1250
4242
1572
5065
6660
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

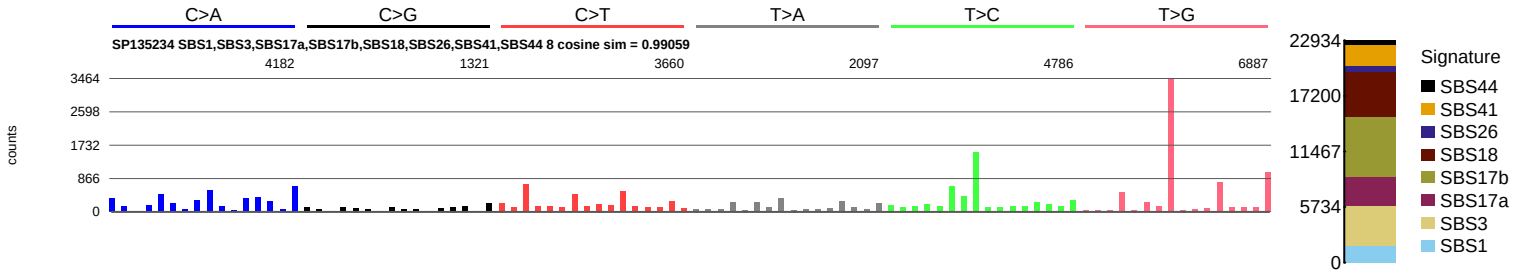

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.99068
4145
1250
4246
1573
5066
6660
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.99066
4195
1339
3681
2123
4718
6875
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99063
4175
1324
3676
2091
4783
6889
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.99062
4151
1173
4291
1595
5050
6671
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99062
4181
1379
3644
2139
4692
6895
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.99059
4182
1321
3660
2097
4786
6887
counts
3464
2598
1732
866
0
22934
17200
11467
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SBS44
SBS41
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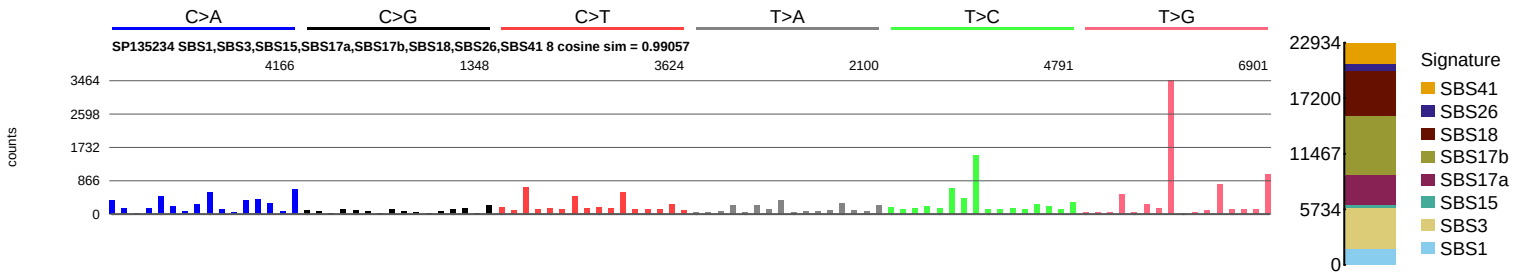
SBS17b
SBS17a
SBS3
SBS1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C>G
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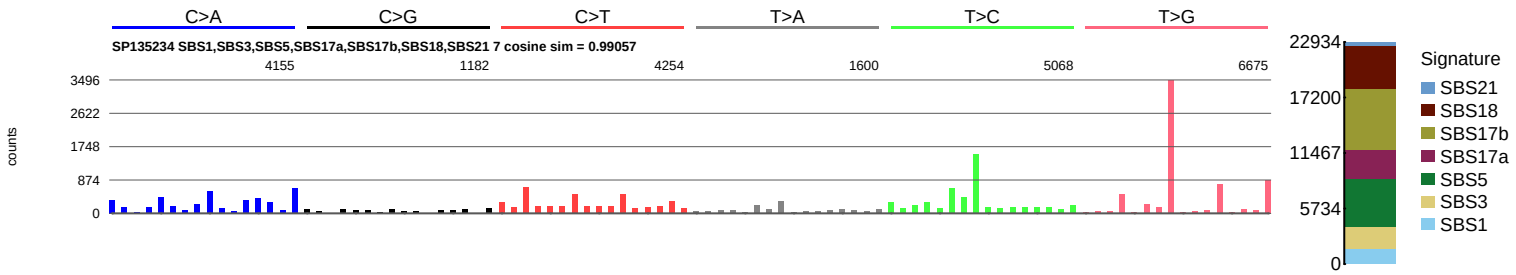
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99057
4166
1348
3624
2100
4791
6901
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
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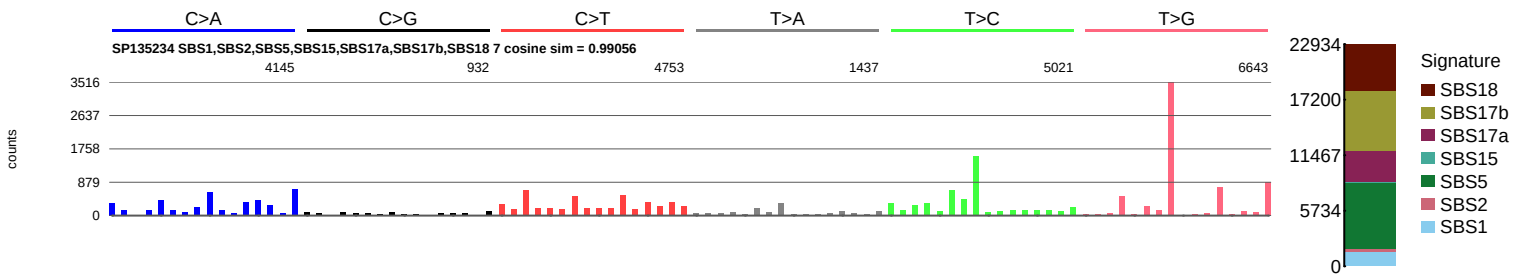
SP135234 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.99057
4155
1182
4254
1600
5068
6675
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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C>G
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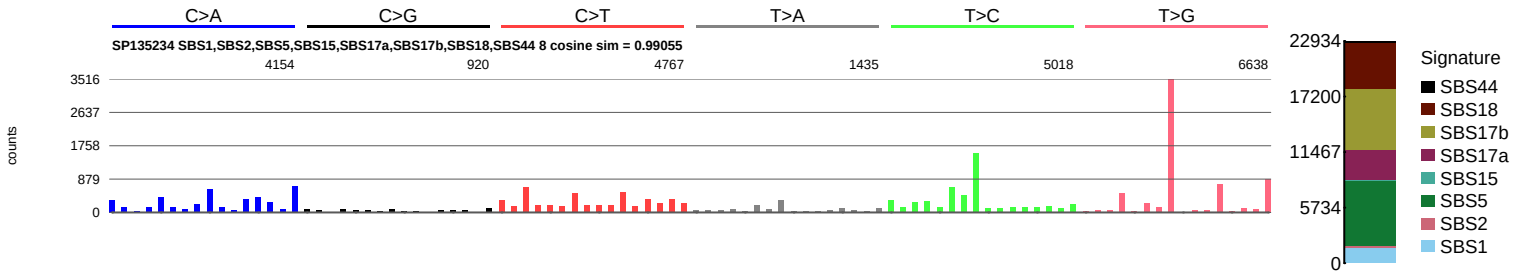
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.99056
4145
932
4753
1437
5021
6643
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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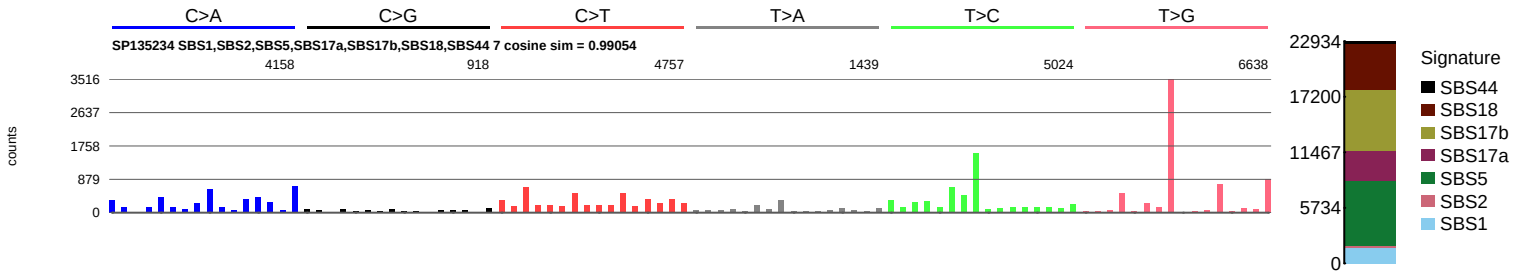
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.99055
4154
920
4767
1435
5018
6638
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
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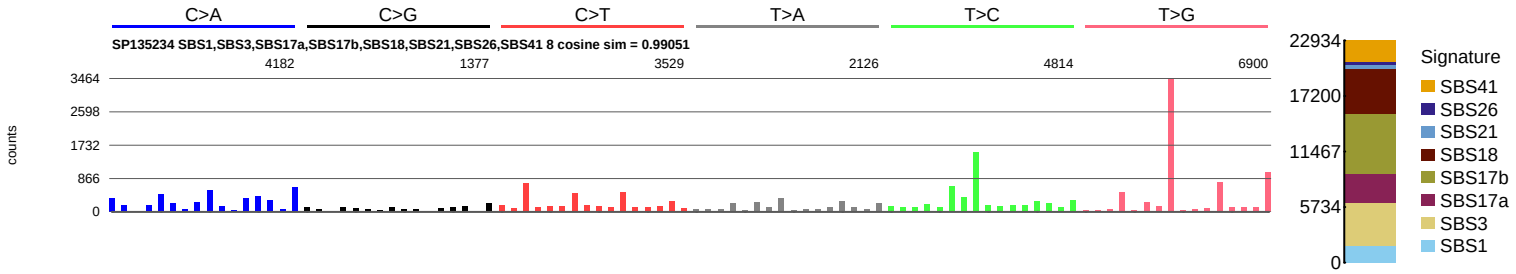
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.99054
4158
918
4757
1439
5024
6638
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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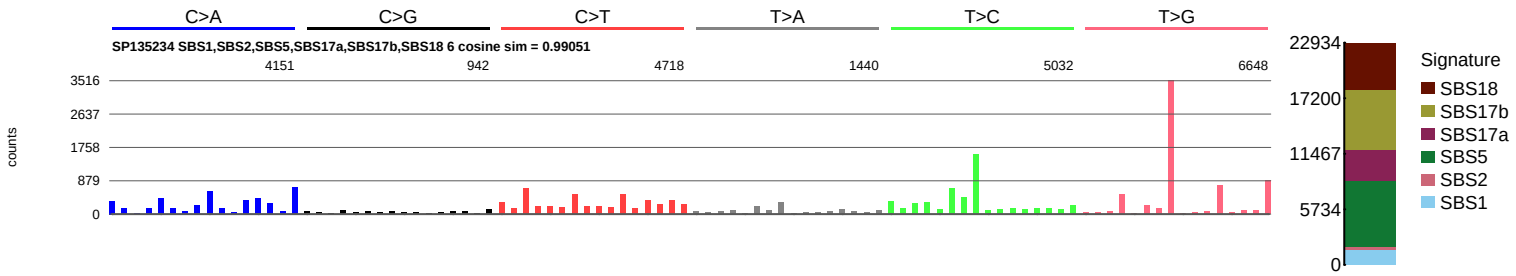
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.99051
4182
1377
3529
2126
4814
6900
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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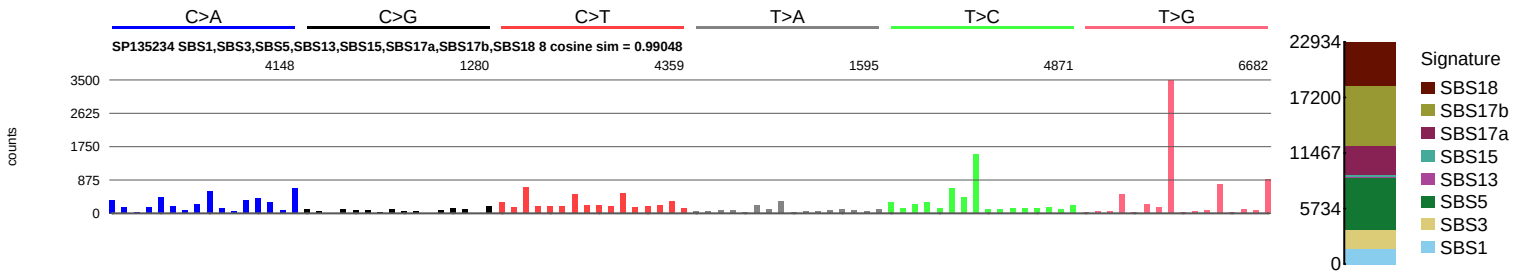
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.99051
4151
942
4718
1440
5032
6648
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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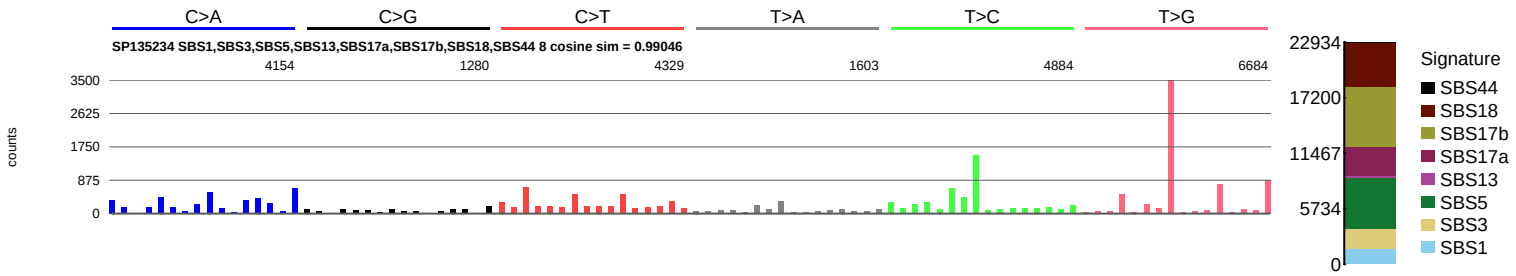
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.99048
4148
1280
4359
1595
4871
6682
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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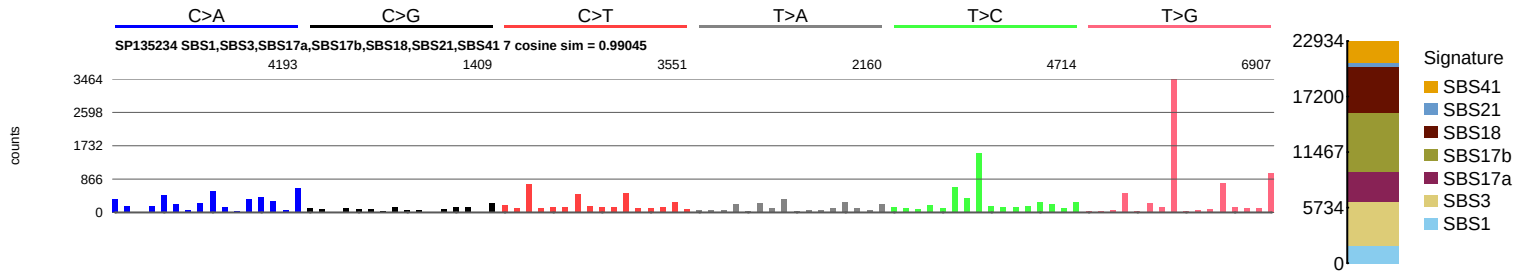
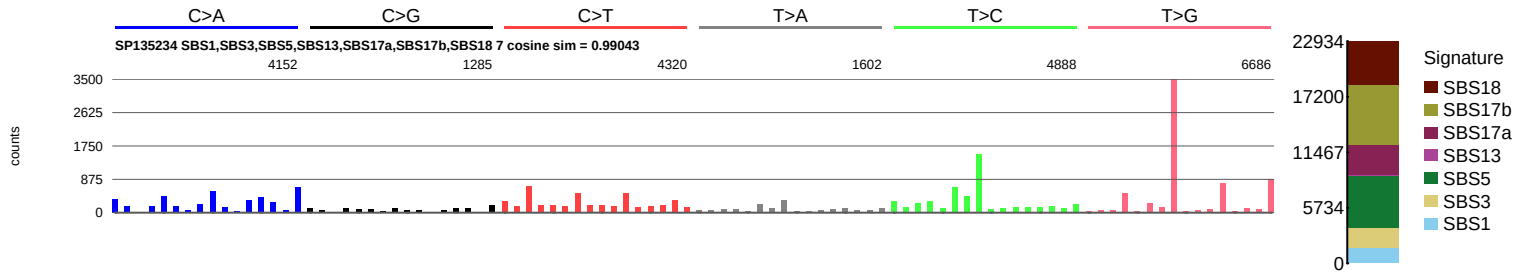
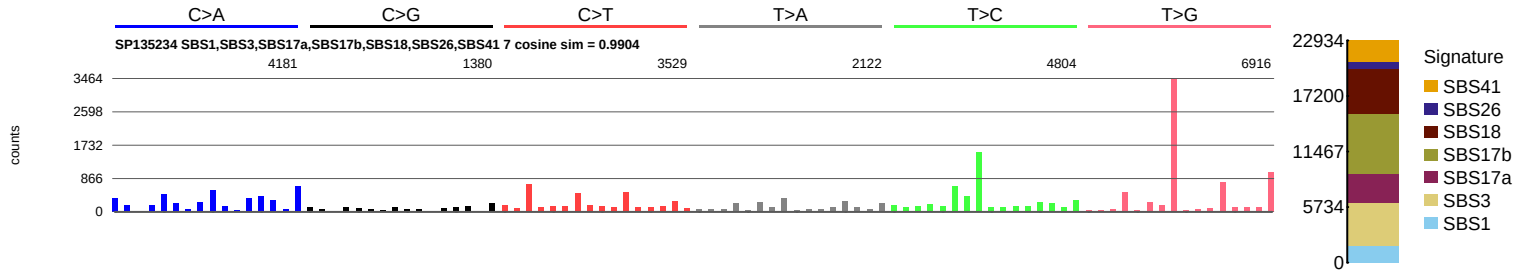
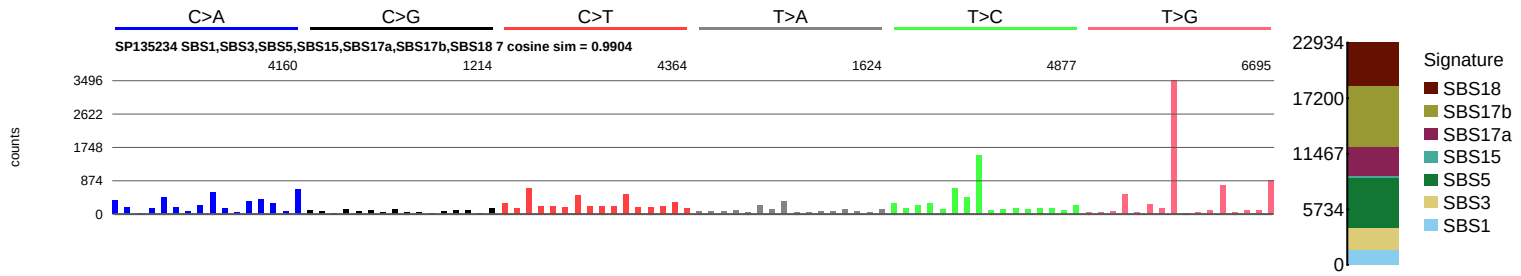
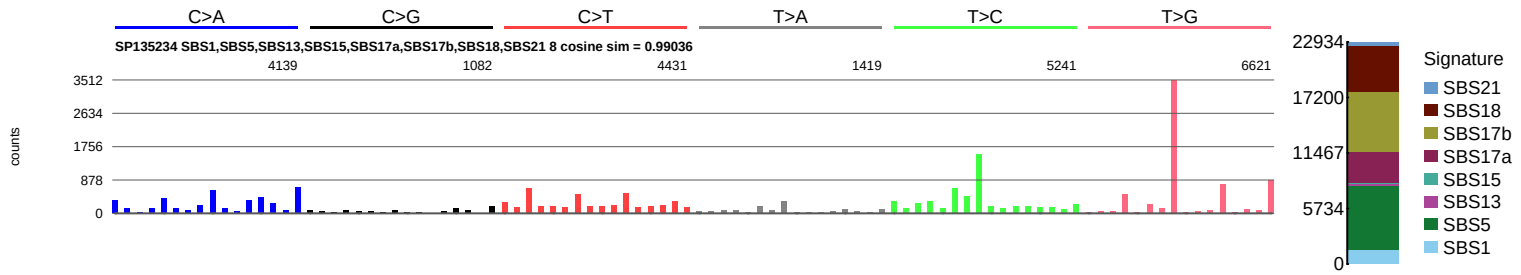
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.99046
4154
1280
4329
1603
4884
6684
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

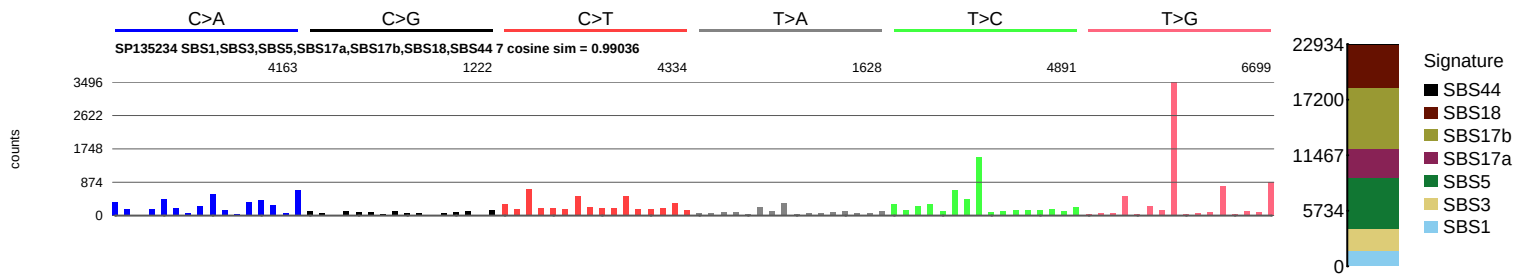

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99045
4193
1409
3551
2160
4714
6907
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.99043
4152
1285
4320
1602
4888
6686
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.9904
4181
1380
3529
2122
4804
6916
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.9904
4160
1214
4364
1624
4877
6695
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.99036
4139
1082
4431
1419
5241
6621
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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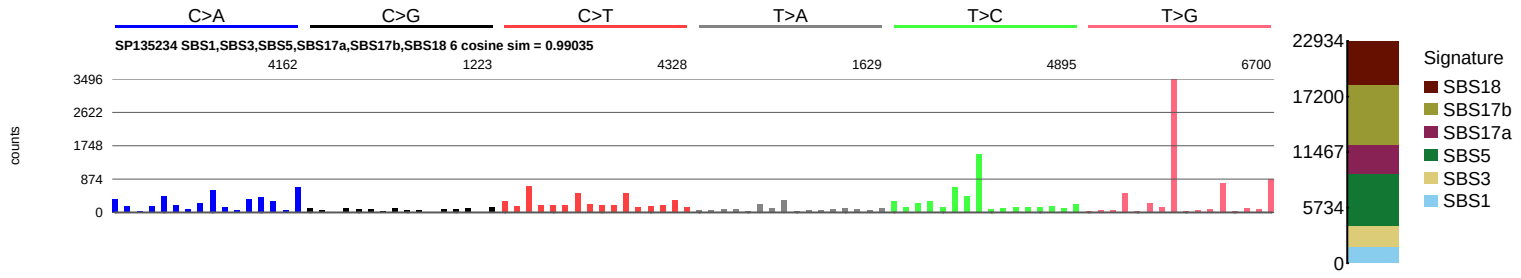
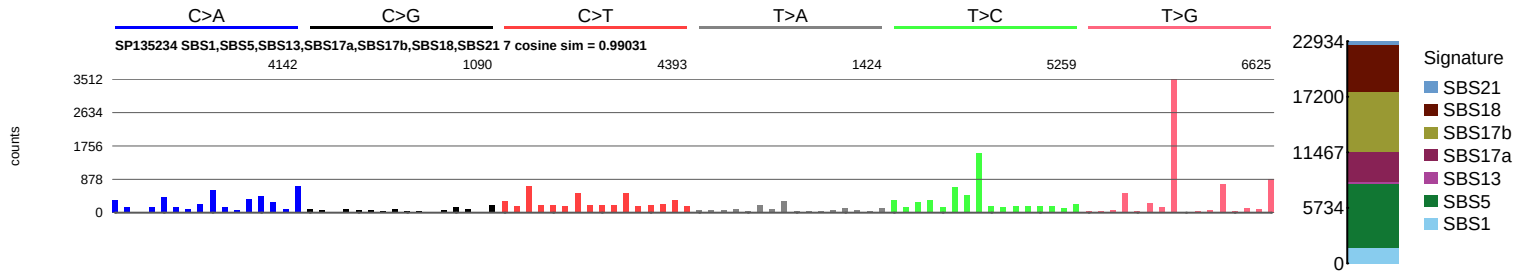
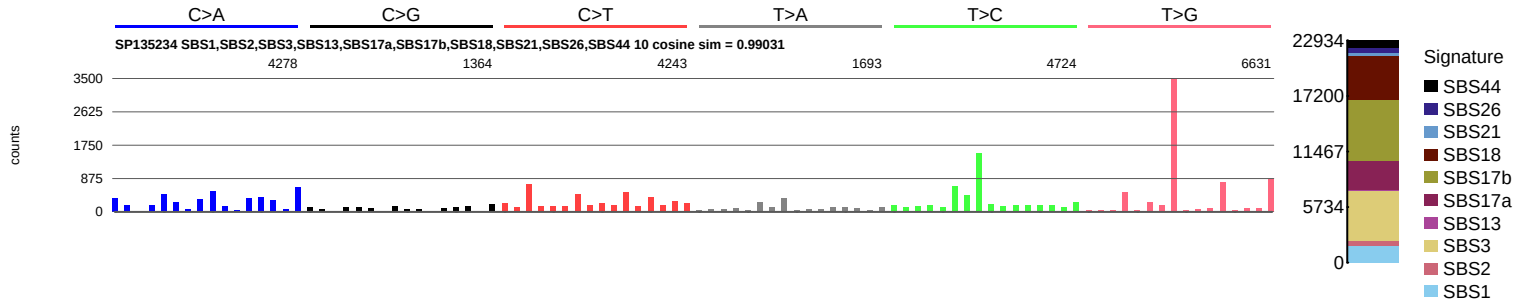
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.99036
4163
1222
4334
1628
4891
6699
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.99035
4162
1223
4328
1629
4895
6700
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.99031
4142
1090
4393
1424
5259
6625
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.99031
4278
1364
4243
1693
4724
6631
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
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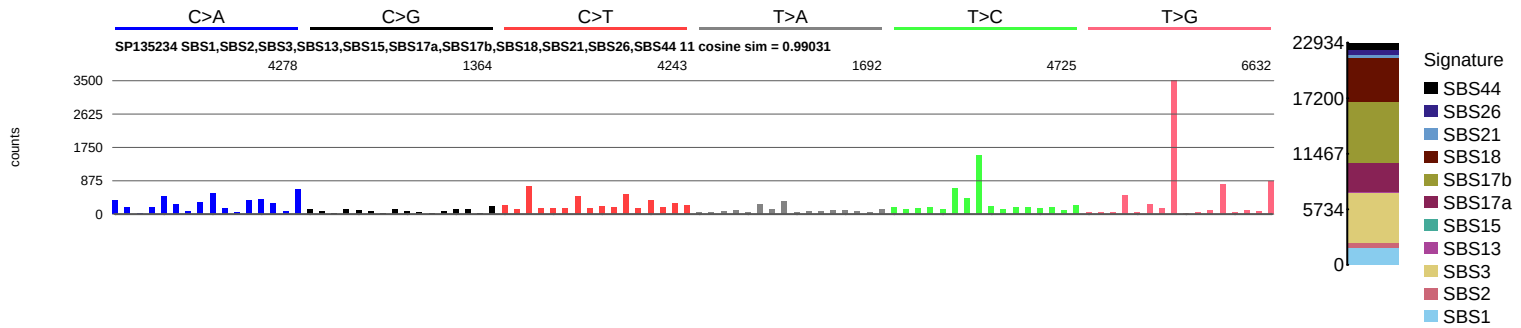
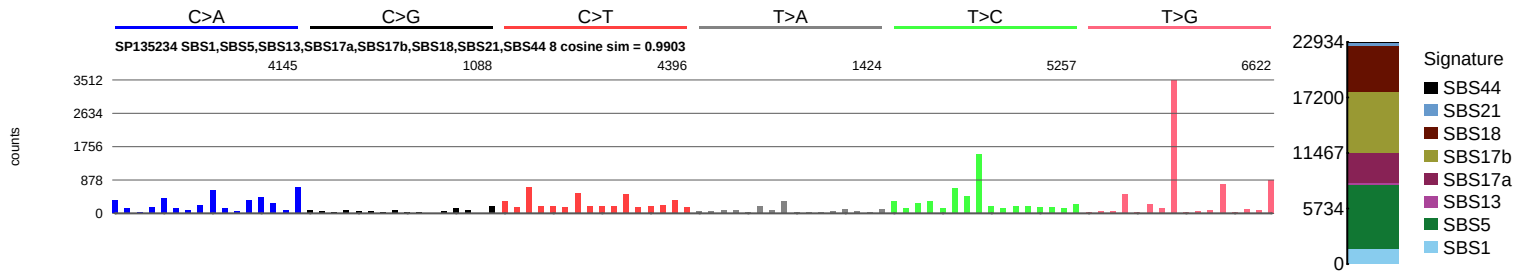
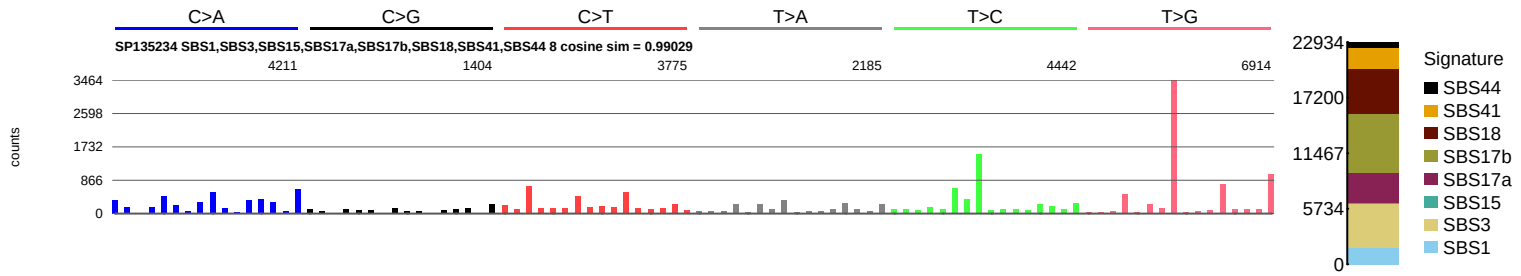
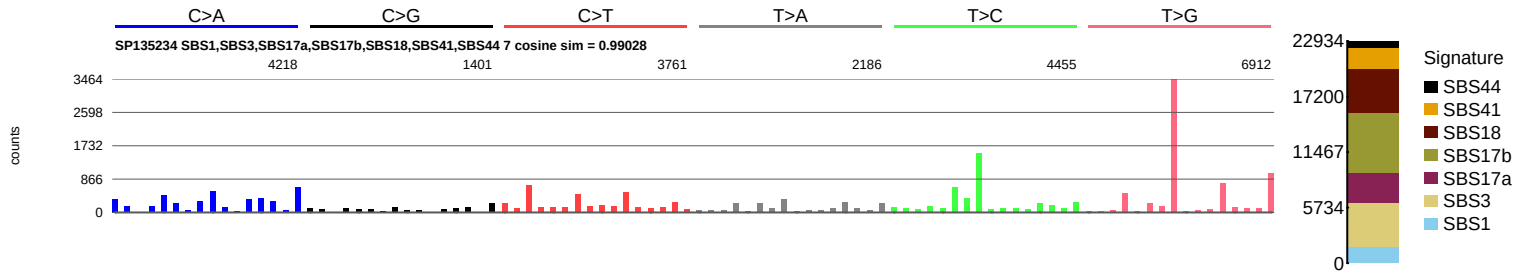
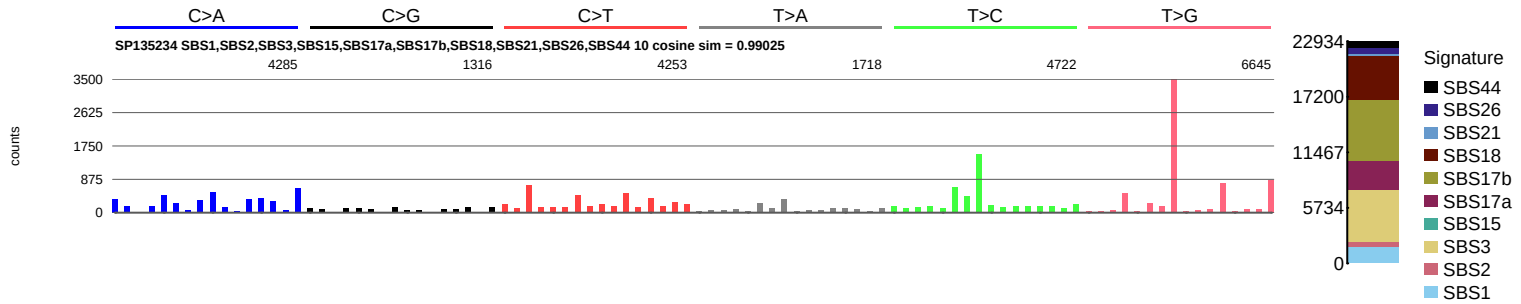
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

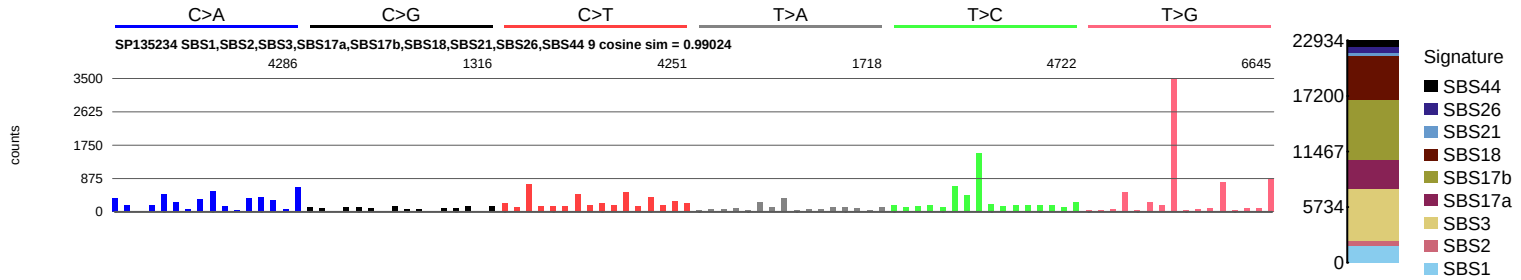

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 11 cosine sim = 0.99031
4278
1364
4243
1692
4725
6632
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

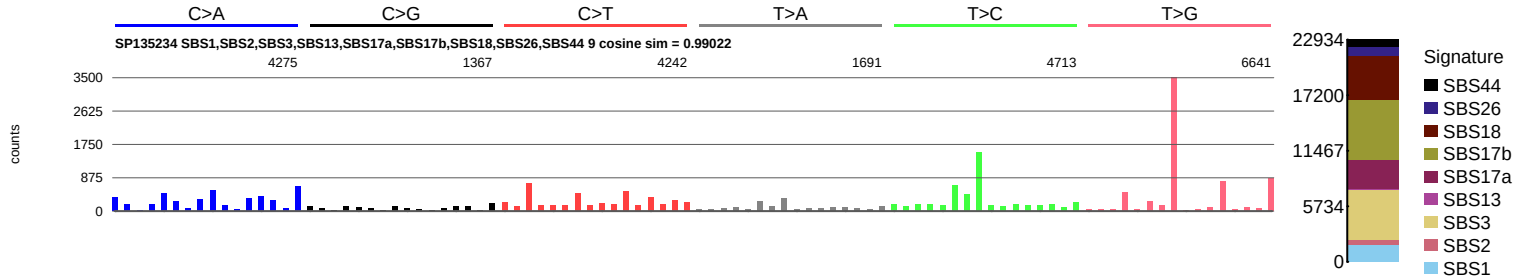

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.9903
4145
1088
4396
1424
5257
6622
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99029
4211
1404
3775
2185
4442
6914
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.99028
4218
1401
3761
2186
4455
6912
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.99025
4285
1316
4253
1718
4722
6645
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.99024
4286
1316
4251
1718
4722
6645
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.99022
4275
1367
4242
1691
4713
6641
counts
3500
2625
1750
875
0
22934
17200
11467
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SBS44
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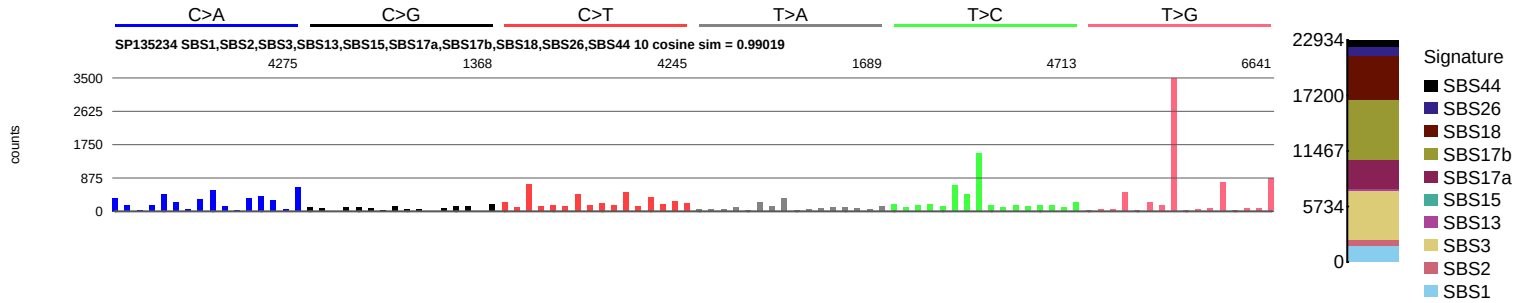
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SP135234 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.99019
4275
1368
4245
1689
4713
6641
counts
3500
2625
1750
875
0
22934
17200
11467
5734
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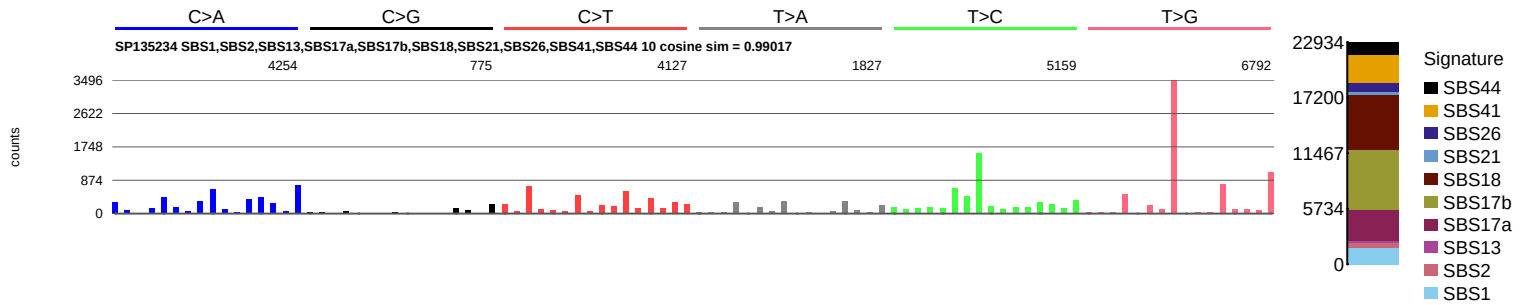
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SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99017
4254
775
4127
1827
5159
6792
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
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SBS21
SBS18
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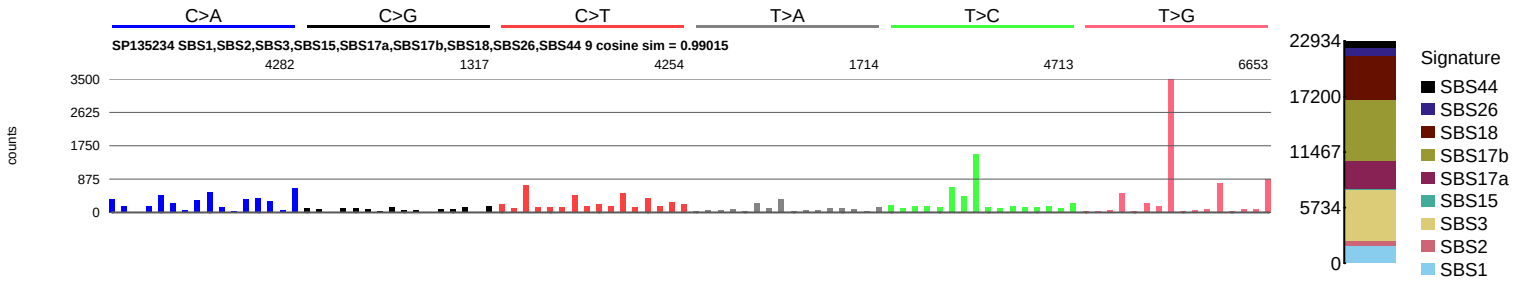
SBS17a
SBS13
SBS2
SBS1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4282
1317
4254
1714
4713
6653
counts
3500
2625
1750
875
0
22934
17200
11467
5734
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SBS26
SBS18
SBS17b
SBS17a
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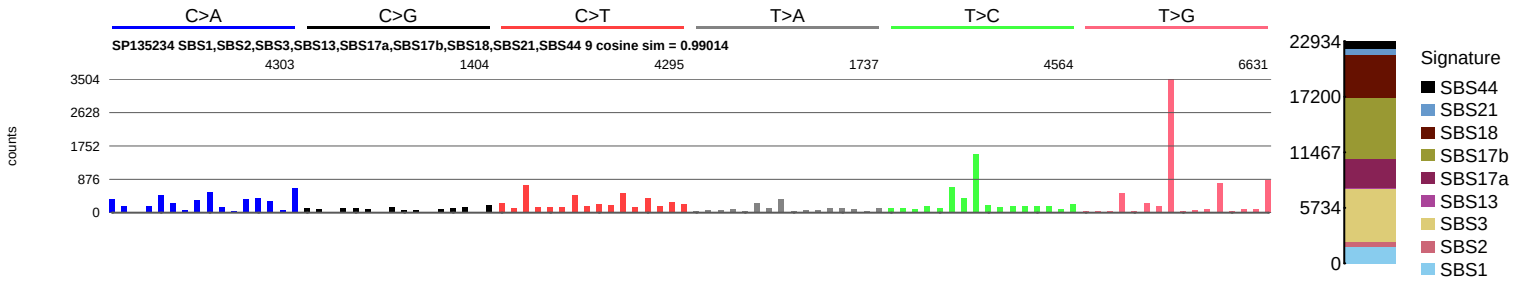
SBS2
SBS1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SP135234 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.99014
4303
1404
4295
1737
4564
6631
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
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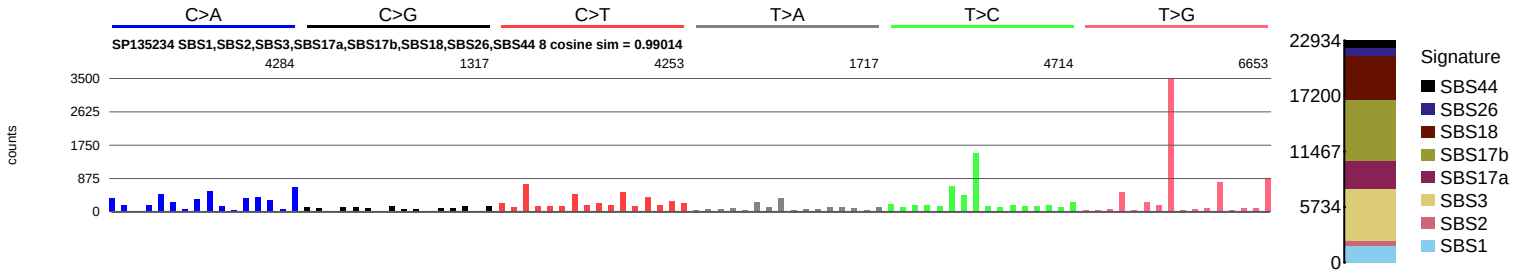
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
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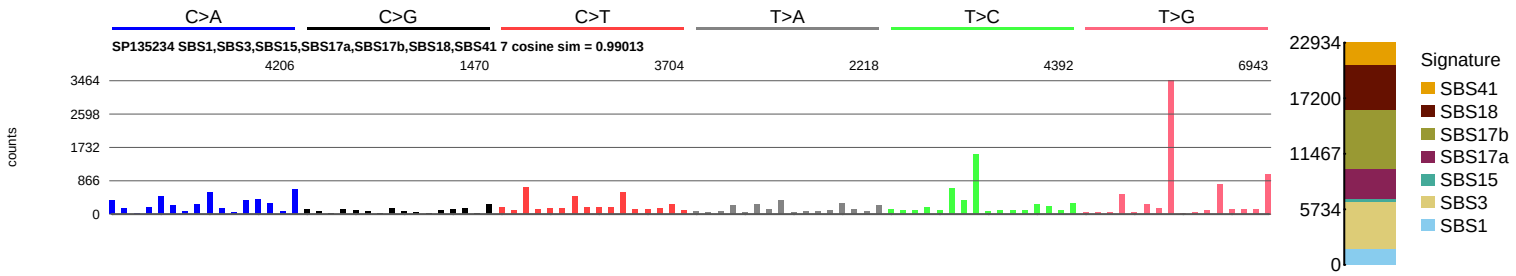
T>G
SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.99014
4284
1317
4253
1717
4714
6653
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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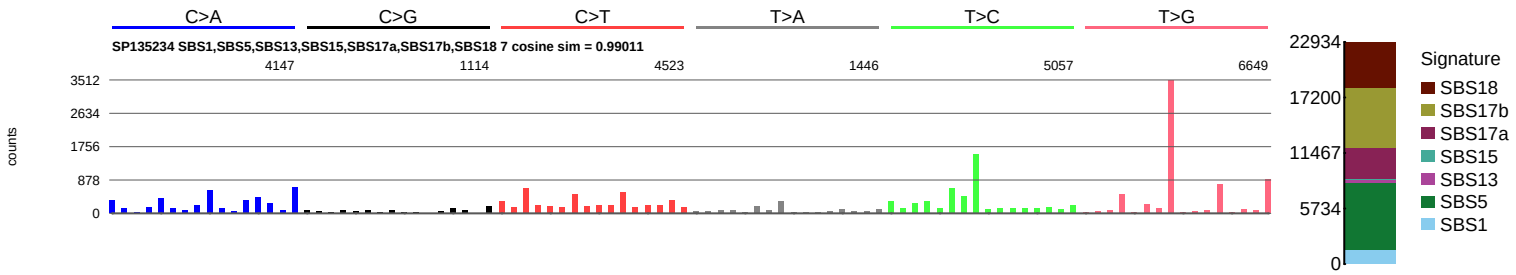
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99013
4206
1470
3704
2218
4392
6943
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
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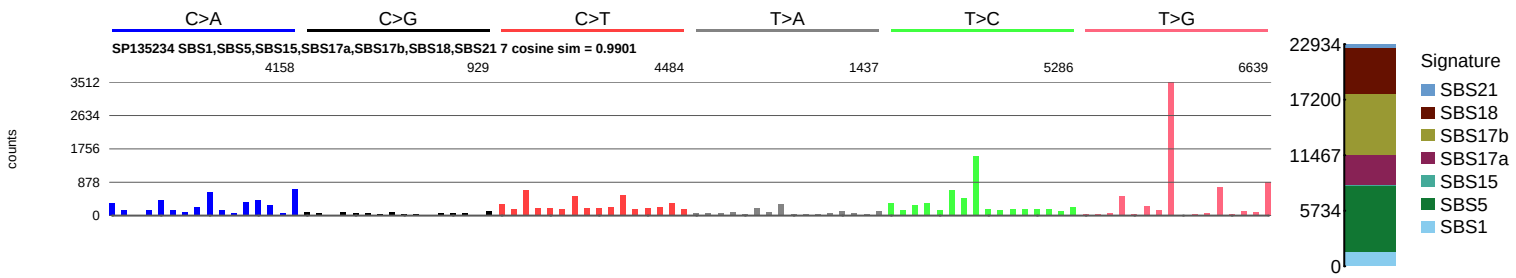
T>C
T>G
SP135234 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.99011
4147
1114
4523
1446
5057
6649
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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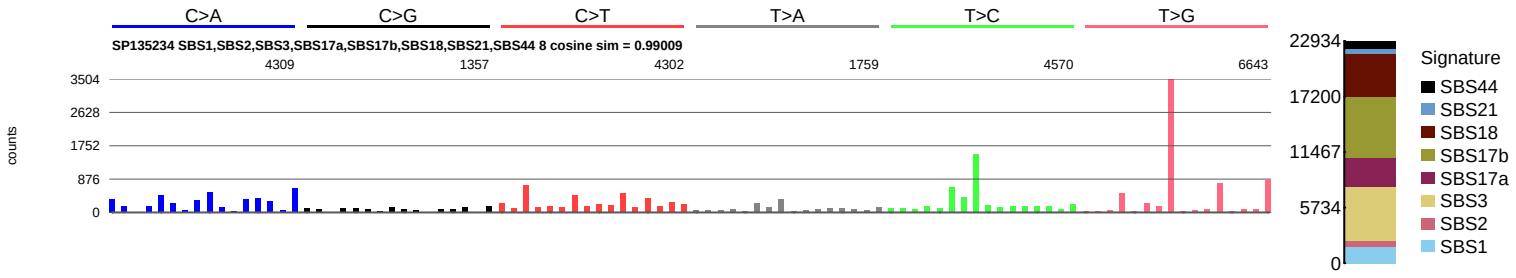
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.9901
4158
929
4484
1437
5286
6639
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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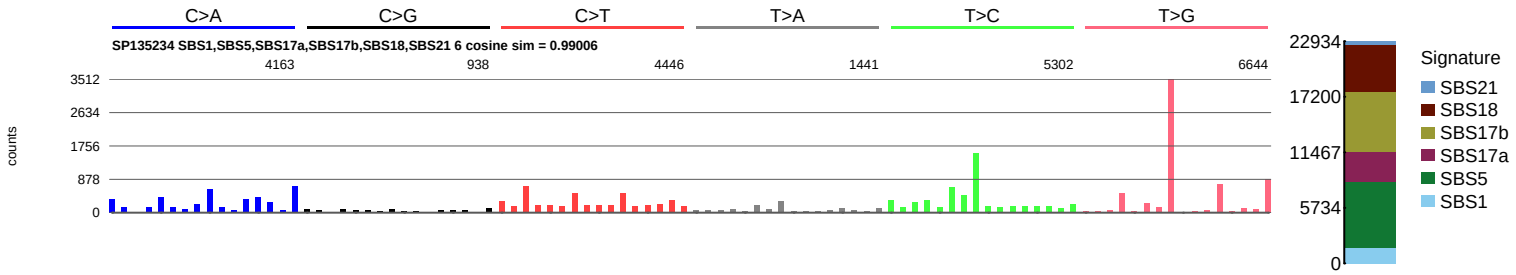
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.99009
4309
1357
4302
1759
4570
6643
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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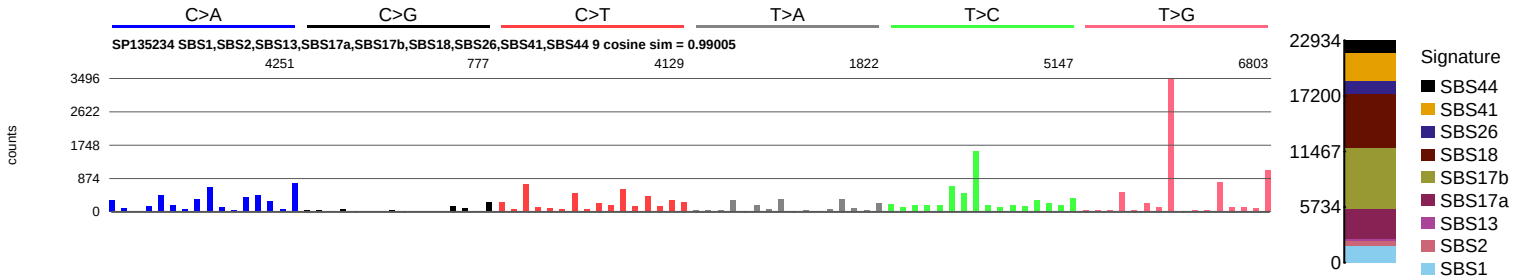
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.99006
4163
938
4446
1441
5302
6644
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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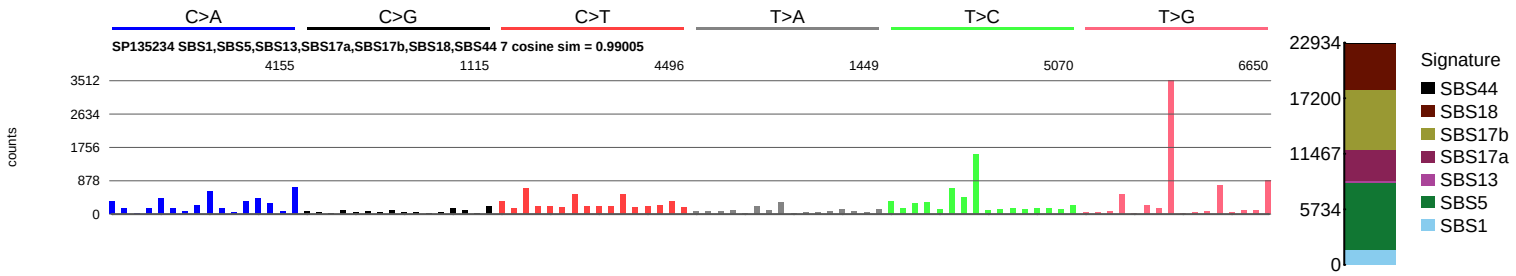
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99005
4251
777
4129
1822
5147
6803
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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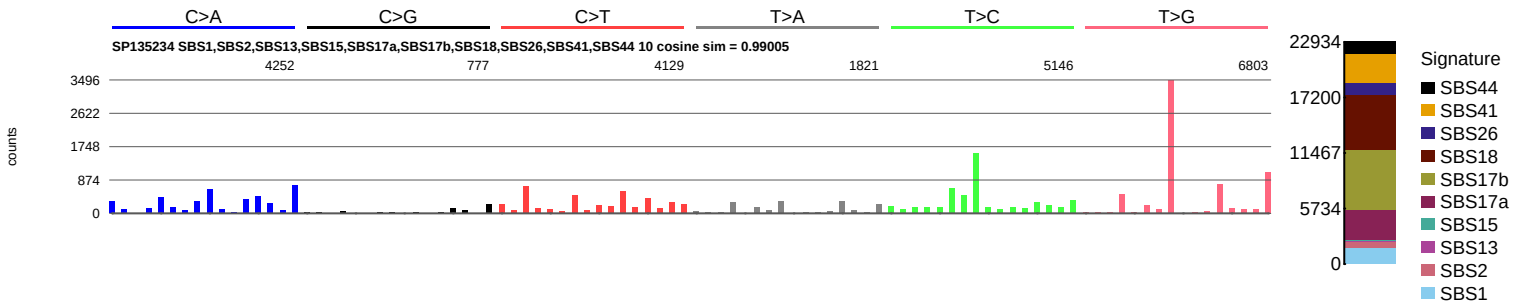
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.99005
4155
1115
4496
1449
5070
6650
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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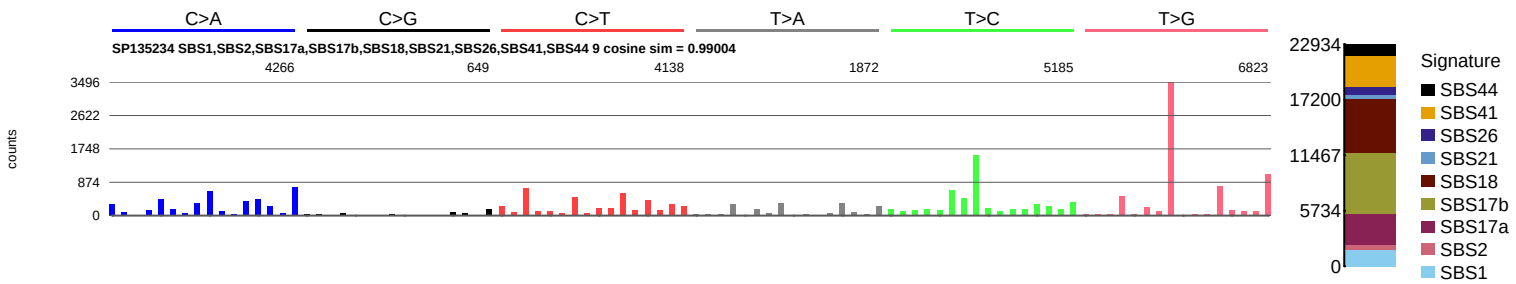
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.99005
4252
777
4129
1821
5146
6803
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.99004
4266
649
4138
1872
5185
6823
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
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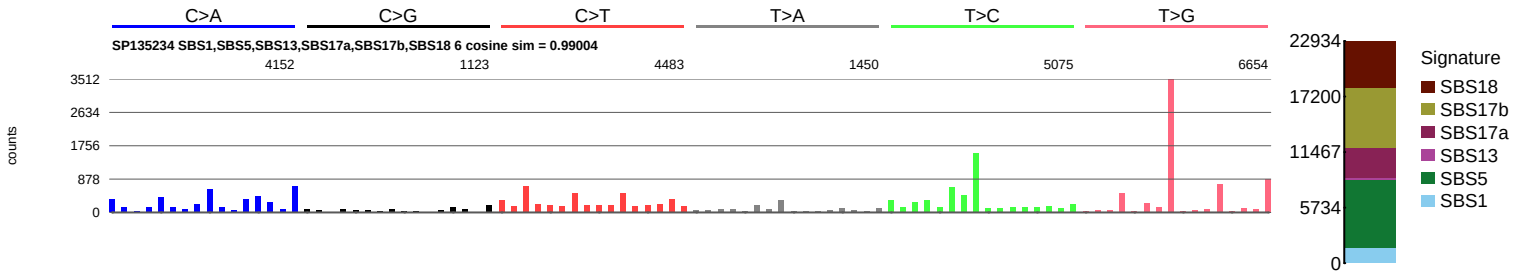
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.99004
4152
1123
4483
1450
5075
6654
counts
3512
2634
1756
878
0
22934
17200
11467
5734
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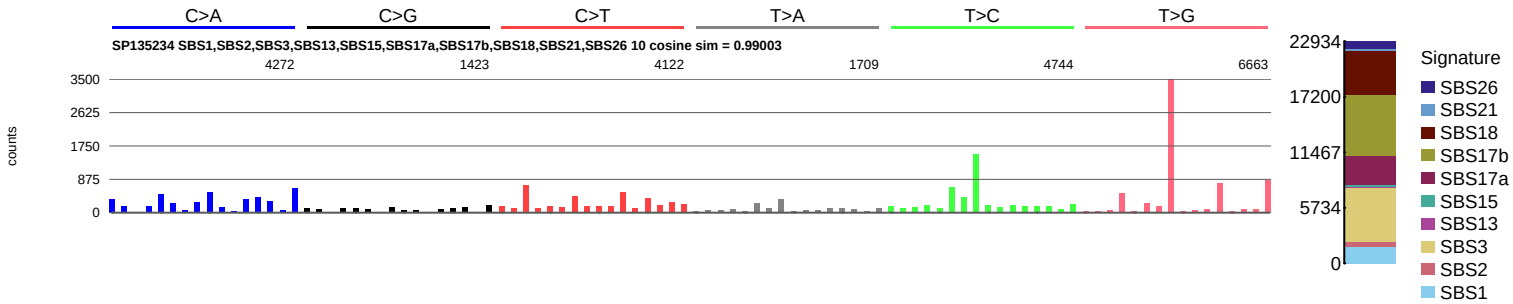
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.99003
4272
1423
4122
1709
4744
6663
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
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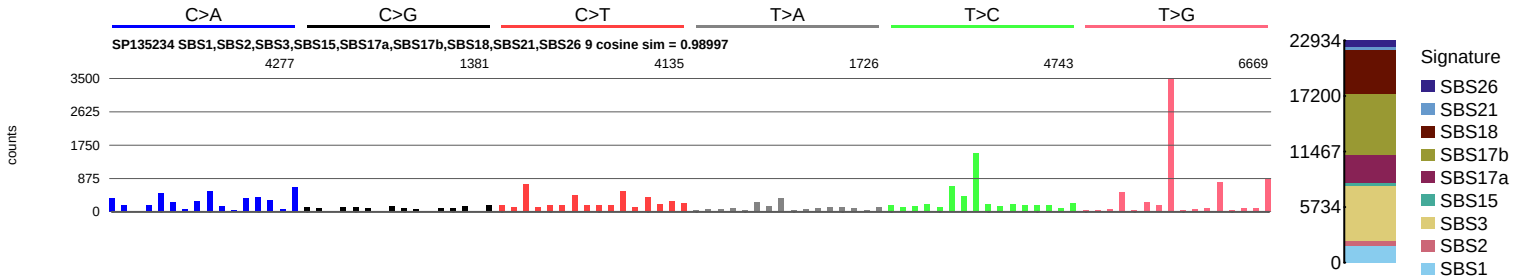
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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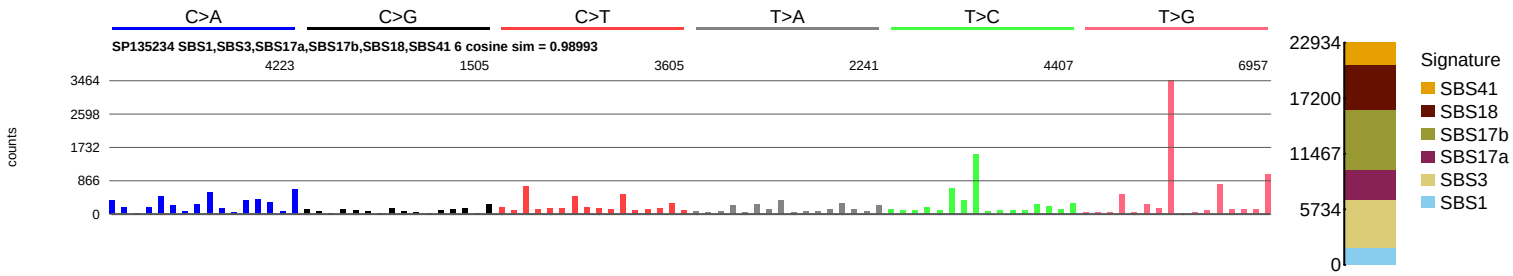
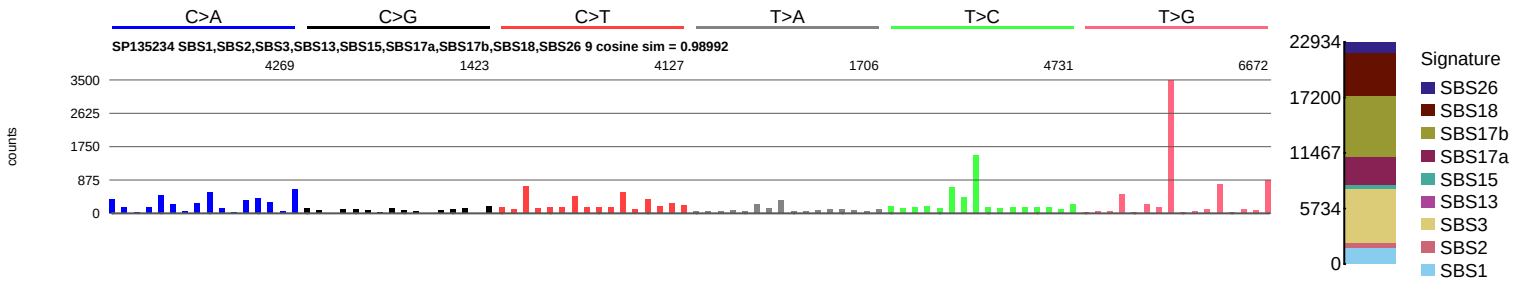
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98997
4277
1381
4135
1726
4743
6669
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98993
4223
1505
3605
2241
4407
6957
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1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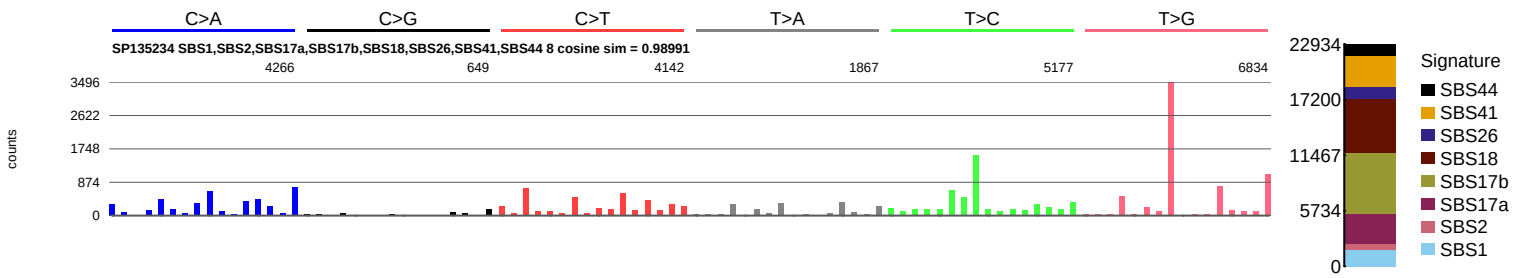
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98992
4269
1423
4127
1706
4731
6672
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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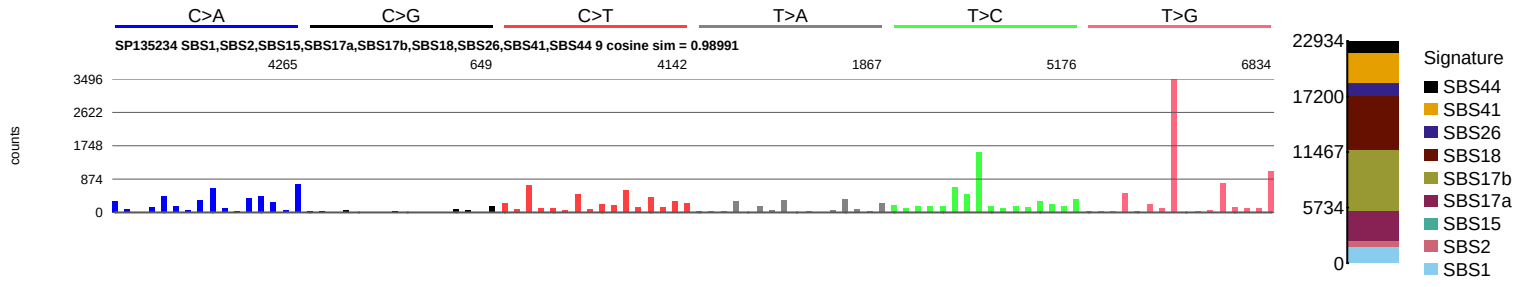
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98991
4266
649
4142
1867
5177
6834
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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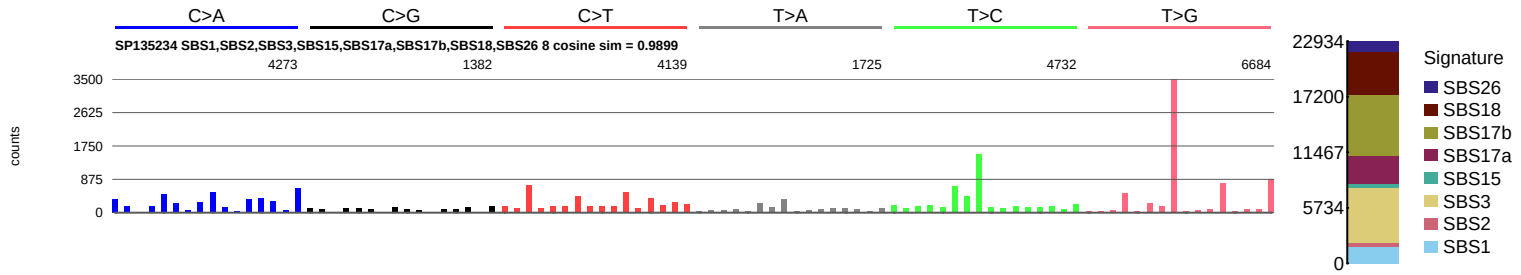
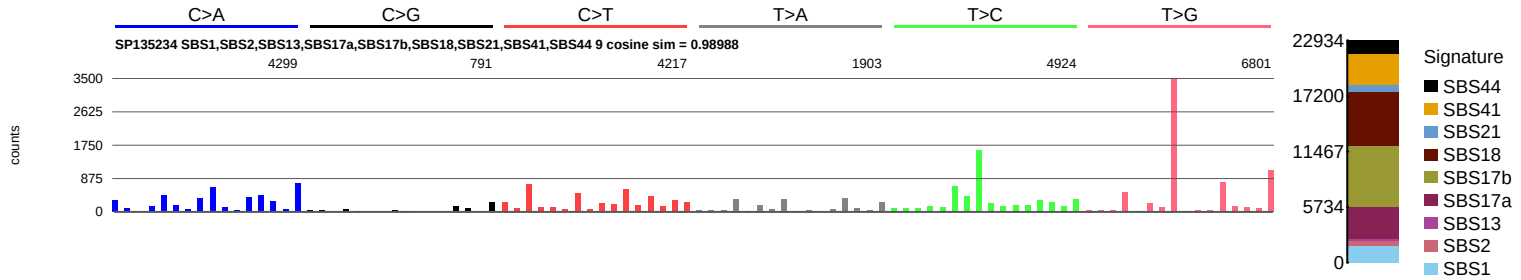
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98991
4265
649
4142
1867
5176
6834
counts
3496
2622
1748
874
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.9899
4273
1382
4139
1725
4732
6684
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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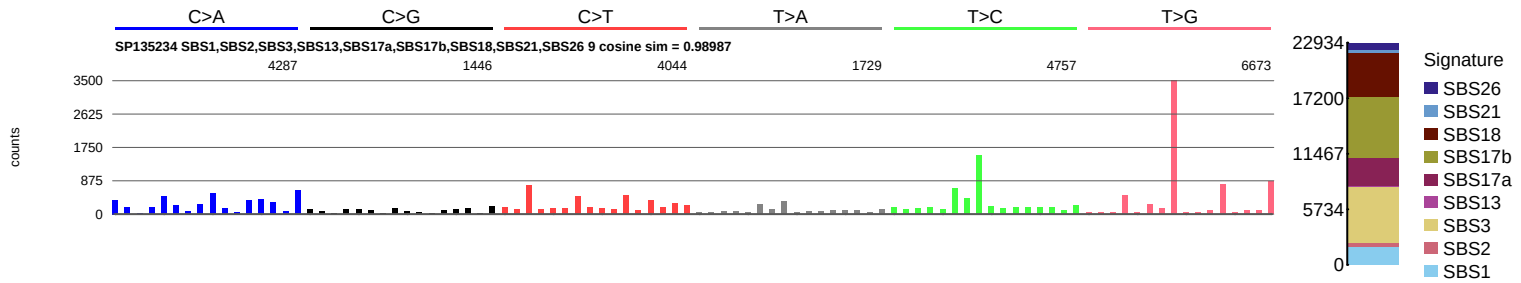
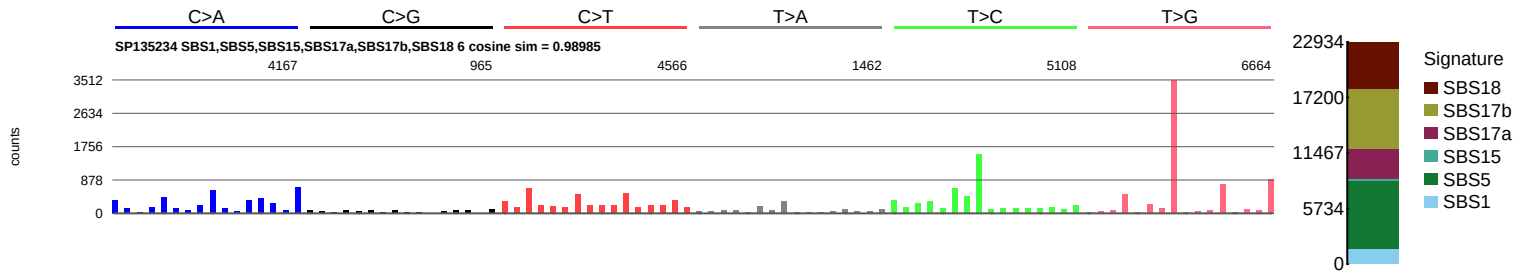
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98988
4299
791
4217
1903
4924
6801
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98987
4287
1446
4044
1729
4757
6673
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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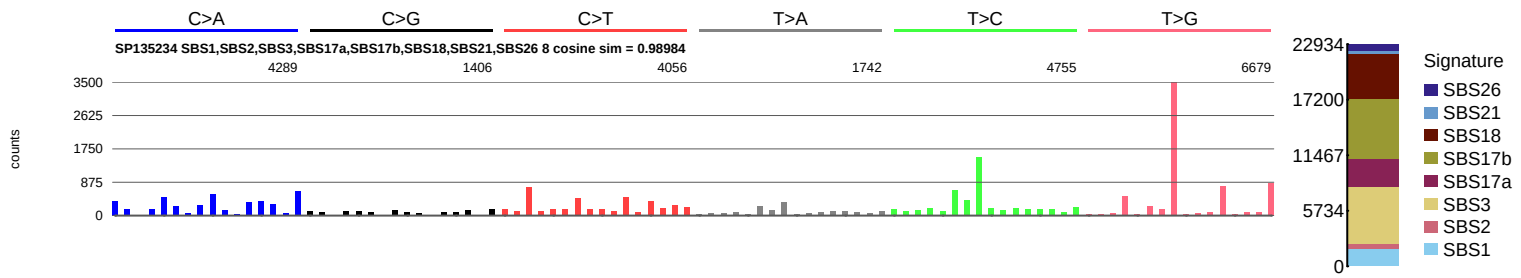
SP135234 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98985
4167
965
4566
1462
5108
6664
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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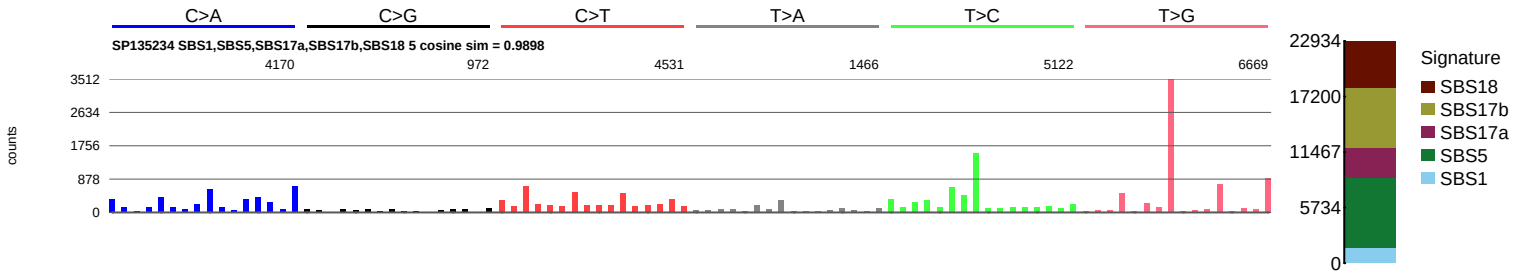
SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98984
4289
1406
4056
1742
4755
6679
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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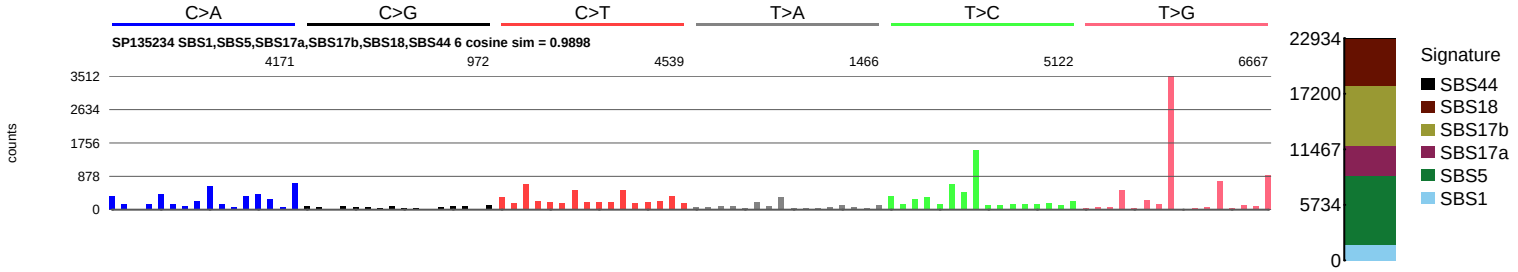
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.9898
4170
972
4531
1466
5122
6669
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS1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C>G
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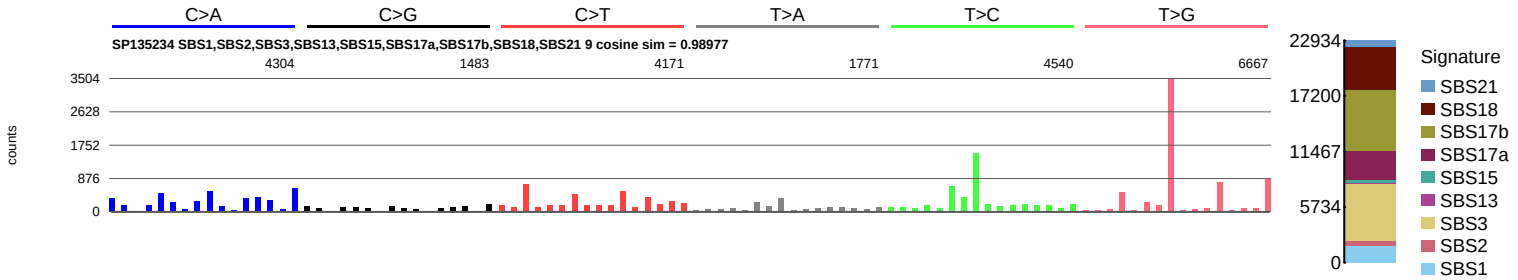
T>G
SP135234 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.9898
4171
972
4539
1466
5122
6667
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS1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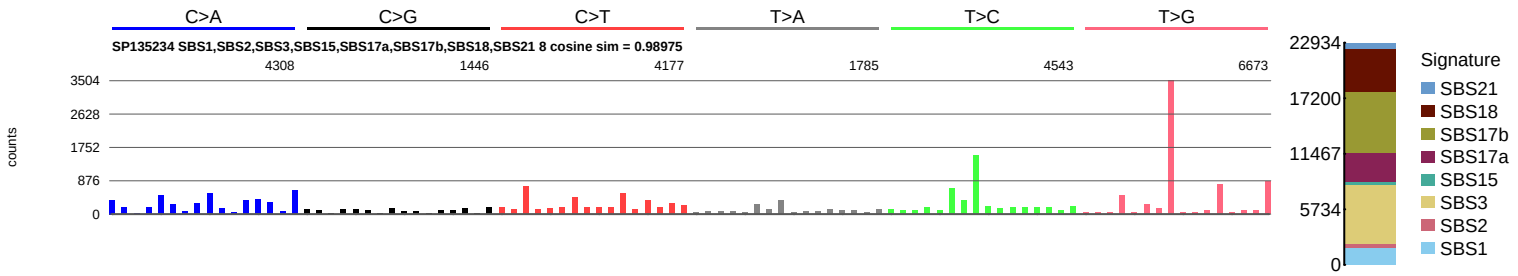
SP135234 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98977
4304
1483
4171
1771
4540
6667
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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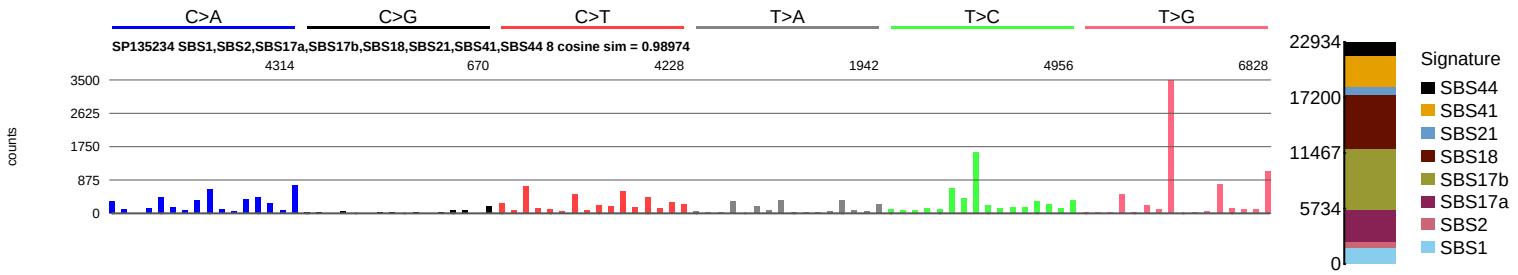
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98975
4308
1446
4177
1785
4543
6673
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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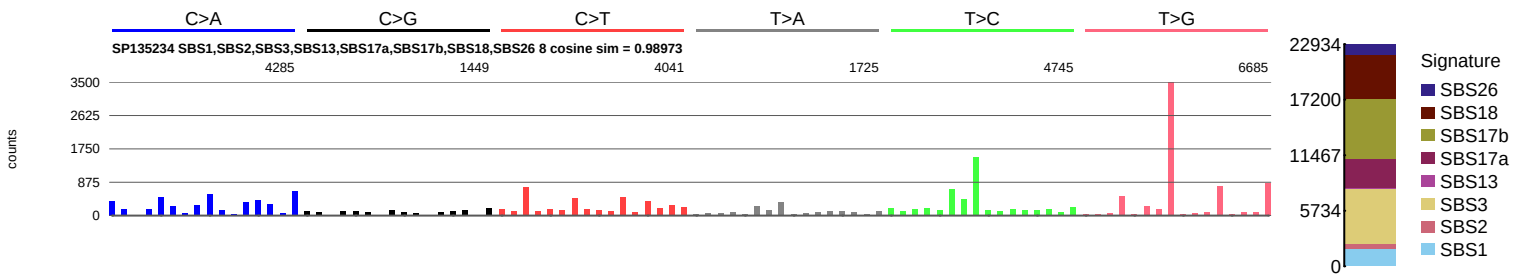
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98974
4314
670
4228
1942
4956
6828
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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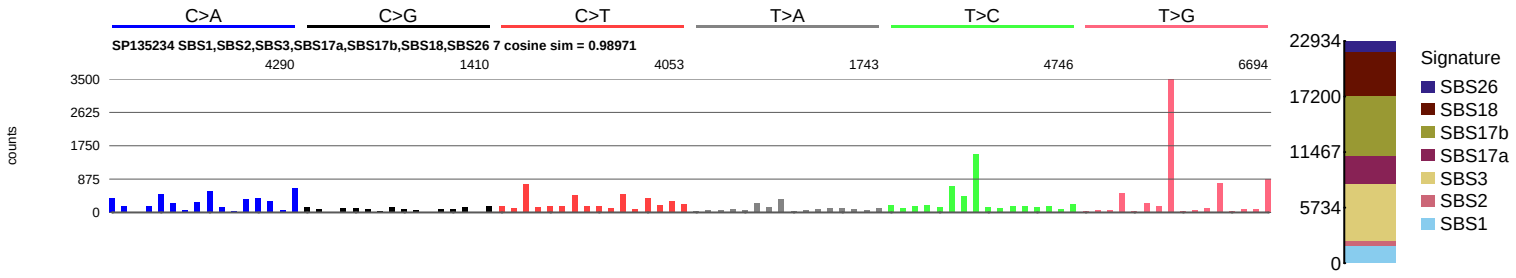
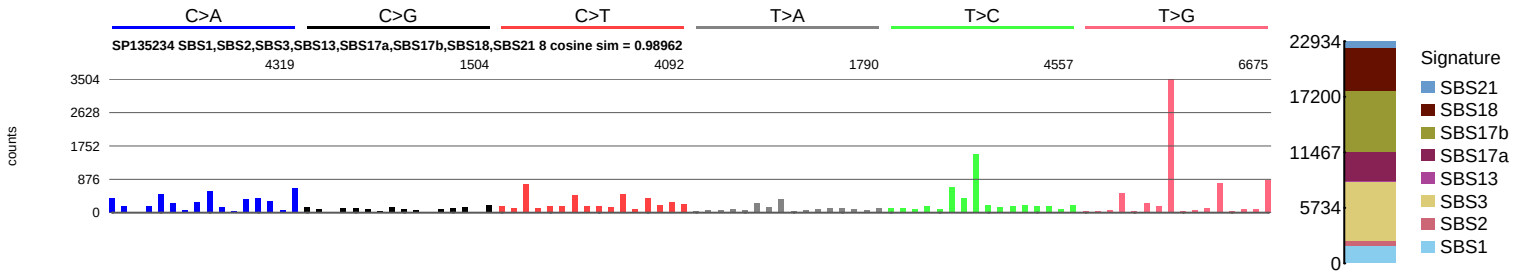
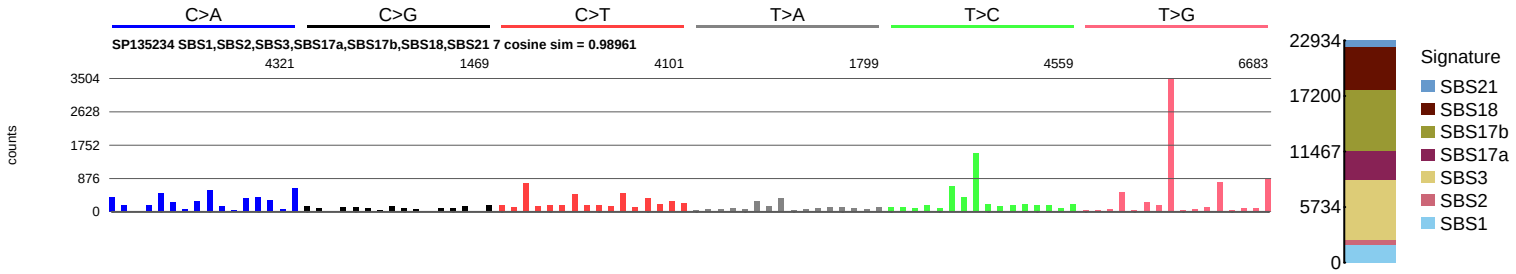
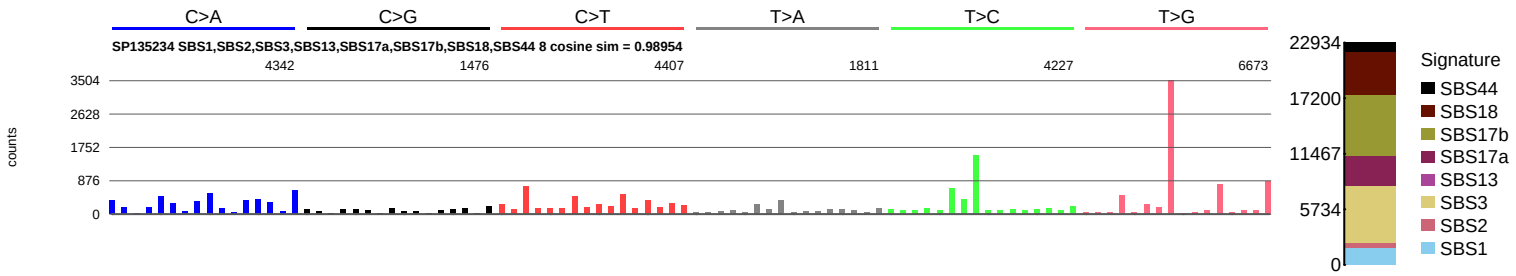
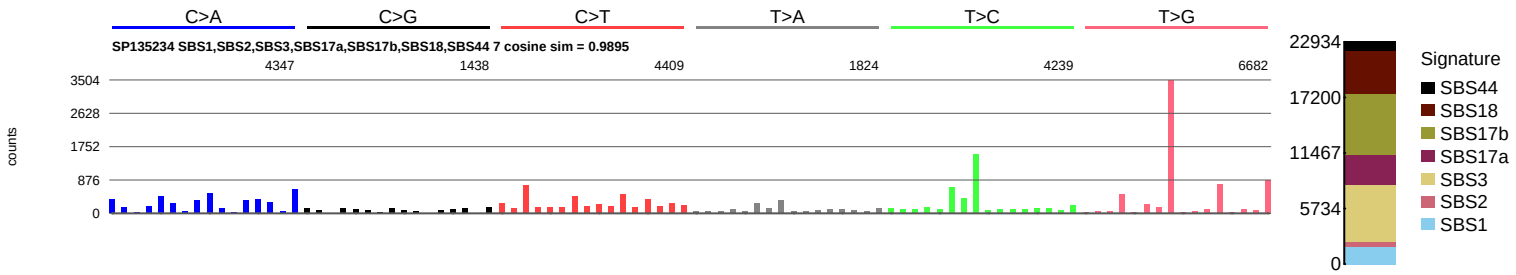
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98973
4285
1449
4041
1725
4745
6685
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

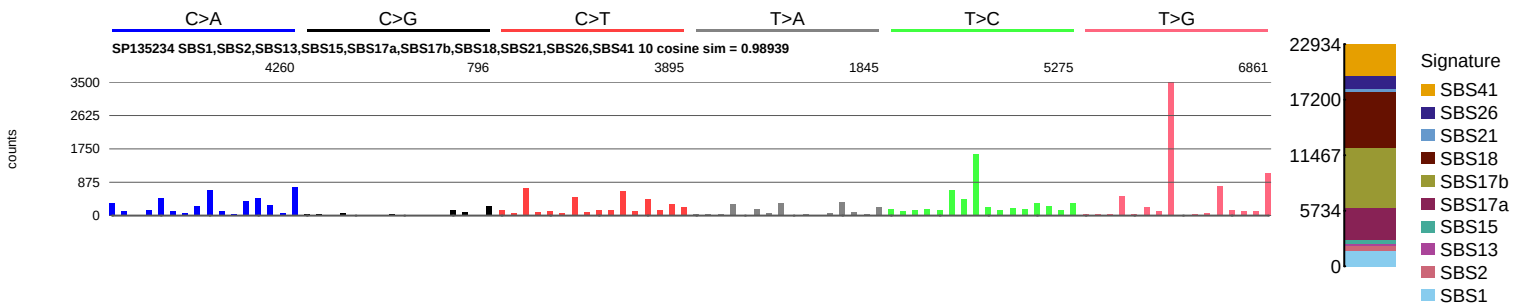

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98971
4290
1410
4053
1743
4746
6694
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

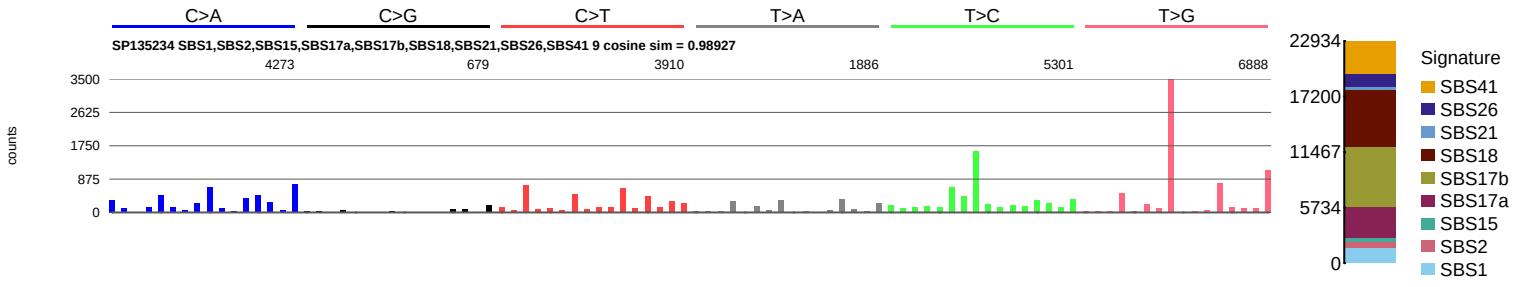

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98962
4319
1504
4092
1790
4557
6675
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

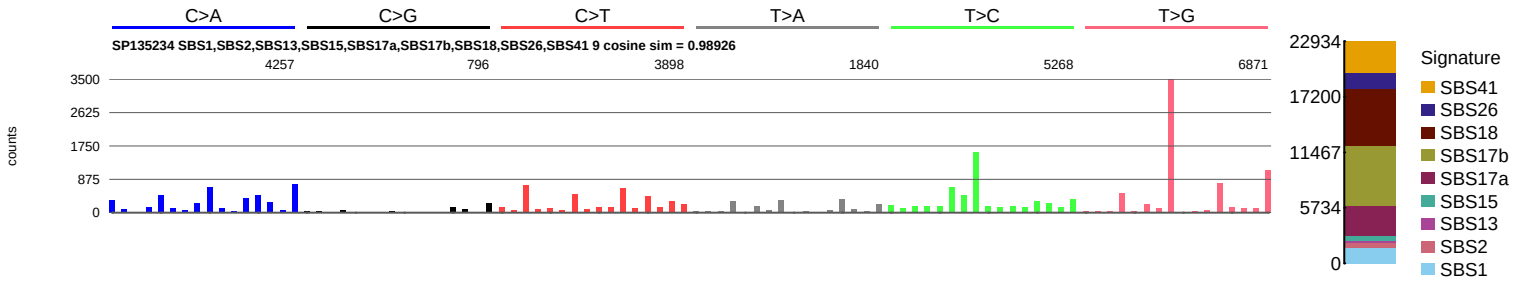

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98961
4321
1469
4101
1799
4559
6683
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98954
4342
1476
4407
1811
4227
6673
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.9895
4347
1438
4409
1824
4239
6682
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98939
4260
796
3895
1845
5275
6861
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98927
4273
679
3910
1886
5301
6888
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98926
4257
796
3898
1840
5268
6871
counts
3500
2625
1750
875
0
22934
17200
11467
5734
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Signature
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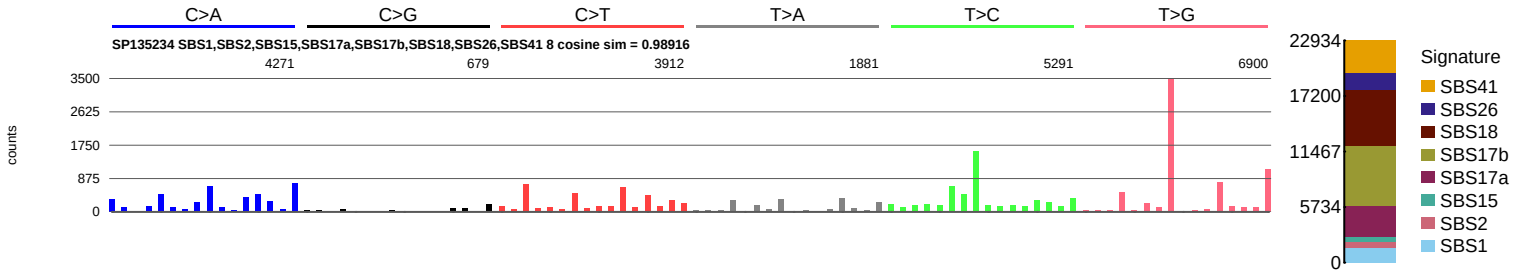
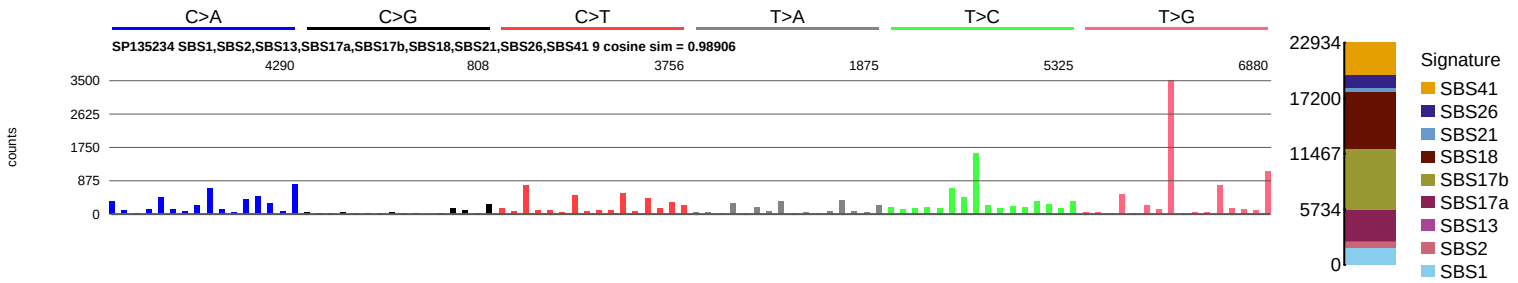
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98916
4271
679
3912
1881
5291
6900
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
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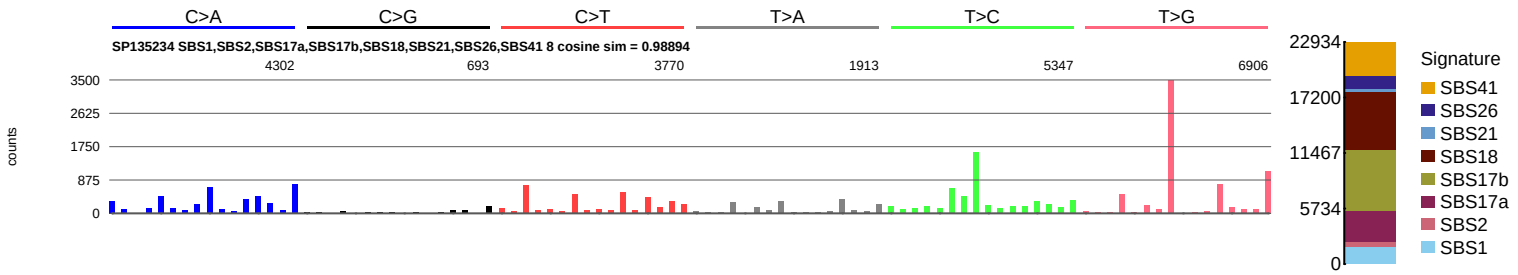
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98906
4290
808
3756
1875
5325
6880
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
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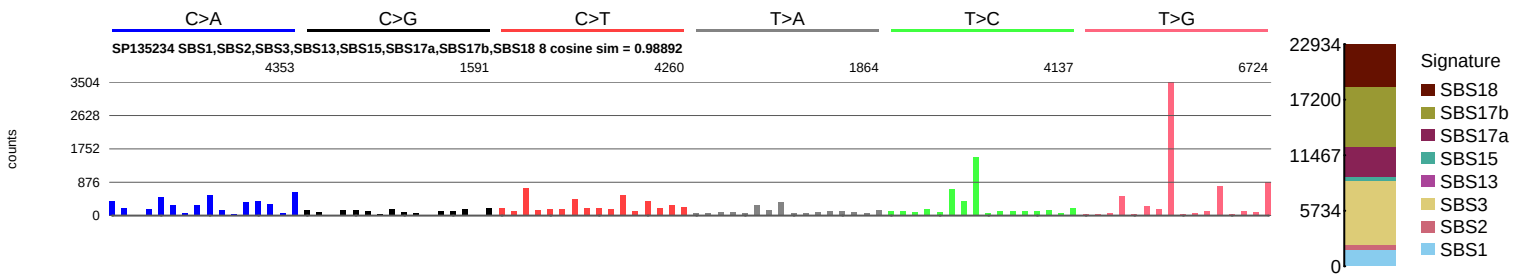
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98894
4302
693
3770
1913
5347
6906
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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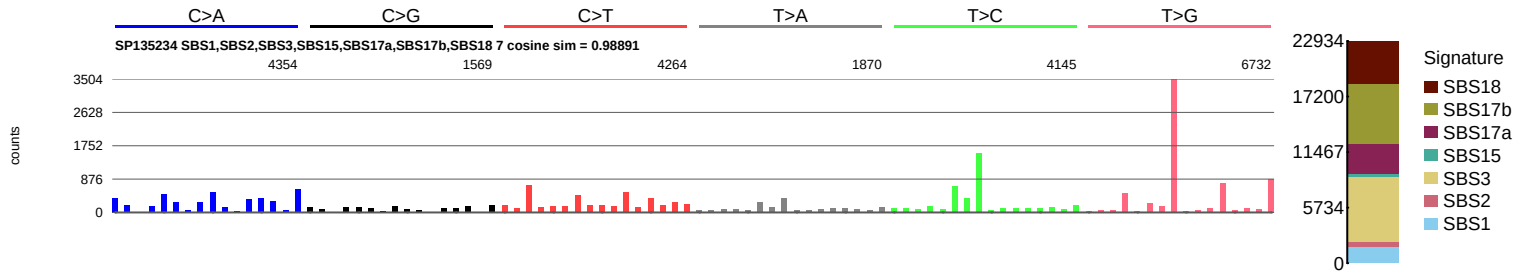
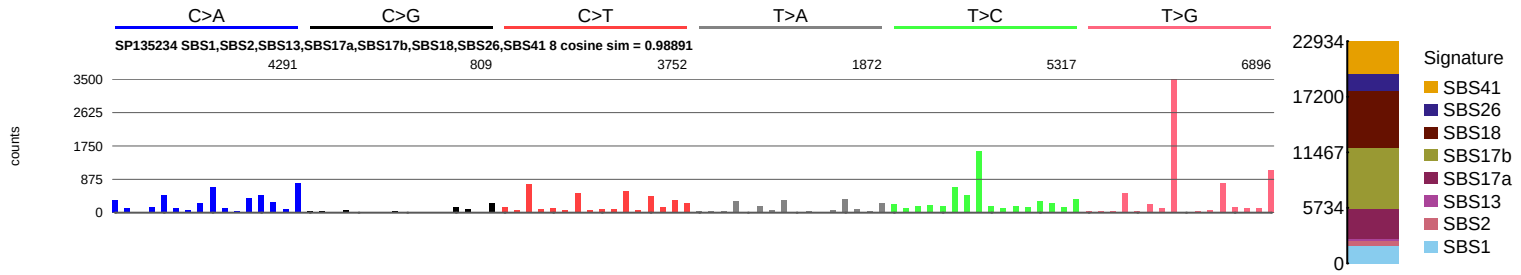
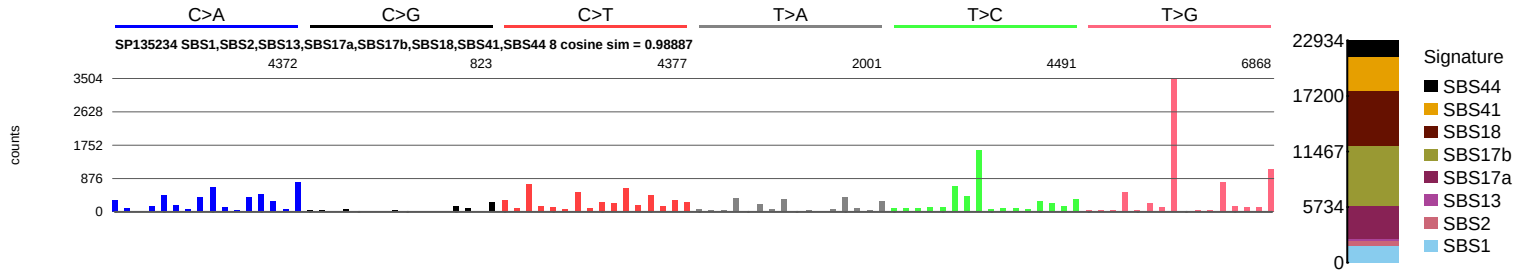
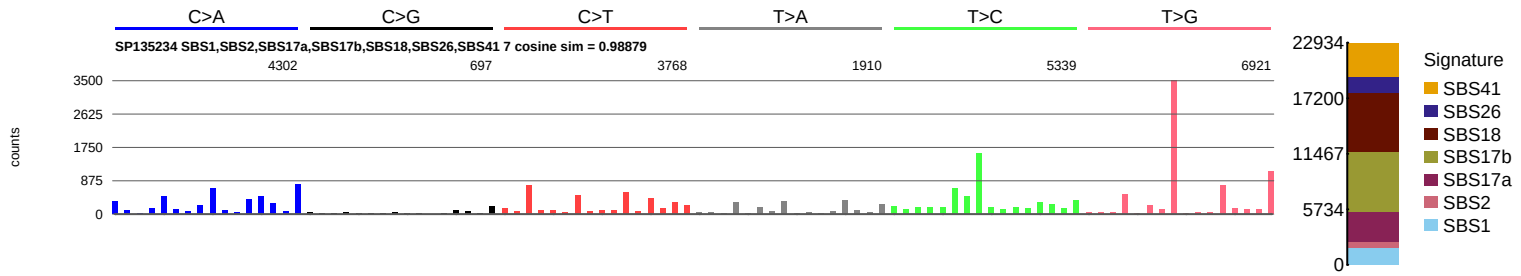
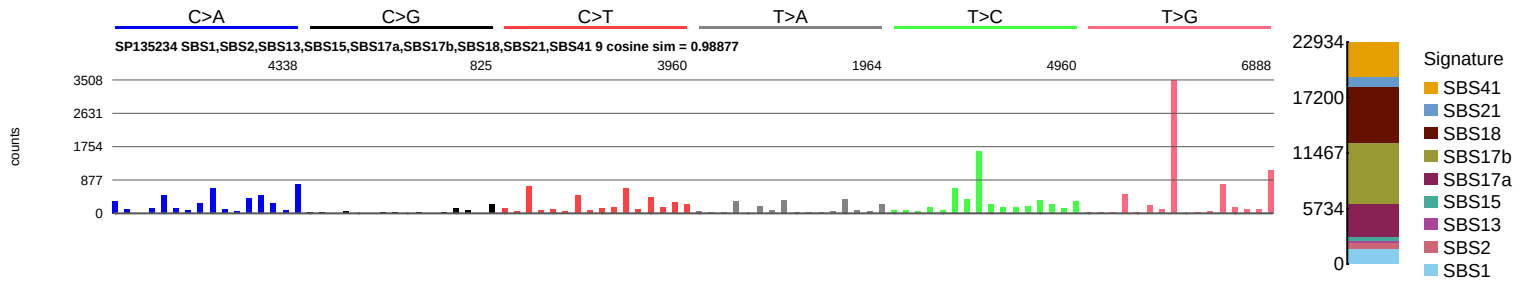
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98892
4353
1591
4260
1864
4137
6724
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98891
4354
1569
4264
1870
4145
6732
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98891
4291
809
3752
1872
5317
6896
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98887
4372
823
4377
2001
4491
6868
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98879
4302
697
3768
1910
5339
6921
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98877
4338
825
3960
1964
4960
6888
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
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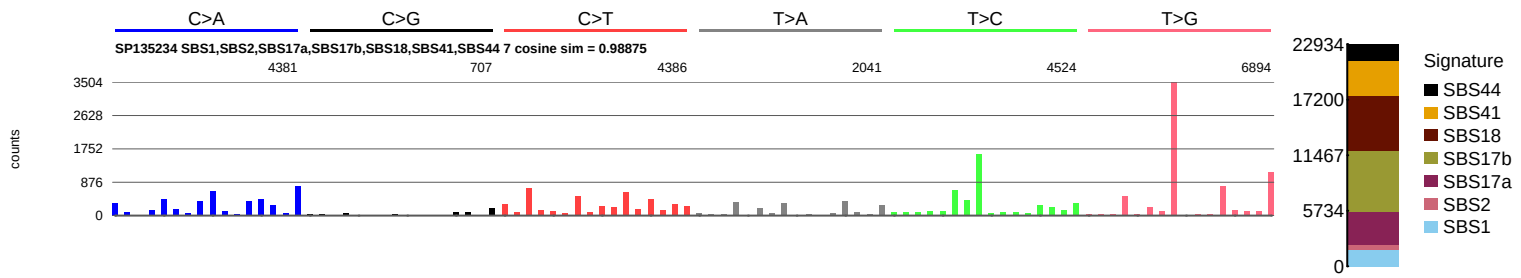
SBS2
SBS1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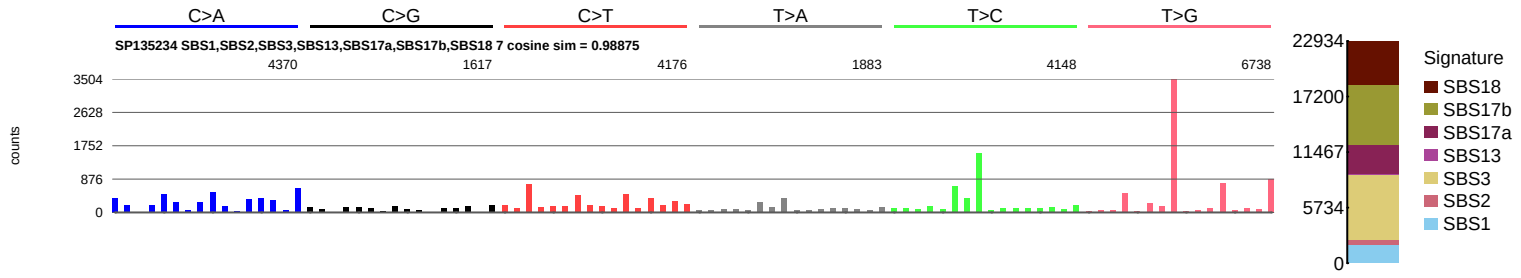
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98875
4381
707
4386
2041
4524
6894
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1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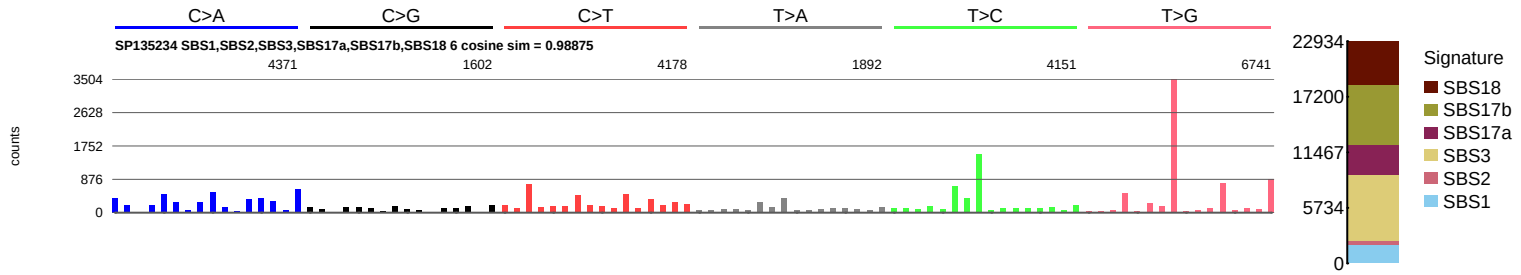
SP135234 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98875
4370
1617
4176
1883
4148
6738
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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SP135234 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98875
4371
1602
4178
1892
4151
6741
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
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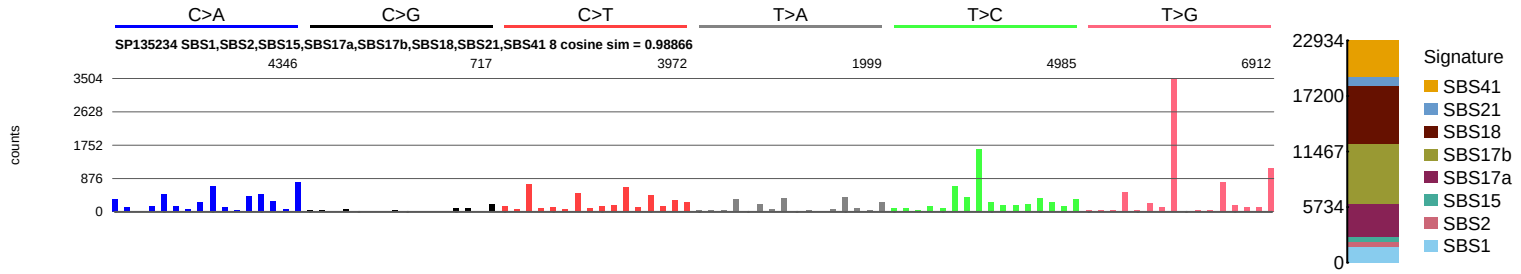
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98866
4346
717
3972
1999
4985
6912
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
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SBS41
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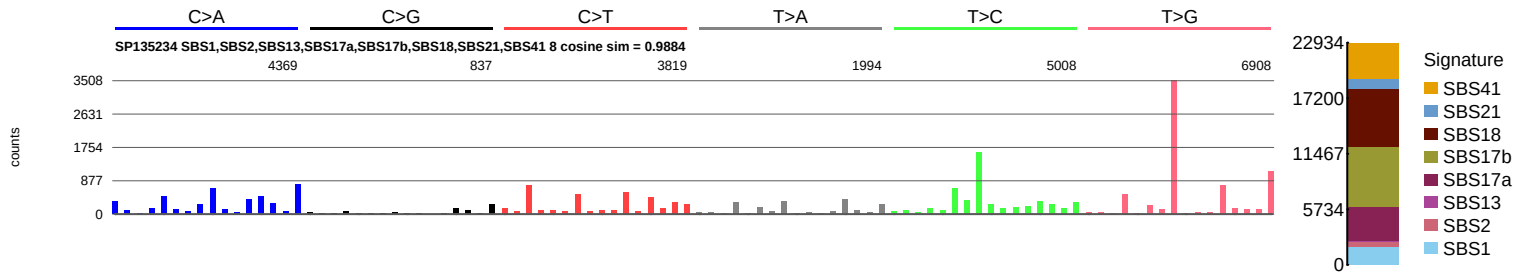
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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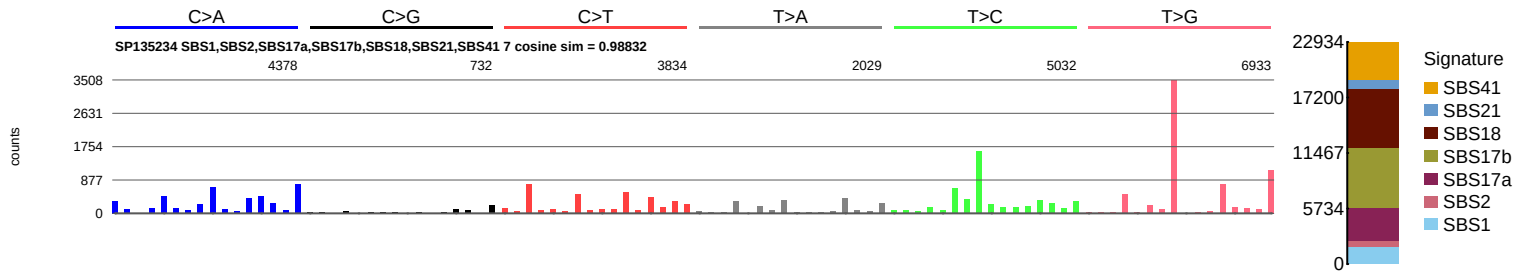
T>G
SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.9884
4369
837
3819
1994
5008
6908
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
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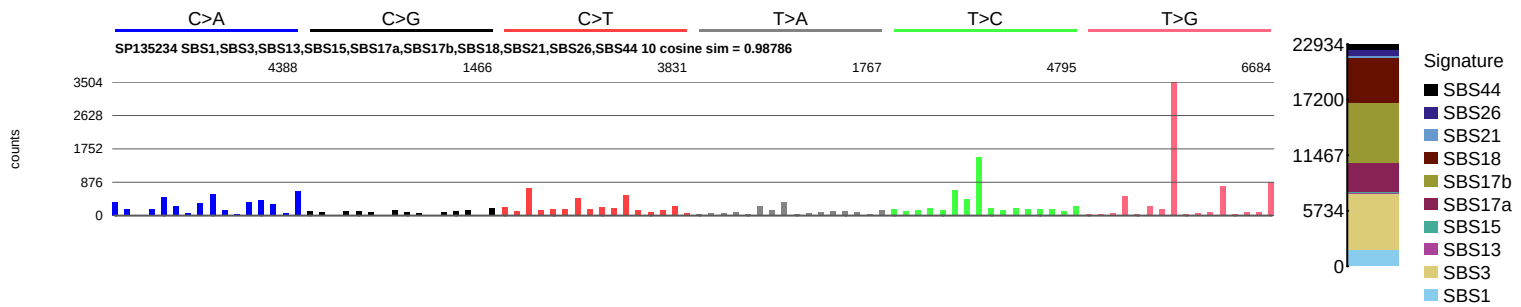
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98832
4378
732
3834
2029
5032
6933
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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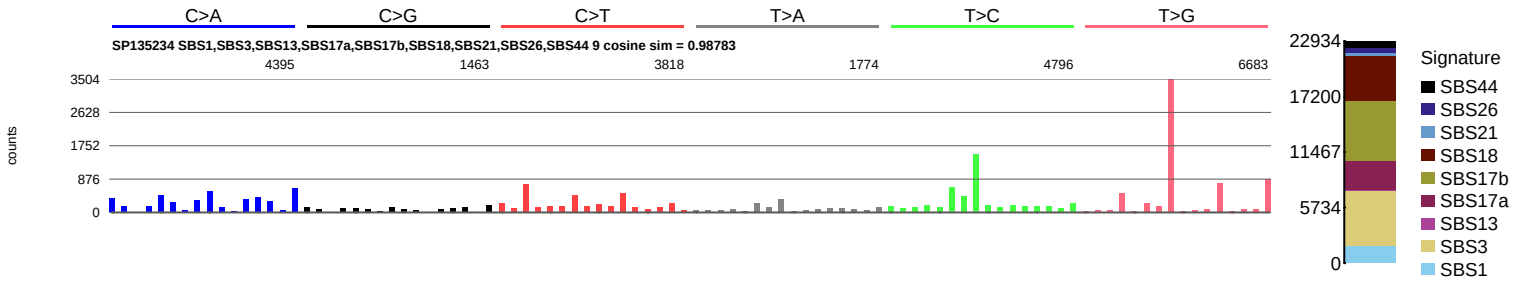
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98786
4388
1466
3831
1767
4795
6684
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98783
4395
1463
3818
1774
4796
6683
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
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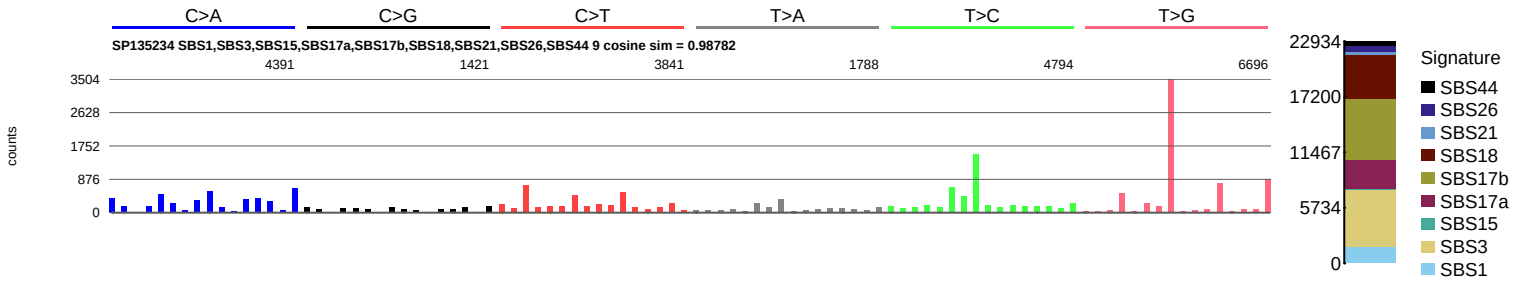
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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C>G
C>T
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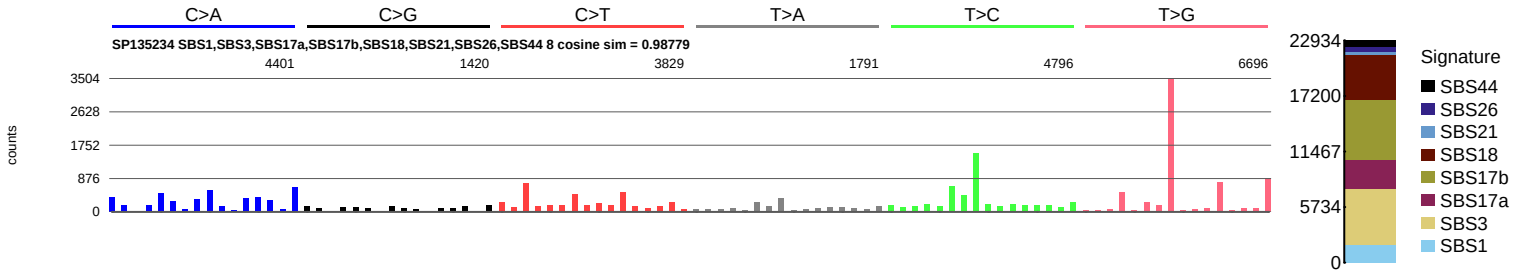
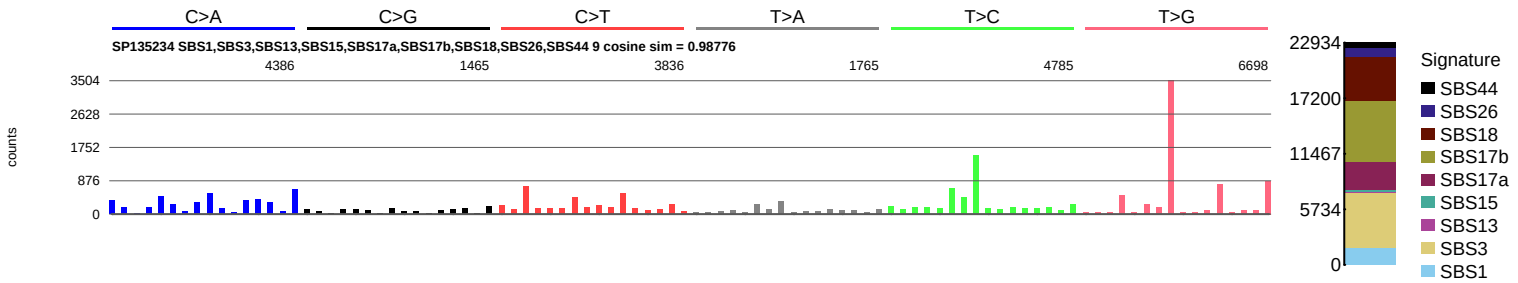
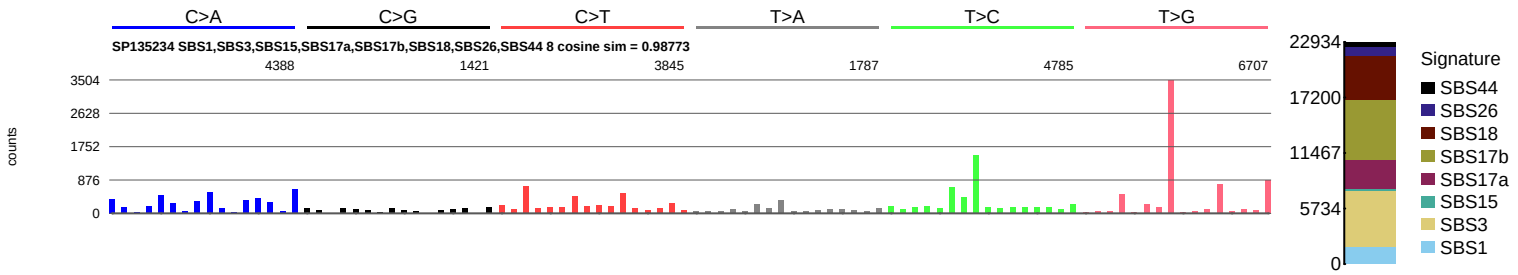
T>C
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98782
4391
1421
3841
1788
4794
6696
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.98779
4401
1420
3829
1791
4796
6696
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98776
4386
1465
3836
1765
4785
6698
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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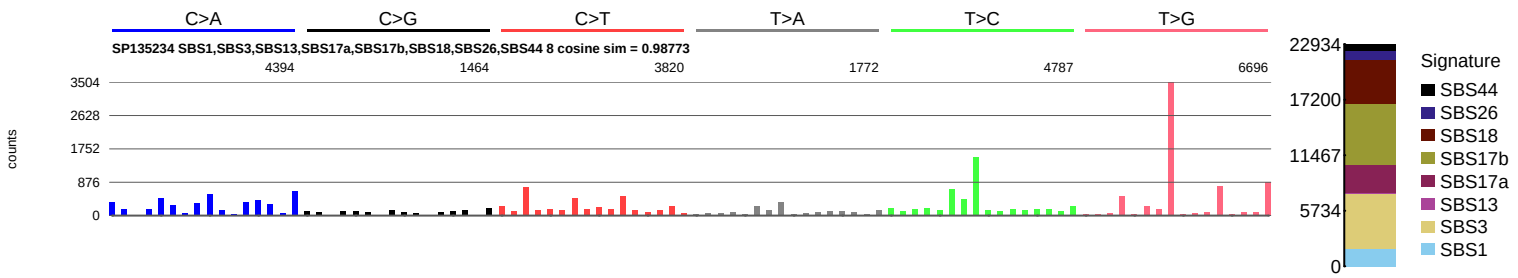
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98773
4388
1421
3845
1787
4785
6707
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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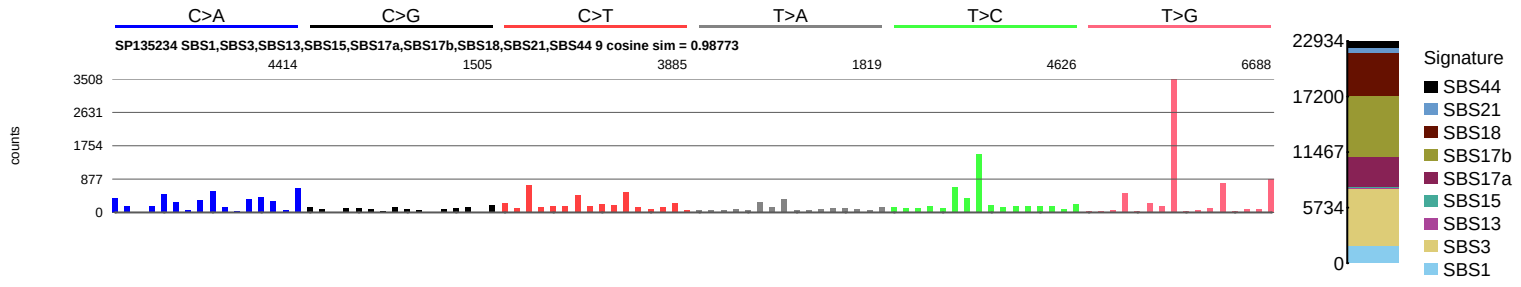
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98773
4394
1464
3820
1772
4787
6696
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98773
4414
1505
3885
1819
4626
6688
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
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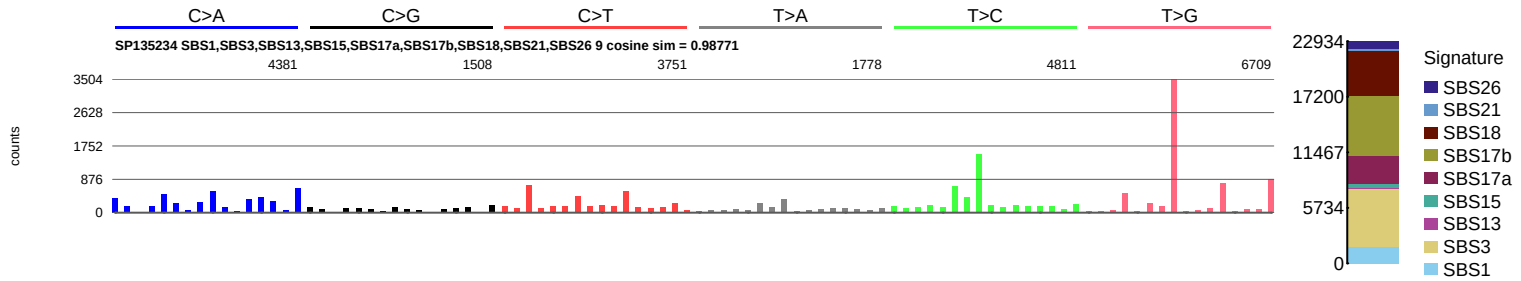
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98771
4381
1508
3751
1778
4811
6709
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
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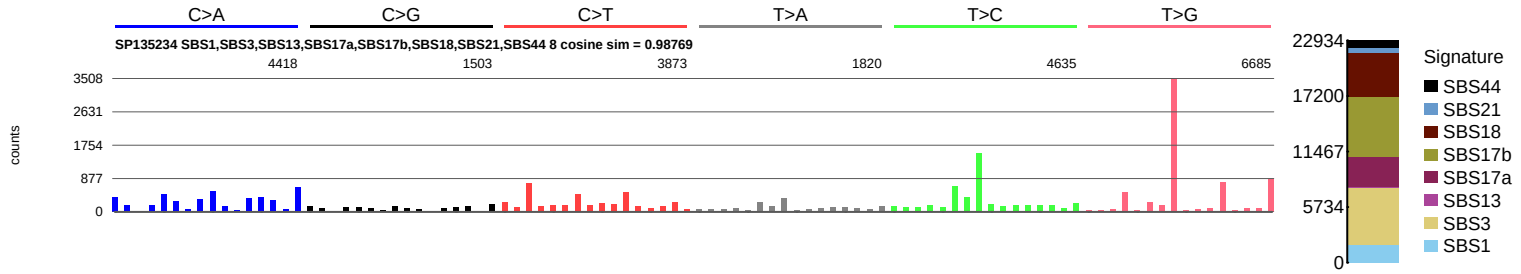
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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C>G
C>T
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T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98769
4418
1503
3873
1820
4635
6685
counts
3508
2631
1754
877
0
22934
17200
11467
5734
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Signature
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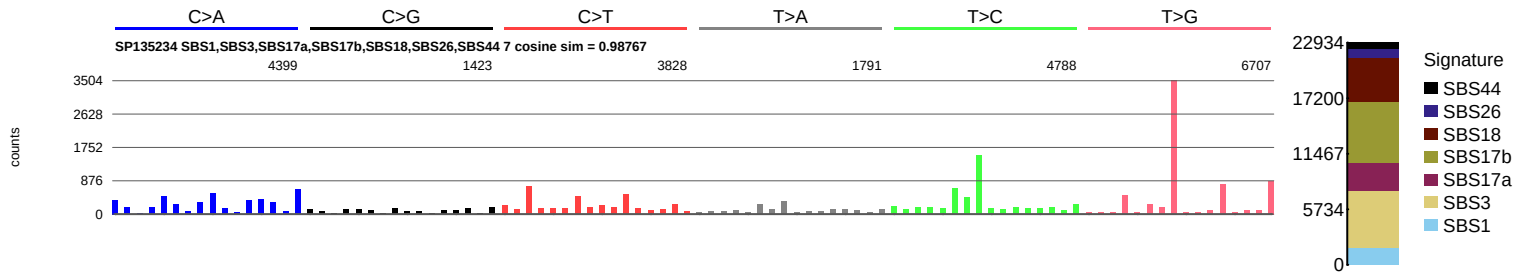
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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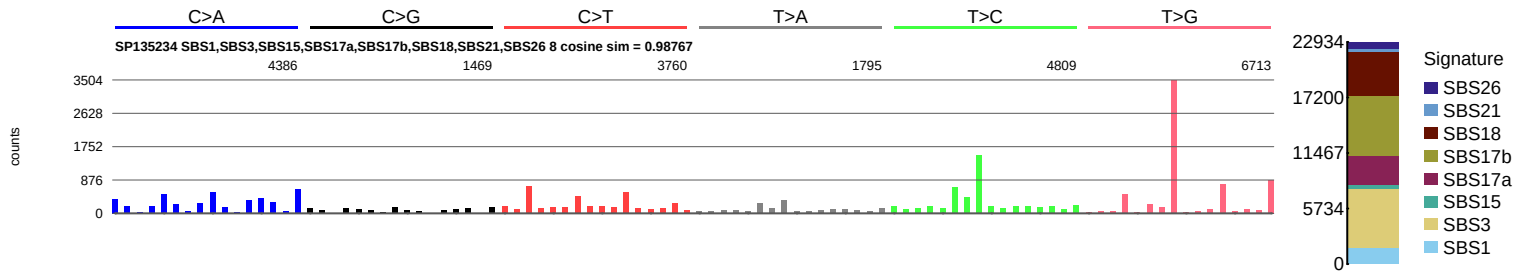
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98767
4399
1423
3828
1791
4788
6707
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1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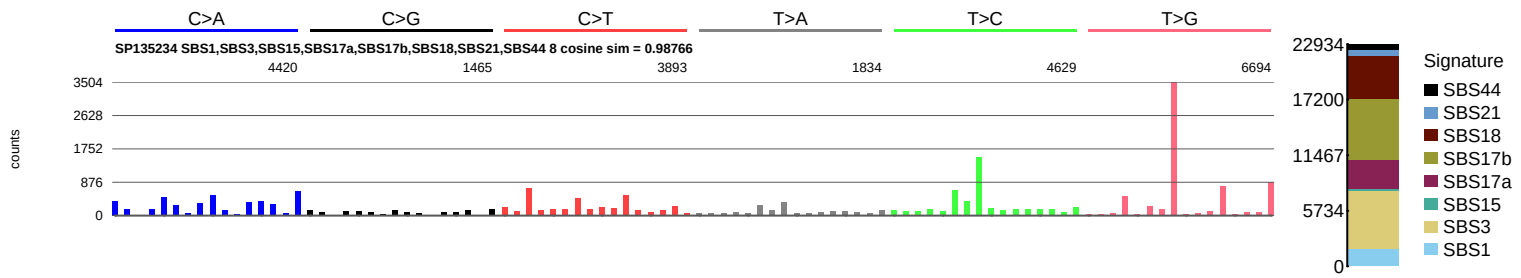
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98767
4386
1469
3760
1795
4809
6713
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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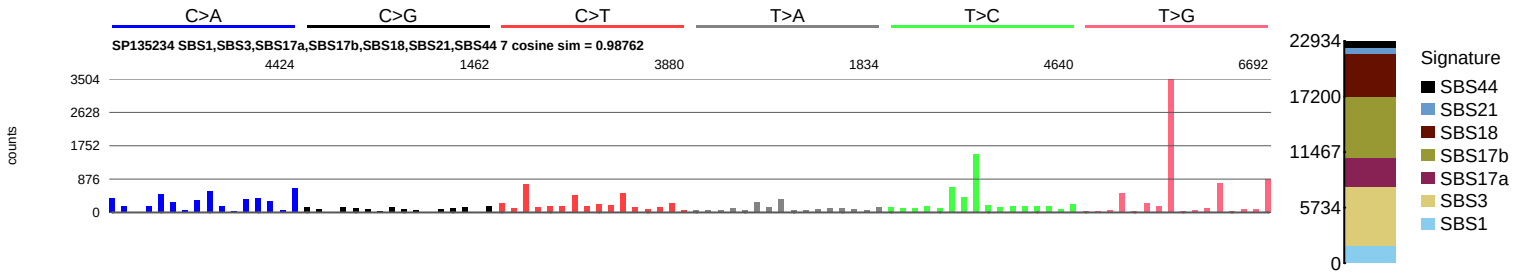
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98766
4420
1465
3893
1834
4629
6694
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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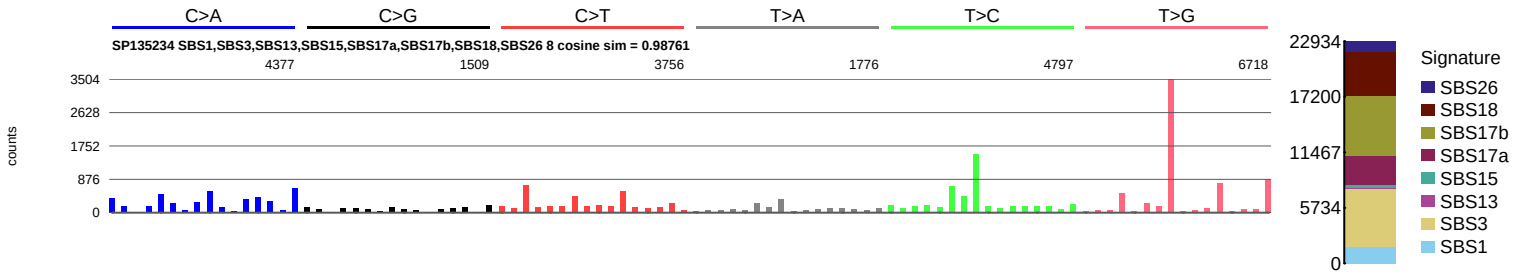
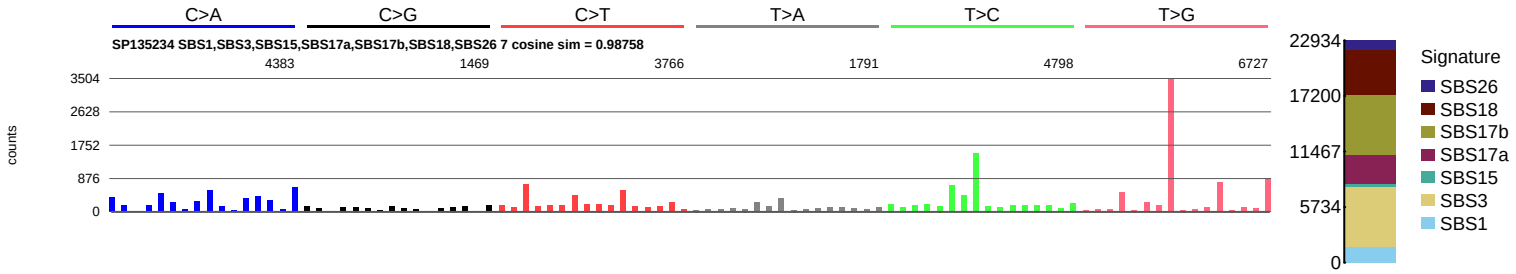
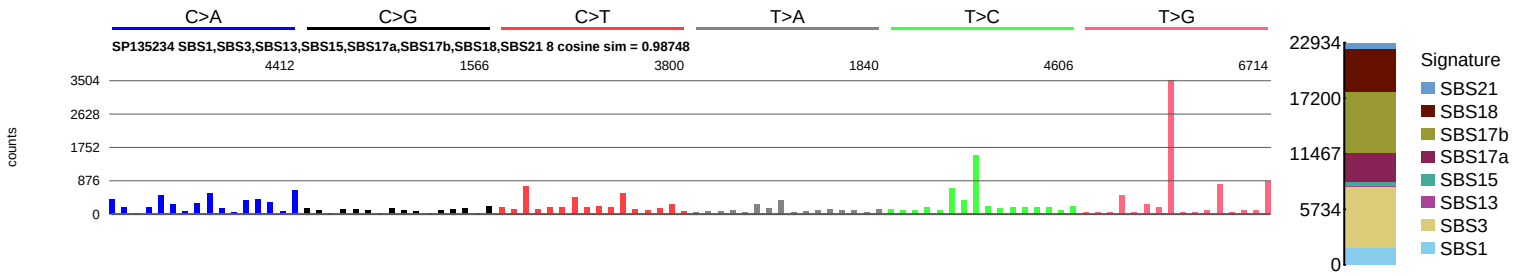
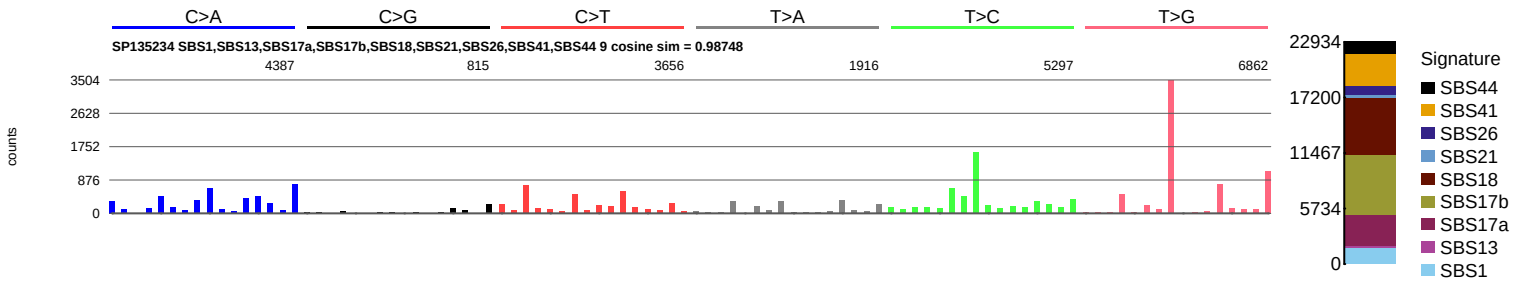
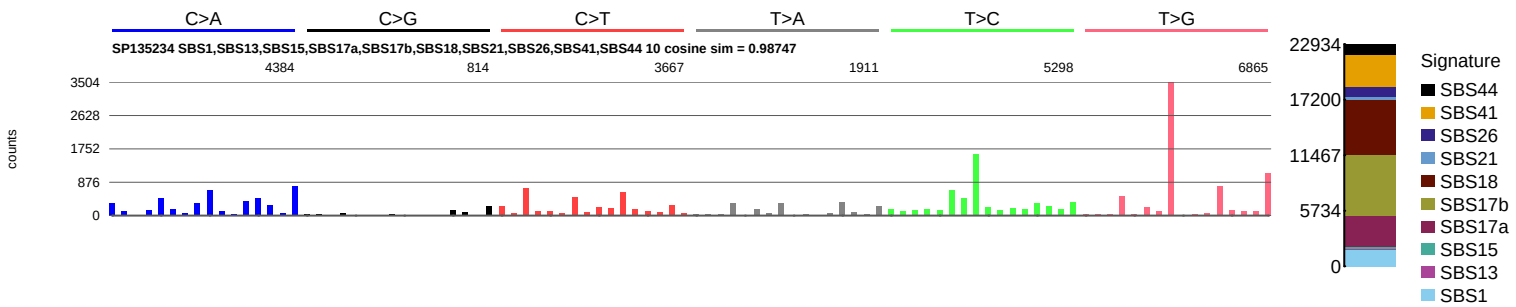
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98762
4424
1462
3880
1834
4640
6692
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98761
4377
1509
3756
1776
4797
6718
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98758
4383
1469
3766
1791
4798
6727
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98748
4412
1566
3800
1840
4606
6714
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98748
4387
815
3656
1916
5297
6862
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98747
4384
814
3667
1911
5298
6865
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
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SBS41
SBS26
SBS21
SBS18
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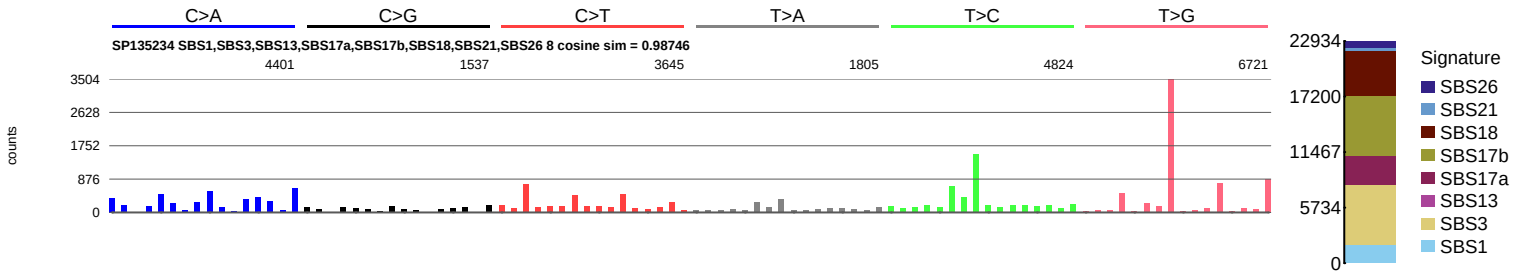
SBS17a
SBS15
SBS13
SBS1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SP135234 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98746
4401
1537
3645
1805
4824
6721
counts
3504
2628
1752
876
0
22934
17200
11467
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SBS26
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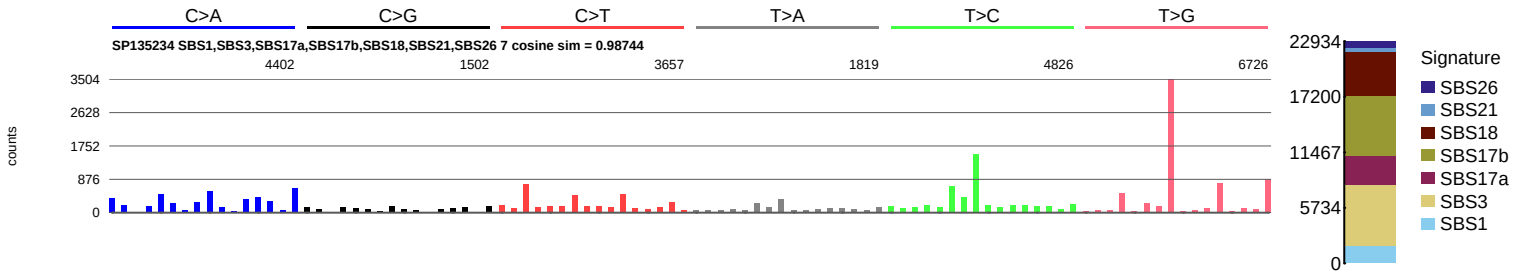
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26 7 cosine sim = 0.98744
4402
1502
3657
1819
4826
6726
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
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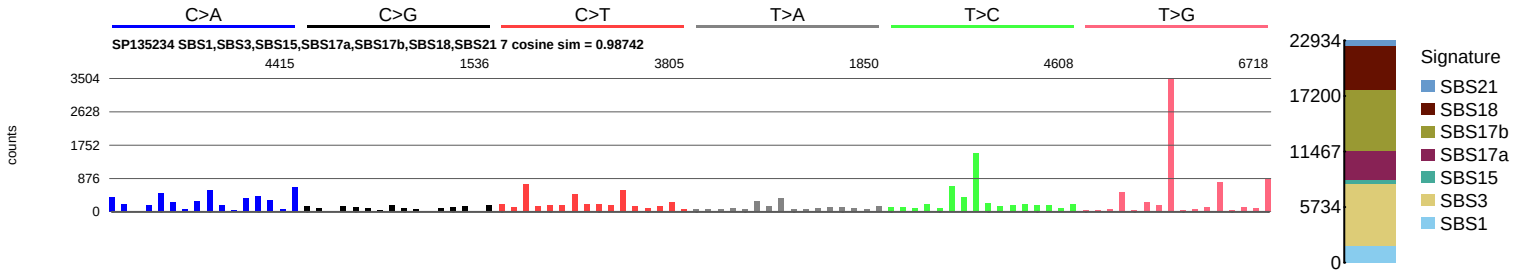
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98742
4415
1536
3805
1850
4608
6718
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
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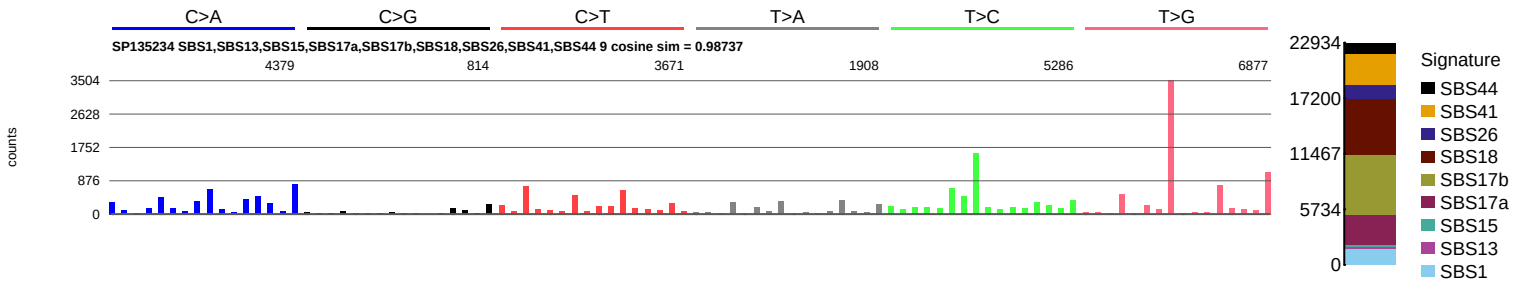
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
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T>A
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SP135234 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98737
4379
814
3671
1908
5286
6877
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
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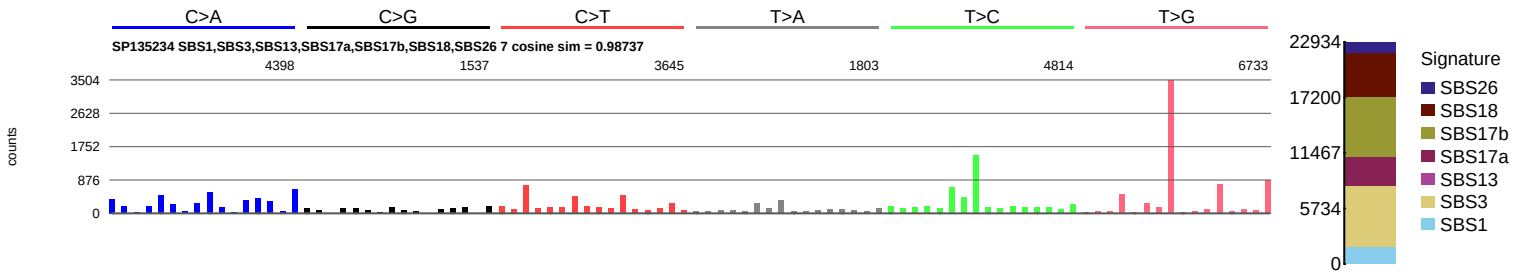
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98737
4398
1537
3645
1803
4814
6733
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
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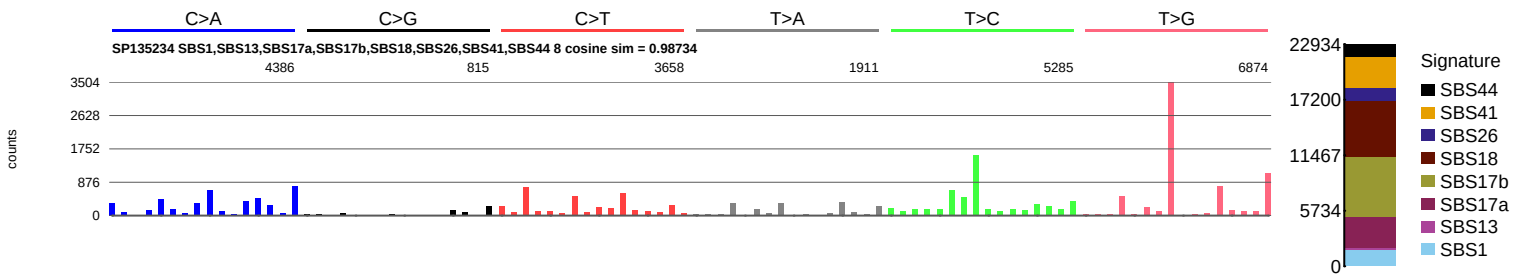
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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C>G
C>T
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T>G
SP135234 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98734
4386
815
3658
1911
5285
6874
counts
3504
2628
1752
876
0
22934
17200
11467
5734
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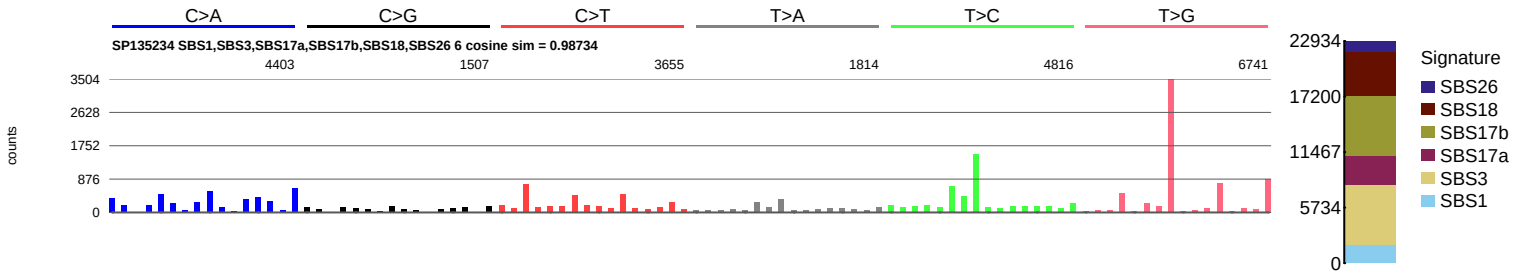
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1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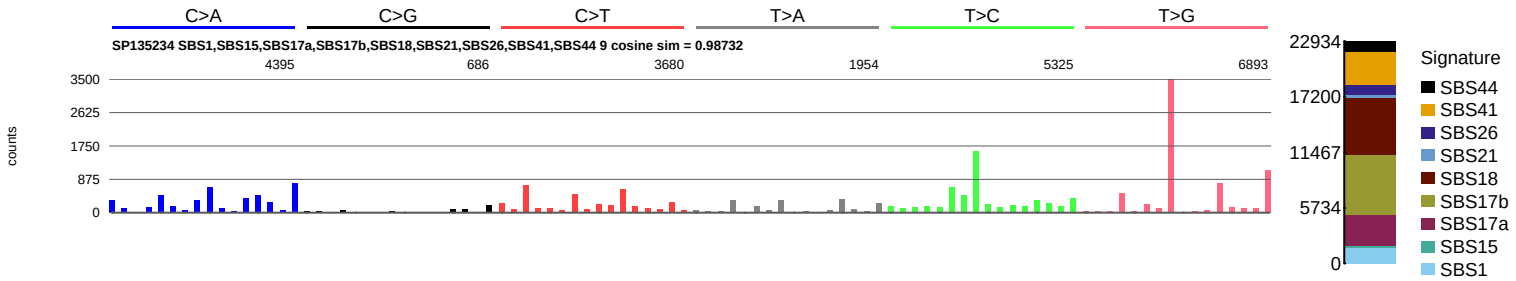
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98734
4403
1507
3655
1814
4816
6741
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98732
4395
686
3680
1954
5325
6893
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
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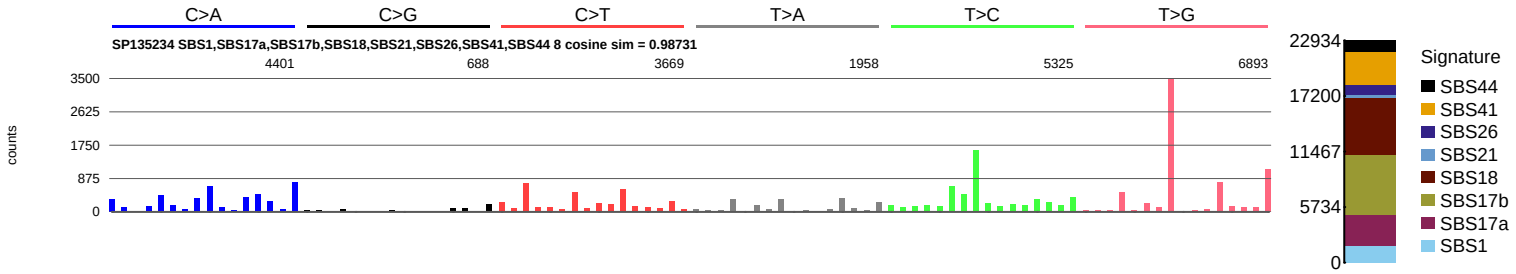
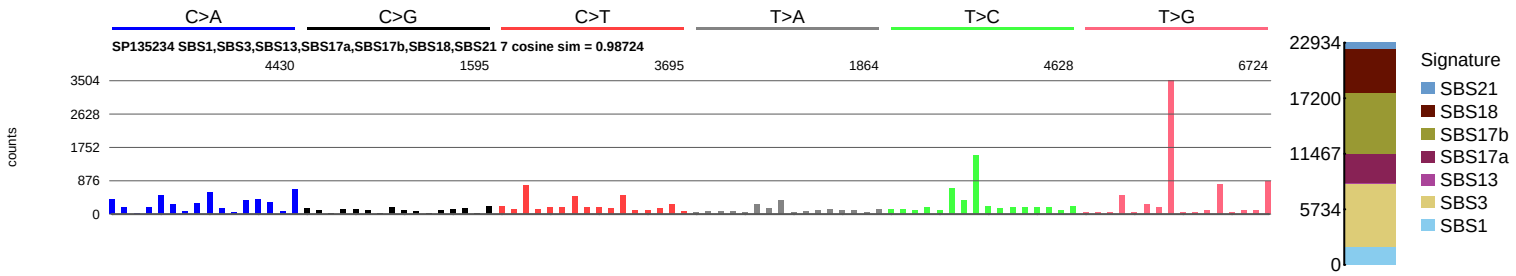
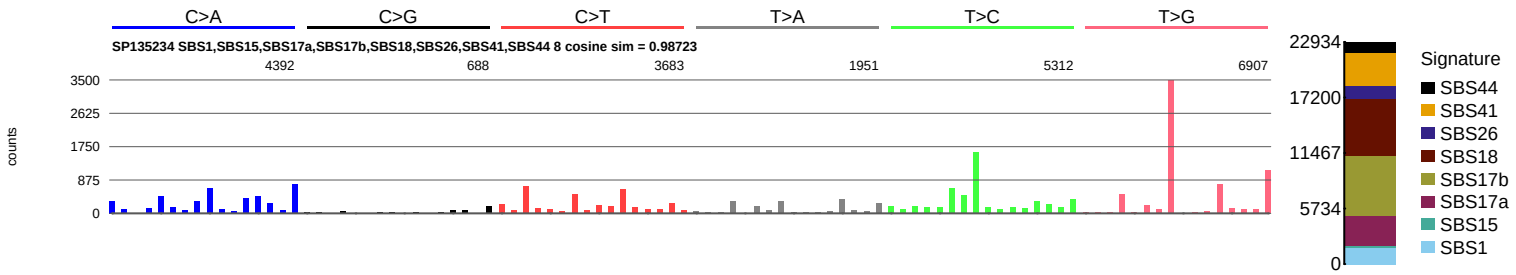
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98731
4401
688
3669
1958
5325
6893
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98724
4430
1595
3695
1864
4628
6724
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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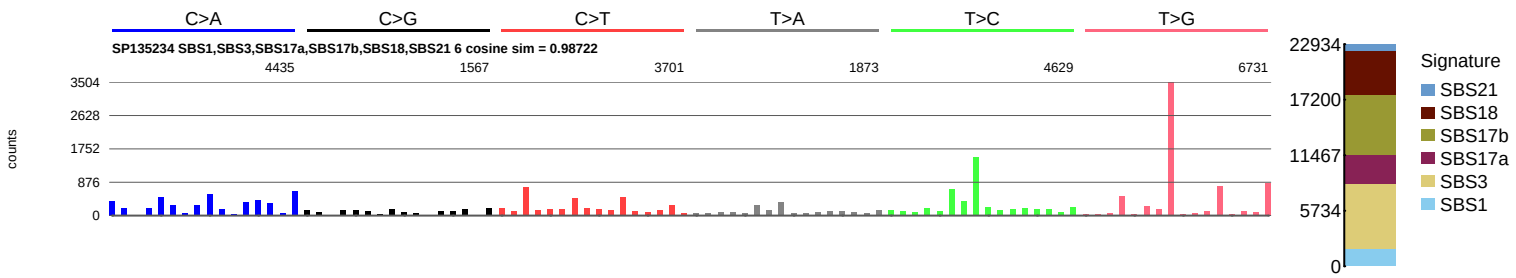
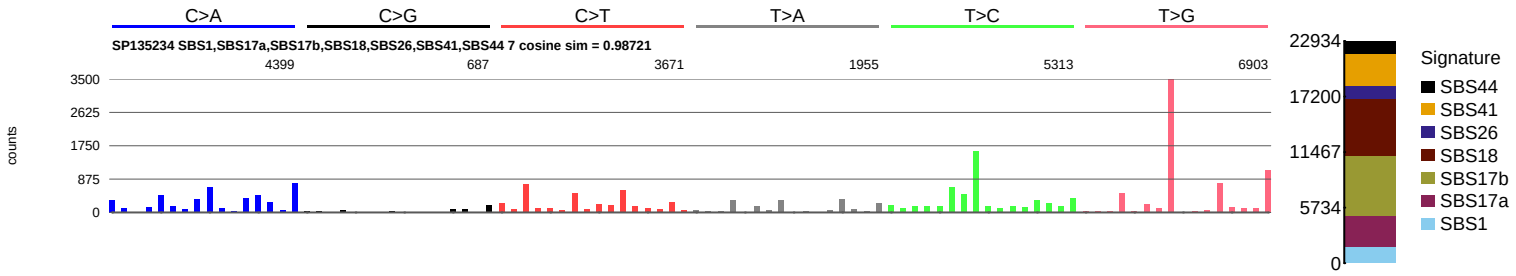
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98723
4392
688
3683
1951
5312
6907
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98722
4435
1567
3701
1873
4629
6731
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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C>G
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T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98721
4399
687
3671
1955
5313
6903
counts
3500
2625
1750
875
0
22934
17200
11467
5734
0
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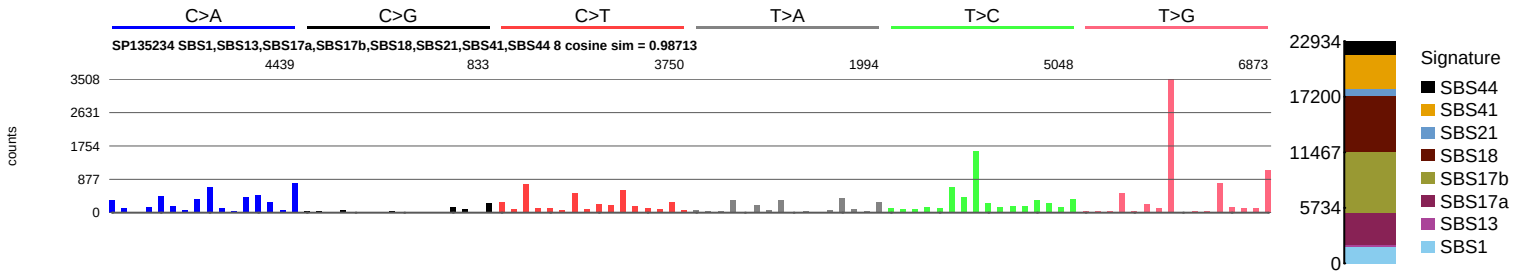
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1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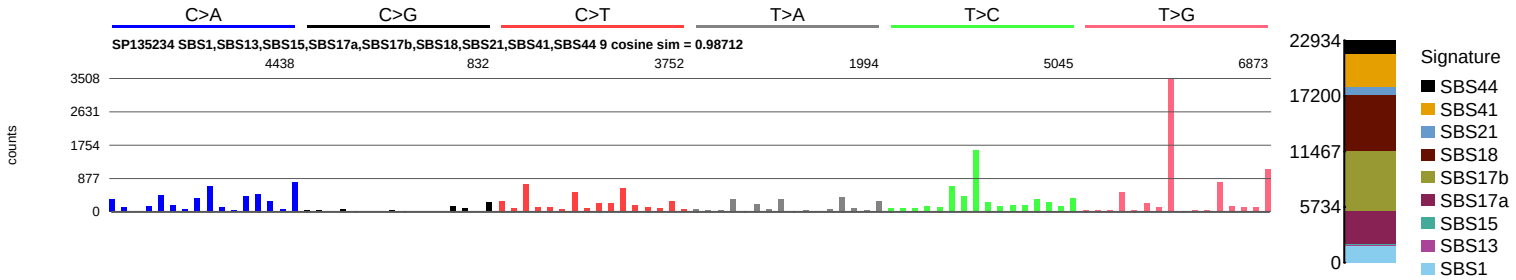
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98713
4439
833
3750
1994
5048
6873
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1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C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98712
4438
832
3752
1994
5045
6873
counts
3508
2631
1754
877
0
22934
17200
11467
5734
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Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
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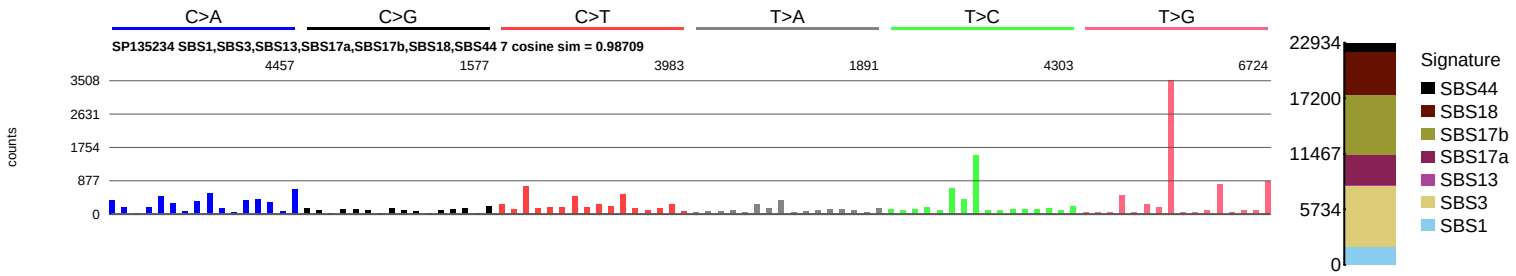
SBS15
SBS13
SBS1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C>G
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T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98709
4457
1577
3983
1891
4303
6724
counts
3508
2631
1754
877
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SBS17a
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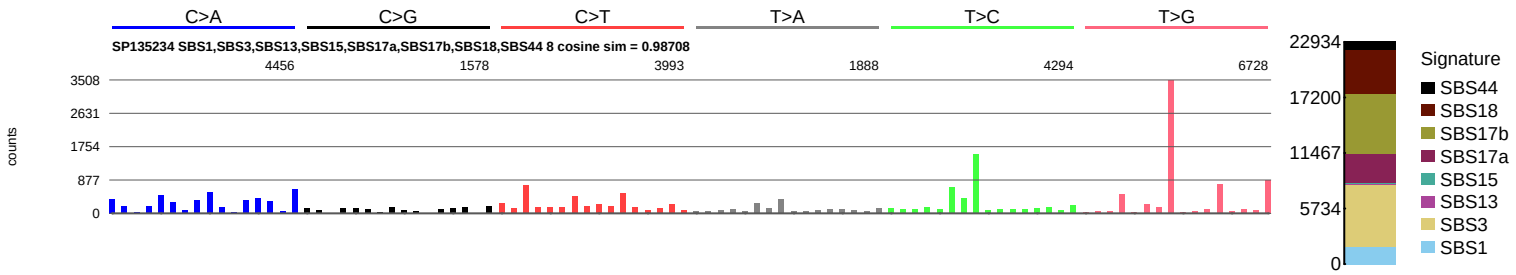
SBS3
SBS1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T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98708
4456
1578
3993
1888
4294
6728
counts
3508
2631
1754
877
0
22934
17200
11467
5734
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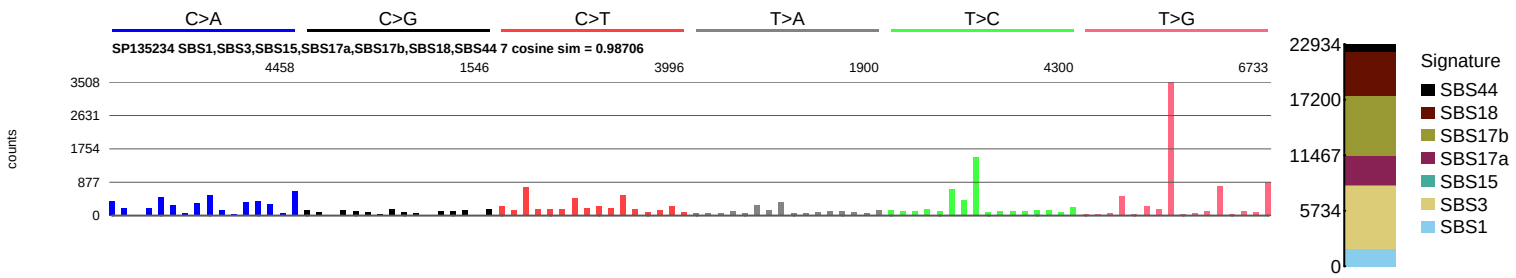
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
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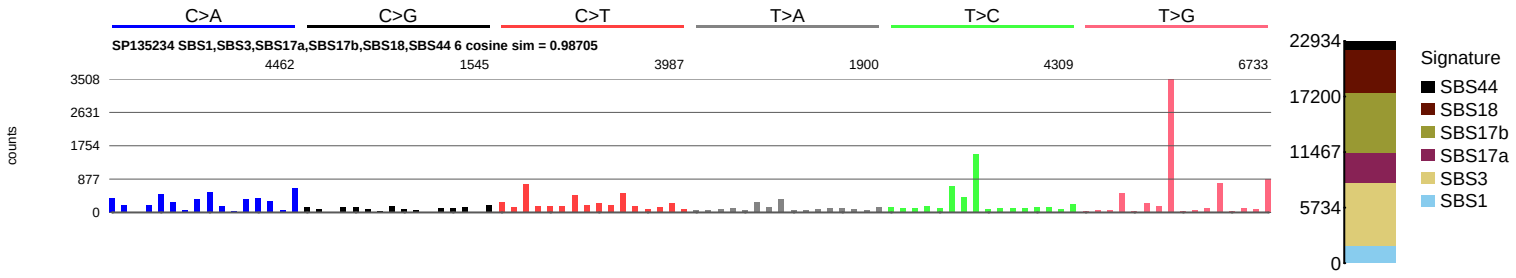
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98706
4458
1546
3996
1900
4300
6733
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
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T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98705
4462
1545
3987
1900
4309
6733
counts
3508
2631
1754
877
0
22934
17200
11467
5734
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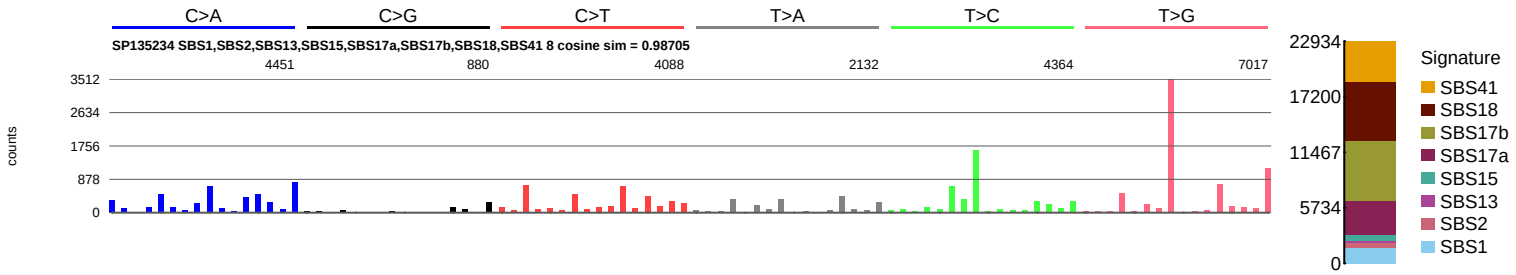
SBS18
SBS17b
SBS17a
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98705
4451
880
4088
2132
4364
7017
counts
3512
2634
1756
878
0
22934
17200
11467
5734
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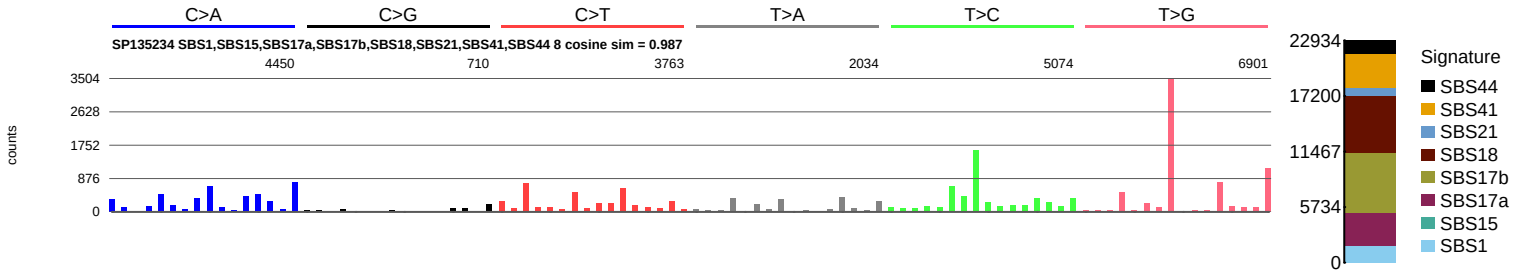
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.987
4450
710
3763
2034
5074
6901
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
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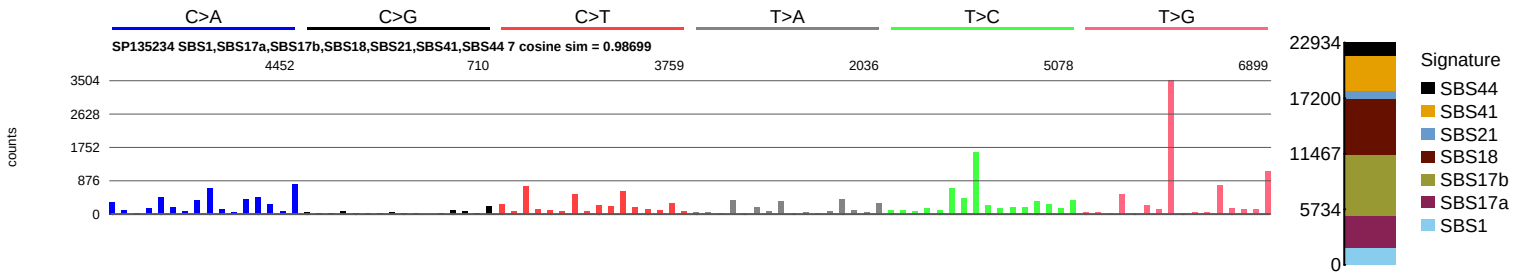
SBS15
SBS1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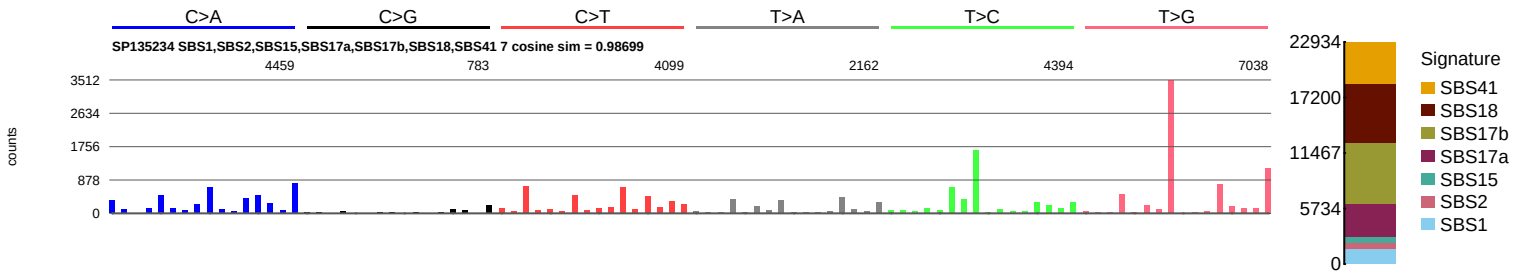
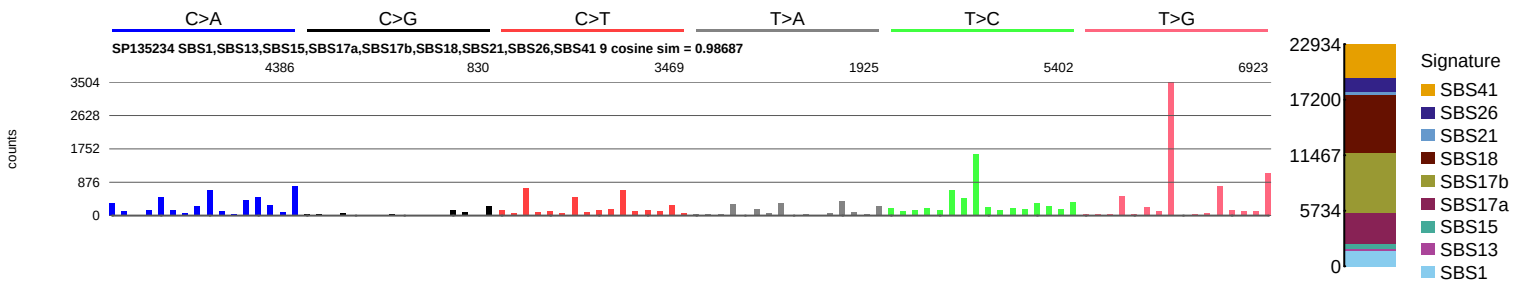
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98699
4452
710
3759
2036
5078
6899
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98699
4459
783
4099
2162
4394
7038
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98687
4386
830
3469
1925
5402
6923
counts
3504
2628
1752
876
0
22934
17200
11467
5734
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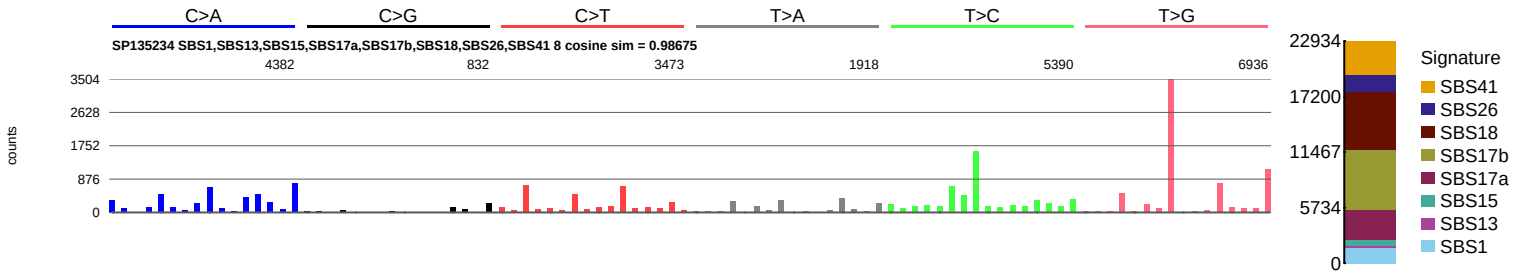
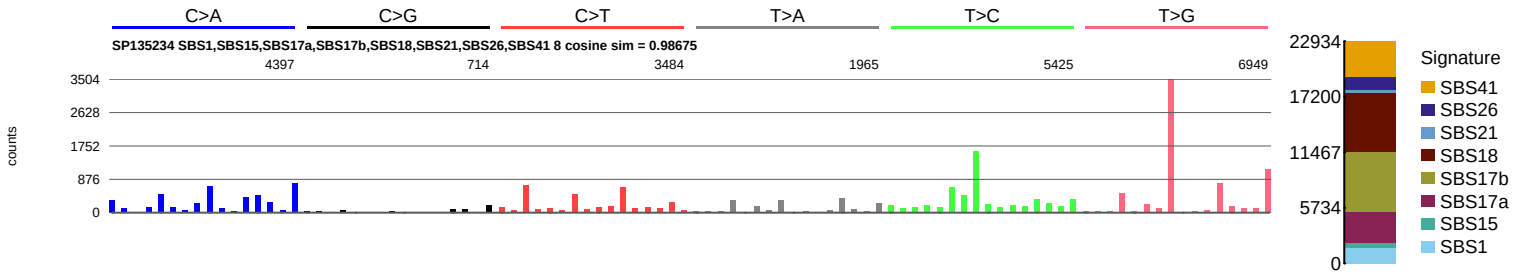
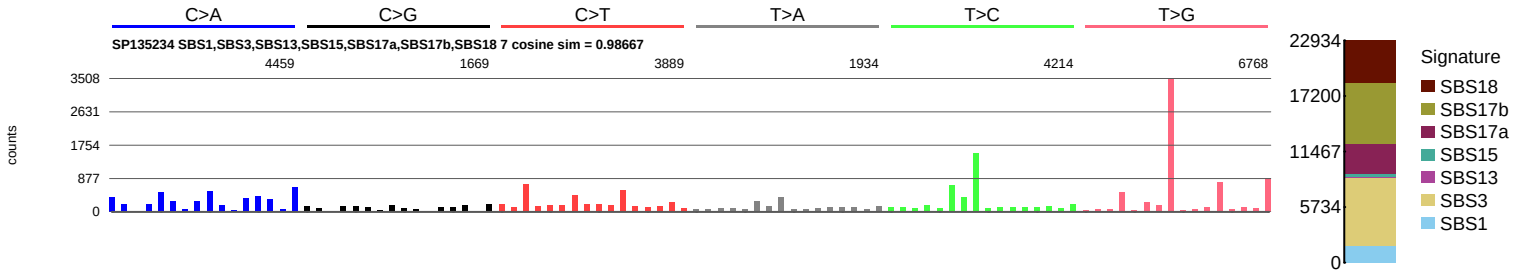
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98675
4382
832
3473
1918
5390
6936
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98675
4397
714
3484
1965
5425
6949
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
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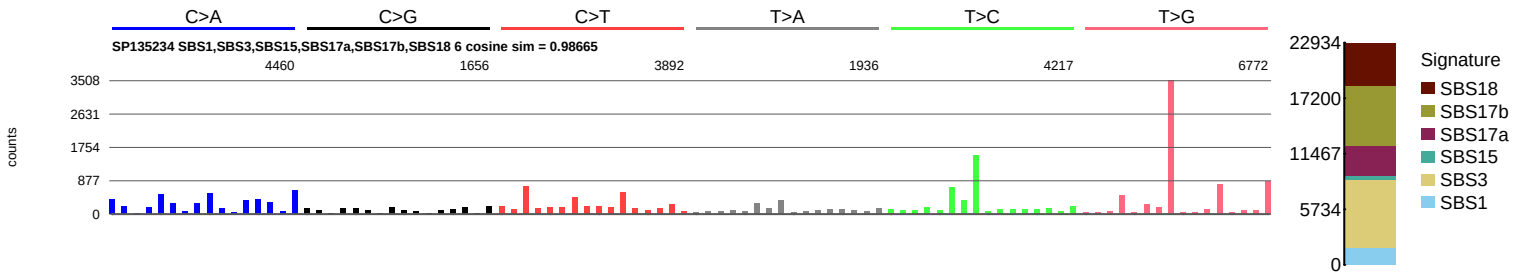
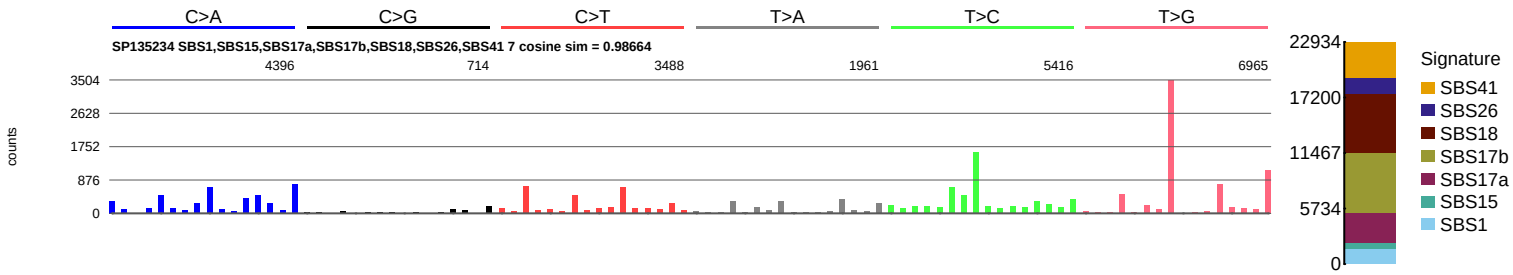
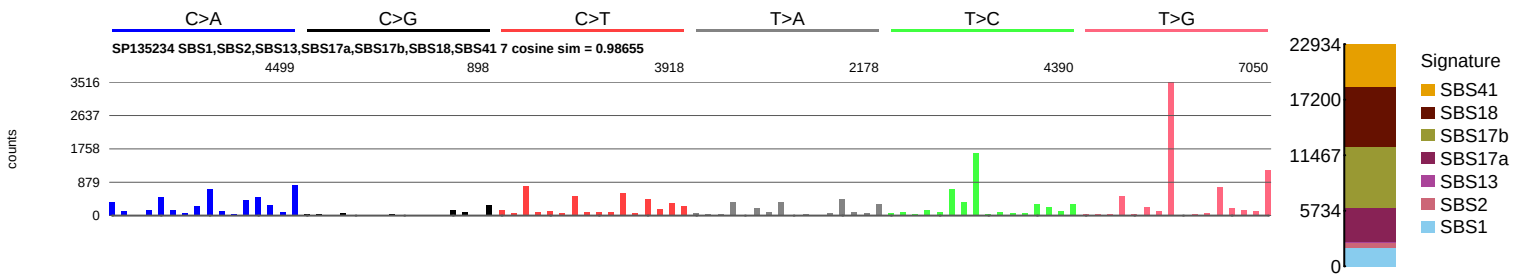
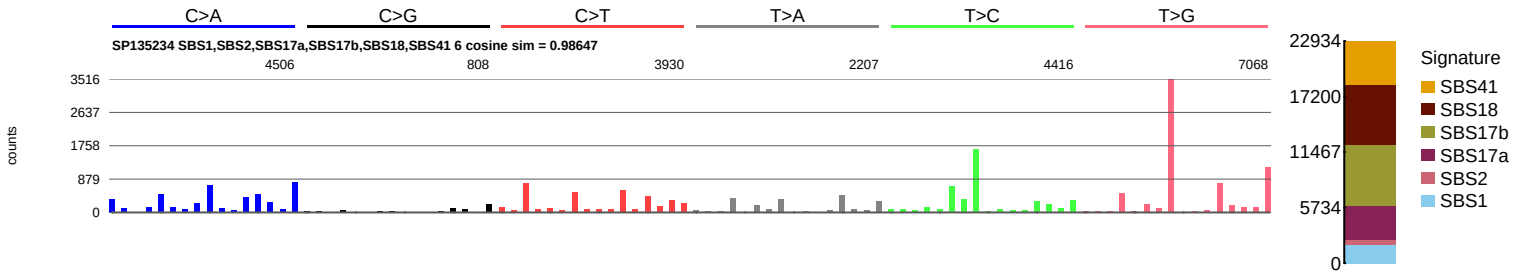
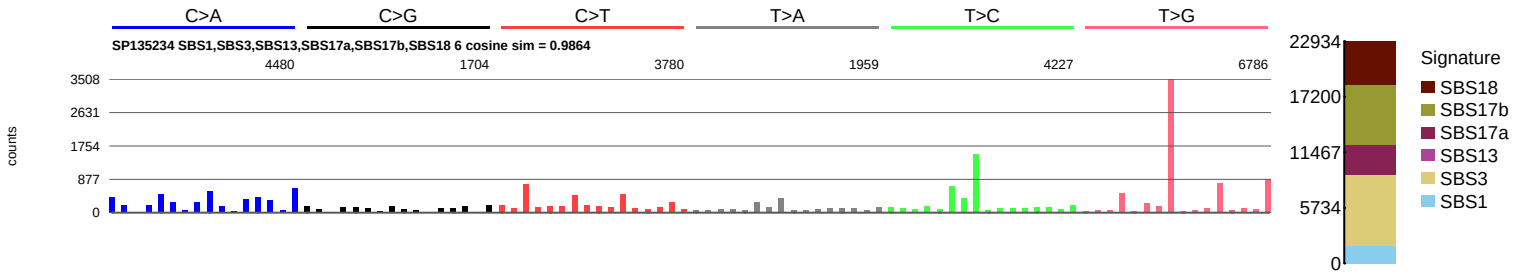
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98667
4459
1669
3889
1934
4214
6768
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

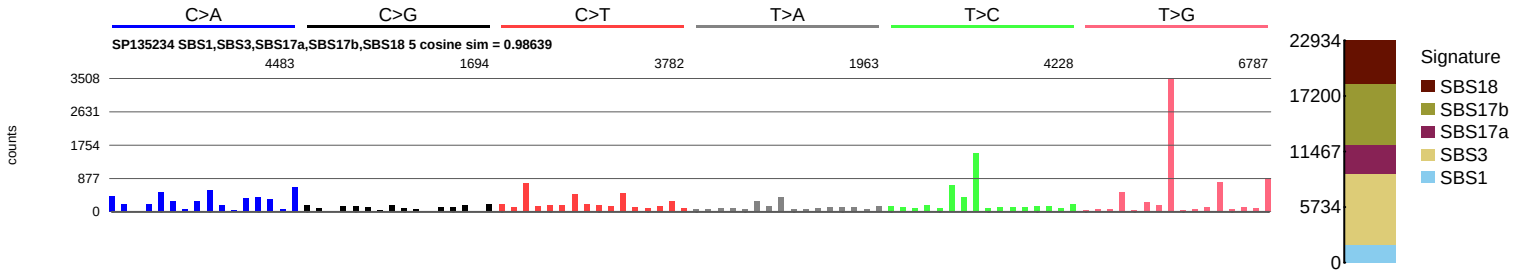

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98665
4460
1656
3892
1936
4217
6772
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

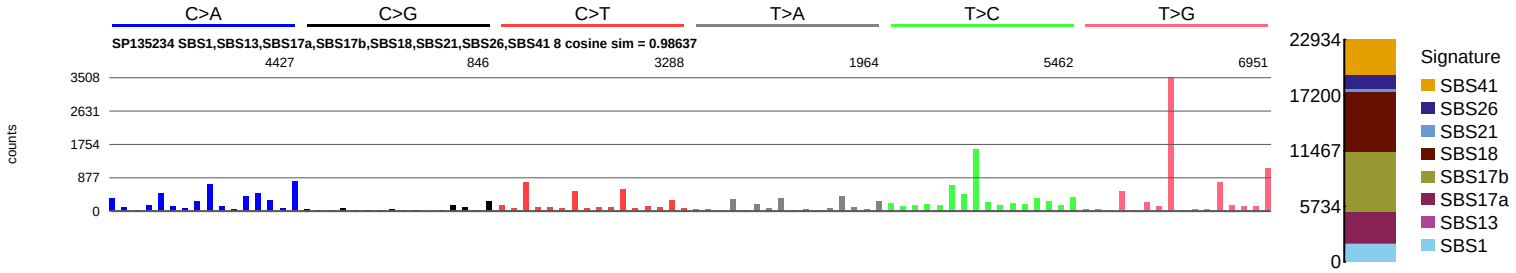

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98664
4396
714
3488
1961
5416
6965
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

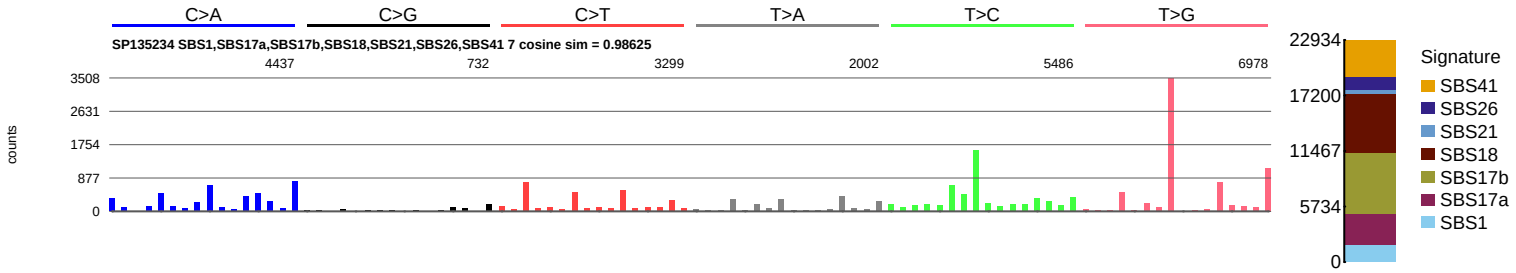

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98655
4499
898
3918
2178
4390
7050
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

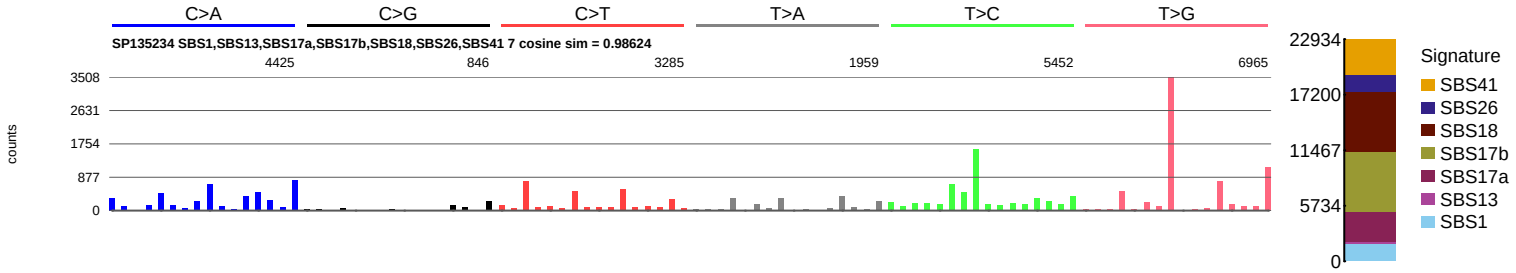

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98647
4506
808
3930
2207
4416
7068
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1

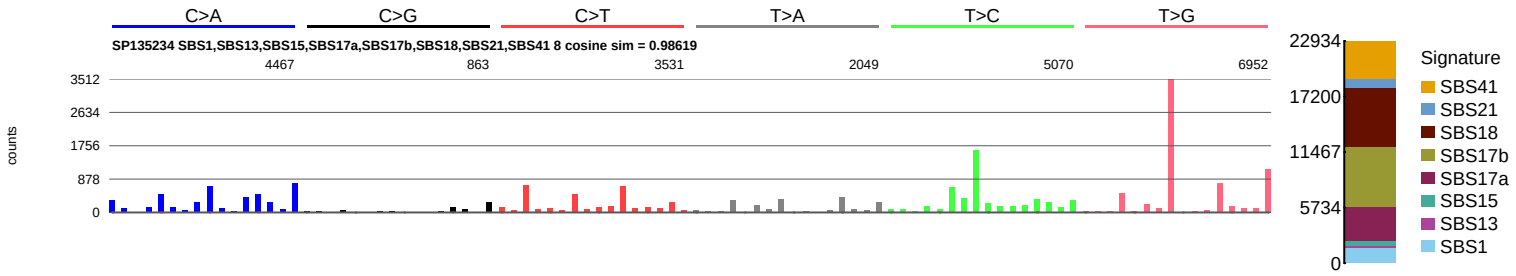

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.9864
4480
1704
3780
1959
4227
6786
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

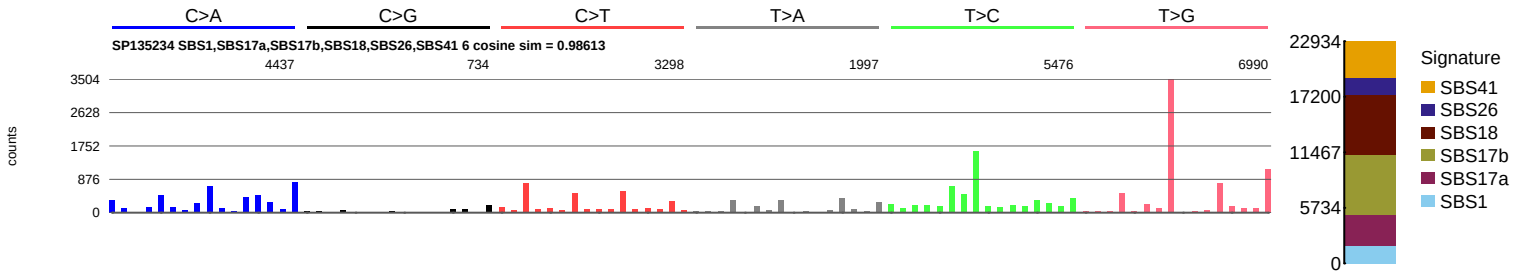

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS3,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98639
4483
1694
3782
1963
4228
6787
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS1

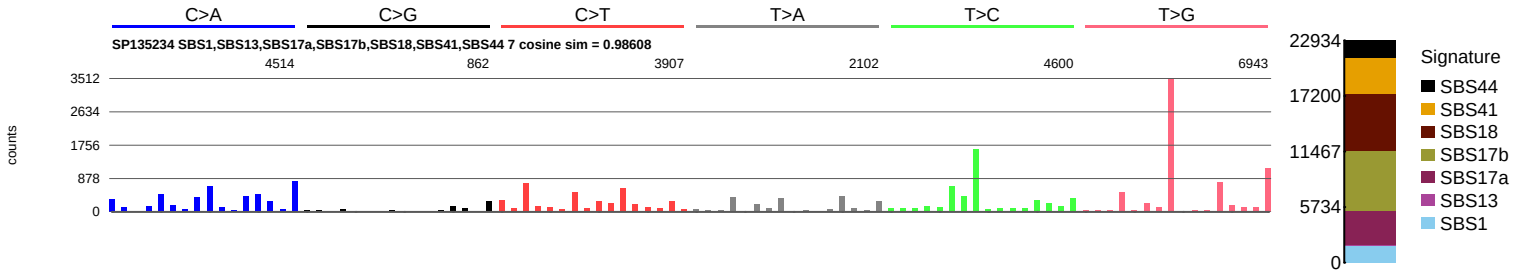

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98637
4427
846
3288
1964
5462
6951
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

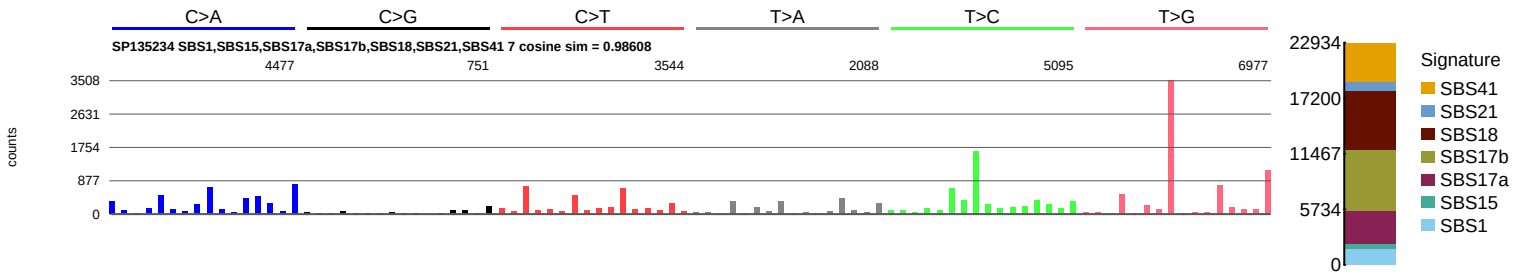

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.98625
4437
732
3299
2002
5486
6978
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1

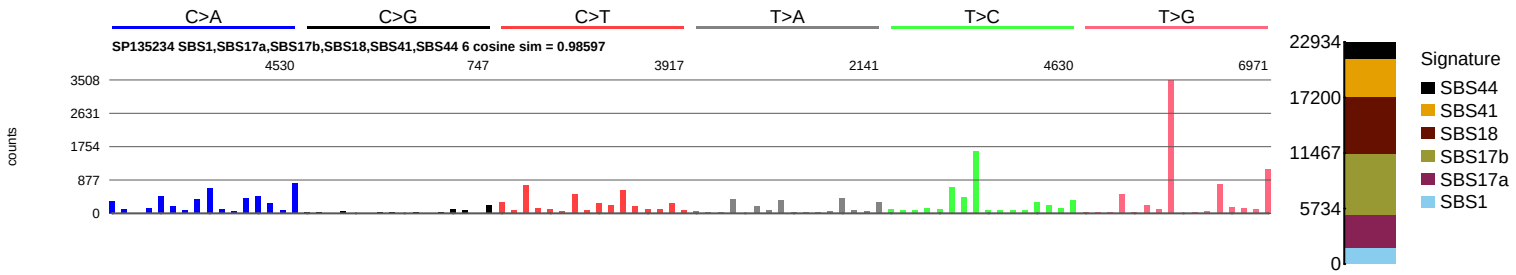

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98624
4425
846
3285
1959
5452
6965
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

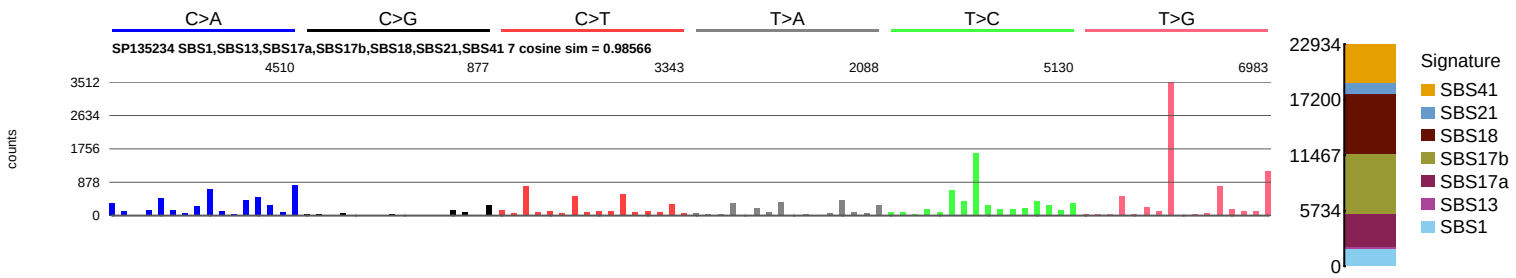

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98619
4467
863
3531
2049
5070
6952
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

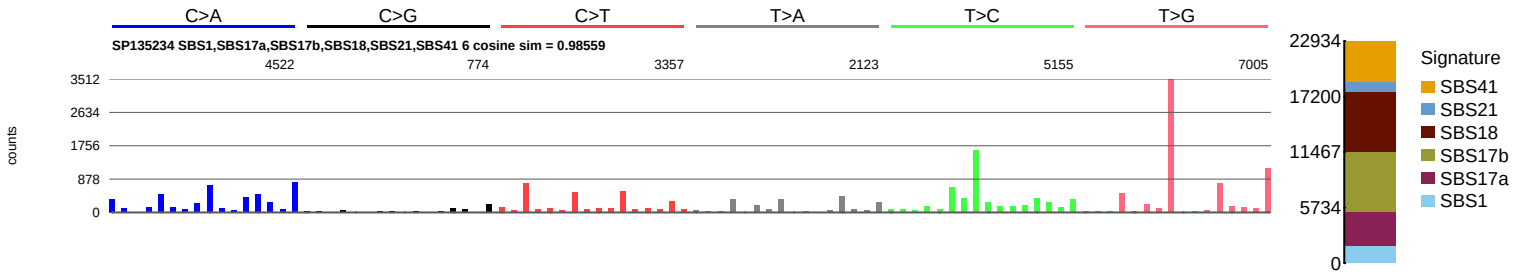

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98613
4437
734
3298
1997
5476
6990
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1

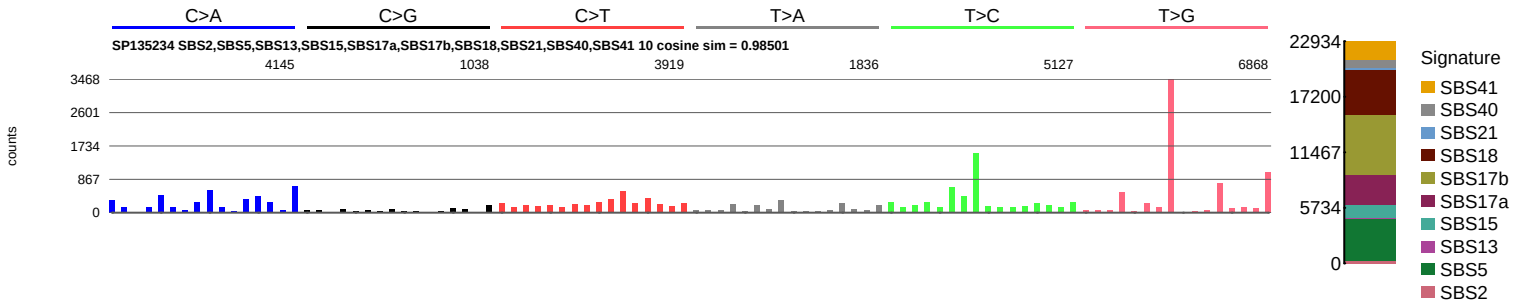

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98608
4514
862
3907
2102
4600
6943
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98608
4477
751
3544
2088
5095
6977
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98597
4530
747
3917
2141
4630
6971
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98566
4510
877
3343
2088
5130
6983
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98559
4522
774
3357
2123
5155
7005
counts
3512
2634
1756
878
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98501
4145
1038
3919
1836
5127
6868
counts
3468
2601
1734
867
0
22934
17200
11467
5734
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Signature
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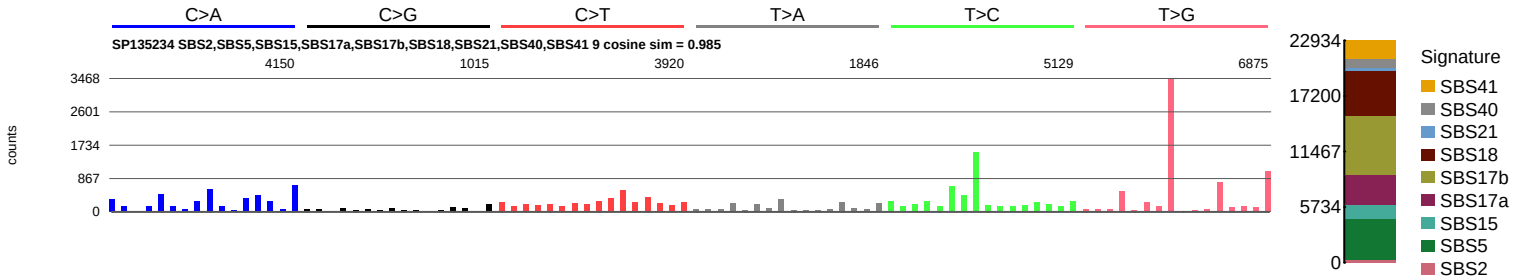
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.985
4150
1015
3920
1846
5129
6875
counts
3468
2601
1734
867
0
22934
17200
11467
5734
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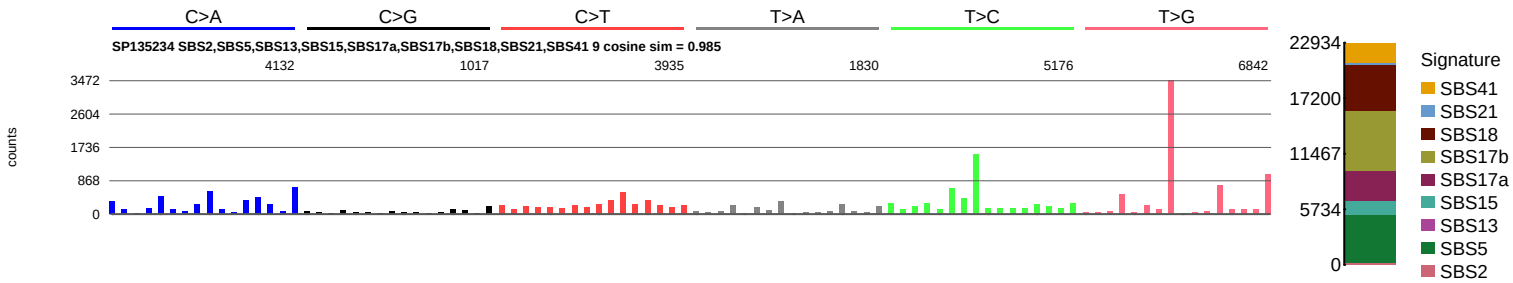
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.985
4132
1017
3935
1830
5176
6842
counts
3472
2604
1736
868
0
22934
17200
11467
5734
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Signature
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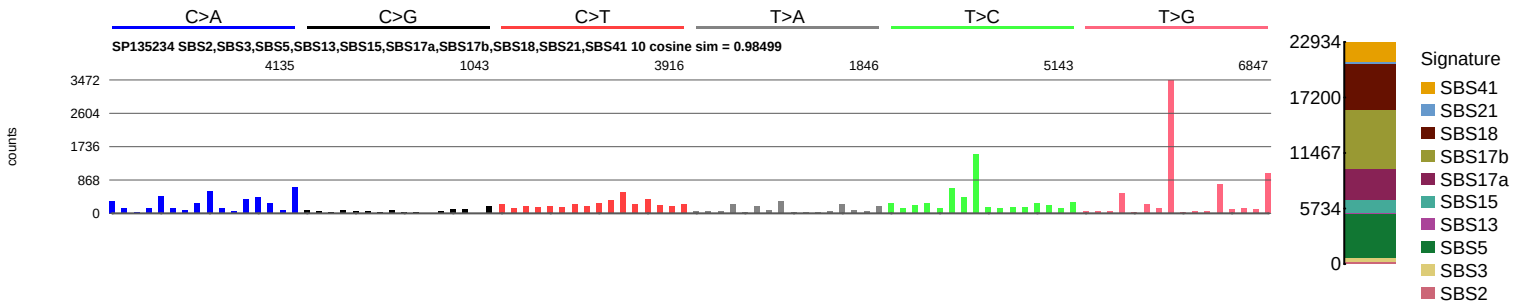
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98499
4135
1043
3916
1846
5143
6847
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
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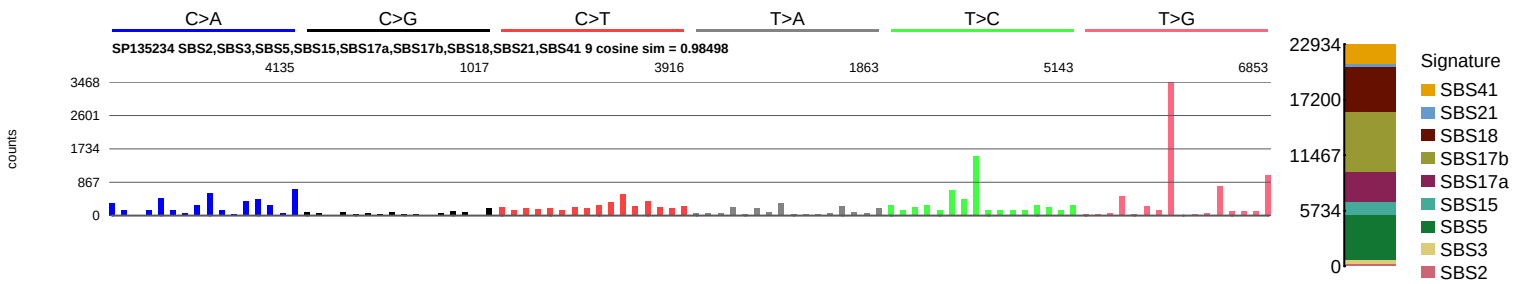
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98498
4135
1017
3916
1863
5143
6853
counts
3468
2601
1734
867
0
22934
17200
11467
5734
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Signature
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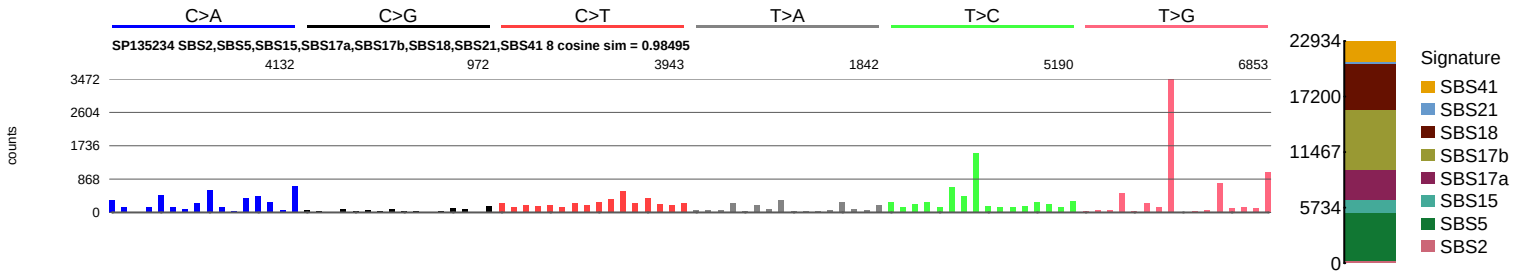
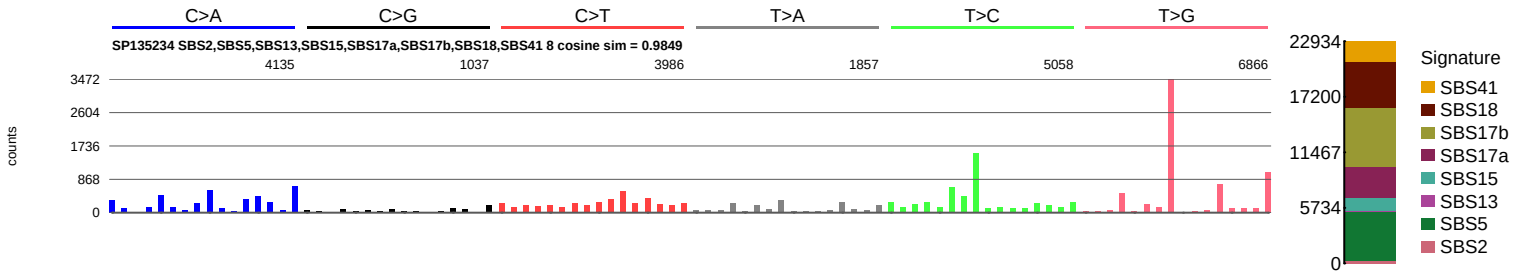
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98495
4132
972
3943
1842
5190
6853
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9849
4135
1037
3986
1857
5058
6866
counts
3472
2604
1736
868
0
22934
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SBS41
SBS18
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SBS17a
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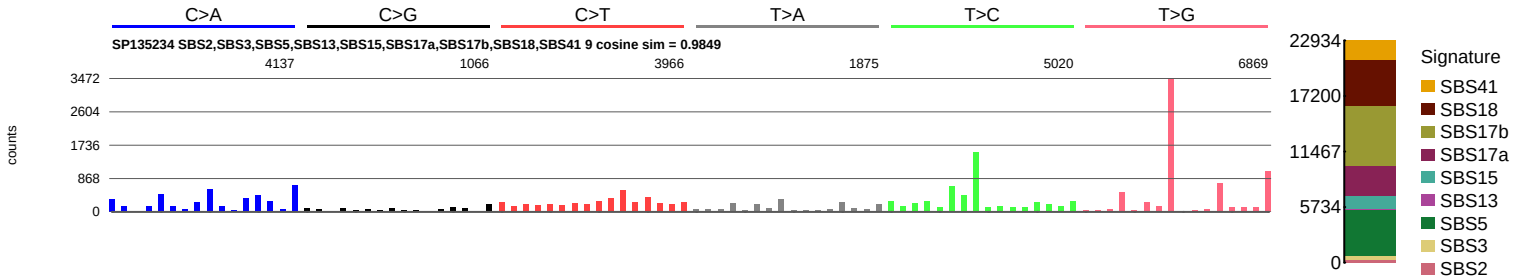
SBS5
SBS2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SP135234 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.9849
4137
1066
3966
1875
5020
6869
counts
3472
2604
1736
868
0
22934
17200
11467
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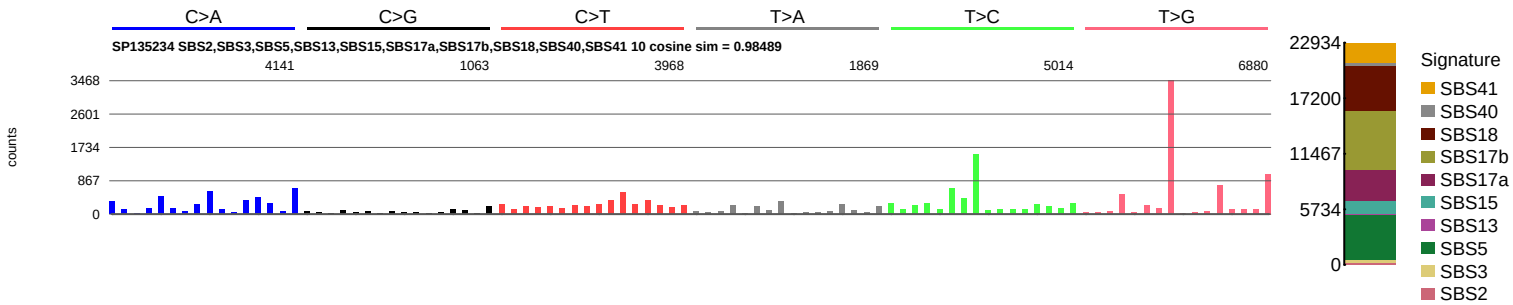
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SP135234 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98489
4141
1063
3968
1869
5014
6880
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
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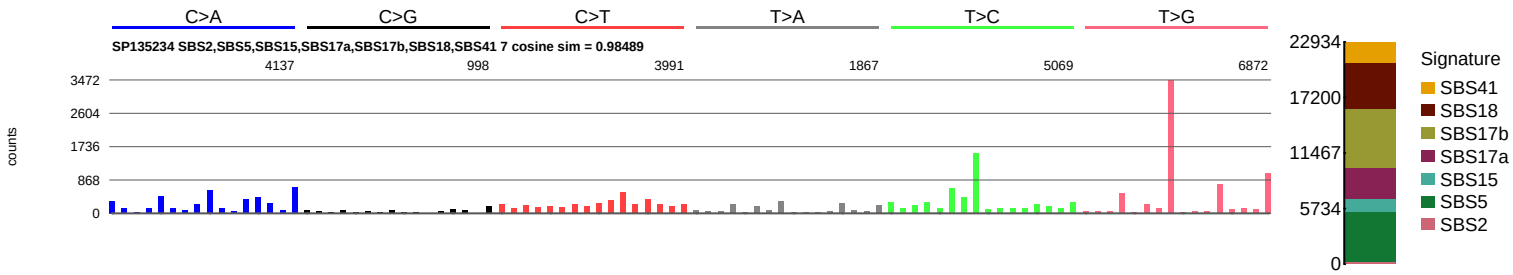
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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T>G
SP135234 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98489
4137
998
3991
1867
5069
6872
counts
3472
2604
1736
868
0
22934
17200
11467
5734
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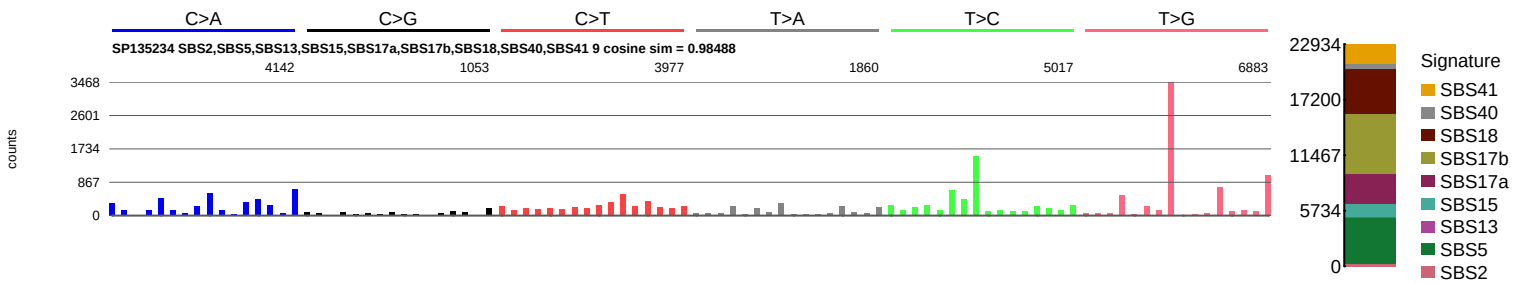
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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SP135234 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98488
4142
1053
3977
1860
5017
6883
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
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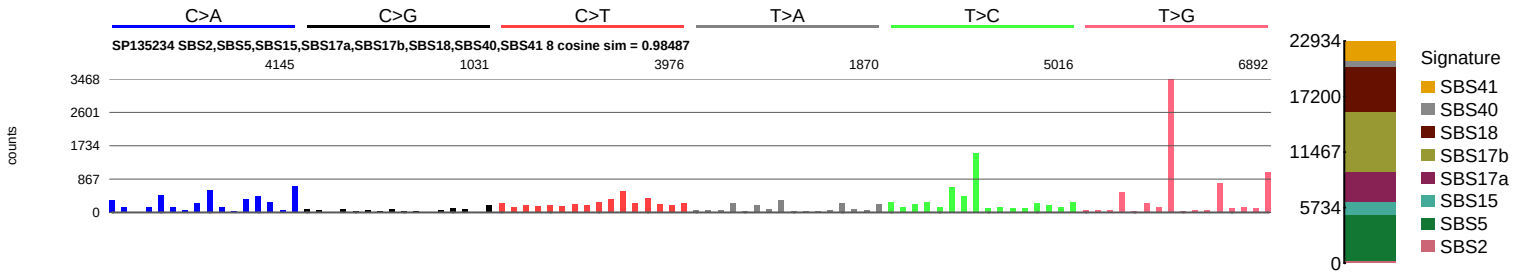
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2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C>G
C>T
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SP135234 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98487
4145
1031
3976
1870
5016
6892
counts
3468
2601
1734
867
0
22934
17200
11467
5734
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Signature
SBS41
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SBS15
SBS5
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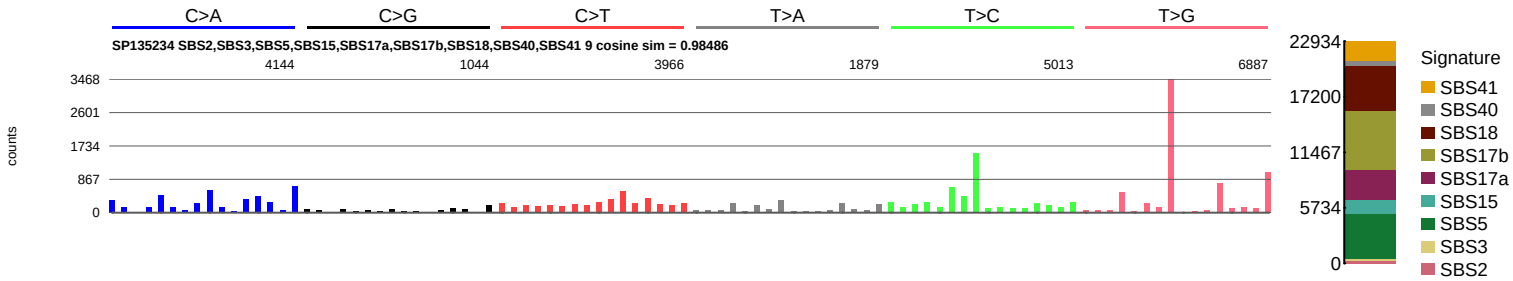
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C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98486
4144
1044
3966
1879
5013
6887
counts
3468
2601
1734
867
0
22934
17200
11467
5734
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SBS18
SBS17b
SBS17a
SBS15
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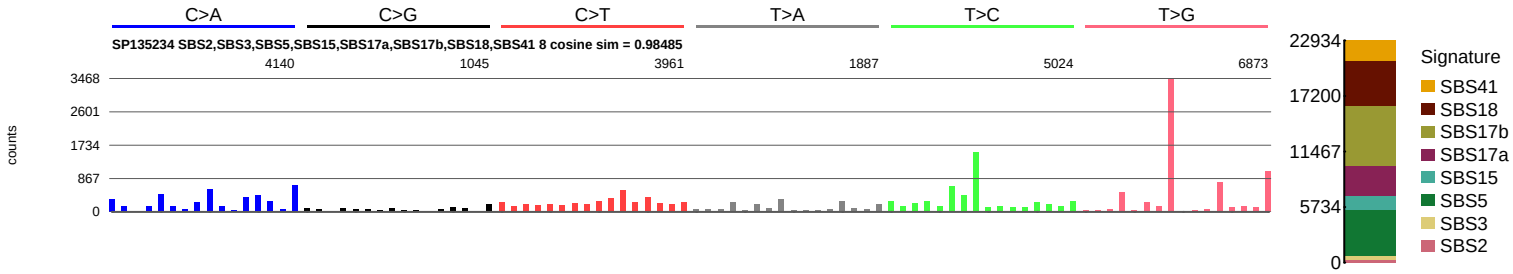
SBS3
SBS2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C>G
C>T
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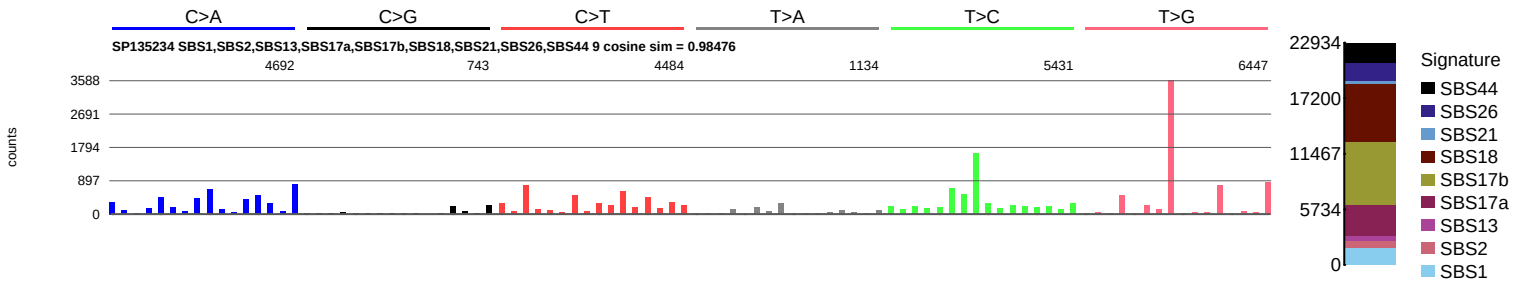
T>C
T>G
SP135234 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98485
4140
1045
3961
1887
5024
6873
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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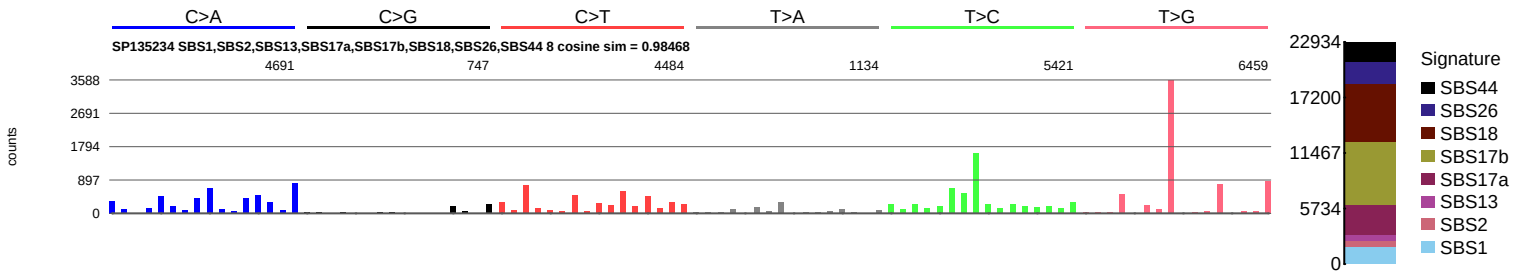
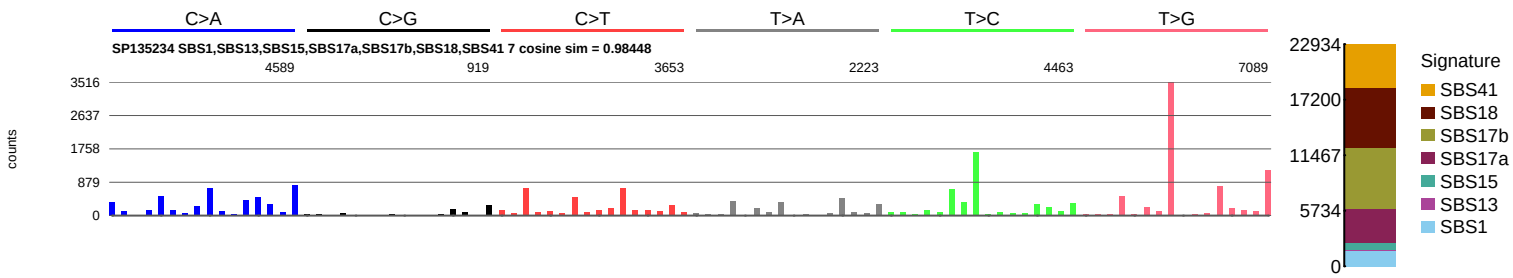
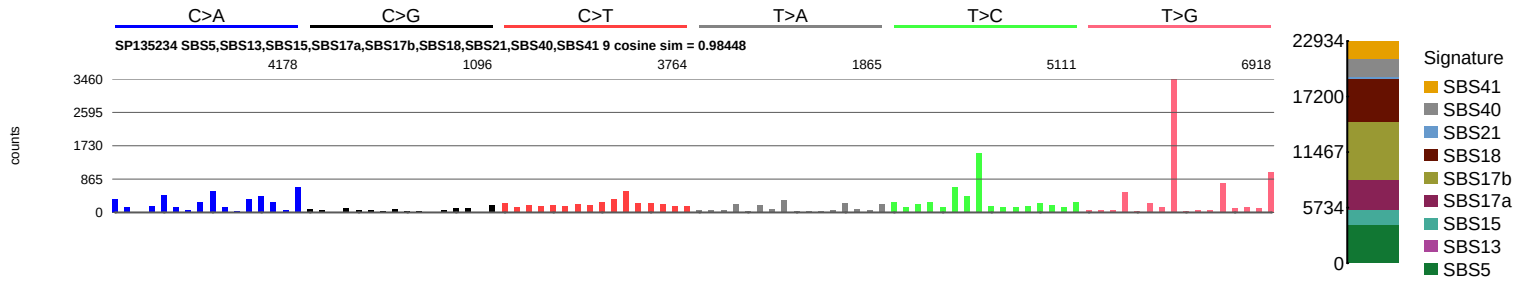
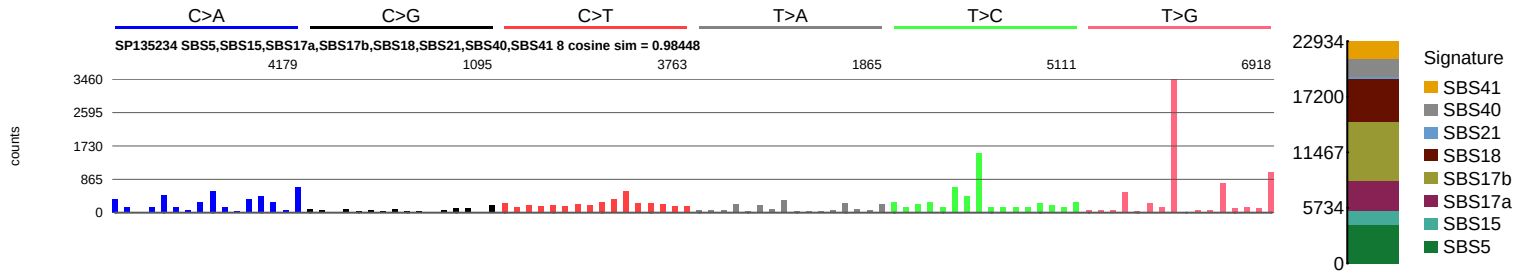
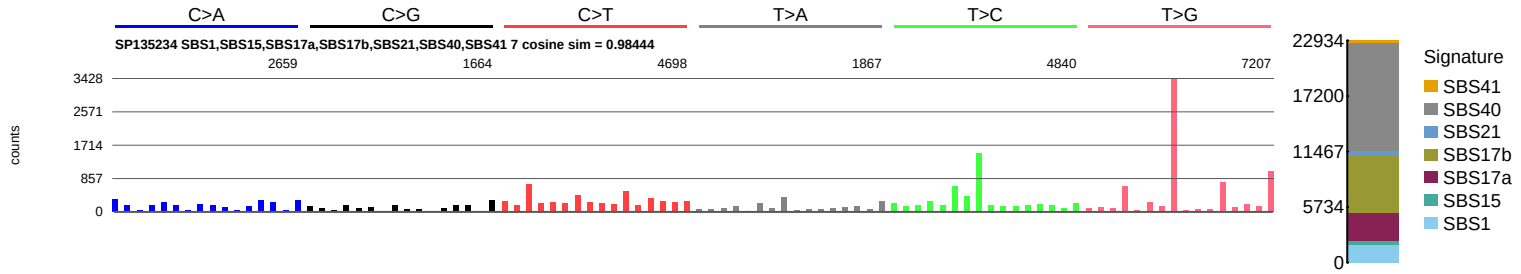
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98476
4692
743
4484
1134
5431
6447
counts
3588
2691
1794
897
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98468
4691
747
4484
1134
5421
6459
counts
3588
2691
1794
897
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98448
4589
919
3653
2223
4463
7089
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98448
4178
1096
3764
1865
5111
6918
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98448
4179
1095
3763
1865
5111
6918
counts
3460
2595
1730
865
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5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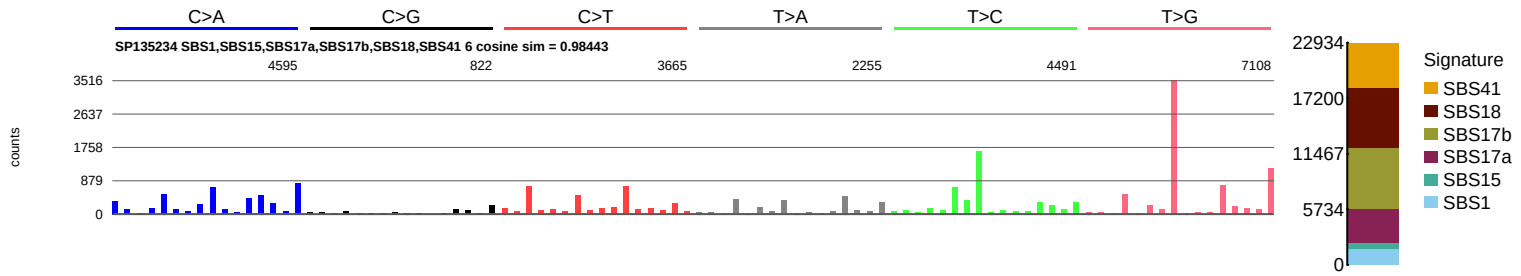
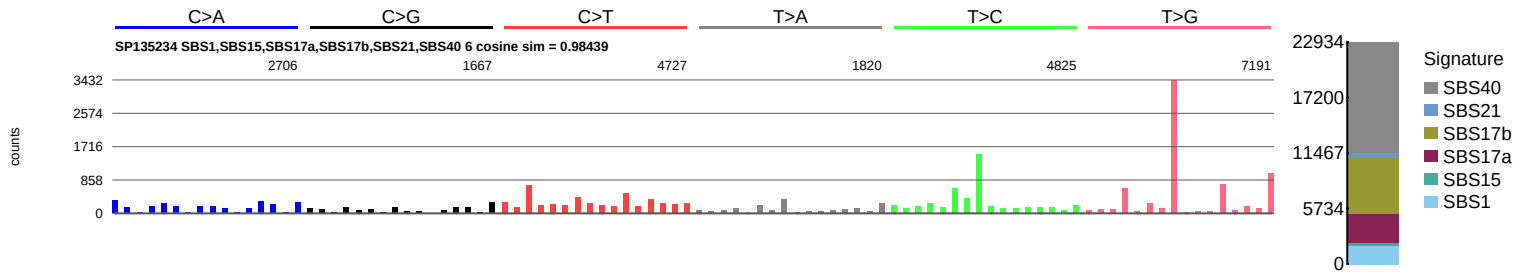
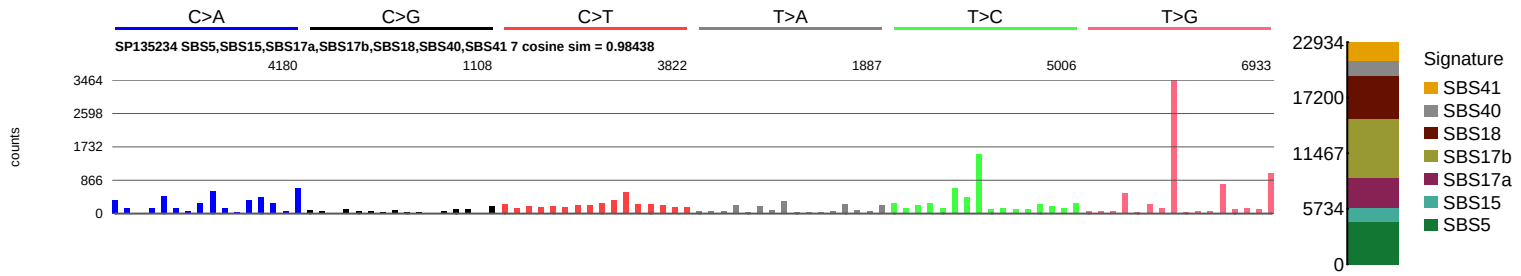
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98444
2659
1664
4698
1867
4840
7207
counts
3428
2571
1714
857
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98443
4595
822
3665
2255
4491
7108
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98439
2706
1667
4727
1820
4825
7191
counts
3432
2574
1716
858
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1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C>G
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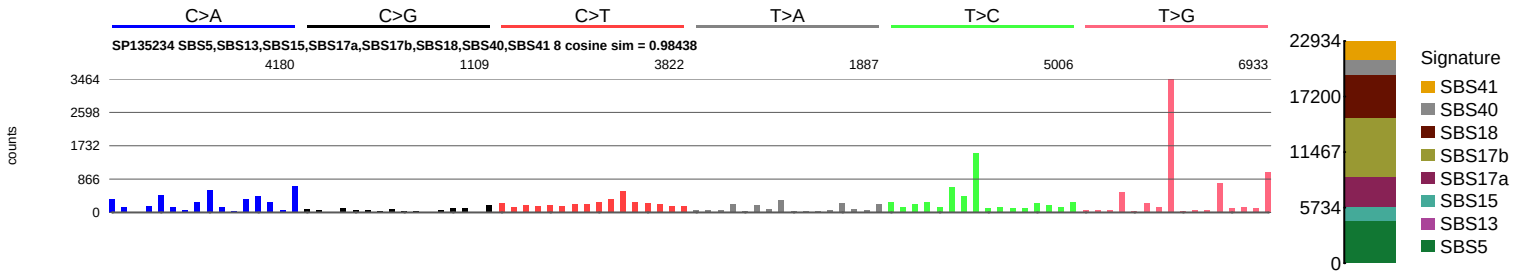
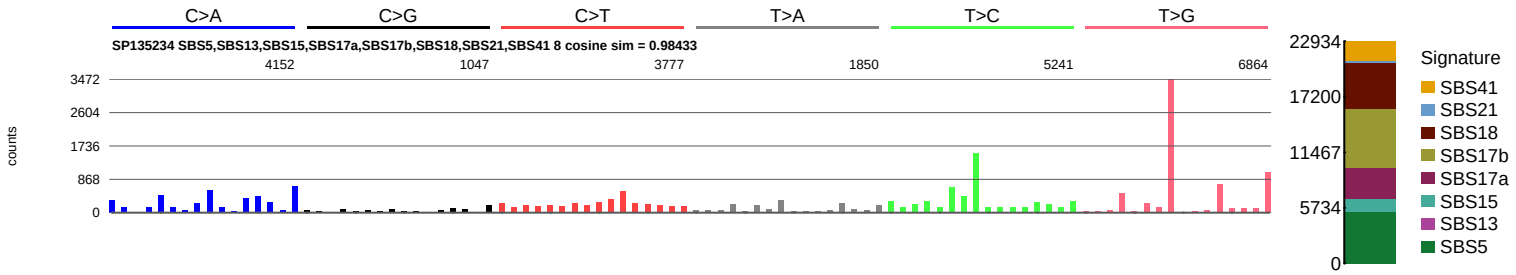
T>A
T>C
T>G
SP135234 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98438
4180
1108
3822
1887
5006
6933
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98438
4180
1109
3822
1887
5006
6933
counts
3464
2598
1732
866
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98433
4152
1047
3777
1850
5241
6864
counts
3472
2604
1736
868
0
22934
17200
11467
5734
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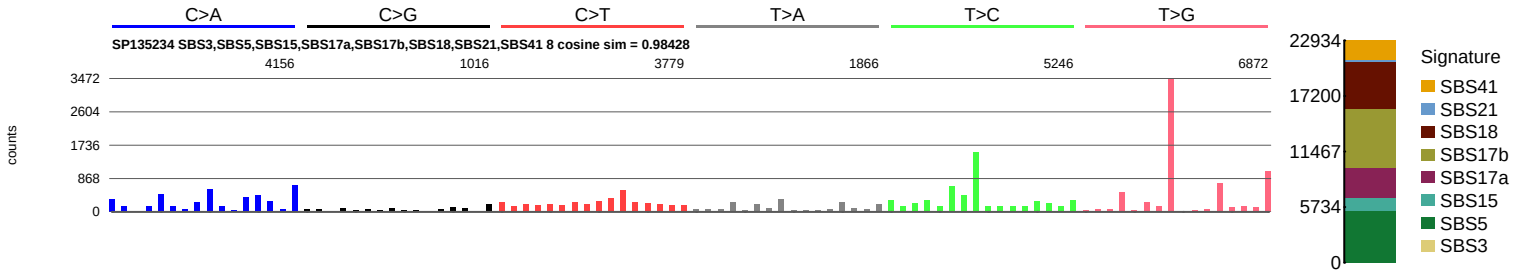
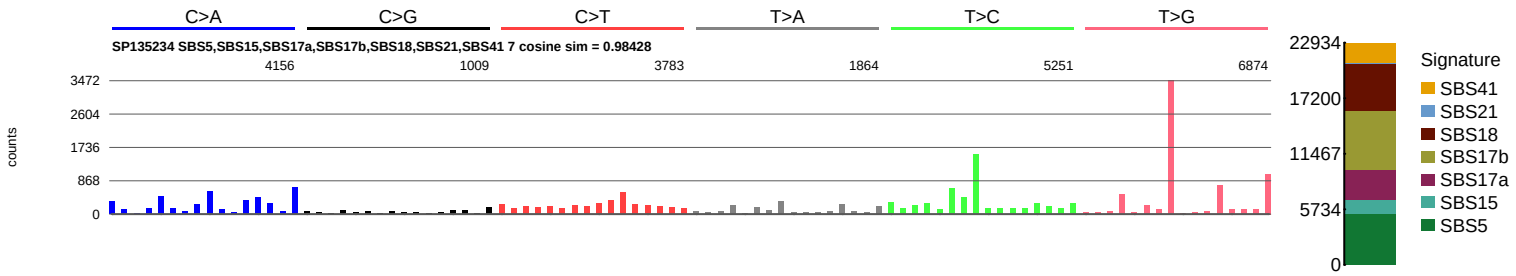
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98428
4156
1016
3779
1866
5246
6872
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98428
4156
1009
3783
1864
5251
6874
counts
3472
2604
1736
868
0
22934
17200
11467
5734
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SBS41
SBS21
SBS18
SBS17b
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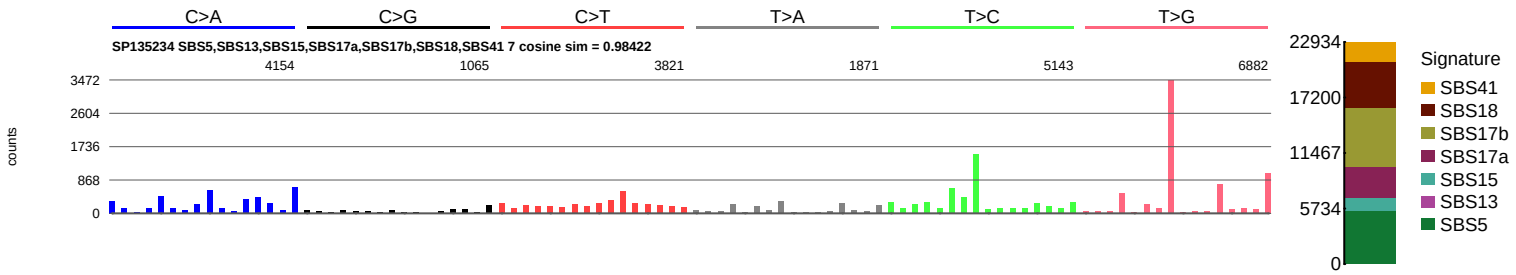
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98422
4154
1065
3821
1871
5143
6882
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
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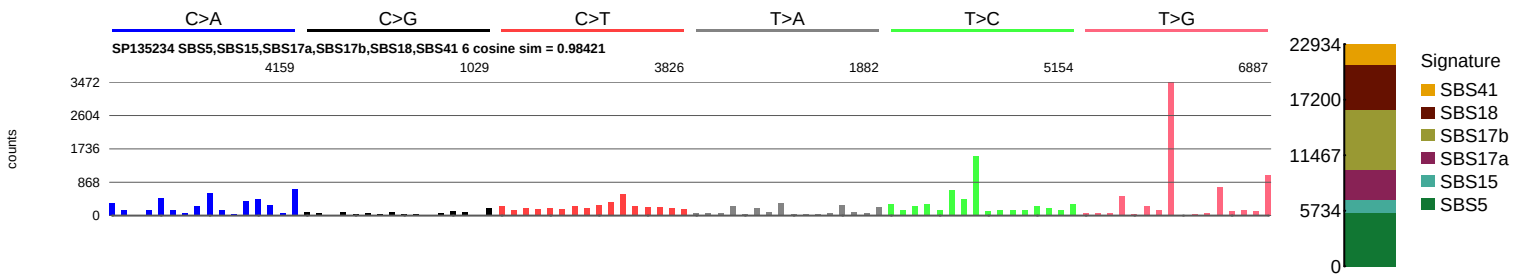
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98421
4159
1029
3826
1882
5154
6887
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
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SBS41
SBS18
SBS17b
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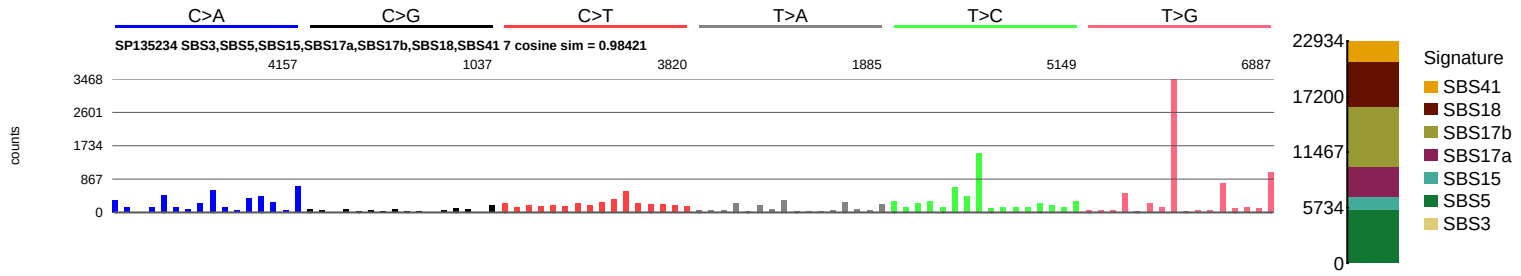
SBS15
SBS5


C>A
C>G
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T>C
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SP135234 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98421
4157
1037
3820
1885
5149
6887
counts
3468
2601
1734
867
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22934
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11467
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SBS41
SBS18
SBS17b
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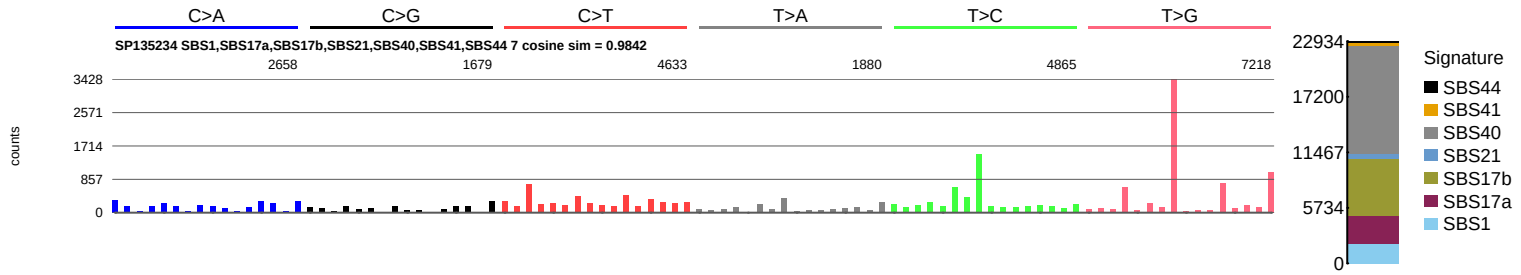
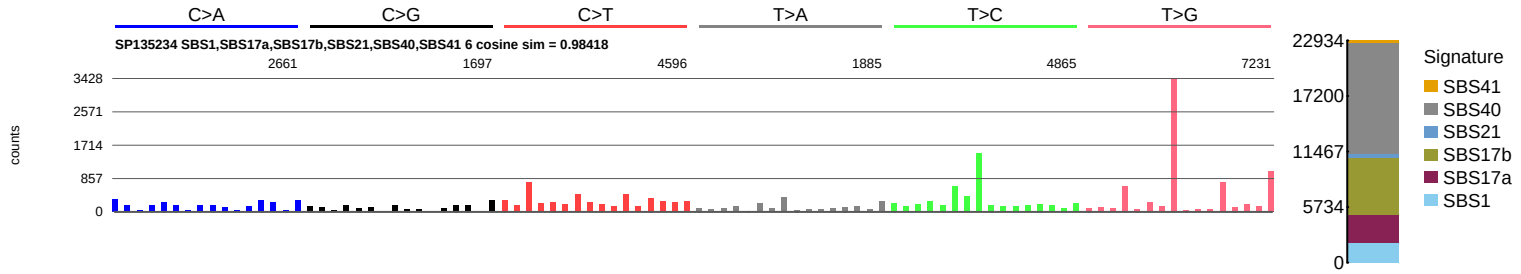
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.9842
2658
1679
4633
1880
4865
7218
counts
3428
2571
1714
857
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
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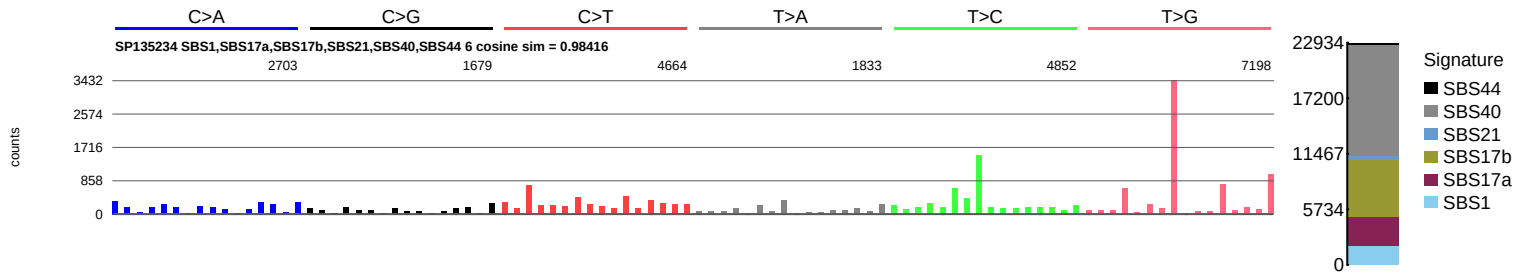
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98418
2661
1697
4596
1885
4865
7231
counts
3428
2571
1714
857
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
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T>G
SP135234 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98416
2703
1679
4664
1833
4852
7198
counts
3432
2574
1716
858
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22934
17200
11467
5734
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Signature
SBS44
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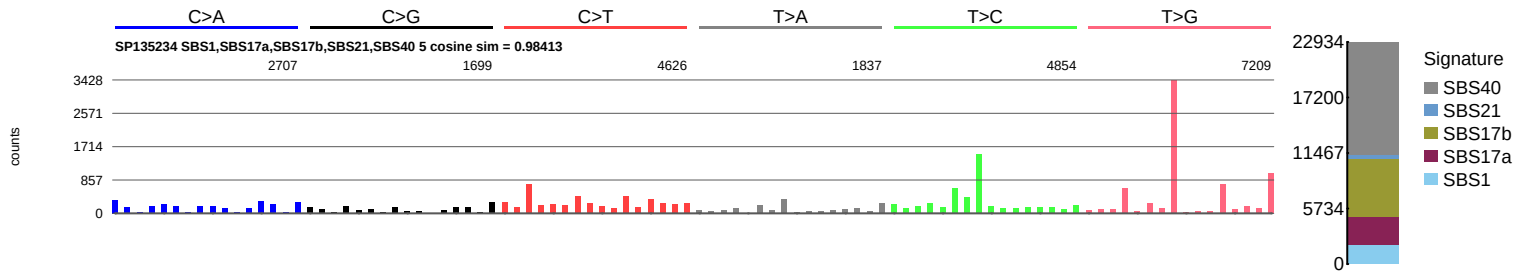
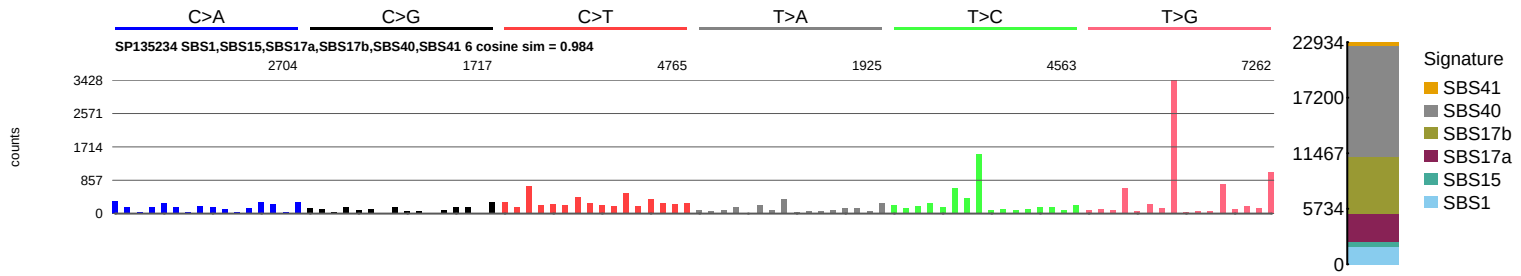
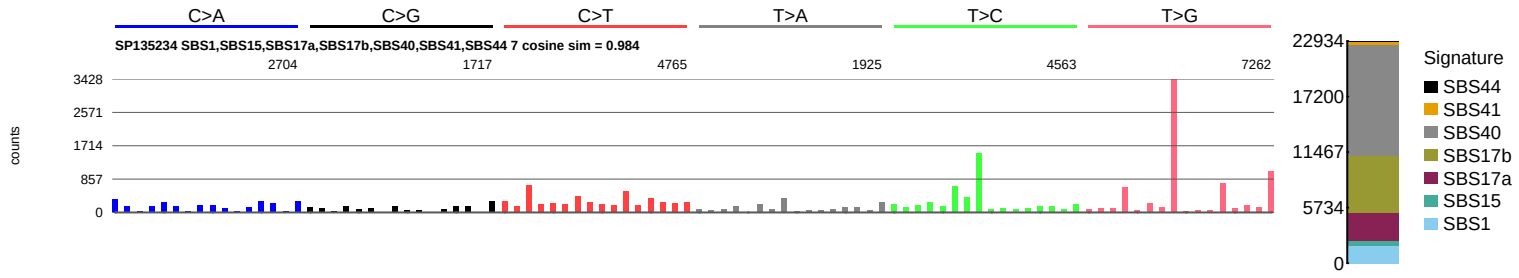
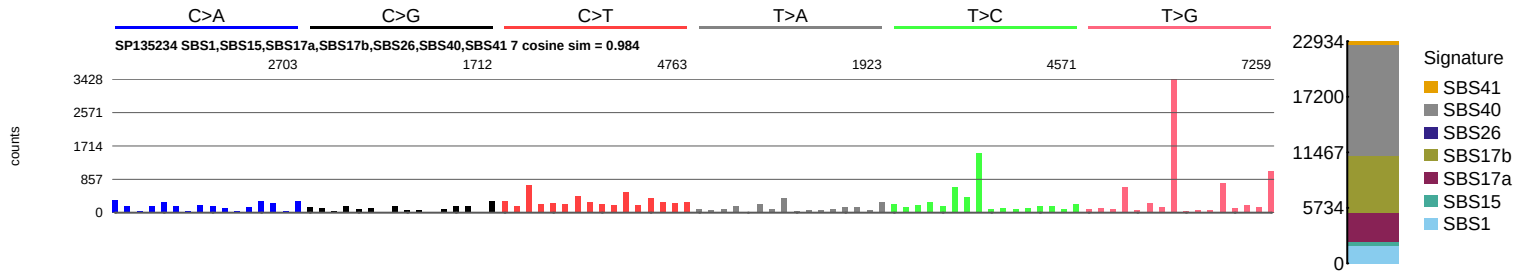
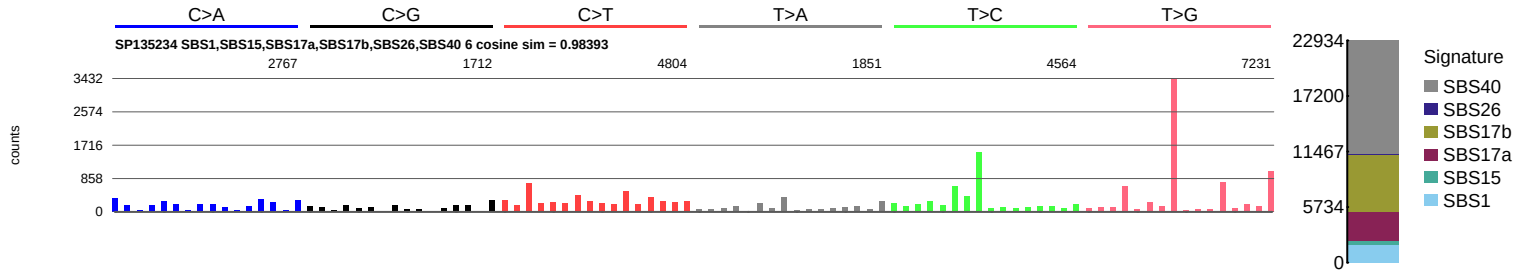
SBS21
SBS17b
SBS17a
SBS1

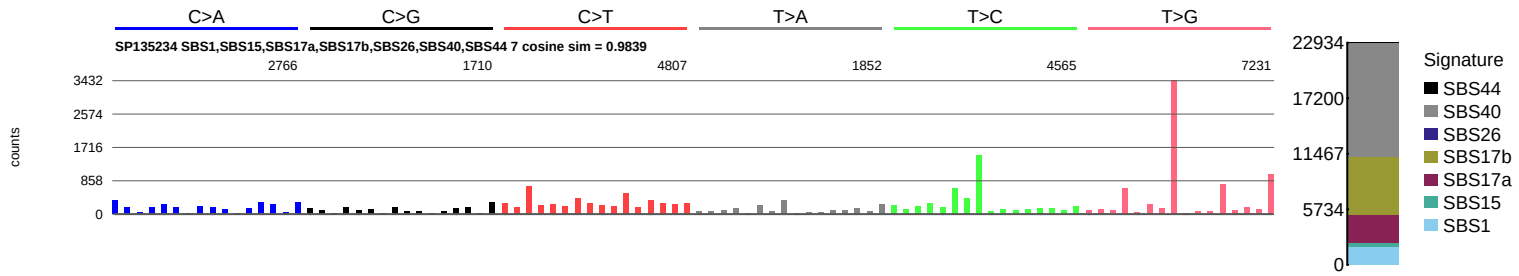

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98413
2707
1699
4626
1837
4854
7209
counts
3428
2571
1714
857
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS1

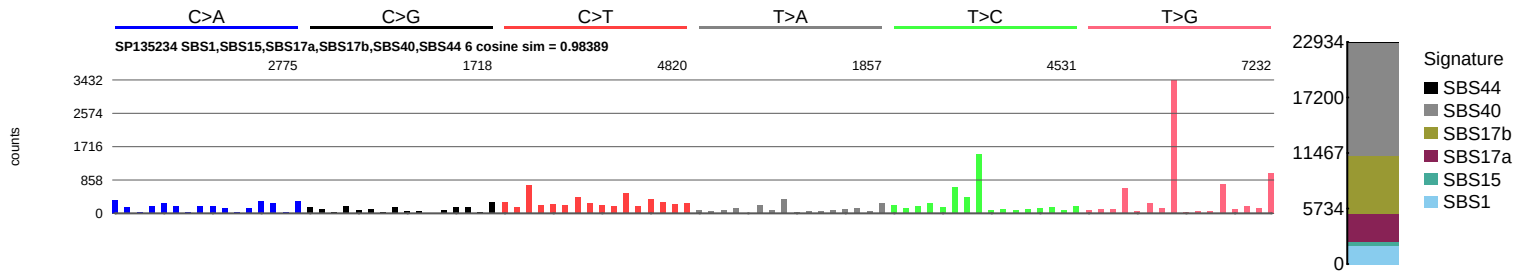

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.984
2704
1717
4765
1925
4563
7262
counts
3428
2571
1714
857
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1

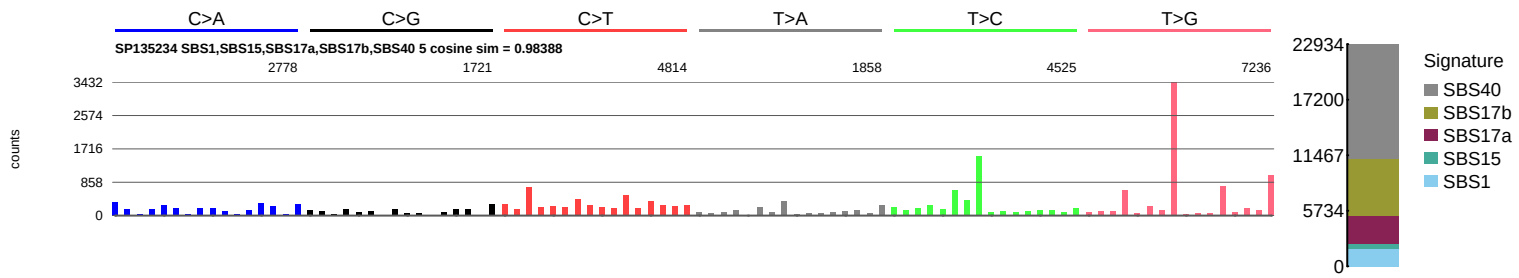

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.984
2704
1717
4765
1925
4563
7262
counts
3428
2571
1714
857
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1

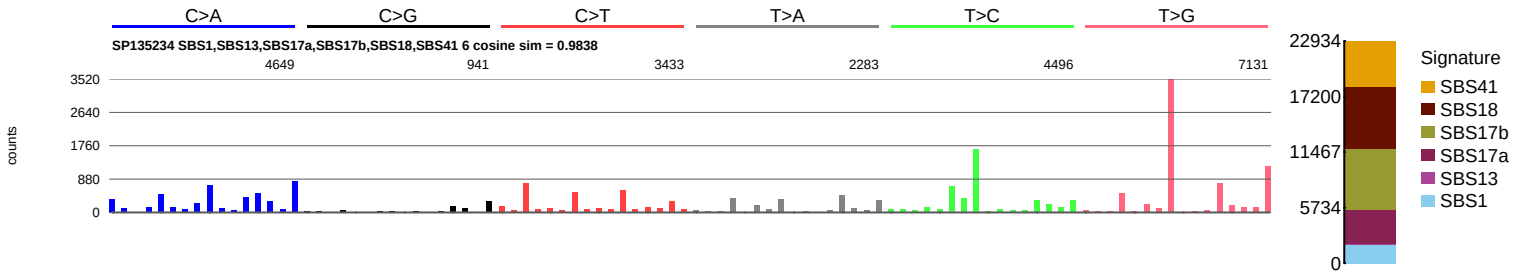

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.984
2703
1712
4763
1923
4571
7259
counts
3428
2571
1714
857
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1

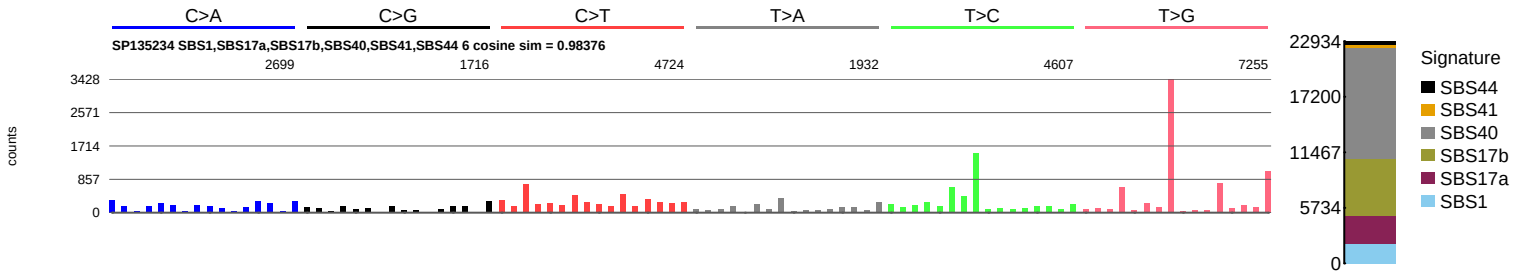

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98393
2767
1712
4804
1851
4564
7231
counts
3432
2574
1716
858
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1

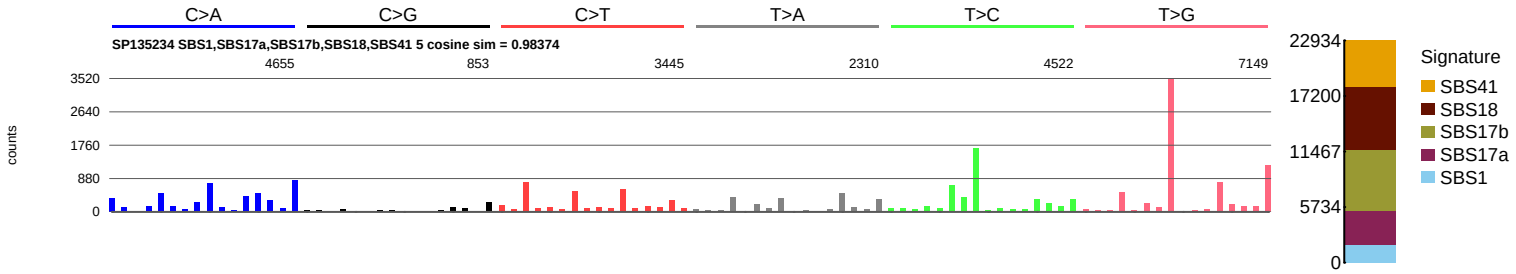

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.9839
2766
1710
4807
1852
4565
7231
counts
3432
2574
1716
858
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1

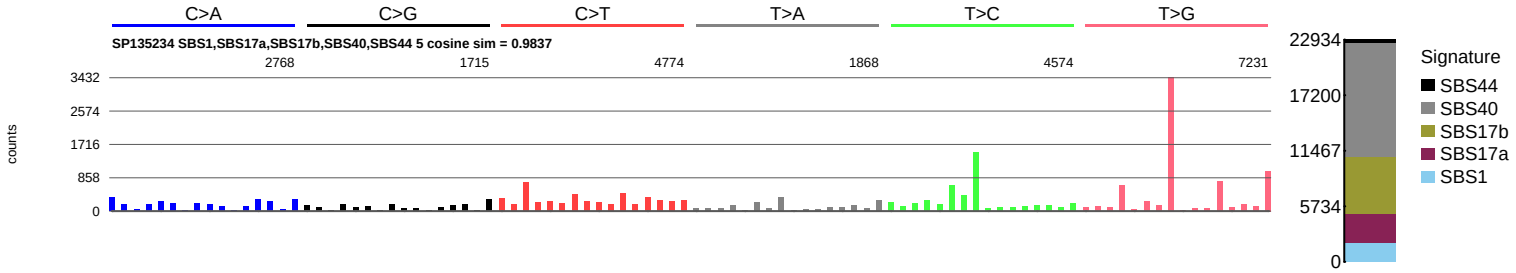

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98389
2775
1718
4820
1857
4531
7232
counts
3432
2574
1716
858
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS1

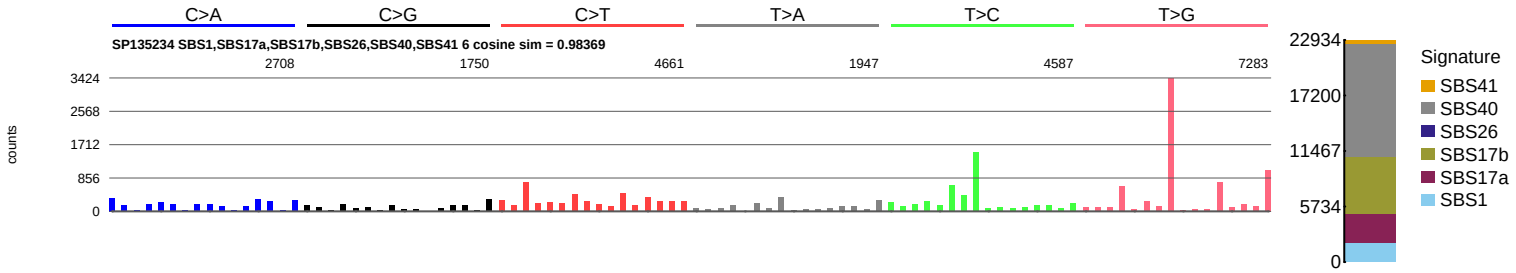

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98388
2778
1721
4814
1858
4525
7236
counts
3432
2574
1716
858
0
22934
17200
11467
5734
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS1

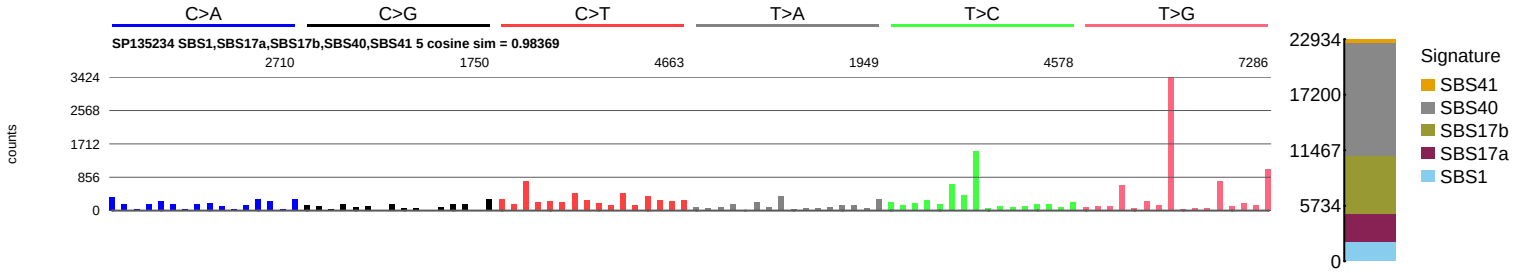

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.9838
4649
941
3433
2283
4496
7131
counts
3520
2640
1760
880
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS1

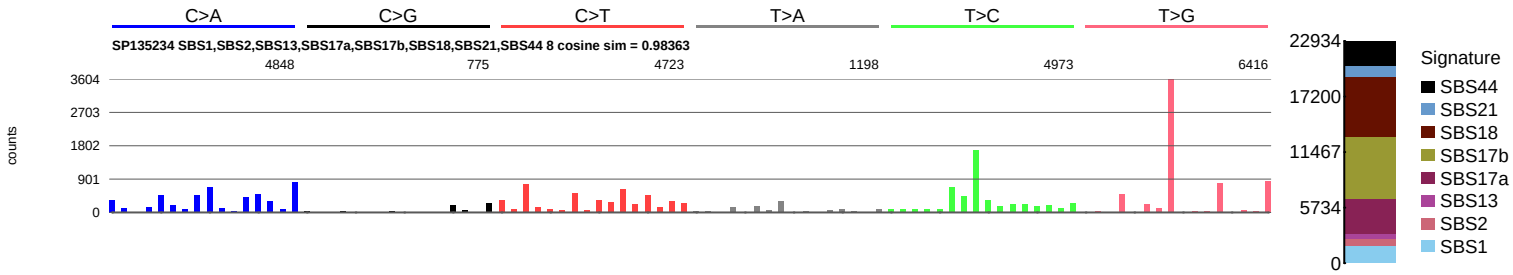

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.98376
2699
1716
4724
1932
4607
7255
counts
3428
2571
1714
857
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS1

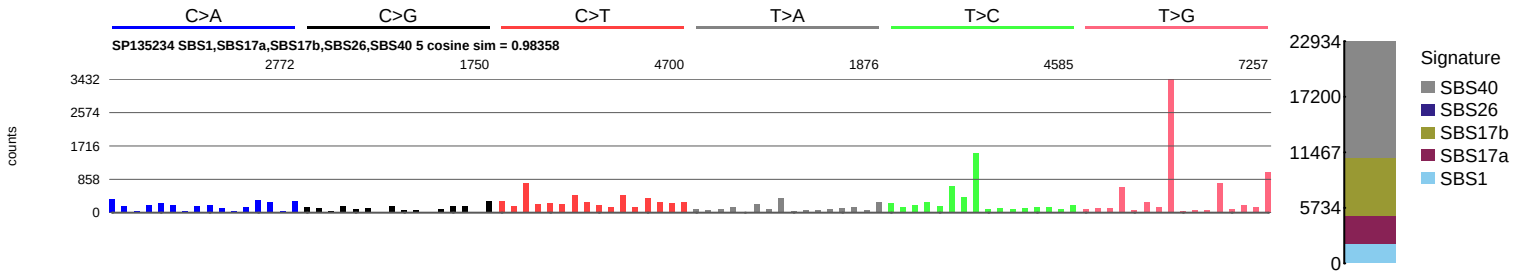

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98374
4655
853
3445
2310
4522
7149
counts
3520
2640
1760
880
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS1

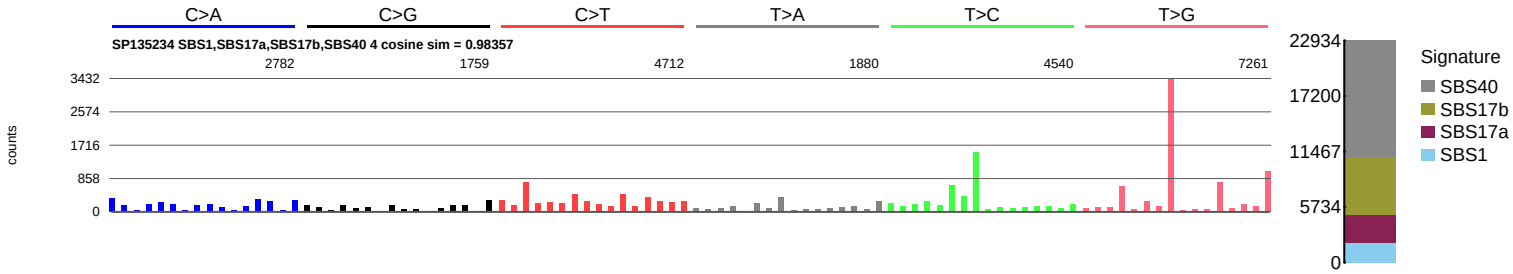

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.9837
2768
1715
4774
1868
4574
7231
counts
3432
2574
1716
858
0
22934
17200
11467
5734
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS1

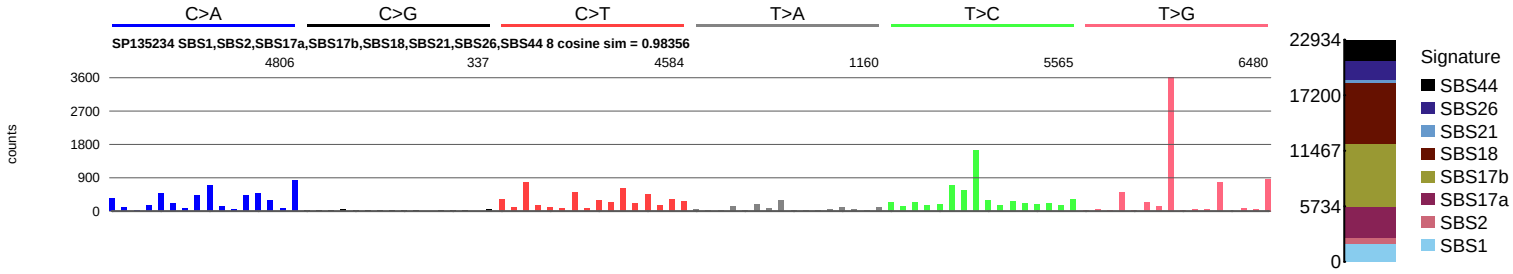

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98369
2708
1750
4661
1947
4587
7283
counts
3424
2568
1712
856
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1

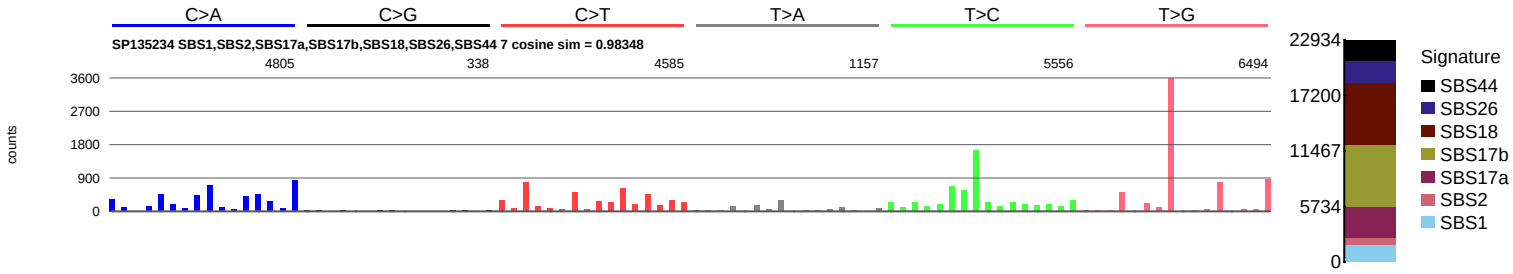

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98369
2710
1750
4663
1949
4578
7286
counts
3424
2568
1712
856
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS1

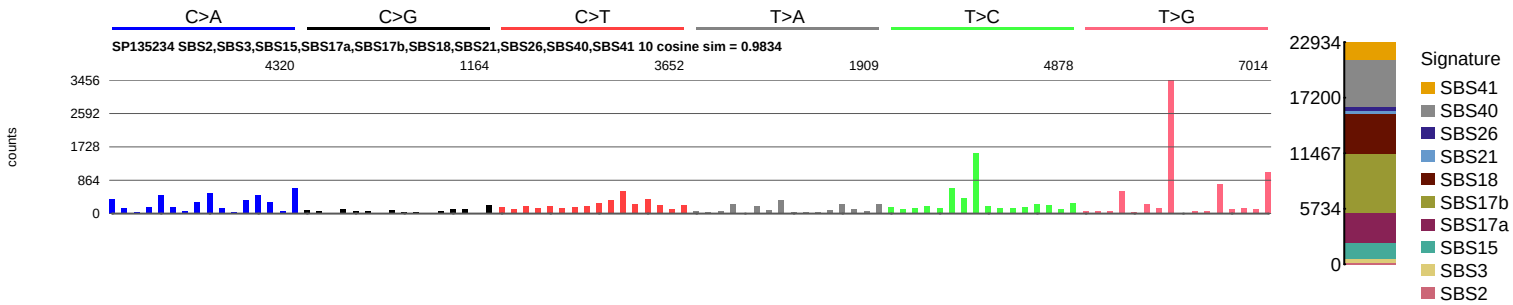

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98363
4848
775
4723
1198
4973
6416
counts
3604
2703
1802
901
0
22934
17200
11467
5734
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

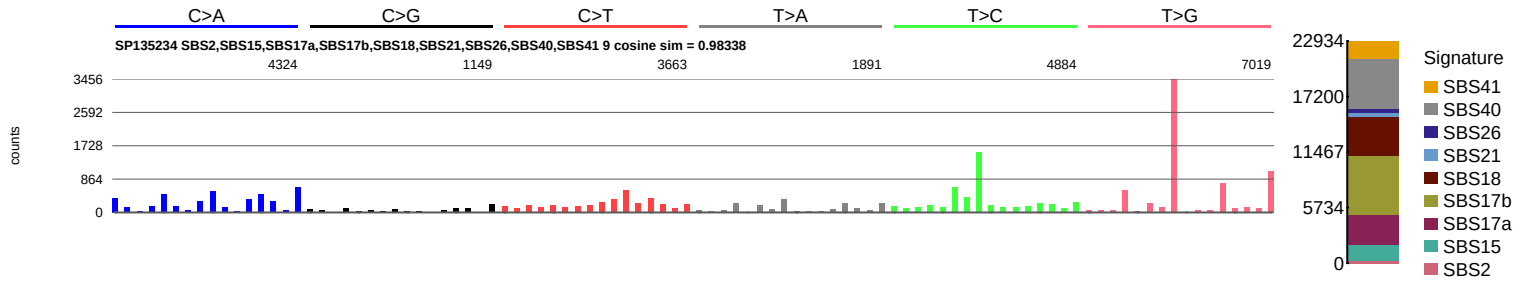

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98358
2772
1750
4700
1876
4585
7257
counts
3432
2574
1716
858
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS1

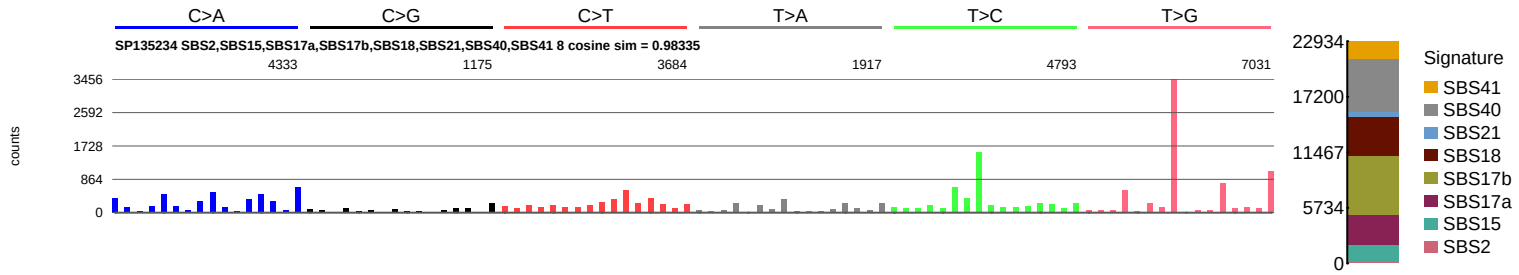

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98357
2782
1759
4712
1880
4540
7261
counts
3432
2574
1716
858
0
22934
17200
11467
5734
0
Signature
SBS40
SBS17b
SBS17a
SBS1

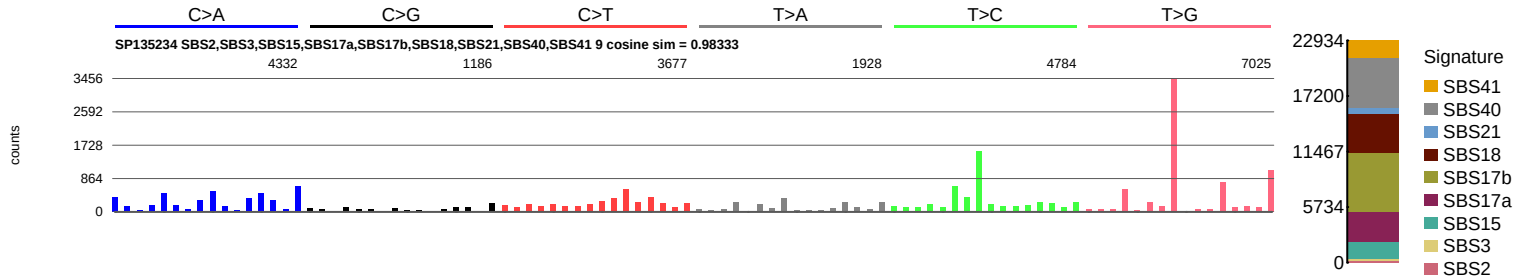

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.98356
4806
337
4584
1160
5565
6480
counts
3600
2700
1800
900
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

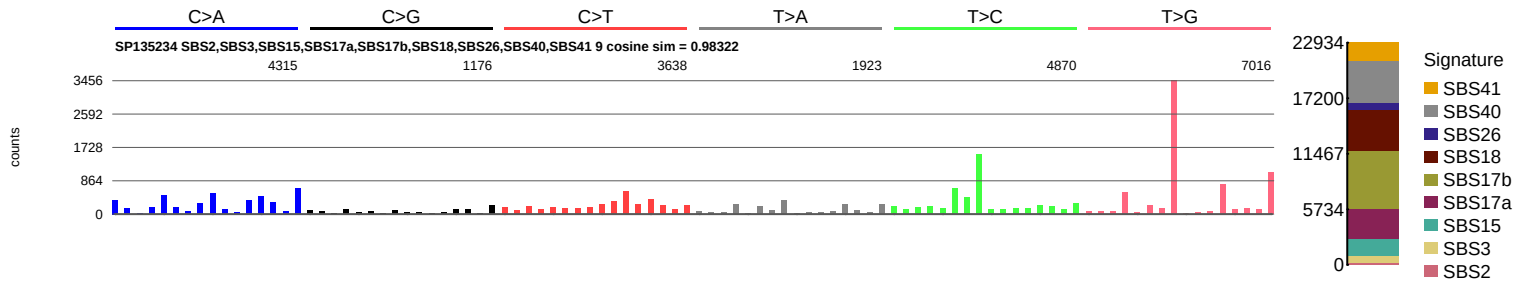

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98348
4805
338
4585
1157
5556
6494
counts
3600
2700
1800
900
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

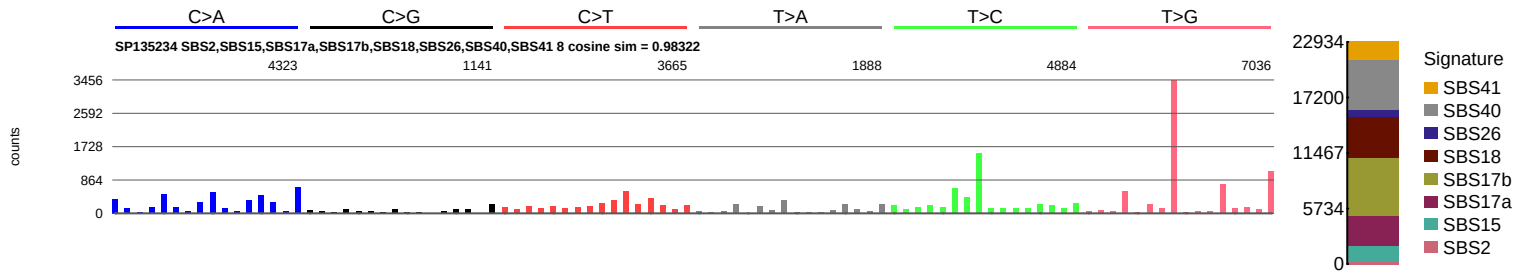

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.9834
4320
1164
3652
1909
4878
7014
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98338
4324
1149
3663
1891
4884
7019
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98335
4333
1175
3684
1917
4793
7031
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98333
4332
1186
3677
1928
4784
7025
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98322
4315
1176
3638
1923
4870
7016
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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C>G
C>T
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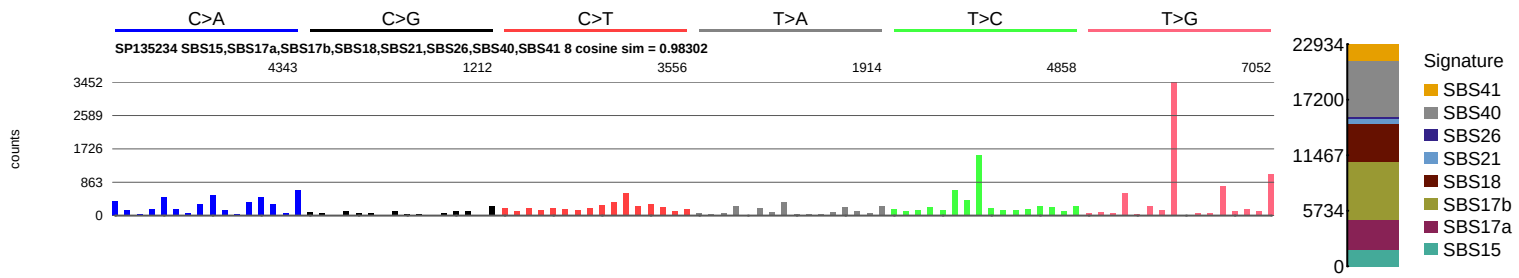
T>G
SP135234 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98322
4323
1141
3665
1888
4884
7036
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2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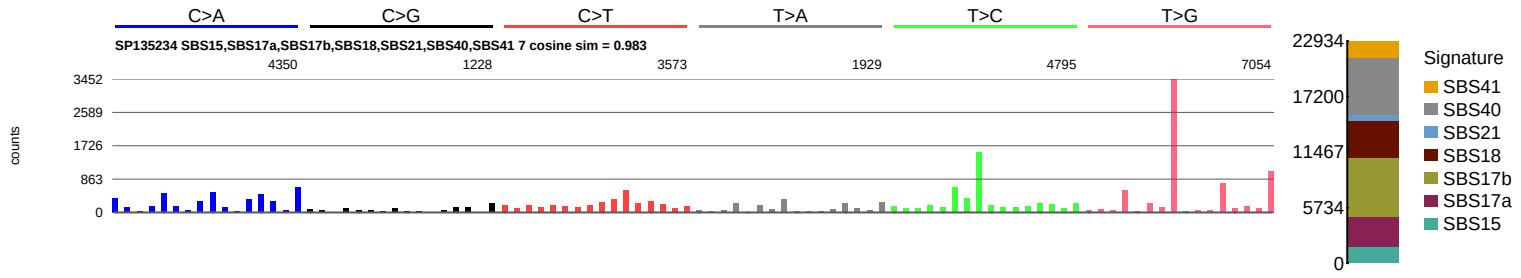
C>G
C>T
T>A
T>C
T>G
SP135234 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98302
4343
1212
3556
1914
4858
7052
counts
3452
2589
1726
863
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15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C>G
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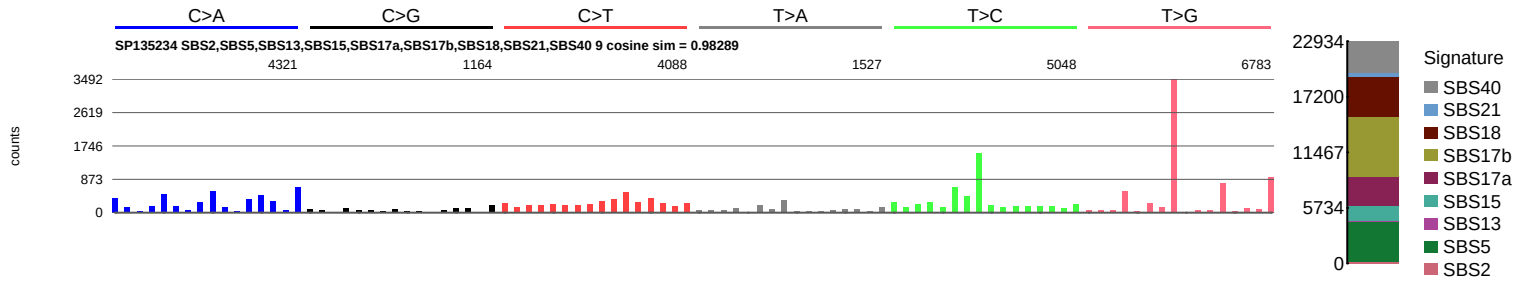
T>C
T>G
SP135234 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.983
4350
1228
3573
1929
4795
7054
counts
3452
2589
1726
863
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15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C>G
C>T
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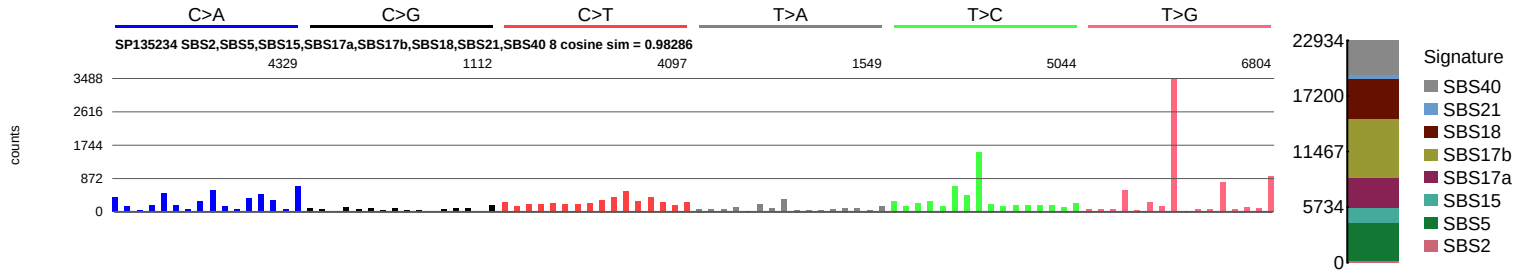
T>G
SP135234 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98289
4321
1164
4088
1527
5048
6783
counts
3492
2619
1746
873
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2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C>G
C>T
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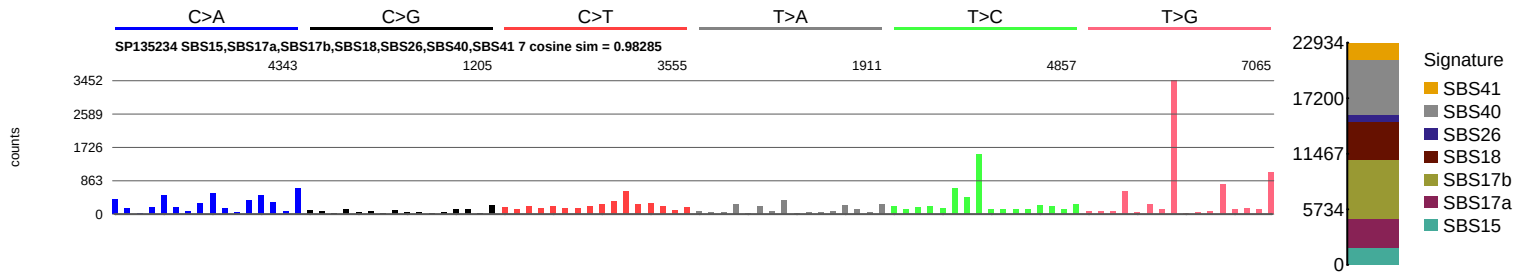
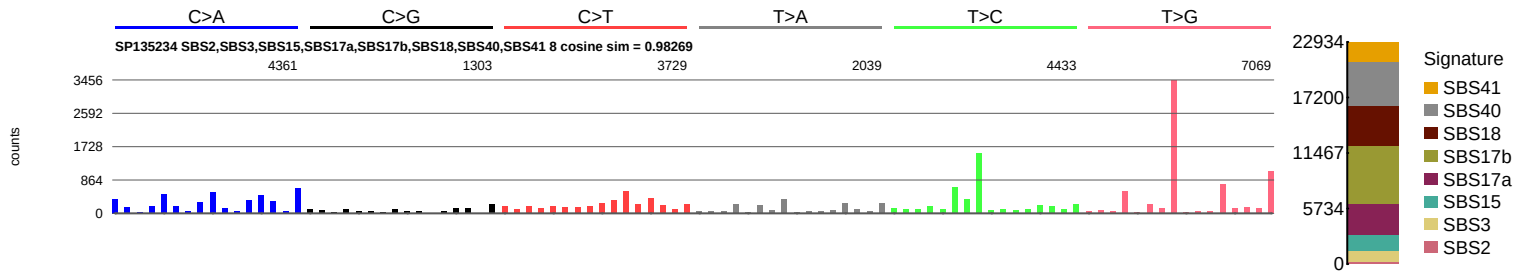
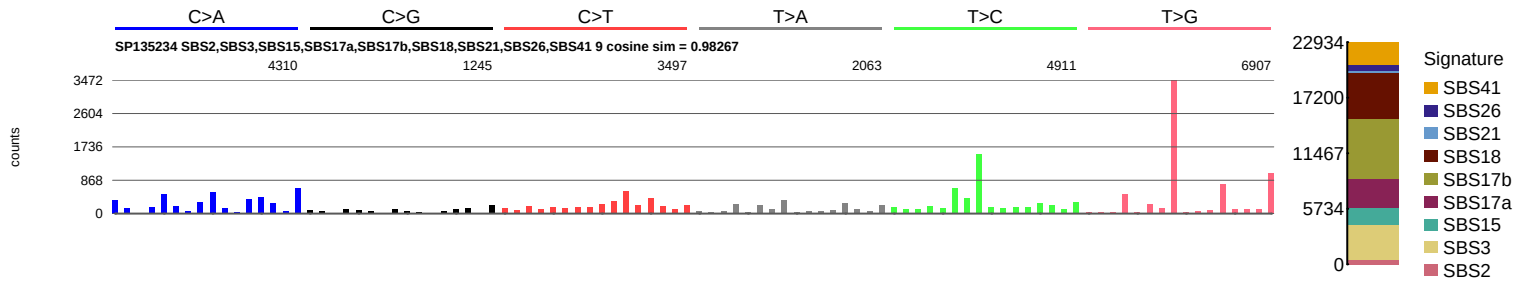
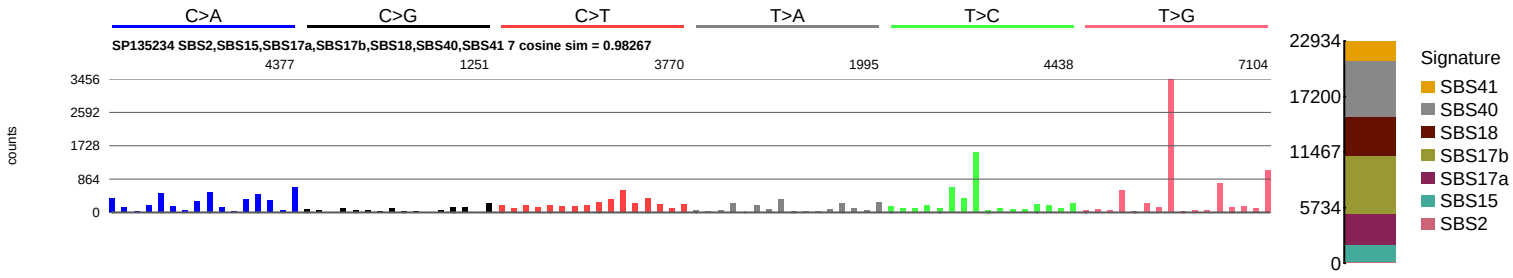
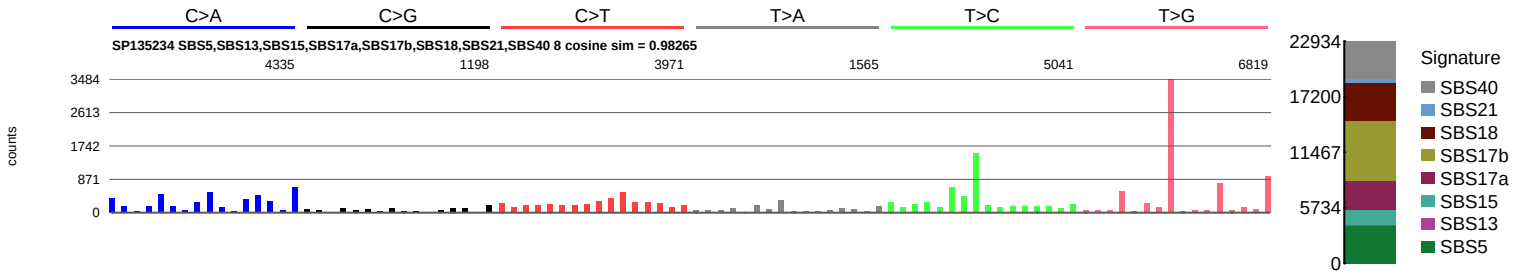
T>C
T>G
SP135234 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98286
4329
1112
4097
1549
5044
6804
counts
3488
2616
1744
872
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

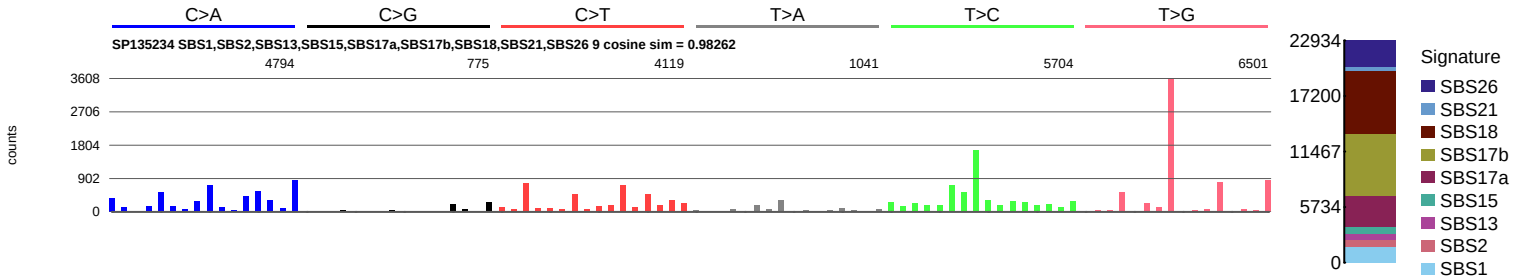

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98285
4343
1205
3555
1911
4857
7065
counts
3452
2589
1726
863
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15

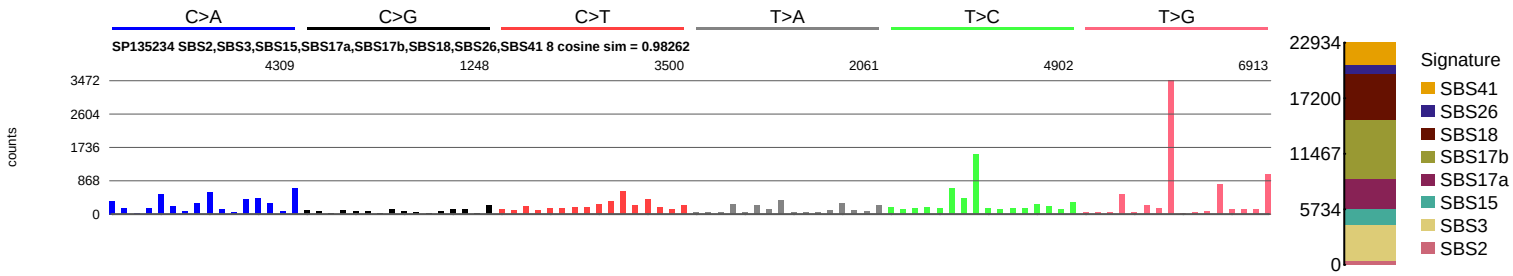

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98269
4361
1303
3729
2039
4433
7069
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

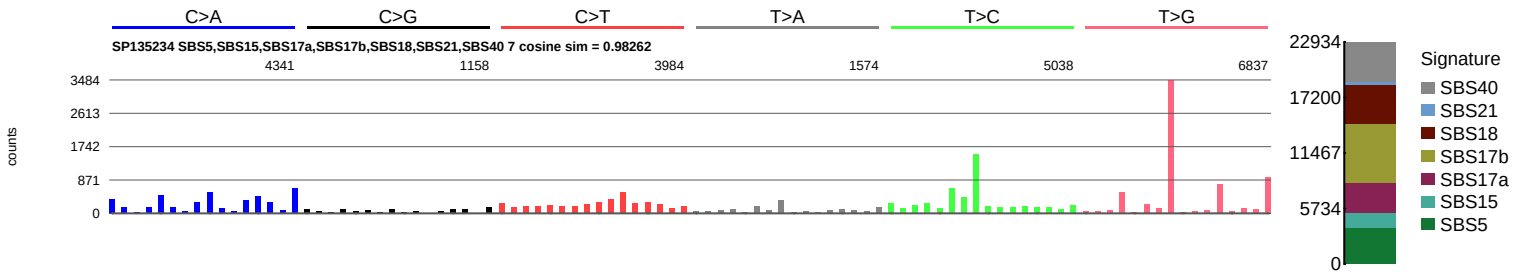

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98267
4310
1245
3497
2063
4911
6907
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

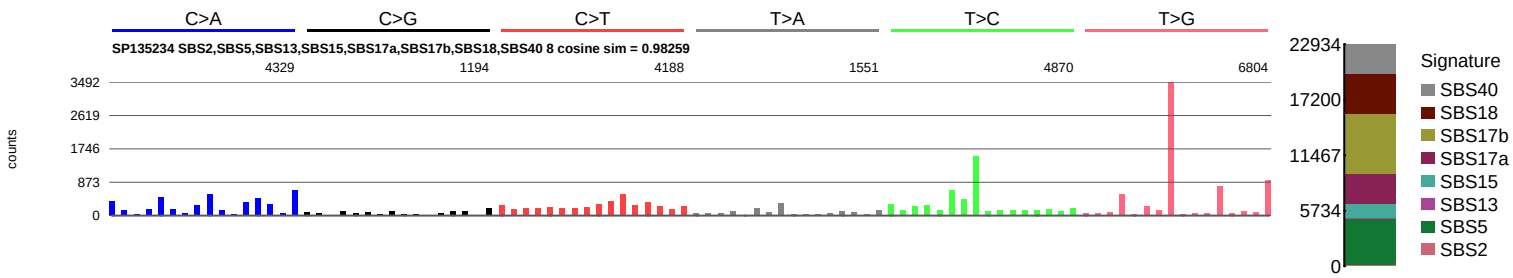

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98267
4377
1251
3770
1995
4438
7104
counts
3456
2592
1728
864
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2

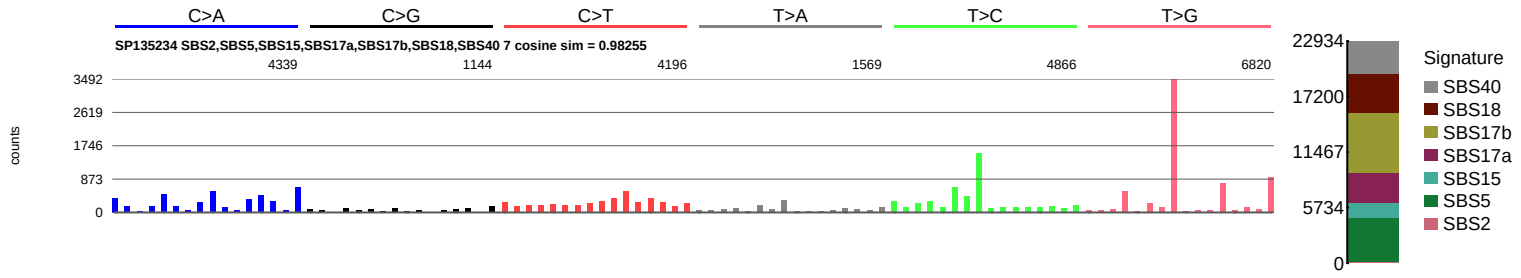

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98265
4335
1198
3971
1565
5041
6819
counts
3484
2613
1742
871
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

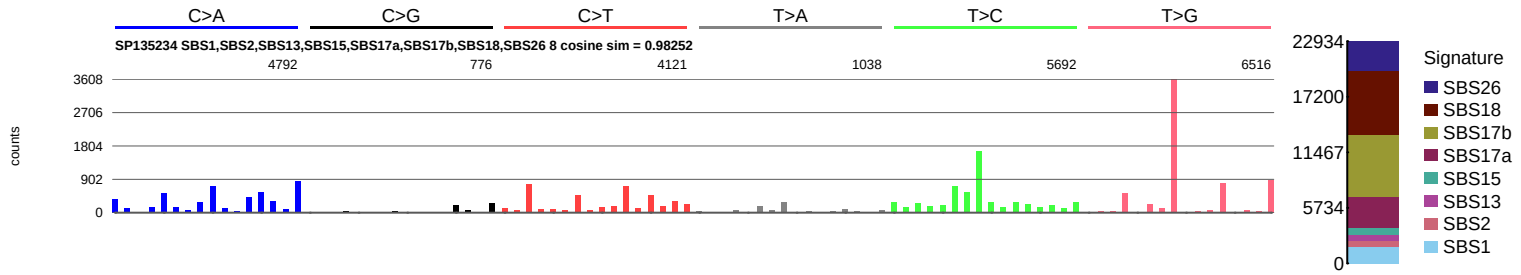

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98262
4794
775
4119
1041
5704
6501
counts
3608
2706
1804
902
0
22934
17200
11467
5734
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

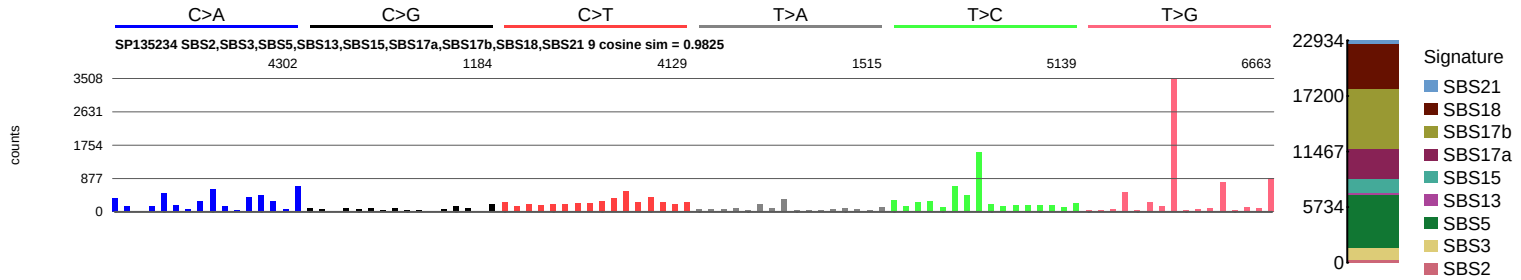

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98262
4309
1248
3500
2061
4902
6913
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

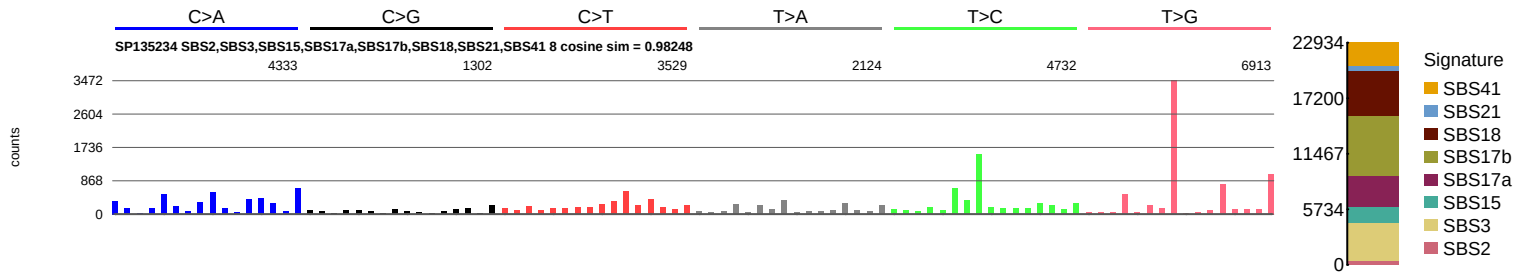

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98262
4341
1158
3984
1574
5038
6837
counts
3484
2613
1742
871
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5

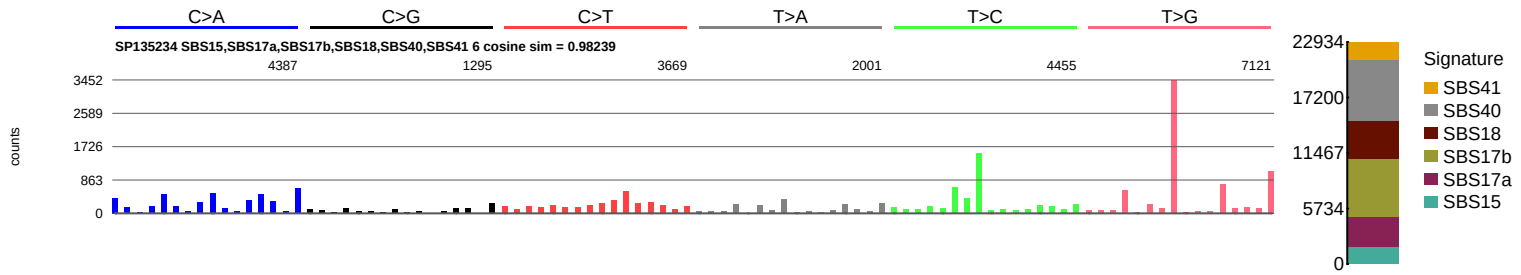

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98259
4329
1194
4188
1551
4870
6804
counts
3492
2619
1746
873
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

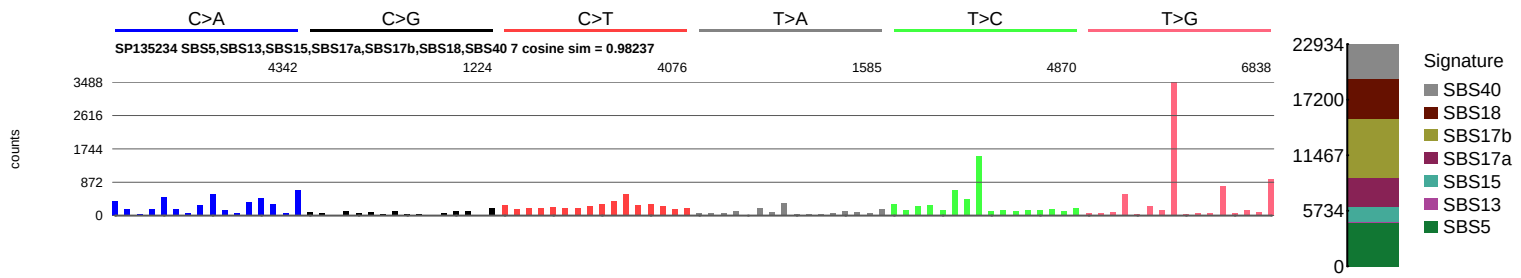

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98255
4339
1144
4196
1569
4866
6820
counts
3492
2619
1746
873
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

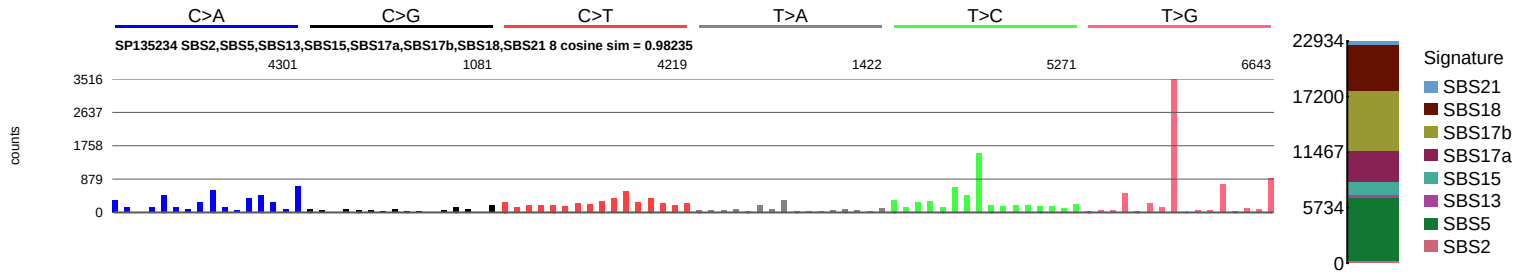

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98252
4792
776
4121
1038
5692
6516
counts
3608
2706
1804
902
0
22934
17200
11467
5734
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

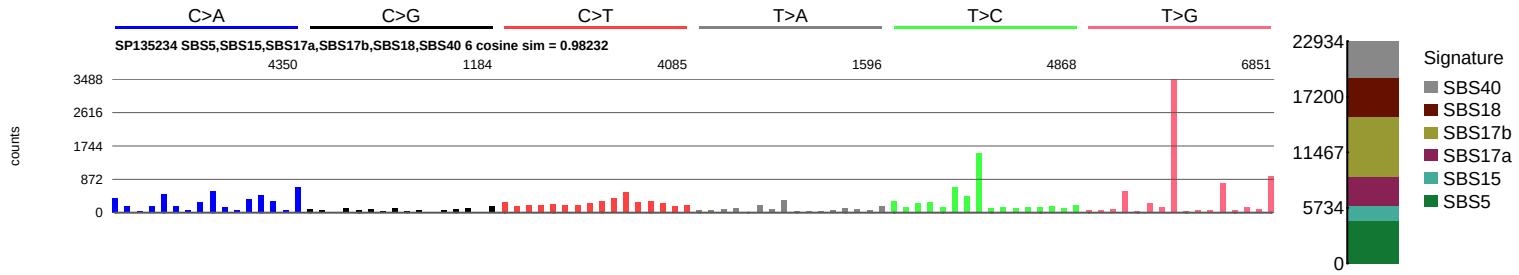

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.9825
4302
1184
4129
1515
5139
6663
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

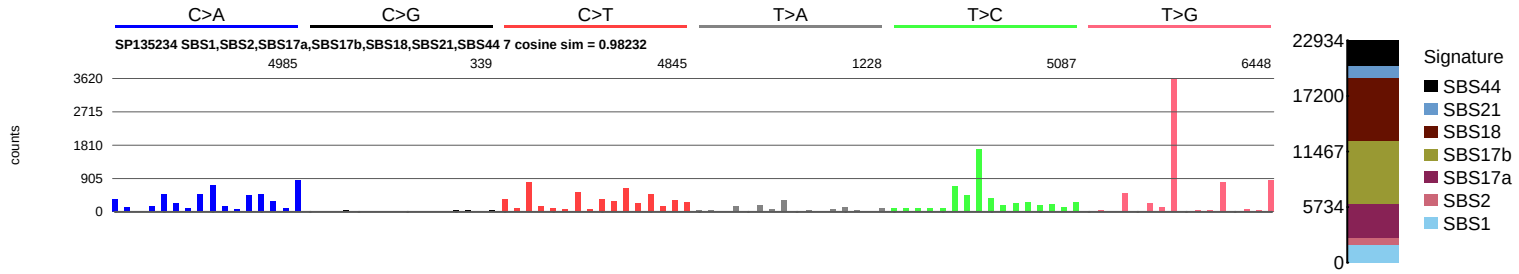

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98248
4333
1302
3529
2124
4732
6913
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

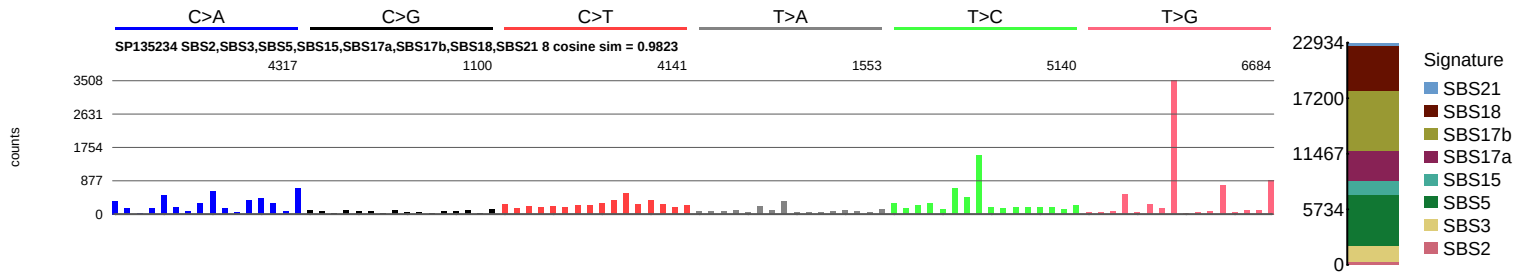

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98239
4387
1295
3669
2001
4455
7121
counts
3452
2589
1726
863
0
22934
17200
11467
5734
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15

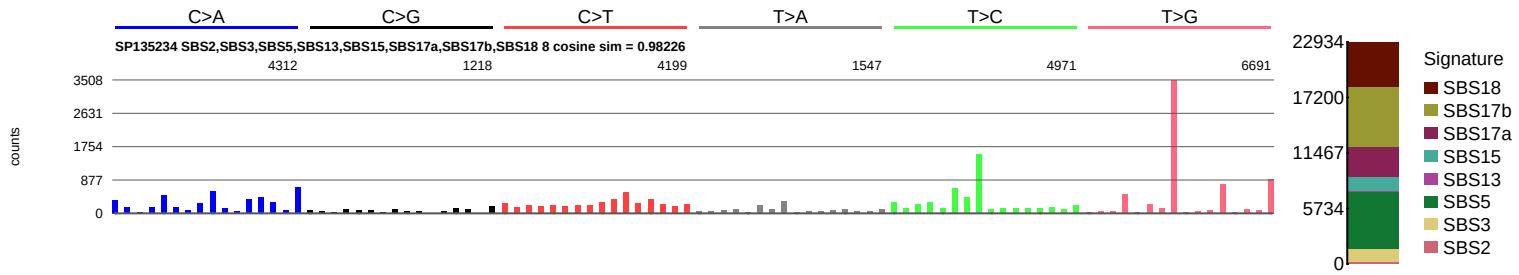

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98237
4342
1224
4076
1585
4870
6838
counts
3488
2616
1744
872
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98235
4301
1081
4219
1422
5271
6643
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98232
4350
1184
4085
1596
4868
6851
counts
3488
2616
1744
872
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98232
4985
339
4845
1228
5087
6448
counts
3620
2715
1810
905
0
22934
17200
11467
5734
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.9823
4317
1100
4141
1553
5140
6684
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98226
4312
1218
4199
1547
4971
6691
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
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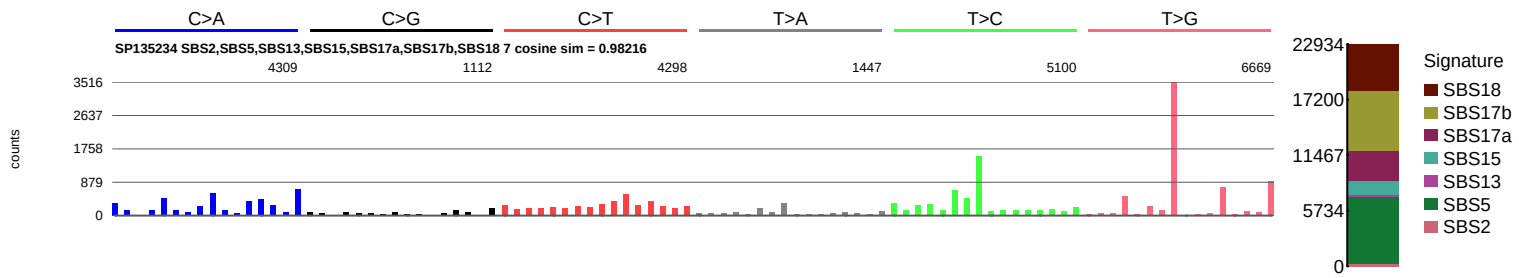
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SP135234 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98216
4309
1112
4298
1447
5100
6669
counts
3516
2637
1758
879
0
22934
17200
11467
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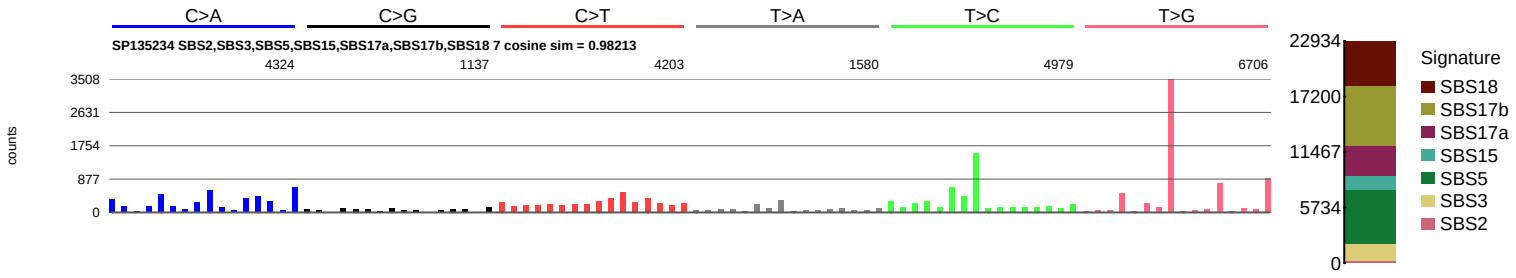
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SP135234 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98213
4324
1137
4203
1580
4979
6706
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
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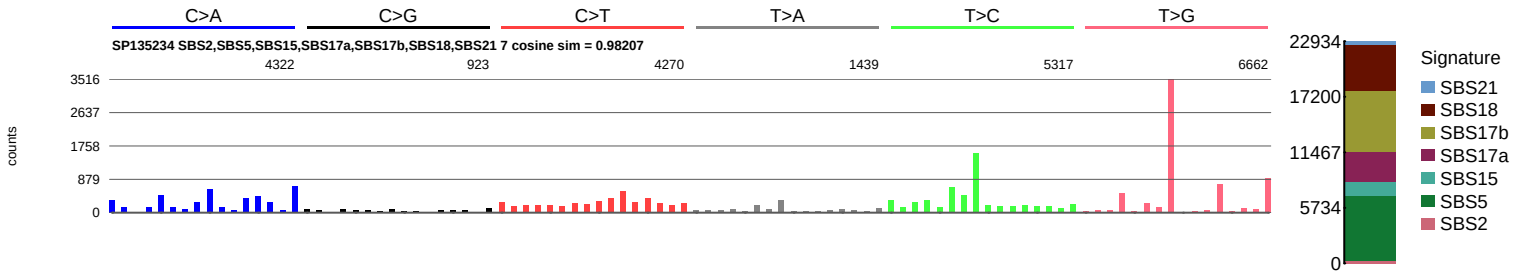
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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SP135234 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98207
4322
923
4270
1439
5317
6662
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
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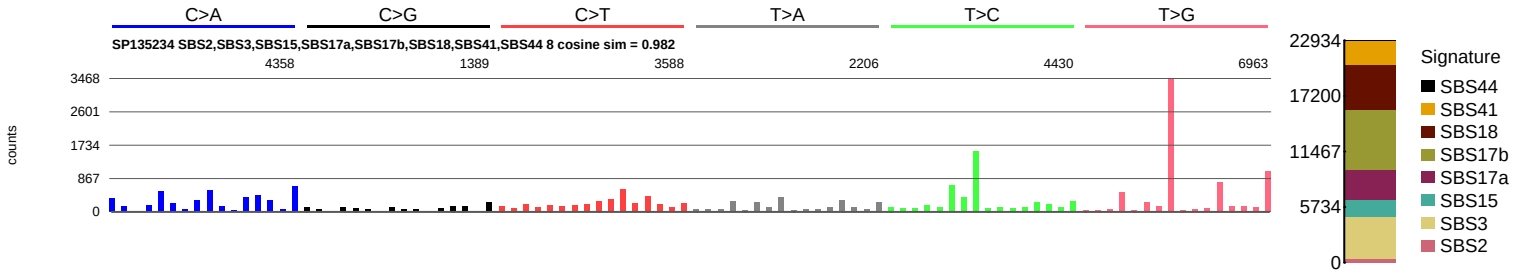
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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SP135234 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.982
4358
1389
3588
2206
4430
6963
counts
3468
2601
1734
867
0
22934
17200
11467
5734
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Signature
SBS44
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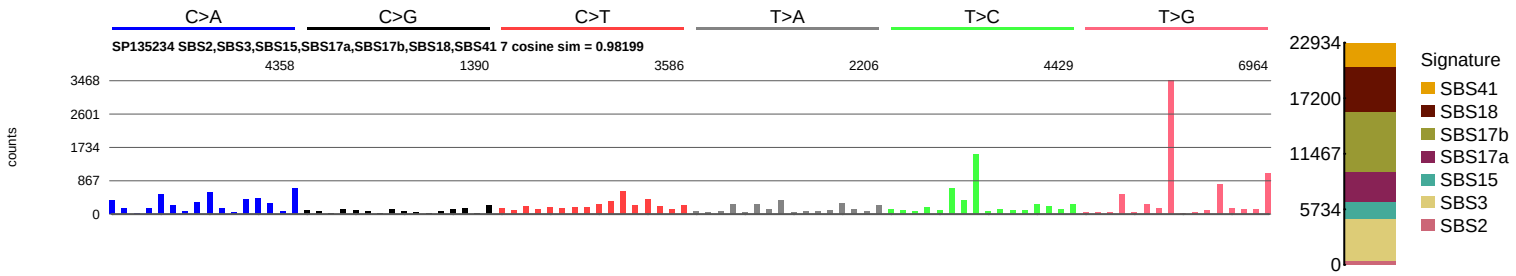
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
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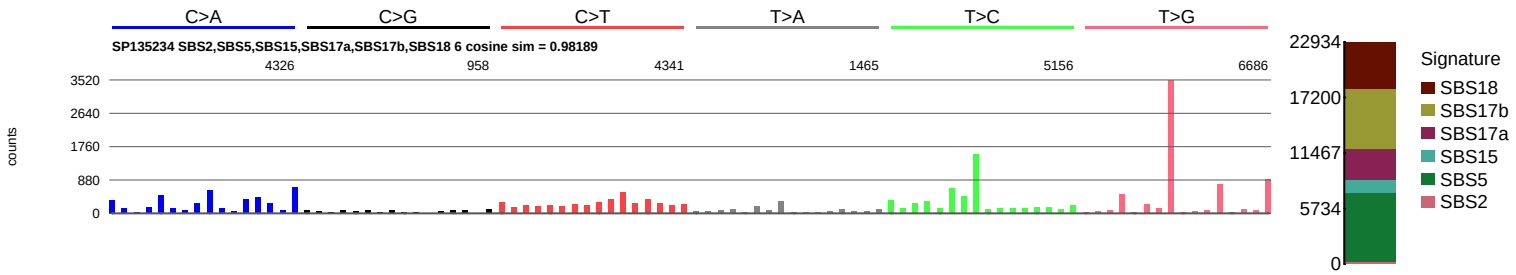
T>C
T>G
SP135234 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98199
4358
1390
3586
2206
4429
6964
counts
3468
2601
1734
867
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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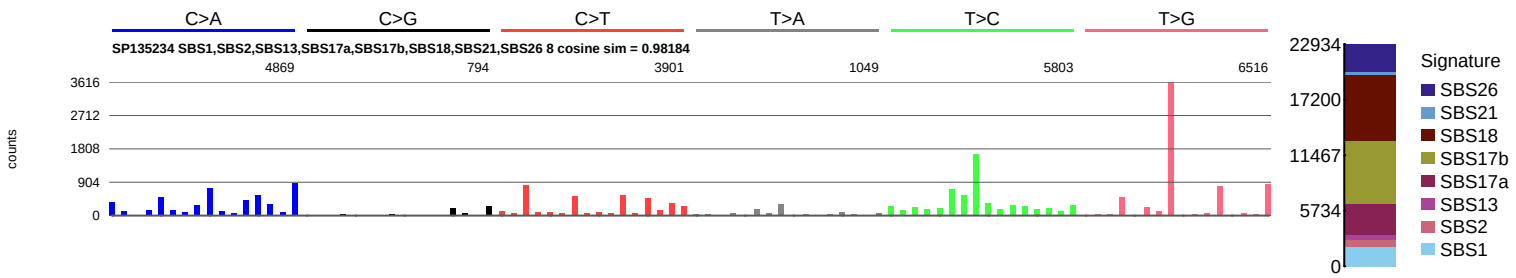
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98189
4326
958
4341
1465
5156
6686
counts
3520
2640
1760
880
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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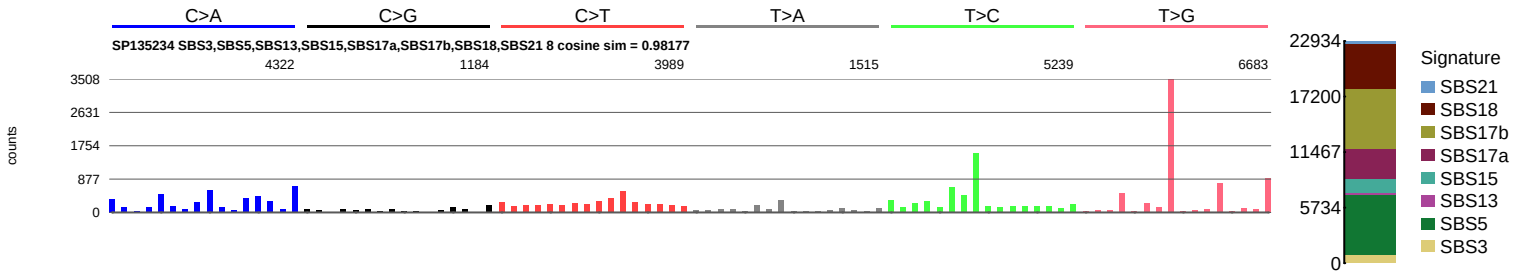
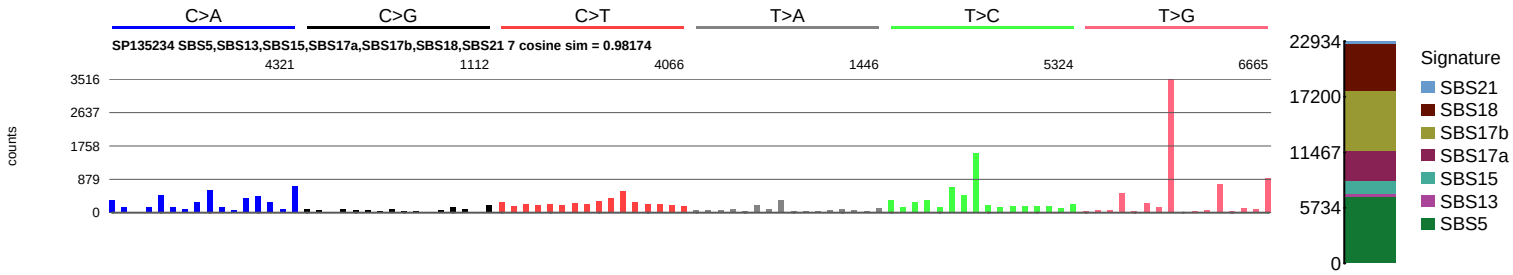
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98184
4869
794
3901
1049
5803
6516
counts
3616
2712
1808
904
0
22934
17200
11467
5734
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98177
4322
1184
3989
1515
5239
6683
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3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C>G
C>T
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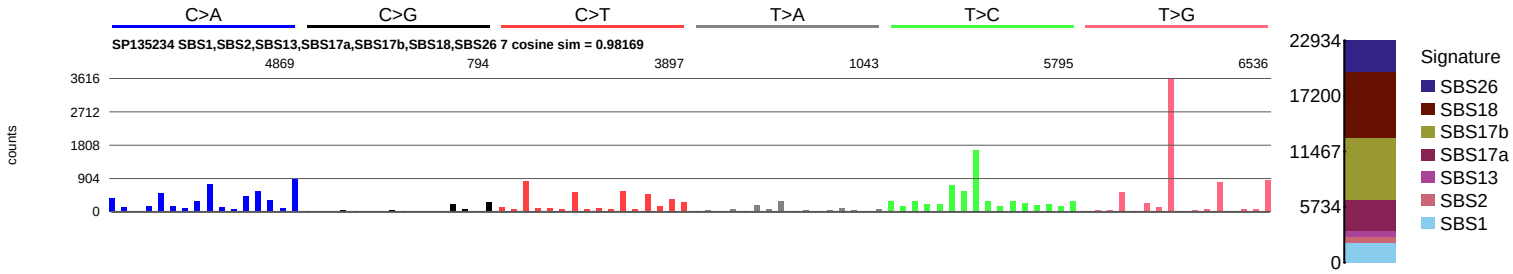
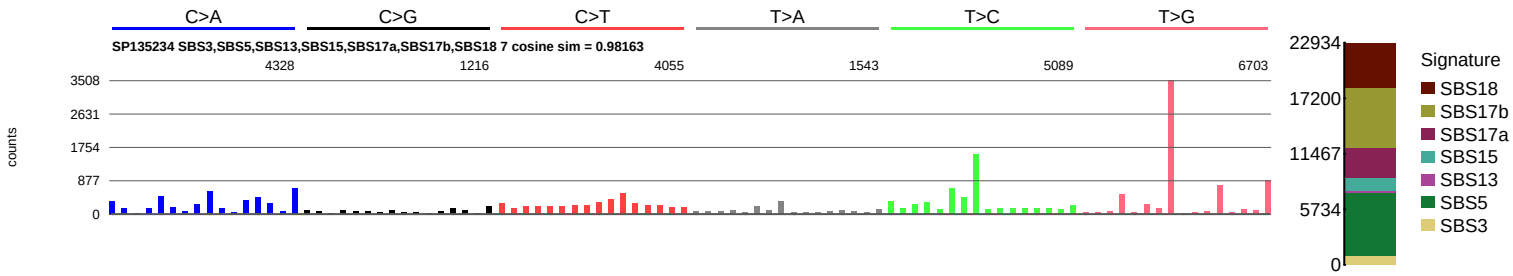
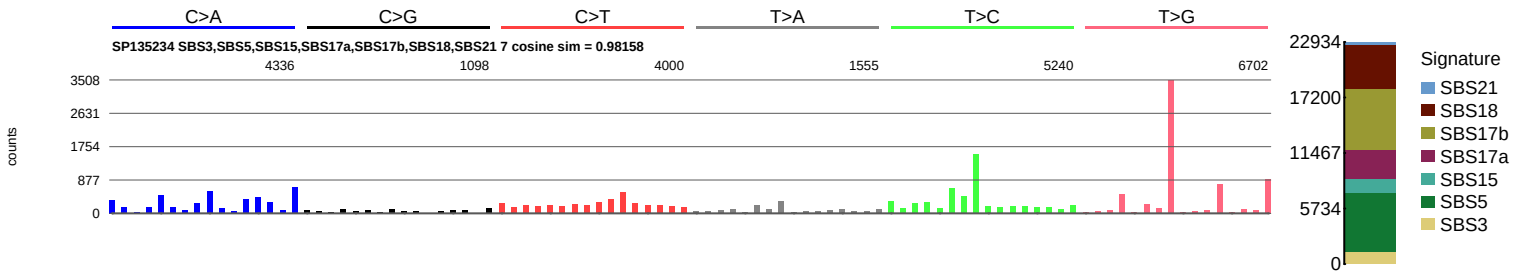
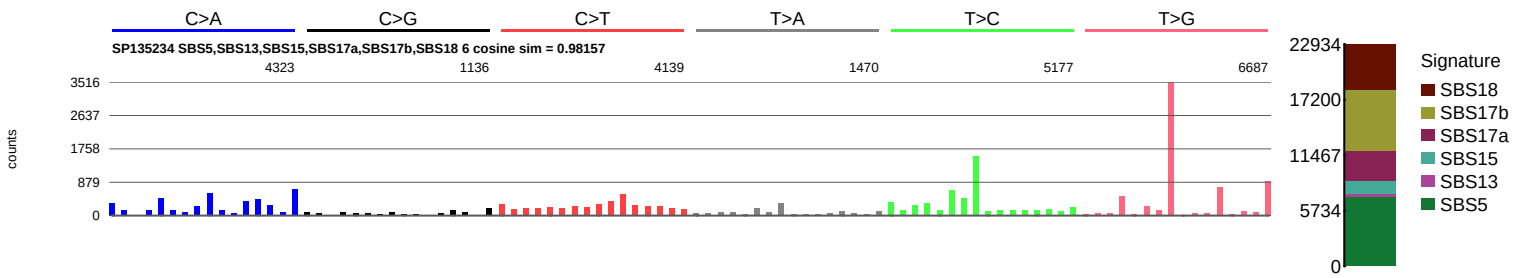
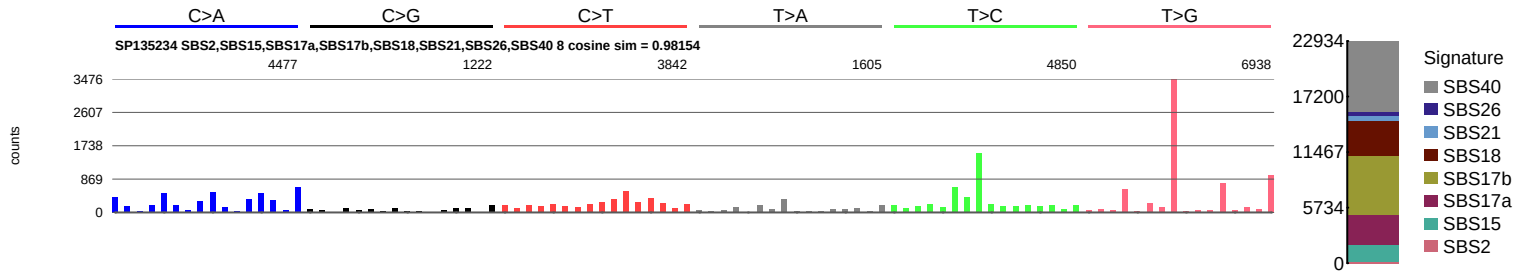
T>C
T>G
SP135234 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98174
4321
1112
4066
1446
5324
6665
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

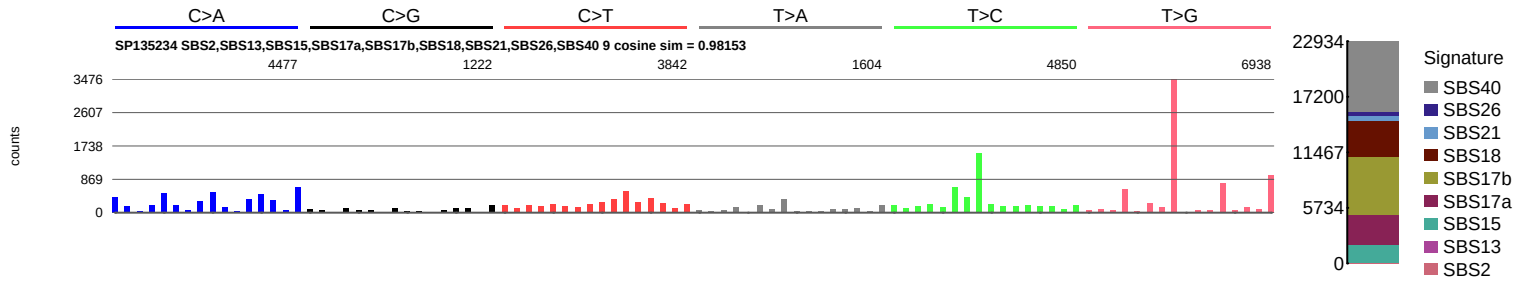

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98169
4869
794
3897
1043
5795
6536
counts
3616
2712
1808
904
0
22934
17200
11467
5734
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

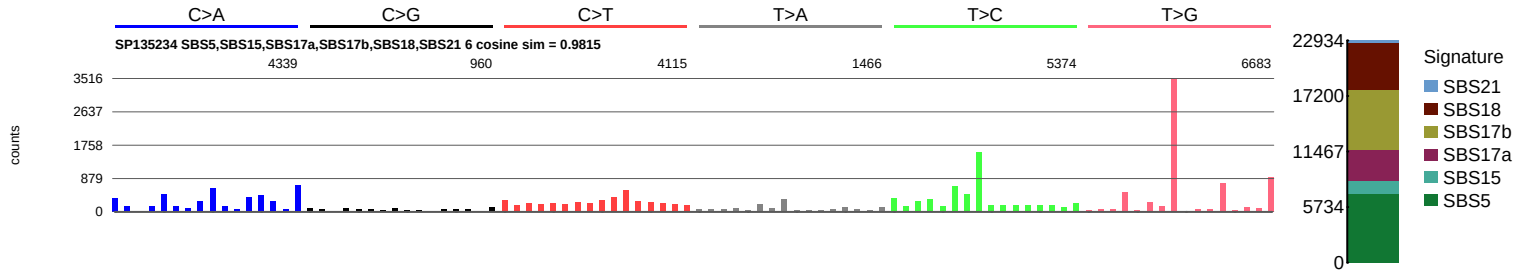

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98163
4328
1216
4055
1543
5089
6703
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

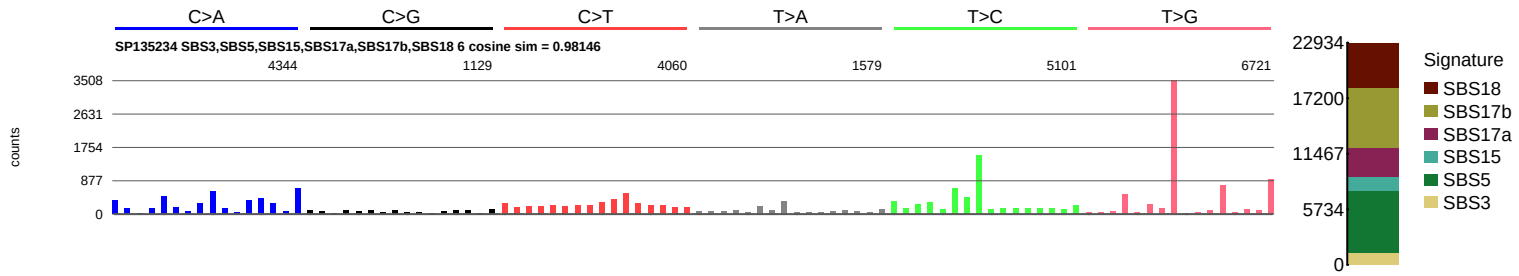

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98158
4336
1098
4000
1555
5240
6702
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

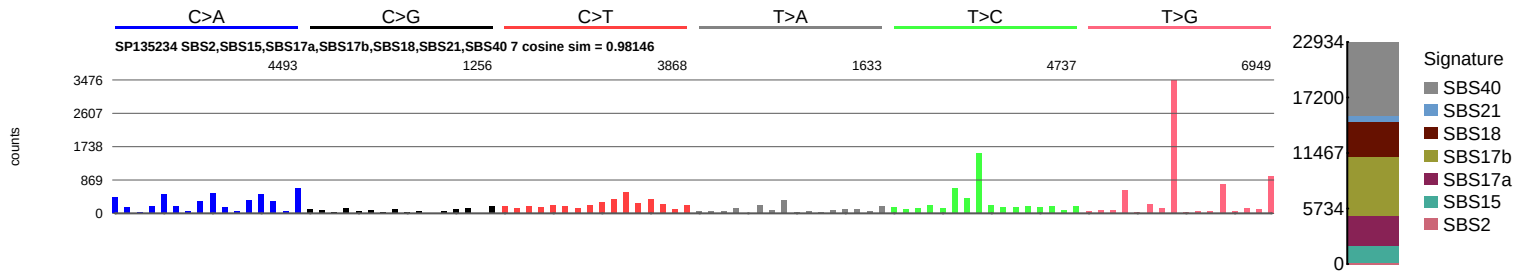

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98157
4323
1136
4139
1470
5177
6687
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

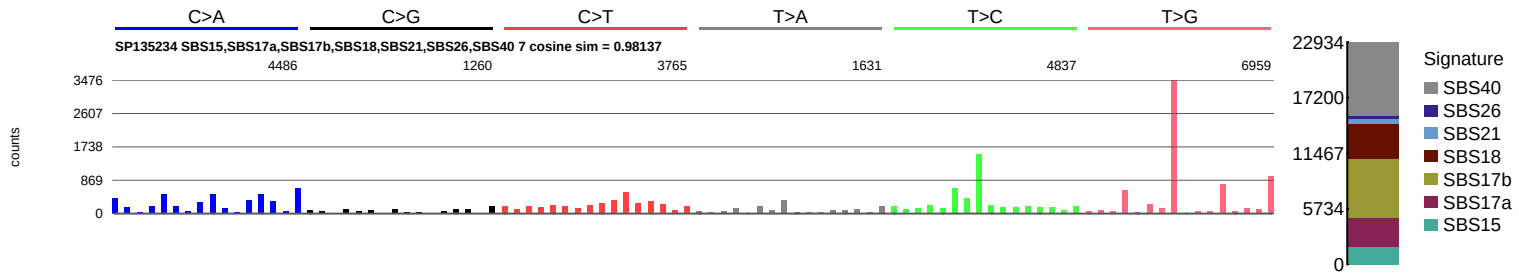

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98154
4477
1222
3842
1605
4850
6938
counts
3476
2607
1738
869
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98153
4477
1222
3842
1604
4850
6938
counts
3476
2607
1738
869
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.9815
4339
960
4115
1466
5374
6683
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98146
4344
1129
4060
1579
5101
6721
counts
3508
2631
1754
877
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98146
4493
1256
3868
1633
4737
6949
counts
3476
2607
1738
869
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98137
4486
1260
3765
1631
4837
6959
counts
3476
2607
1738
869
0
22934
17200
11467
5734
0
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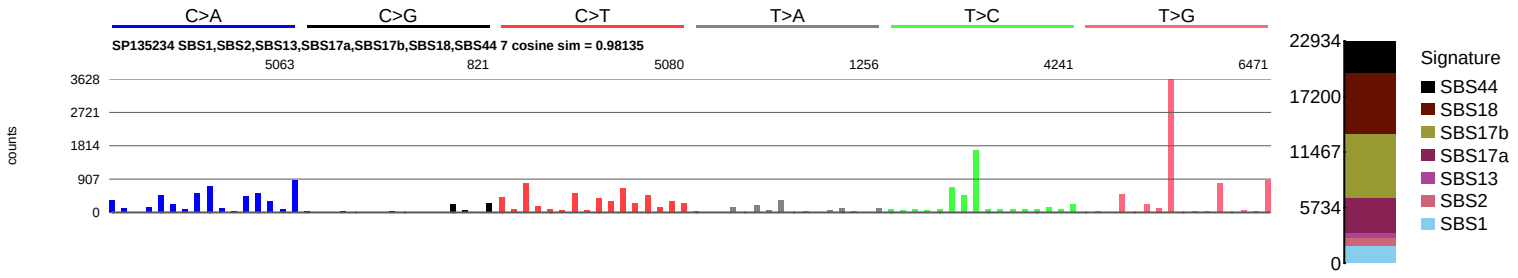
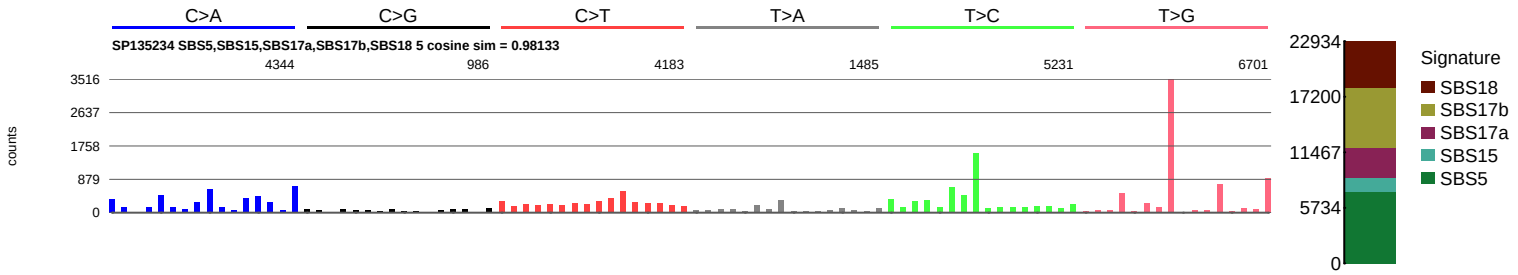
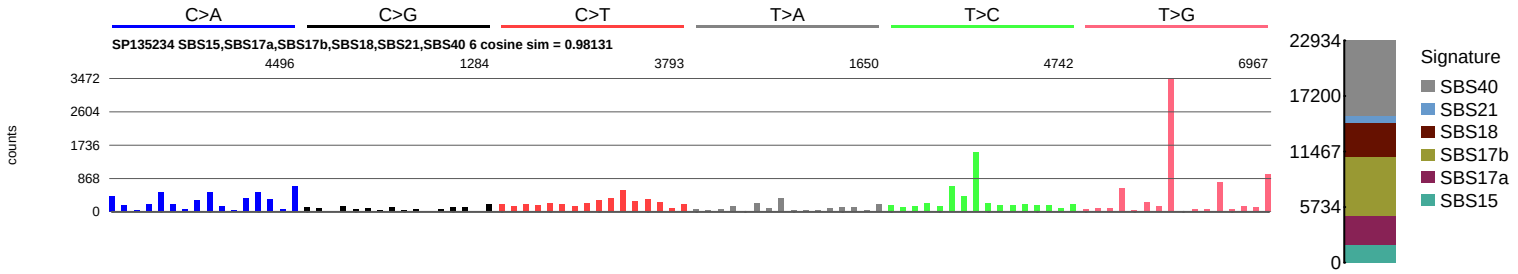
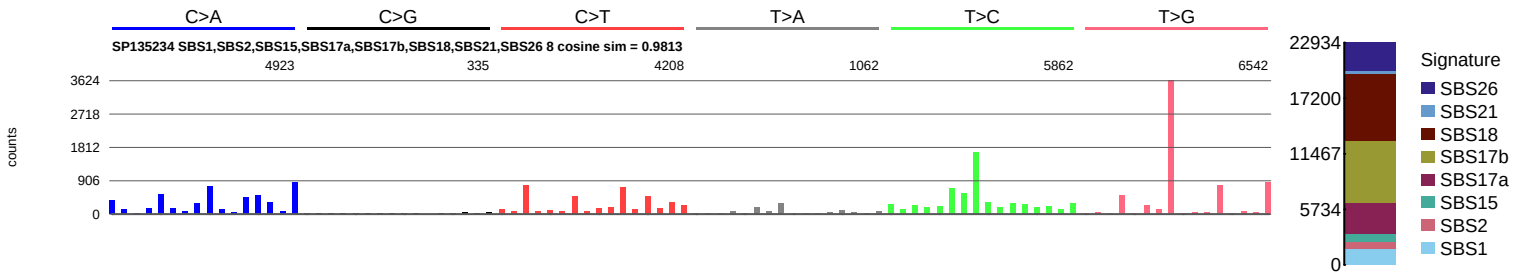
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98135
5063
821
5080
1256
4241
6471
counts
3628
2721
1814
907
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS5,SBS15,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98133
4344
986
4183
1485
5231
6701
counts
3516
2637
1758
879
0
22934
17200
11467
5734
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98131
4496
1284
3793
1650
4742
6967
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.9813
4923
335
4208
1062
5862
6542
counts
3624
2718
1812
906
0
22934
17200
11467
5734
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Signature
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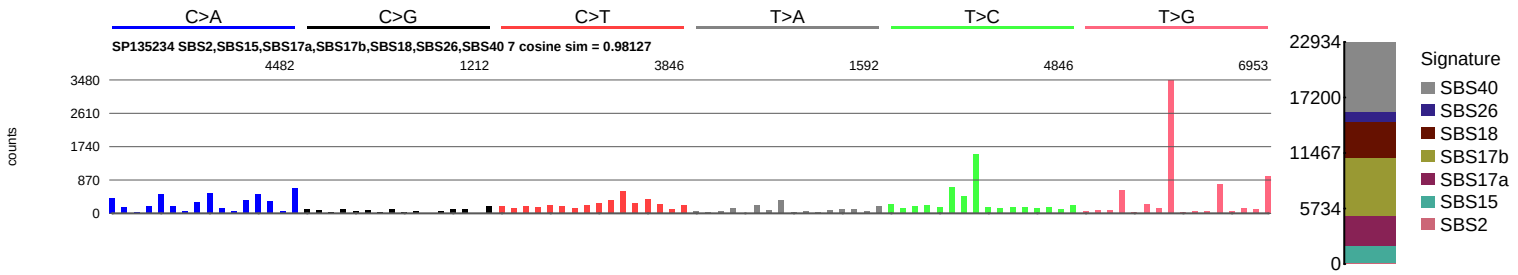
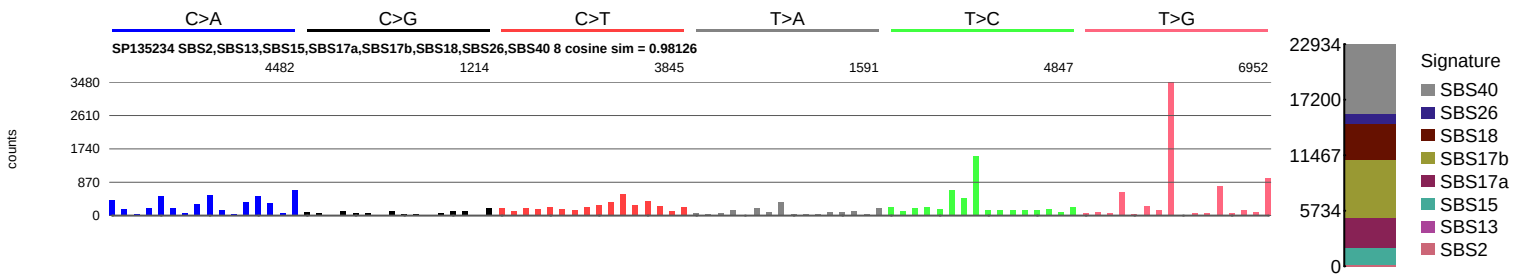
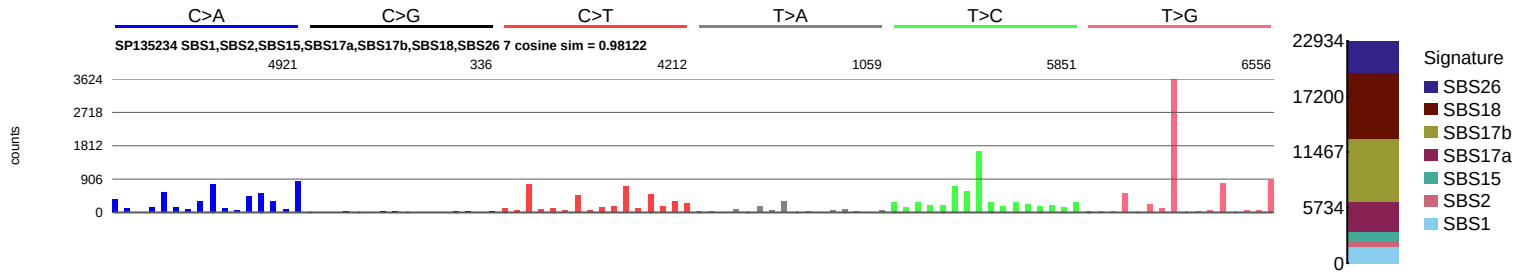
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98127
4482
1212
3846
1592
4846
6953
counts
3480
2610
1740
870
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98126
4482
1214
3845
1591
4847
6952
counts
3480
2610
1740
870
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2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C>G
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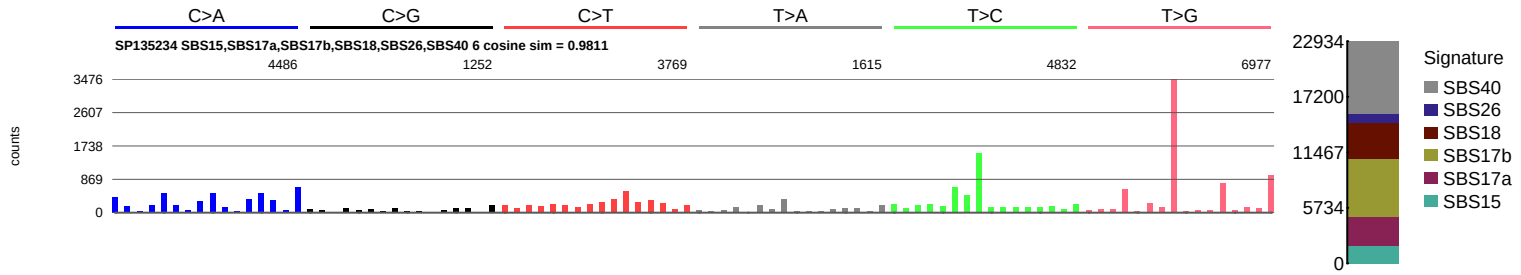
T>G
SP135234 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98122
4921
336
4212
1059
5851
6556
counts
3624
2718
1812
906
0
22934
17200
11467
5734
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
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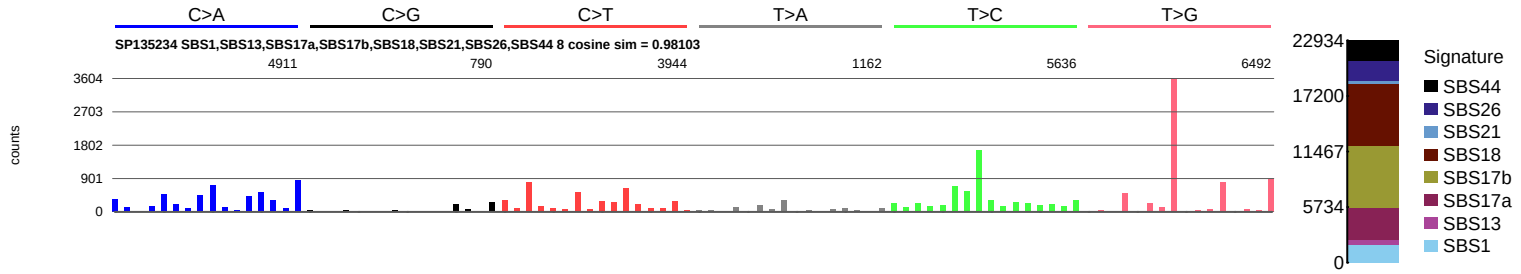
T>G
SP135234 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.9811
4486
1252
3769
1615
4832
6977
counts
3476
2607
1738
869
0
22934
17200
11467
5734
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15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C>T
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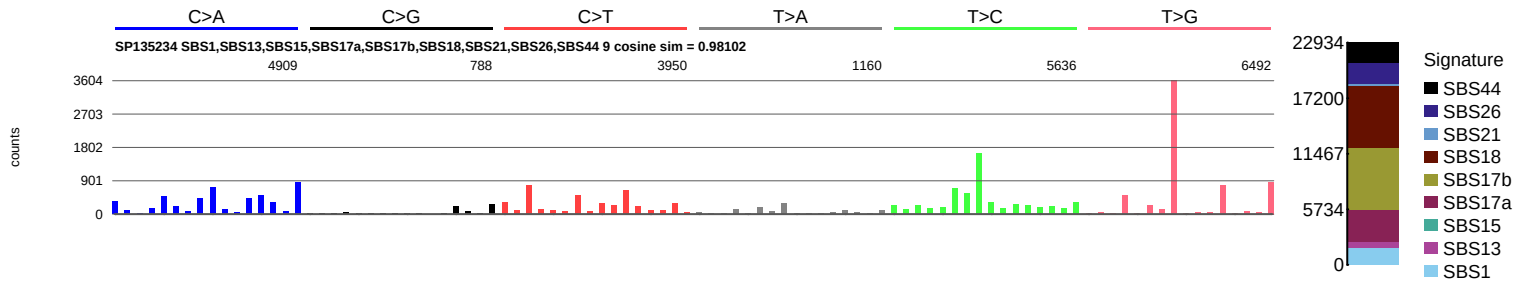
T>C
T>G
SP135234 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.98103
4911
790
3944
1162
5636
6492
counts
3604
2703
1802
901
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1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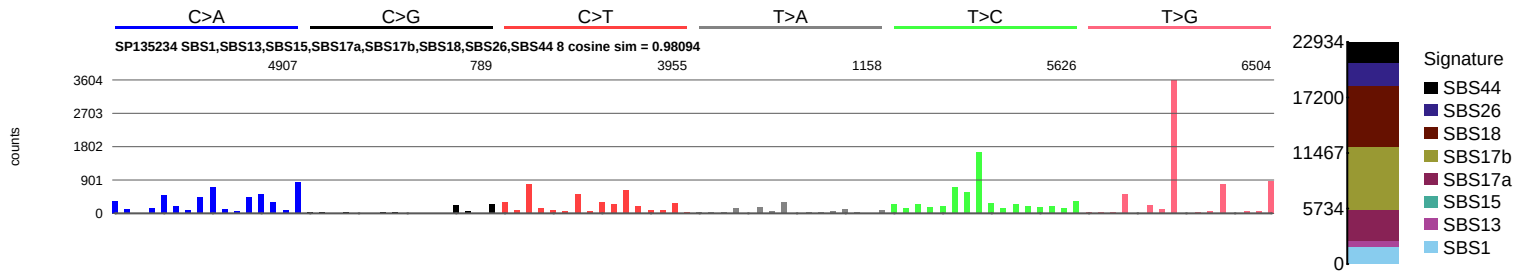
T>C
T>G
SP135234 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98102
4909
788
3950
1160
5636
6492
counts
3604
2703
1802
901
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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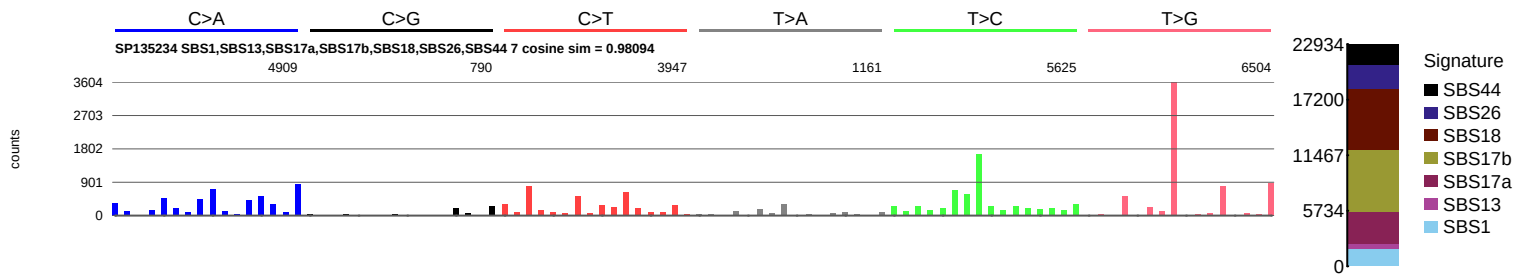
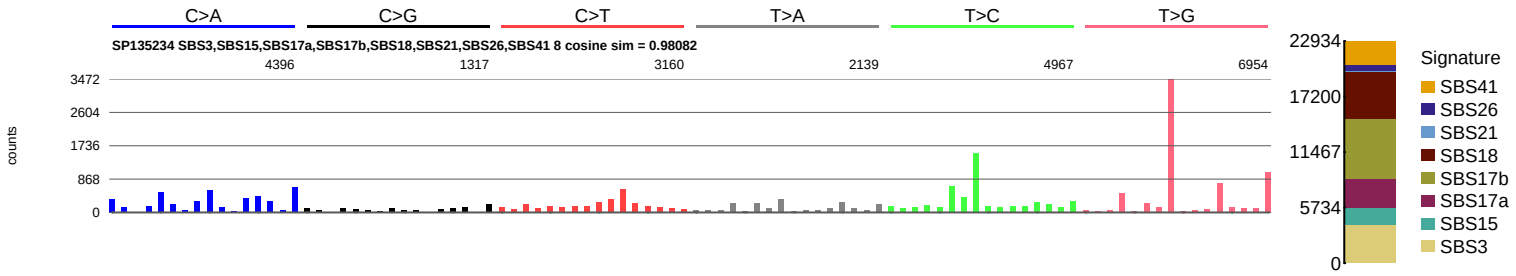
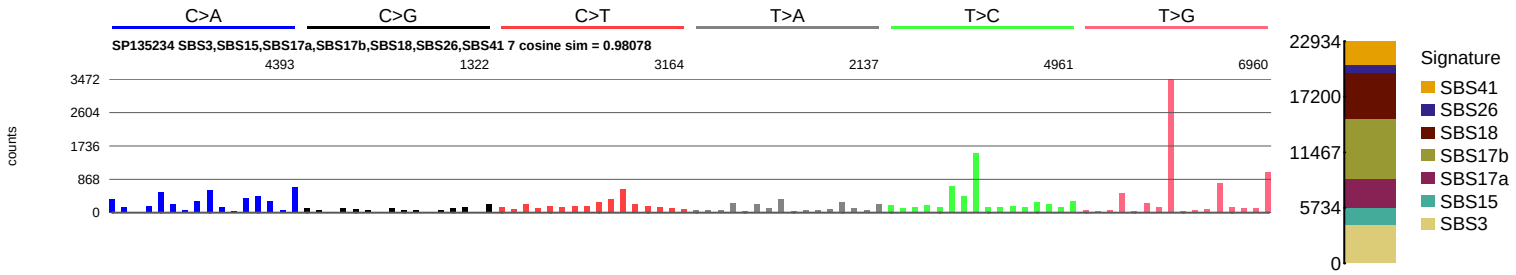
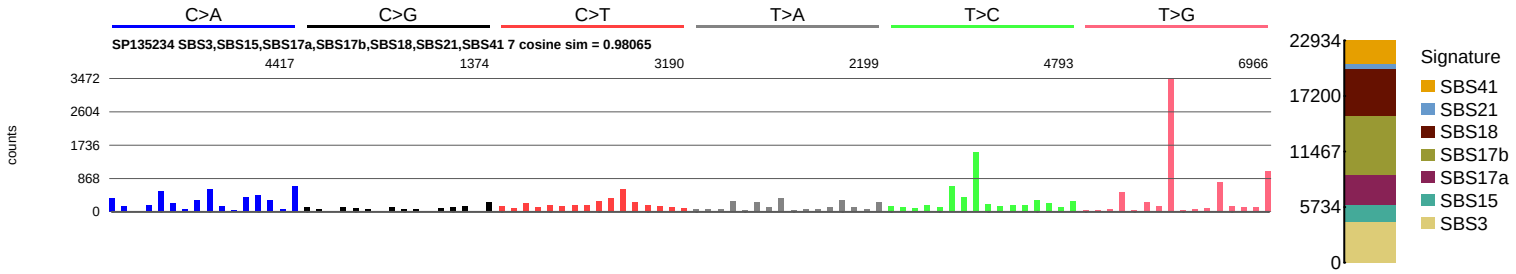
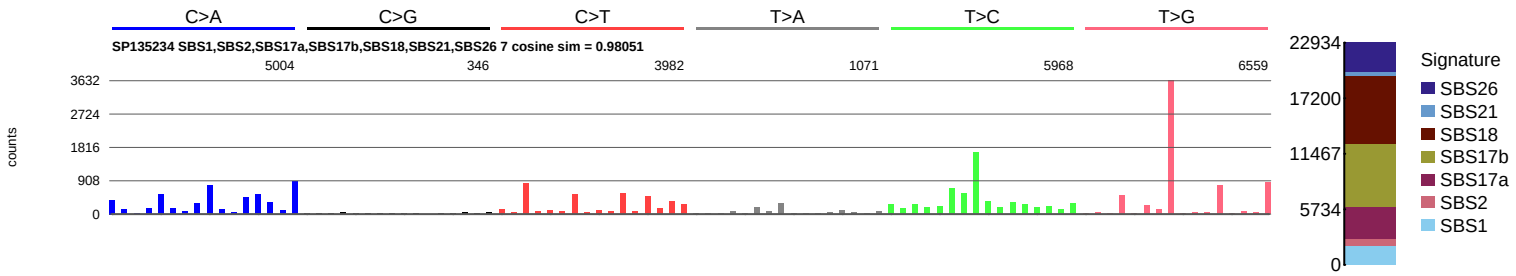
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98094
4907
789
3955
1158
5626
6504
counts
3604
2703
1802
901
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

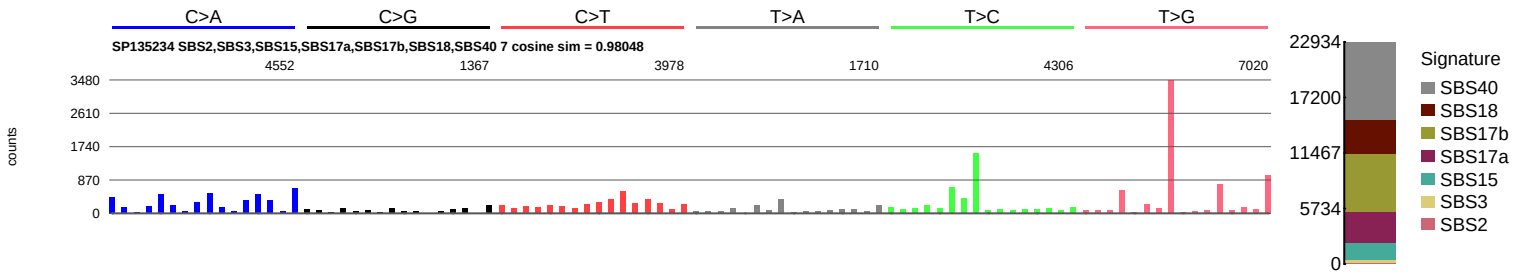

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98094
4909
790
3947
1161
5625
6504
counts
3604
2703
1802
901
0
22934
17200
11467
5734
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

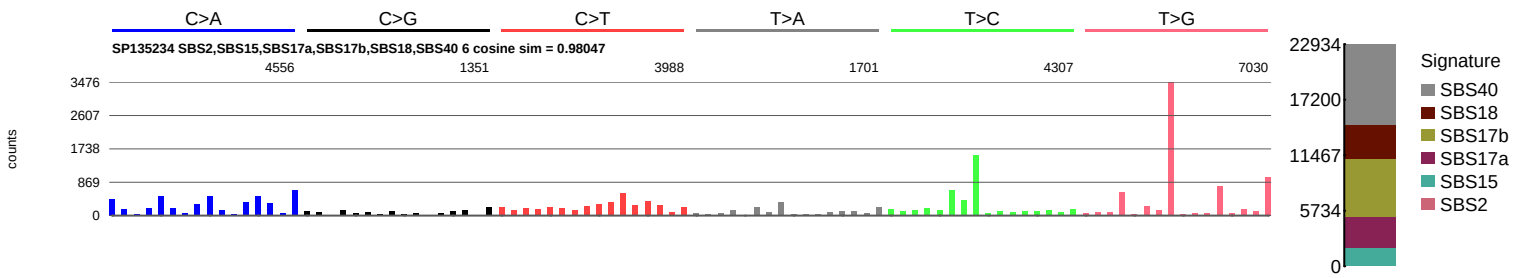

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98082
4396
1317
3160
2139
4967
6954
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3

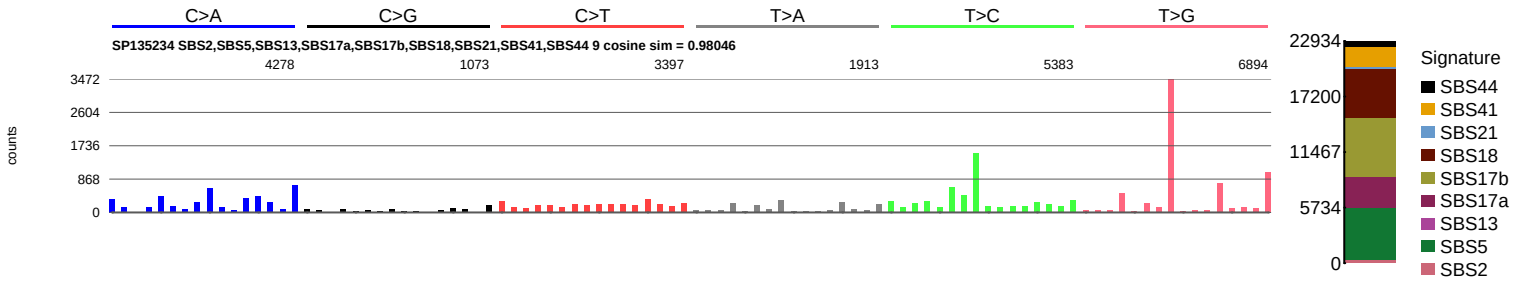

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98078
4393
1322
3164
2137
4961
6960
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3

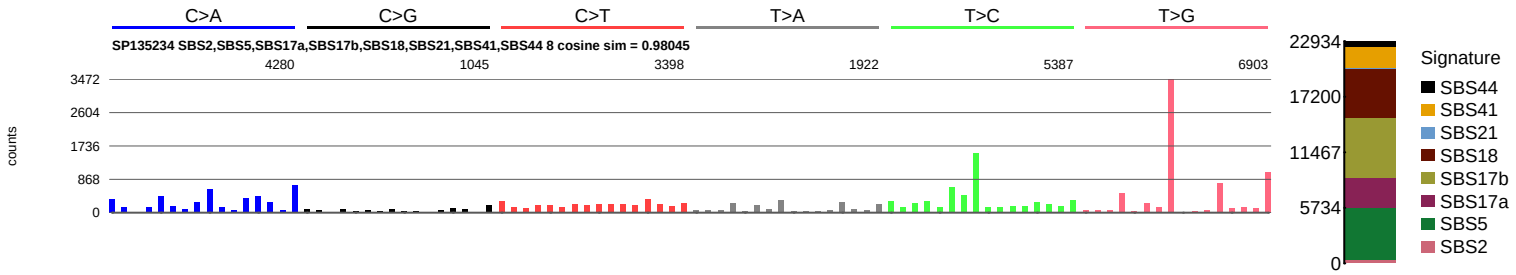

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98065
4417
1374
3190
2199
4793
6966
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3

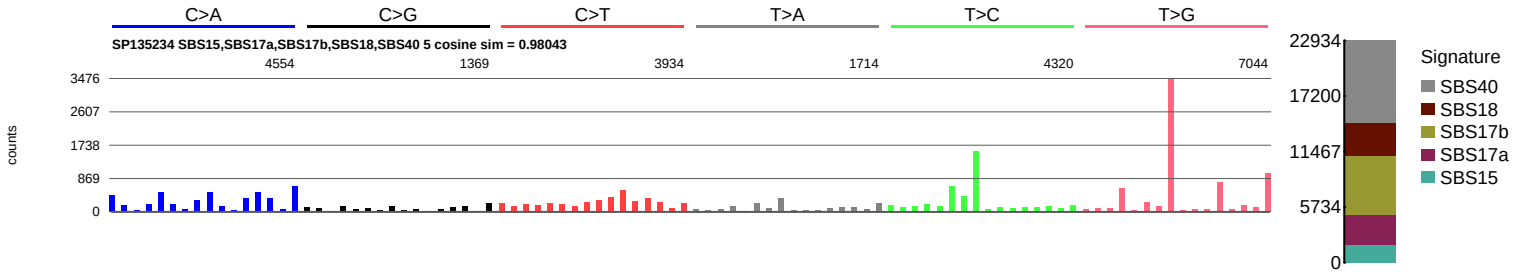

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26 7 cosine sim = 0.98051
5004
346
3982
1071
5968
6559
counts
3632
2724
1816
908
0
22934
17200
11467
5734
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

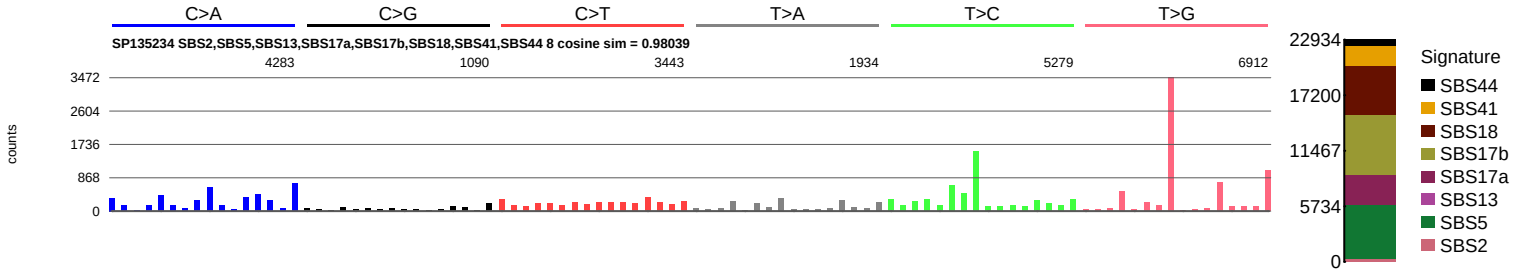

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98048
4552
1367
3978
1710
4306
7020
counts
3480
2610
1740
870
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

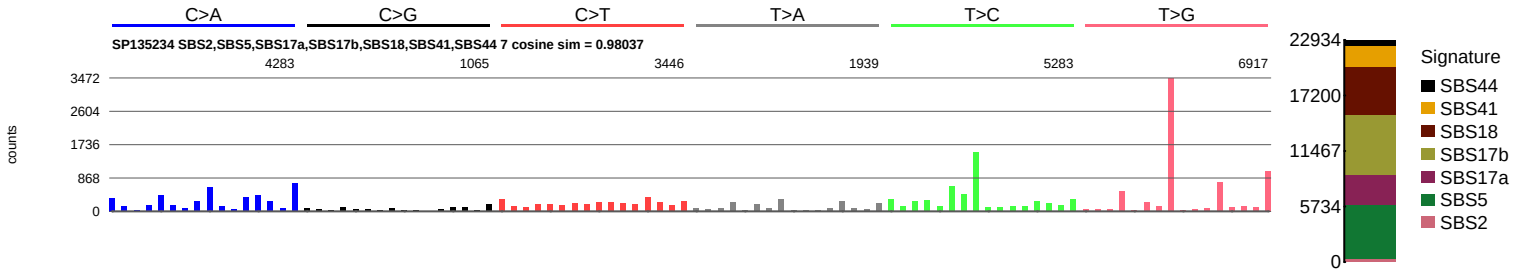

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98047
4556
1351
3988
1701
4307
7030
counts
3476
2607
1738
869
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2

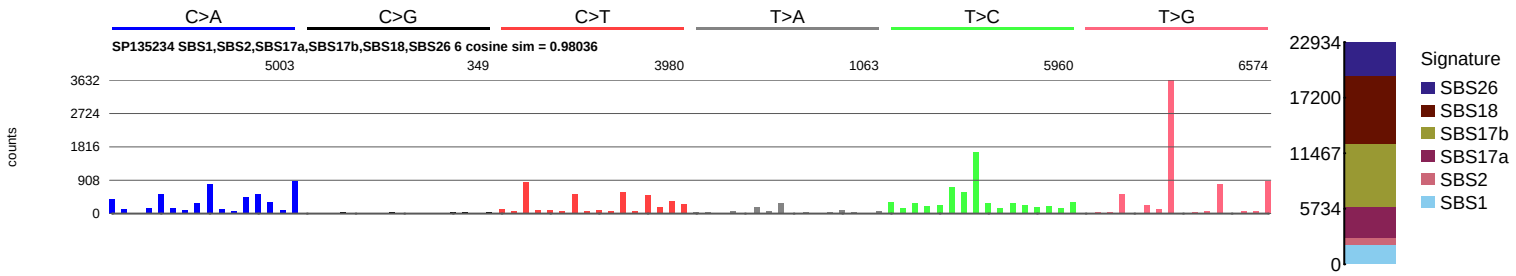

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98046
4278
1073
3397
1913
5383
6894
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

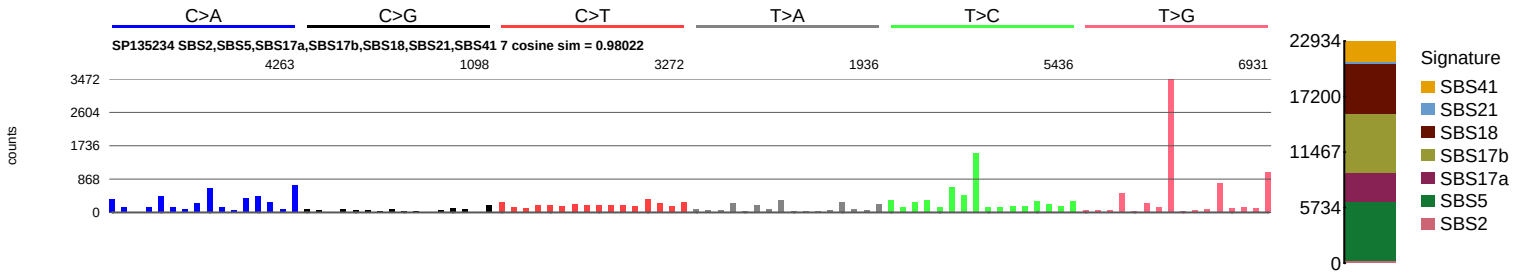

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98045
4280
1045
3398
1922
5387
6903
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

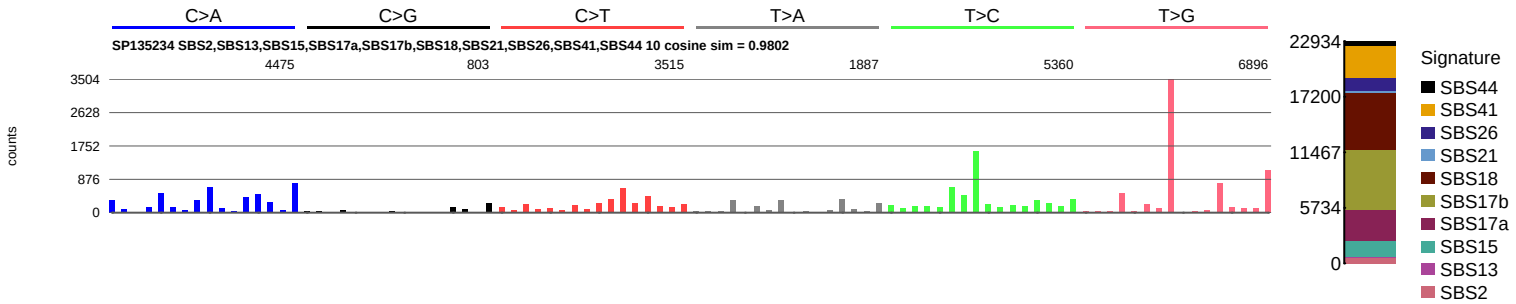

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS15,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98043
4554
1369
3934
1714
4320
7044
counts
3476
2607
1738
869
0
22934
17200
11467
5734
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15

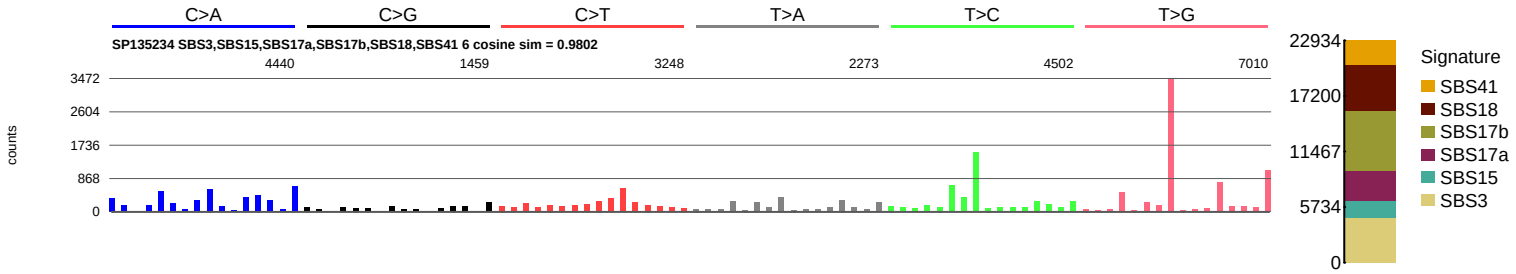

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98039
4283
1090
3443
1934
5279
6912
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

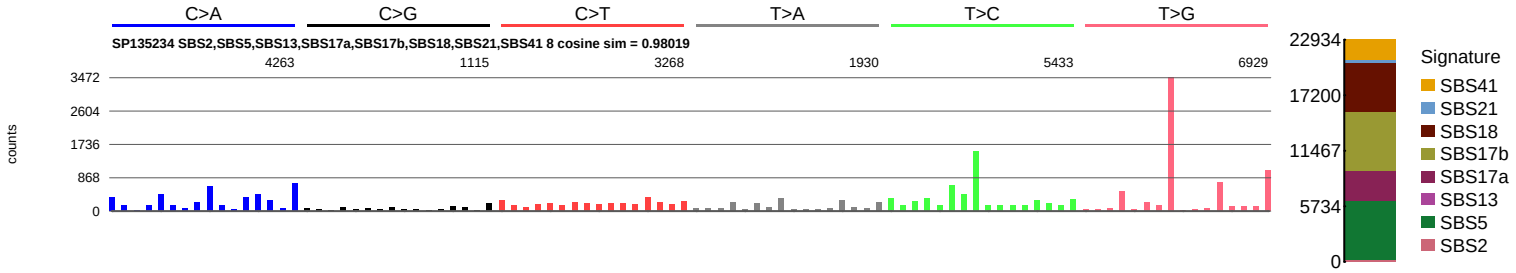

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98037
4283
1065
3446
1939
5283
6917
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2

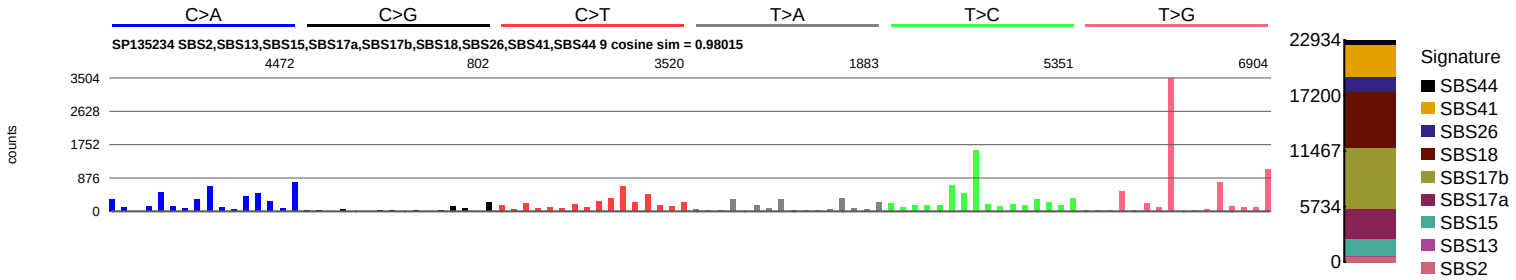

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98036
5003
349
3980
1063
5960
6574
counts
3632
2724
1816
908
0
22934
17200
11467
5734
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

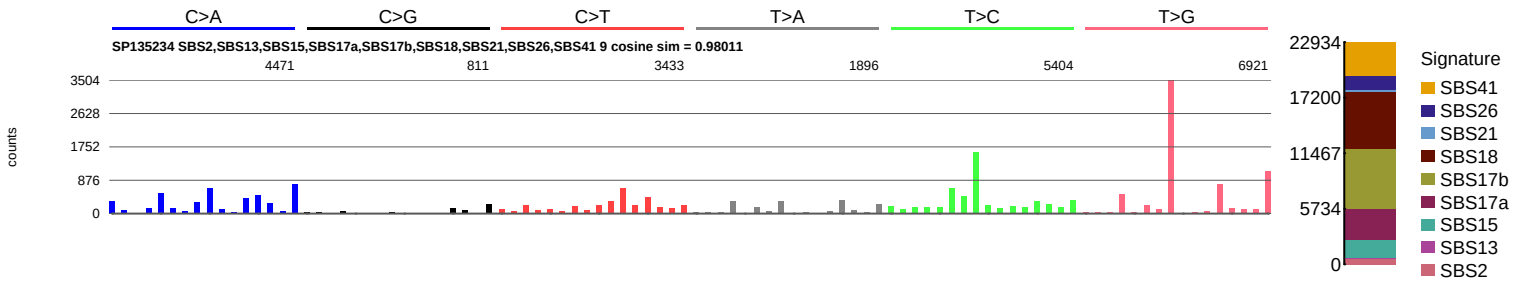

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98022
4263
1098
3272
1936
5436
6931
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

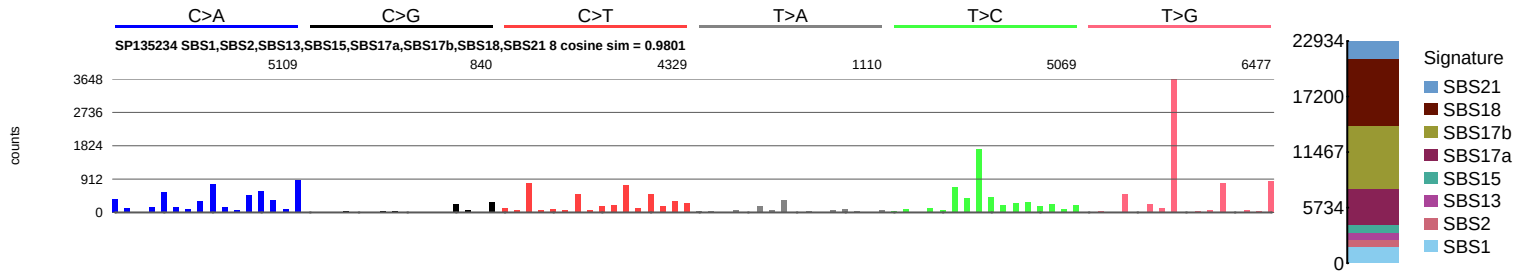

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.9802
4475
803
3515
1887
5360
6896
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

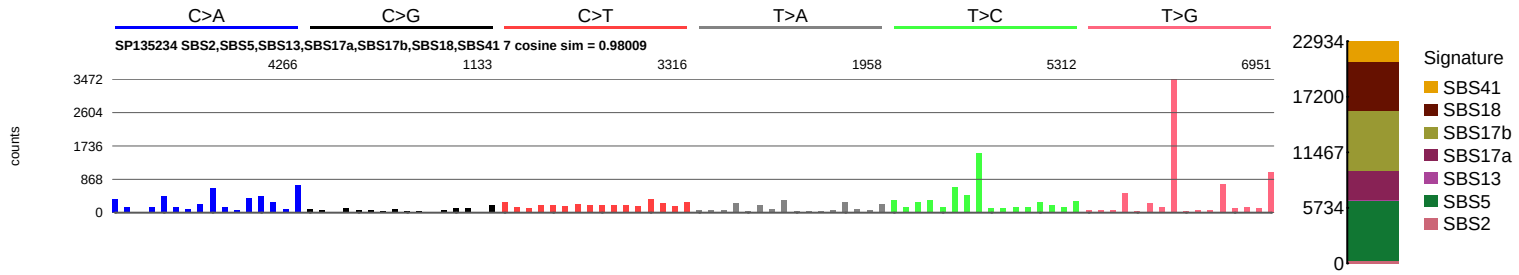

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.9802
4440
1459
3248
2273
4502
7010
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3

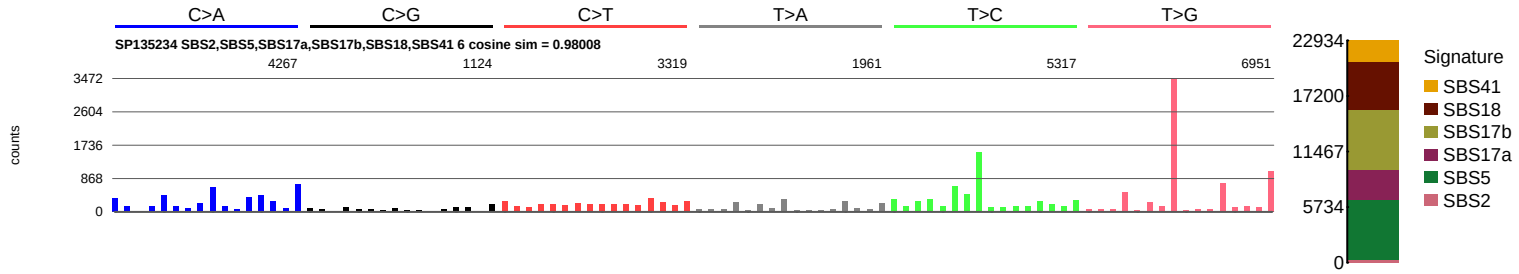

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98019
4263
1115
3268
1930
5433
6929
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

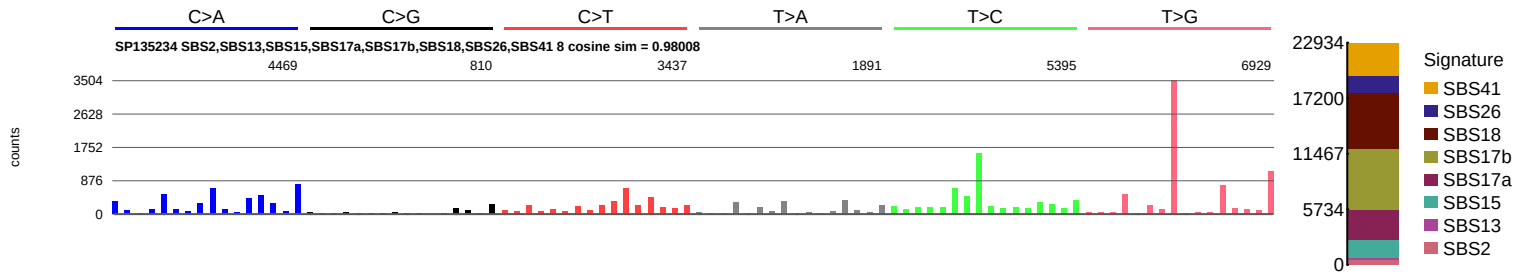

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98015
4472
802
3520
1883
5351
6904
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

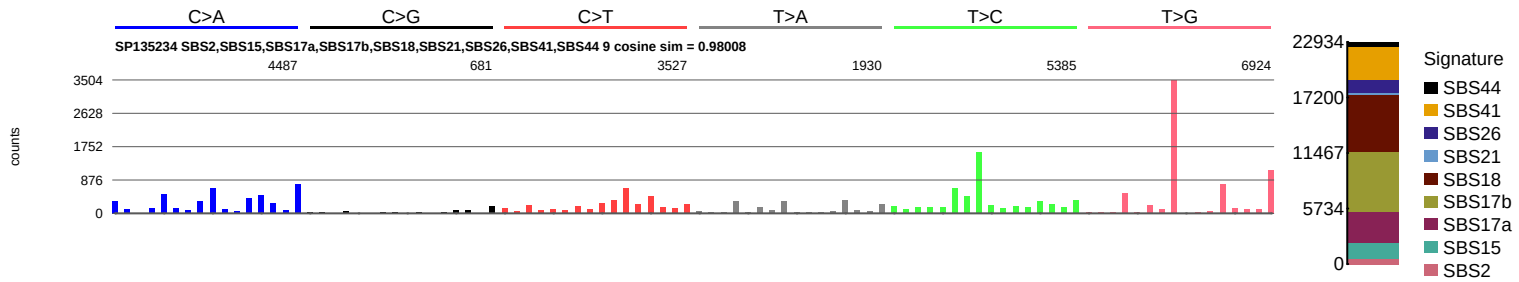

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98011
4471
811
3433
1896
5404
6921
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

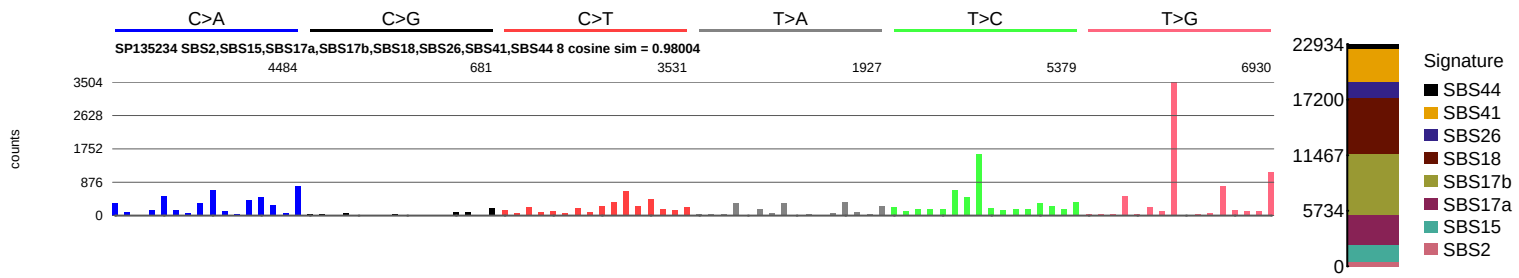

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.9801
5109
840
4329
1110
5069
6477
counts
3648
2736
1824
912
0
22934
17200
11467
5734
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

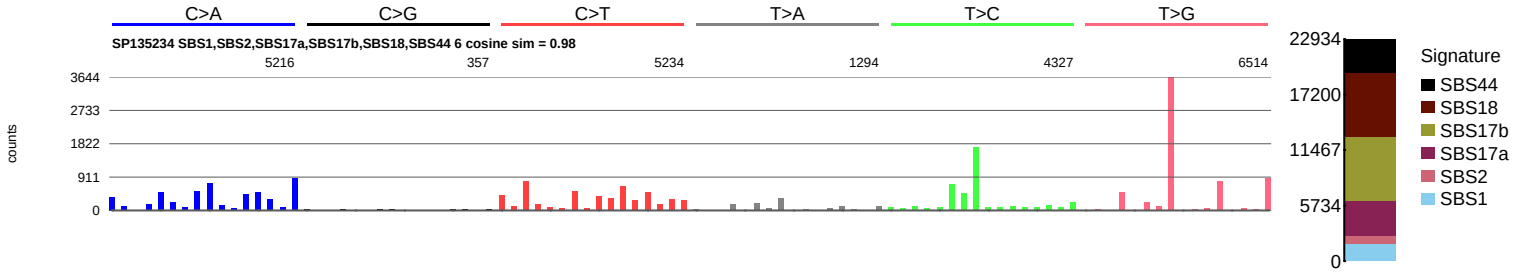

C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98009
4266
1133
3316
1958
5312
6951
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98008
4267
1124
3319
1961
5317
6951
counts
3472
2604
1736
868
0
22934
17200
11467
5734
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98008
4469
810
3437
1891
5395
6929
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98008
4487
681
3527
1930
5385
6924
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98004
4484
681
3531
1927
5379
6930
counts
3504
2628
1752
876
0
22934
17200
11467
5734
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135234 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98
5216
357
5234
1294
4327
6514
counts
3644
2733
1822
911
0
22934
17200
11467
5734
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS2
SBS1